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OF

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THE present volume contains 69 papers numbered 417 to 485, published for the most part in the years 1866 to 1872; they include a series of astronomical papers published in the *Memoirs* and *Monthly Notices* of the Royal Astronomical Society.

The Portrait in Volume VI. is from the Painting in Oil by Mr Lowes Dickinson in the year 1874, presented by the Subscribers to Trinity College, Cambridge, and now in the Hall of the College: the portrait in the present volume is a photograph of a pencil sketch by Mr Lowes Dickinson in the year 1893.

The Table for the seven volumes is

Vol. I.	Numbers 1 to 100.
Vol. II.	Numbers 101 to 158.
Vol. III.	Numbers 159 to 222.
Vol. IV.	Numbers 223 to 299.
Vol. V.	Numbers 300 to 383.
Vol. VI.	Numbers 384 to 416.
Vol. VII.	Numbers 417 to 485.

CONTENTS

Advertisement	2
417. On the Locus of the Foci of the Conics which pass through Four Given Points	4
418. A Remark on Differential Equations	7
419. A Theorem on Differential Operators	9
420. On Riccati's Equation	10
421. Note on the Solvibility of Equations by means of Radicals	13
422. On the Geodesic Lines on an Oblate Spheroid	15

[417]

On the Locus of the Foci of the Conics which pass through Four Given Points.

[From the Philosophical Magazine, vol. XXXII. (1866), pp. 362–365.]

THE curve which is the locus of the foci of the conics which pass through four given points is, as appears from a general theorem of M. Chasles, a sextic curve having a double point at each of the circular points at infinity; and Professor Sylvester, in his “Supplemental Note on the Analogues in Space to the Cartesian Ovals *in plano*” (Phil. Mag., May 1866), has further remarked that the lines (eight in all) joining the circular points at infinity with any one of the four points are all of them double tangents of the curve; whence each of these points is a focus (more accurately a quadruple focus) of the curve. It is to be added that, besides the circular points at infinity, the curve has 6 double points (3 of these are the centres of the quadrangles formed by the 4 points), in all 8 double points; the class is therefore = 14. Hence also the number of tangents to the curve from a circular point at infinity is = 10; viz. these are the 4 double tangents each reckoned twice, and 2 single tangents; and the theoretical number of foci is = 100; viz. we have

16 quadruple foci, or intersections of a double tangent by a double tangent	$16 \times 4 = 64$
16 double foci, or intersections of a double tangent by a single tangent	$16 \times 2 = 32$
4 single foci, or intersections of a single tangent by a single tangent	$\frac{4}{100}$

To verify the foregoing results, consider any two given points I, J , and the series of conics which pass through four given points A, B, C, D ; we have thus a curve the locus of the intersections of the tangents from I and the tangents from J to any conic of the series; which curve, if I, J are the circular points at infinity, is the required curve of foci. Taking $U + \lambda V = 0$ for the equation of a conic of the series, the pair of tangents from I is given by an equation of the form

$$(\lambda, 1)^2(x, y, z)^2 = 0,$$

and the pair of tangents from J by an equation of the like form

$$(\lambda, 1)^2(x, y, z)^2 = 0;$$

and by eliminating λ from these equations, we obtain the equation of the required curve. This in the first instance presents itself as an equation of the eighth order; but it is to be observed that in the series of conics there are two conics each of them touching the line IJ , and that, considering the tangents drawn to either of these conics, the line IJ presents itself as part of the locus; that is, the line IJ twice repeated is part of the locus; and the residual curve is thus of the order $8 - 2 = 6$; that is, the required curve is of the order 6. The consideration of the same two conics shows that each of the points I, J is a double point on the curve. Moreover, by taking for the conic any one of the line-pairs through the four points, it appears that each of the points $(AB.CD), (AC.BD), (AD.BC)$ is a double point on the curve: this establishes the existence of five double points. The two conics of the series which touch the line IA are a single conic taken twice, and the consideration of this conic shows that the line IA is a double tangent to the curve; similarly each of the eight lines $I(A, B, C, D)$ and $J(A, B, C, D)$ is a double tangent to the curve.

Instead of seeking to establish directly the existence of the remaining three double points, the easier course is to show that, besides the four double tangents from I , the number of tangents from I to the curve is $= 2$; for, this being so, the total number of tangents from I to the curve will be $(2 \times 4 + 2 =) 10$; that is, I being a double point, the class of the curve is $= 14$; and assuming that the depression $(6 \times 5 - 14 =) 16$ in the class of the curve is caused by double points, the number of double points will be $= 8$. But observing that in the series of conics there is one conic which passes through J , so that the tangents from J to this conic are the tangent at J twice repeated, then it is easy to see that the tangents from I to this conic, at the points where they meet the tangent at J , touch the required curve, and that these two tangents are in fact (besides the double tangents) the only tangents from I to the curve; that is, the number of tangents from I to the curve is $= 2$.

Considering I, J as the circular points at infinity, and writing A, B, C, D to denote the squared distances of a point P from the four points A, B, C, D respectively, then, as remarked by Professor Sylvester, the equation

$$\lambda\sqrt{A} + \mu\sqrt{B} + \nu\sqrt{C} + \pi\sqrt{D} = 0$$

(where λ, μ, ν, π are constants) is in general a curve of the order 8; but the ratios $\lambda : \mu : \nu : \pi$ may be so determined that the order of the curve in question shall be $= 6$; the resulting curve of the order 6 is (not one of a group of curves, but the very curve) the locus of the foci of the conics through the four points. And the determination of the ratios $\lambda : \mu : \nu : \pi$ is in fact quite simple; for writing

$$\begin{aligned} A &= (x - a)^2 + (y - \alpha)^2 \\ &= \rho^2 - 2(ax + \alpha y) + \&c., \quad (\text{if } \rho^2 = x^2 + y^2), \end{aligned}$$

and therefore

$$\sqrt{A} = \rho \sqrt{1 - \dots},$$

with similar values for $\sqrt{B}, \sqrt{C}, \sqrt{D}$, it is easy to see that, considering λ, μ, ν, π as standing for $\pm\sqrt{A}, \pm\sqrt{B}, \pm\sqrt{C}, \pm\sqrt{D}$ respectively, the conditions for the reduction to the order 6 are

$$\begin{aligned} \lambda + \mu + \nu + \pi &= 0, \\ \lambda a + \mu b + \nu c + \pi d &= 0, \\ \lambda \alpha_1 + \mu b_1 + \nu c_1 + \pi d_1 &= 0, \end{aligned}$$

and hence that the required equation of the curve of foci is

$$\Sigma \begin{vmatrix} 1, & 1, & 1 \\ b, & c, & d \\ b_1, & c_1, & d_1 \end{vmatrix} \sqrt{(x - a)^2 + (y - \alpha_1)^2} = 0,$$

or, as this may also be written,

$$\Sigma \pm (B, C, D) \sqrt{A} = 0,$$

where $(B, C, D), \&c.$ are the areas of the triangles $B, C, D, \&c.$

I remark, in conclusion, that the number of conditions to be satisfied in order that a curve may have for double points two given points I, J , may have besides six double points, and may have for double tangents eight given lines, is $(3 + 3 + 6 + 16 =) 28$; the number of constants contained in the general equation of the order 6 is $= 27$. The conditions that a curve of the order 6 shall have for double points two given points I, J , shall besides have six double points, and shall have for double tangents four given lines through I and four given lines through J ,

are more than sufficient for the determination of the sextic curve; and the existence of a sextic curve satisfying these conditions is therefore a theorem.

In the case where the points I, J lie on a conic of the series, the consideration of this conic shows that the curve has a ninth double point, the pole of the line IJ in regard to the conic in question: in this case the sextic curve, as is known, breaks up into two cubic curves. [It need not do so, for a proper sextic curve may have nine (or indeed ten) double points.]

P.S. In general the curve $\lambda\sqrt{A} + \mu\sqrt{B} + \nu\sqrt{C} + \pi\sqrt{D} = 0$ has (exclusively of multiple points at infinity) six double points; viz. these are situate at the intersections of the pairs of circles,

$$\begin{aligned}(\lambda\sqrt{A} + \mu\sqrt{B} = 0, \quad \nu\sqrt{C} + \pi\sqrt{D} = 0), \\(\lambda\sqrt{A} + \nu\sqrt{C} = 0, \quad \mu\sqrt{B} + \pi\sqrt{D} = 0), \\(\lambda\sqrt{A} + \pi\sqrt{D} = 0, \quad \mu\sqrt{B} + \nu\sqrt{C} = 0).\end{aligned}$$

In the case of the curve of foci, the first, second, and third pairs of circles intersect respectively in the points $(AB.CD)$, $(AC.BD)$, $(AD.BC)$, which, as mentioned above, are double points on the curve; and they besides intersect in three other points, which are the other three double points mentioned above.

Professor Sylvester reminds me that he mentioned to me in conversation that he had himself obtained the foregoing equation $\Sigma \pm (B, C, D)\sqrt{A} = 0$, for the locus of the foci of the conics which pass through the four points A, B, C, D .

Cambridge, October 10, 1866.

[418]

A Remark on Differential Equations.*[From the Philosophical Magazine, vol. XXXII. (1866), pp. 379–381.]*

CONSIDER a differential equation $f(x, y, p) = 0$, of the first order, but of the degree n , where f is a rational and integral function of (x, y, p) not rationally decomposable into factors: the integral equation contains an arbitrary constant c , and represents therefore a system of curves, for any one of which curves the differential equation is satisfied: the differential equation is assumed to be such that the curves are algebraical curves. The curves in question may be considered as undecomposable curves; in fact, if the curve $U^\alpha V^\beta W^\gamma \dots = 0$ (composed of the undecomposable curves $U = 0, V = 0, W = 0, \dots$) satisfies the differential equation, then either the curves $U = 0, V = 0, W = 0, \dots$ each satisfy the differential equation, and instead of the curve $U^\alpha V^\beta W^\gamma \dots = 0$ we have thus the undecomposable curves $U = 0, V = 0, W = 0, \dots$ each satisfying the differential equation; or if any of these curves, for instance $W = 0$, &c., do not satisfy the differential equation, then W^γ , &c. are mere extraneous factors which may and ought to be rejected, and instead of the original curve $U^\alpha V^\beta W^\gamma \dots = 0$, we have the undecomposable curves $U = 0, V = 0$ satisfying the differential equation. Assuming, as above, the existence of an algebraical solution, this may always be expressed in the form $\phi(x, y, c) = 0$, where ϕ is a rational and integral function of (x, y, c) , of the degree n as regards the arbitrary constant c : this appears by the consideration that for given values (x_0, y_0) of (x, y) the differential equation and the integral equation must each of them give the same number of values of p . It is to be observed that ϕ regarded as a function of (x, y, c) cannot be rationally decomposable into factors; for if the equation were $\phi = \Phi\Psi \dots = 0$, Φ, Ψ , &c. being each of them rational and integral functions of (x, y, c) , then the differential equation would be satisfied by at least one of the equations $\Phi = 0, \Psi = 0, \dots$ that is, by an equation of a degree less than n in the arbitrary constant c .

But the equation $\phi(x, y, c) = 0$ is not of necessity the equation of an undecomposable curve, and the undecomposable curve which constitutes the proper solution of the differential equation cannot always be represented by an equation of the form in question. For although ϕ regarded as a function of (x, y, c) is not rationally decomposable into factors, yet it may very well happen that ϕ regarded as a function of (x, y) is rationally decomposable into factors (geometrically the sections by the planes $z = c$ of the undecomposable surface $\phi(x, y, z) = 0$ may each of them be composed of two or more distinct curves); and assuming that the function ϕ is thus decomposed into its prime factors, then each factor equated to zero gives an undecomposable curve satisfying the differential equation, and constituting the proper solution thereof.

It may be observed that, by the foregoing process of decomposition, we sometimes reduce the original equation $\phi(x, y, c) = 0$ into a like equation $\phi_1(x, y, c_1) = 0$ of a more simple form. Thus, for instance, if we have $\phi(x, y, c) = U^2 - c = 0$, U being a rational and integral function of (x, y) , then instead of $U^2 - c = 0$ we have the equations $U + \sqrt{c} = 0, U - \sqrt{c} = 0$, each of which is an equation of the form $U - c_1 = 0$, or we pass from the original equation $\phi(x, y, c) = U^2 - c = 0$ to the simplified equation

$$\phi_1(x, y, c_1) = U - c_1 = 0.$$

Again, to take a somewhat more complicated instance, if the given integral equation be

$$\phi(x, y, c) = U^4 + c^2 V^4 + (c + 1)^2 W^4 - 2c U^2 V^2 - 2(c + 1) U^2 W^2 - 2c(c + 1) V^2 W^2 = 0,$$

then the equation $U^2 + V^2 \sqrt{c} + W^2 \sqrt{c + 1} = 0$, writing therein $\sqrt{c} = \frac{c_1^2 + 1}{c_1^2 - 1}$, and therefore

$\sqrt{c + 1} = \frac{2c_1}{c_1^2 - 1}$, becomes

$$U(c_1^2 - 1) + V \cdot 2c_1 + W(c_1^2 + 1) = 0;$$

so that we pass from the original equation $\phi(x, y, c)$ to the simplified equation

$$\phi_1(x, y, c_1) = U(c_1^2 - 1) + V \cdot 2c_1 + W(c_1^2 + 1) = 0.$$

But observe that the possibility of the rationalization depends on the form of the radicals $\sqrt{c}$ and $\sqrt{c+1}$; if we had had $\sqrt{c}$ and $\sqrt{c^2+1}$ (or c and $\sqrt{c^4+1}$), the rationalization could not have been effected.

Returning to the case of an integral equation $\phi(x, y, c) = 0$, where ϕ regarded as a function of (x, y) is decomposable into factors, then equating to zero any one of the prime factors of ϕ , we obtain an integral equation $\psi(x, y, c_1, c_2, \dots, c_k) = 0$, where $c_1, c_2, \dots, c_k$ are irrational functions (not of necessity representable by radicals, and without any superior limit to the number of these functions) of c : here ψ regarded as a function of (x, y) is of course undecomposable, and the equation $\psi(x, y, c_1, c_2, \dots, c_k) = 0$ belongs to the undecomposable curve which is the proper solution of the differential equation. The result may be stated under a quasi-geometrical form; viz. regarding $c_1, c_2, \dots, c_k$ as the coordinates of a point in k -dimensional space, then as these are functions of the single parameter c , the point to which they belong is an arbitrary point on a certain curve or $(k-1)$ fold locus C in the k -dimensional space. And this curve must be such that to given values of (x, y) there shall correspond n points on the curve; that is, treating (x, y) as constants, the surface or onefold locus $\psi(x, y, c_1, c_2, \dots, c_k) = 0$, and the curve or $(k-1)$ fold locus C , shall meet in n points. The conclusion stated in the foregoing quasi-geometrical form is, that the solution of the differential equation may be exhibited in the form $\psi(x, y, c_1, c_2, \dots, c_k) = 0$; viz. ψ is a rational and integral function of $(x, y, c_1, c_2, \dots, c_k)$, where $(c_1, c_2, \dots, c_k)$ are the coordinates of an arbitrary or variable point on a curve or $(k-1)$ fold locus C in a k -dimensional space, which curve meets the surface or onefold locus $\psi(x, y, c_1, c_2, \dots, c_k) = 0$ in n points, and where ψ regarded as a function of (x, y) is not rationally decomposable into factors.

Cambridge, October 13, 1866.

[419]

A Theorem on Differential Operators.

[From a paper by Prof. Sylvester, “Note on the Test Operators which occur in the Calculus of Invariants, &c.,” *Philosophical Magazine*, vol. XXXII. (1866), pp. 461–472, see p. 471.]

THE paper concludes with an Observation from Professor Cayley as follows:

“In the case of two variables, if

$$P_1 = (ax + by) \frac{d}{dx} + (cx + dy) \frac{d}{dy},$$

then in the notation of matrices,

$$P_1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} (x, y) \begin{pmatrix} \frac{d}{dx} & \frac{d}{dy} \end{pmatrix},$$

$$P_2 = \frac{1}{2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^2 (x, y) \begin{pmatrix} \frac{d}{dx} & \frac{d}{dy} \end{pmatrix},$$

$$P_3 = \frac{1}{6} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^3 (x, y) \begin{pmatrix} \frac{d}{dx} & \frac{d}{dy} \end{pmatrix};$$

whence also

$$P \cdot P_1 = P_1 \cdot P + P_2 = \frac{1}{2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^2 (x, y) \begin{pmatrix} \frac{d}{dx} & \frac{d}{dy} \end{pmatrix} = 3P_2,$$

which accords with your theorem,

$$E_1 * E_2 * = E_2 * E_1 * = E_3 E_0 * + 3E_2 *."$$

I have taken the liberty of writing in the above $\frac{d}{dx}$, $\frac{d}{dy}$ for δ_x , δ_y , and P for δ in the original. It will be useful to bear in mind that in any operator such as E_1* or E_2* , the asterisk forms an integral part of the symbol. Thus $E_1 * E_2*$, if we choose, may be written under the form of E_1* multiplied by E_2* , i.e. $(E_1*) \times (E_2*)$, where the cross is the sign of ordinary algebraical multiplication.

[420]

On Riccati's Equation.

[From the *Philosophical Magazine*, vol. XXXVI. (1868), pp. 348–351.]

THE following is, it appears to me, the proper form in which to present the solution of Riccati's equation.

The equation may be written

$$\frac{dy}{dx} + y^2 = x^{2q-2},$$

which is integrable by algebraic and exponential functions if $(2i+1)q = \pm 1$, i being zero, or a positive integer. To effect the integration, writing $y = \frac{1}{u} \frac{du}{dx}$, we have

$$\frac{d^2u}{dx^2} = x^{2q-2} u.$$

The peculiar advantage of this well-known transformation has not (so far as I am aware) been explicitly stated; it puts in evidence the form under which the sought-for function y contains the constant of integration. In fact if $u = P$, $u = Q$ be two particular solutions of the equation in u , then the general solution is $u = CP + DQ$; and denoting by P' , Q' the derived functions, the value of y is

$$y = \frac{CP' + DQ'}{CP + DQ},$$

showing the form under which the constant of integration $C \div D$ is contained in y .

To complete the solution, assume

$$u = ze^{\frac{1}{q}x^q};$$

we find

$$\frac{d^2z}{dx^2} + 2x^{q-1} \frac{dz}{dx} + (q-1)x^{q-2}z = 0;$$

considering first the particular integral of the form

$$z = A + Bx^q + Cx^{2q} + Dx^{3q} + \&c.,$$

we find that the equation will be satisfied if

$$\begin{aligned} (q-1)A + q(q-1)B &= 0, \\ (3q-1)B + 2q(2q-1)C &= 0, \\ (5q-1)C + 3q(3q-1)D &= 0, \\ (7q-1)D + 4q(4q-1)E &= 0, \\ &\&c. \end{aligned}$$

If $A = 1$, this is

$$\begin{aligned} A &= 1, \\ B &= -\frac{q-1}{q(q-1)}, \\ C &= +\frac{(q-1)(3q-1)}{q(q-1)2q(2q-1)}, \\ D &= -\frac{(q-1)(3q-1)(5q-1)}{q(q-1)2q(2q-1)3q(3q-1)}, \\ &\&c., \end{aligned}$$

where it is to be noticed that the series may be considered to stop so soon as there is in the numerator a factor = 0. For instance, if $5q - 1 = 0$, then if the particular integral had been assumed to be $z = A + Bx^q + Cx^{2q}$, the only conditions to be satisfied by the coefficients are the first and second equations giving the foregoing values of A, B, C . It is immaterial that the analytical expressions of F and the subsequent coefficients contain in the denominators the evanescent factor $5q - 1$; the coefficients after C do not ever come into consideration.

Thus if $(2i + 1)q = +1$, the series terminates, and we have for u the finite particular solution

$$u = P = \left(1 - \frac{q-1}{q(q-1)} x^q + \frac{(q-1)(3q-1)}{q(q-1)2q(2q-1)} x^{2q} - \&c.\right) e^{\frac{1}{q}x^q},$$

and it is easy to see that we may herein change the sign of x^q , thereby obtaining another finite particular solution,

$$u = Q = \left(1 + \frac{q-1}{q(q-1)} x^q + \frac{(q-1)(3q-1)}{q(q-1)2q(2q-1)} x^{2q} + \&c.\right) e^{-\frac{1}{q}x^q}.$$

Reverting to the equation in z , we have next a particular solution of the form

$$z = Ax + Bx^{q+1} + Cx^{2q+1} + Dx^{3q+1} + \&c.,$$

giving between the coefficients the relation

$$\begin{aligned} (q+1)A + (q+1)qB &= 0, \\ (3q+1)B + (2q+1)2qC &= 0, \\ (5q+1)C + (3q+1)3qD &= 0, \\ (7q+1)D + (4q+1)4qE &= 0, \\ &\&c. \end{aligned}$$

If $A = 1$, we have

$$\begin{aligned} A &= 1, \\ B &= -\frac{q+1}{(q+1)q}, \\ C &= +\frac{(q+1)(3q+1)}{(q+1)q(2q+1)2q}, \\ D &= -\frac{(q+1)(3q+1)(5q+1)}{(q+1)q(2q+1)2q(3q+1)3q}, \\ &\&c., \end{aligned}$$

where, as in the former case, the series is considered to terminate as soon as there is an evanescent factor in the numerator, without any regard to the subsequent coefficients which contain in the denominators the same evanescent factor. [In particular, $q = -1$, we have the solution $z = x$.]

Hence if we have $(2i + 1)q = -1$, the series terminates, and we have for u the finite particular solution,

$$u = P = x \left(1 - \frac{q+1}{(q+1)q} x^q + \frac{(q+1)(3q+1)}{(q+1)q(2q+1)2q} x^{2q} - \&c.\right) e^{\frac{1}{q}x^q},$$

from which, changing the sign of x^q , we deduce the other finite particular solution,

$$u = Q = x \left(1 + \frac{q+1}{(q+1)q} x^q + \frac{(q+1)(3q+1)}{(q+1)q(2q+1)2q} x^{2q} + \&c.\right) e^{-\frac{1}{q}x^q}.$$

Hence, in the equation

$$\frac{dy}{dx} + y^2 = x^{2q-2},$$

where $q(2i+1) = \pm 1$, we have (writing $D = 1$)

$$y = \frac{CP' + Q'}{CP + Q},$$

where C is the constant of integration, P, Q are finite series as above, and P', Q' are the derived functions of P and Q . Writing successively $i = 0, i = 1, i = 2$, &c., we may tabulate the solutions:

$$\begin{array}{llll} q + 2i = 1, & \frac{dy}{dx} + y^2 = 1, & P = e^x, & Q = e^{-x}, \\ & \frac{dy}{dx} + y^2 = x^{-4}, & P = xe^{-\frac{1}{x}}, & Q = xe^{\frac{1}{x}}, \\ & \frac{dy}{dx} + y^2 = x^{-\frac{8}{3}}, & P = (1 - 3x^{\frac{1}{3}})e^{3x^{\frac{1}{3}}}, & Q = (1 + 3x^{\frac{1}{3}})e^{-3x^{\frac{1}{3}}}, \\ & \frac{dy}{dx} + y^2 = x^{-\frac{4}{3}}, & P = (1 - 5x^{\frac{1}{5}} + \frac{25}{3}x^{\frac{2}{5}})e^{5x^{\frac{1}{5}}}, & Q = (1 + 5x^{\frac{1}{5}} + \frac{25}{3}x^{\frac{2}{5}})e^{-5x^{\frac{1}{5}}}, \\ & & & \text{\&c.} \end{array}$$

It is hardly necessary to make the final step of calculating P' and Q' and substituting in y ; but, as an example, take the above equation $\frac{dy}{dx} + y^2 = x^{-\frac{8}{3}}$: we have

$$y = \frac{-3x^{-\frac{1}{3}}(Ce^{3x^{\frac{1}{3}}} + e^{-3x^{\frac{1}{3}}})}{C(1 - 3x^{\frac{1}{3}})e^{3x^{\frac{1}{3}}} + (1 + 3x^{\frac{1}{3}})e^{-3x^{\frac{1}{3}}}},$$

which is readily identified with the solution, p. 98 of Boole's *Differential Equations* (Cambridge, 1859).

Cambridge, September 29, 1868.

[421]

Note on the Solvibility of Equations by means of Radicals.*[From the Philosophical Magazine, vol. XXXVI. (1868), pp. 386, 387.]*

IN regard to the theorem that the general quintic equation of the n th order is not solvable by radicals, I believe that the proofs which have been given depend, or at any rate that a proof may be given that shall depend, on the following two lemmas:

I. A one-valued (or symmetrical) function of n letters is a perfect k th power, only when the k th root is a one-valued function of the n letters.

There is an exception in the case $k = 2$, whatever be the value of n : viz. the product of the squares of the differences is a one-valued function, a perfect square; but its square root, or the product of the simple differences, is a two-valued function. It is in virtue of this exception that a quadric equation is solvable by radicals; we have the one-valued function $(x_1 - x_2)^2$, the square of a two-valued function $x_1 - x_2$, and thence the two roots are each expressible in the form

$$\frac{1}{2}\{x_1 + x_2 \pm \sqrt{(x_1 - x_2)^2}\}.$$

II. A two-valued function of n letters is a perfect k th power, only when the k th root is a two-valued function of the n letters.

There is an exception in the case $k = 3$, when $n = 3$ or 4 : viz. for $n = 3$ we have $(x_1 + \omega x_2 + \omega^2 x_3)^3$ (ω an imaginary cube root of unity) a two-valued function, and a perfect cube; whereas its cube root is the six-valued function $x_1 + \omega x_2 + \omega^2 x_3$. And similarly for $n = 4$ we have, for instance,

$$\{x_1 x_2 + x_3 x_4 + \omega(x_1 x_3 + x_2 x_4) + \omega^2(x_1 x_4 + x_2 x_3)\}^3$$

a two-valued function, and a perfect cube, whereas its cube root is a six-valued function. And it is in virtue of this exception that a cubic or a quartic equation is solvable by radicals. But I assume that for $n > 4$ the lemma is true without exception.

The course of demonstration would be something as follows: Imagine, if possible, the root of an equation expressed, by means of radicals, in terms of the coefficients; the expression cannot contain any radical such as $\sqrt[p]{X}$, $p > 2$, where X is a one-valued (or rational) function of the coefficients, not a perfect p th power, for the reason that, expressing the coefficients in terms of the roots, such function $\sqrt[p]{X}$ is not a rational function of the roots; if it were so, by lemma I. it would be a one-valued (that is, a symmetrical) function of the roots; consequently a rational function of the coefficients, or X expressed in terms of the coefficients, would be a perfect p th power.

The expression may however contain a radical $\sqrt{X}$, X a one-valued (or rational) function of the coefficients, not a perfect square: viz. X may be any square function multiplied into that function of the coefficients which is equal to the product of the squared differences of the roots, or, say, multiplied into the discriminant; that is, we may have $X = Q^2 \nabla$, or $\sqrt{X} = Q\sqrt{\nabla}$.

We have next to consider whether the expression can contain any radical $\sqrt[p]{X}$, where X , not being a rational function of the coefficients, is a function expressible by radicals. But the foregoing reasoning shows that if this be so, X cannot contain any radical other than the radical $\sqrt{Q^2 \nabla}$ or $Q\sqrt{\nabla}$, as above; that is, X must be $P + Q\sqrt{\nabla}$, where P and Q are rational functions of the coefficients, and where we may assume that $P + Q\sqrt{\nabla}$ is not a perfect p th power of a function of the like form $P' + Q'\sqrt{\nabla}$. But then, expressing the coefficients in terms of the roots, we have $P + Q\sqrt{\nabla}$, a (rational) two-valued function of the roots; and there is no radical $\sqrt[p]{P + Q\sqrt{\nabla}}$, which is a rational function of the roots; for by lemma II., if such radical existed we should have $\sqrt[p]{P + Q\sqrt{\nabla}}$ a (rational) two-valued function of the roots; that is, it would be

$= P' + Q'\sqrt{\nabla}$, P' and Q' one-valued (symmetrical) functions of the roots, consequently rational functions of the coefficients; or $P + Q\sqrt{\nabla}$ would be a perfect p th power $(P' + Q'\sqrt{\nabla})^p$.

The conclusion is that for $n > 4$ there is not (besides the function $P + Q\sqrt{\nabla}$) any function of the coefficients, expressible by means of radicals, which, when the coefficients are expressed in terms of the roots, will be a rational function of the roots, and consequently there is no possibility of expressing the roots in terms of the coefficients by means of radicals.

Cambridge, October 1, 1868.

[422]

On the Geodesic Lines on an Oblate Spheroid.

[From the *Philosophical Magazine*, vol. XL. (1870), pp. 329–340.]

THE theory of the geodesic lines on an oblate spheroid of any excentricity whatever was investigated by Legendre¹; and the general course of them is well known, viz. each geodesic line undulates between two parallels equidistant from the equator (being thus either a closed curve, or a curve of indefinite length, according to the distance between the two parallels): at a point of contact with the parallel the curve is, of course, at right angles to the meridian; say this is V , a vertex of the geodesic line, and let the meridian through V meet the equator in A ; the geodesic line proceeds from V to meet the equator in a point N , the node, where AN is at most $= 90^\circ$; and the undulations are obtained by the repetition of this portion VN of the geodesic line alternately on each side of the equator and of the meridian.

[Figure 1: A diagram of the oblate spheroid showing the vertex V , the meridian VA through V , the equator, and the node N at which the geodesic line meets the equator.]

I consider in the present paper the series of geodesic lines which cut at right angles a given meridian AC , or, say, a series of geodesic normals. It may be remarked that as V passes from the position A on the equator to the pole C , the angular distance AN increases from a certain determinate value (equal, as will appear, to $\frac{C}{A} \cdot 90^\circ$, if C , A are the polar and equatorial axes respectively) up to the value 90° ; and it thus appears that, attending only to their course after they first meet the equator, the geodesic normals have an envelope resembling in its general appearance the evolute of an ellipse (see fig. 1 and also fig. 2), the centre hereof being the point B at the distance $BA = 90^\circ$, and the axes coinciding in direction with the equator BA and meridian BC : this is in fact a real geodesic evolute of the meridian CA . The point α is, it is clear, the intersection of the equator by the geodesic line for which V is consecutive to the point A (so that $\angle BOA = \left(1 - \frac{C}{A}\right) 90^\circ$); and the point γ is the intersection of the meridian CB by the geodesic line for which V is consecutive to the point C ; and its position will be in this way presently determined. I was anxious, with a view to the construction of a drawing and a model, to obtain some numerical results in relation to a spheroid of considerable excentricity, and I selected that for which

$$\frac{C}{A} = \frac{1}{2} \quad (\text{polar axis} = \frac{1}{2} \text{ equatorial}).$$

[Figure 2: A diagram showing the geodesic evolute of the meridian, with the point B , the vertex region, the points α on the equator and γ on the meridian CB .]

Before proceeding further, I remark that Legendre's expression "reduced latitude" is used in what is not, I think, the ordinary sense; and I propose to substitute the term "parametric latitude": viz., in fig. 3, referring the point P on the ellipse by means of the ordinate MPQ to a point Q on the circle, radius $OK (= OA$, fig. 1), and drawing the normal PT , then we have for the point P the three latitudes,

$$\begin{aligned} \lambda &= \angle PTK, & \text{normal latitude,} \\ \lambda'' &= \angle POK, & \text{central latitude,} \\ \lambda' &= \angle QOK, & \text{parametric latitude;} \end{aligned}$$

¹ *Mém. de l'Inst.* 1806; see also the *Exer. de Calcul Intégral*, t. I. (1811), p. 178, and the *Traité des Fonctions Elliptiques*, t. I. (1825), p. 360.

viz. λ' is the parameter most convenient for the expression of the values of the coordinates x, y ($x = A \cos \lambda', y = C \sin \lambda'$) of a point P on the ellipse. The relations between the three latitudes are

$$\tan \lambda'' = \frac{C}{A} \tan \lambda' = \frac{C^2}{A^2} \tan \lambda,$$

so that $\lambda'', \lambda', \lambda$ are in the order of increasing magnitude. I use in like manner l, l', l'' in regard to the vertex V . The course of a geodesic line is determined by the equation

$$\cos \lambda' \sin \alpha = \text{const.},$$

where λ' is the reduced latitude of any point P on the geodesic line, and α is at this point the azimuth of the geodesic line, or its inclination to the meridian. Hence, if l' be the parametric latitude of the vertex V , the equation is

$$\cos \lambda' \sin \alpha = \cos l'$$

(whence also, when $\lambda' = 0, \alpha = 90^\circ - l'$; that is, the geodesic line cuts the equator at an angle $= 90^\circ - l'$, the parametric latitude of the vertex). The equation in question, $\cos \lambda' \sin \alpha = \cos l'$, leads at once to Legendre's other equations: viz. taking, as above, A, C for the equatorial and polar semiaxes respectively, and δ for the excentricity, $\delta = \sqrt{1 - \frac{C^2}{A^2}}$; and to determine the position of P on the meridian, using (instead of the parametric latitude λ') the angle ϕ determined by the equation

$$\cos \phi = \frac{\sin \lambda'}{\sin l'},$$

and writing, moreover, s to denote the geodesic distance VP , and Λ to denote the longitude of P measured from the meridian CA which passes through the vertex V , these are

$$ds = d\phi \sqrt{C^2 + A^2 \delta^2 \sin^2 l' \cos^2 \phi},$$

$$d\Lambda = \frac{\cos l'}{A} \cdot \frac{d\phi \sqrt{C^2 + A^2 \delta^2 \sin^2 l' \cos^2 \phi}}{1 - \sin^2 l' \cos^2 \phi},$$

which differential expressions are to be integrated from $\phi = 0$; and the equations then determine λ', s , and Λ , all in terms of the angle ϕ ,—that is, virtually s and Λ , the length and longitude, in terms of the parametric latitude λ' .

Writing, with Legendre,

$$c^2 = \frac{A^2 \delta^2 \sin^2 l'}{C^2 + A^2 \delta^2 \sin^2 l'}, \quad = \delta^2 \sin^2 l,$$

$$b^2 = 1 - c^2 = \frac{C^2}{C^2 + A^2 \delta^2 \sin^2 l'}, \quad = 1 - \delta^2 \sin^2 l;$$

also

$$n = \tan^2 l', \quad M = \frac{C}{bA \cos l'} = \frac{C}{A \cos l'},$$

then the formulæ become

$$ds = \frac{C}{b} d\phi \sqrt{1 - c^2 \sin^2 \phi},$$

$$d\Lambda = M \frac{d\phi \sqrt{1 - c^2 \sin^2 \phi}}{1 + n \sin^2 \phi}.$$

Hence integrating from $\phi = 0$, and using the notations F , E , Π of elliptic functions, we have

$$s = \frac{C}{b} E(c, \phi),$$

$$\Lambda = \frac{M}{n} \{(n + c^2) \Pi(n, c, \phi) - c^2 F(c, \phi)\};$$

viz. these belong to any point P whatever on the geodesic line, parametric latitude of vertex $= l'$; and if we write herein $\phi = 90^\circ$, then they will refer to the node N , or point of intersection with the equator.

The position of the point α is at once obtained by writing $l' = 0$: viz. this gives $c = 0$, $b = 1$, $M = \frac{C}{A}$, $n = 0$: the differential expressions are $ds = C d\phi$, $d\Lambda = \frac{C}{A} d\phi$. Or integrating from $\phi = 0$ to $\phi = \frac{1}{2}\pi$, we have $s = A \cdot \frac{C}{A} \cdot \frac{1}{2}\pi$, $\Lambda = \frac{C}{A} \cdot \frac{1}{2}\pi$, agreeing with each other, and giving longitude of $\alpha = \frac{C}{A} \cdot \frac{1}{2}\pi$; or, what is the same thing, $\angle \alpha OB = \frac{1}{2}\pi \left(1 - \frac{C}{A}\right)$.

Writing in the formulæ $l' = 90^\circ$, we have $c = \delta$, $b = \frac{C}{A}$, $\frac{M}{n} = 0$; whence $d\Lambda = 0$, or $\Lambda = \text{const.} = \frac{1}{2}\pi$, since the geodesic line here coincides with the meridian CB ; and moreover $s = A E(\delta, \phi)$; viz. this is merely the expression of the distance from C of a point P on the meridian CB . But we do not thus obtain the position of the point γ .

To find it we must consider a position of V consecutive to C , say, $l' = \frac{1}{2}\pi - \varepsilon$, where ε is indefinitely small; n is thus indefinitely large, and the integral $\Pi(n, c, \phi)$ is not conveniently dealt with. But it may be replaced by an expression depending on $\Pi\left(\frac{c^2}{n}, c, \phi\right)$, where $\frac{c^2}{n}$ is indefinitely small; viz. (Legendre, *Fonct. Ellipt.* vol. I. p. 69) we have

$$\Pi(n, c, \phi) = F(c, \phi) + \frac{1}{\sqrt{a}} \tan^{-1} \frac{\sqrt{a} \tan \phi}{\sqrt{1 - c^2 \sin^2 \phi}} - \Pi\left(\frac{c^2}{n}, c, \phi\right),$$

where

$$a = (1 + n) \left(1 + \frac{c^2}{n}\right).$$

We thus have

$$\Lambda = \frac{M}{n} \left\{ n F(c, \phi) + \frac{\sqrt{n} \sqrt{c^2 + n}}{\sqrt{1 + n}} \tan^{-1} \frac{\sqrt{a} \tan \phi}{\sqrt{1 - c^2 \sin^2 \phi}} - (c^2 + n) \Pi\left(\frac{c^2}{n}, c, \phi\right) \right\},$$

where, $\frac{c^2}{n}$ being small,

$$\begin{aligned} \Pi\left(\frac{c^2}{n}, c, \phi\right) &= \int \frac{d\phi}{\left(1 + \frac{c^2}{n} \sin^2 \phi\right) \sqrt{1 - c^2 \sin^2 \phi}} \\ &= \int \frac{\left(1 - \frac{c^2}{n} \sin^2 \phi\right) d\phi}{\sqrt{1 - c^2 \sin^2 \phi}} = \left(1 - \frac{1}{n}\right) F(c, \phi) + \frac{1}{n} E(c, \phi). \end{aligned}$$

And expanding also the $\tan^{-1}$ term, we thus have

$$\begin{aligned}\Lambda &= \frac{M}{n} \left\{ nF(c, \phi) + \frac{\sqrt{n}\sqrt{c^2+n}}{\sqrt{1+n}} \left[\frac{1}{2}\pi - \frac{\sqrt{1-c^2\sin^2\phi}}{\tan\phi} \cdot \frac{\sqrt{n}}{\sqrt{(1+n)(c^2+n)}} \right] \right. \\ &\quad \left. - (c^2+n) \left[\left(1 - \frac{1}{n}\right) F(c, \phi) + \frac{1}{n} E(c, \phi) \right] \right\} \\ &= \frac{M}{n} \left\{ \left(b^2 + \frac{c^2}{n}\right) F(c, \phi) - \left(1 + \frac{c^2}{n}\right) E(c, \phi) + \frac{\sqrt{n}\sqrt{c^2+n}}{\sqrt{1+n}} \cdot \frac{1}{2}\pi \right. \\ &\quad \left. - \frac{n}{n+1} \cot\phi \sqrt{1-c^2\sin^2\phi} \right\},\end{aligned}$$

which, in the term in $\{\}$ neglecting negative powers of n , becomes

$$\Lambda = \frac{M}{n} \left\{ \sqrt{n} \cdot \frac{1}{2}\pi + b^2 F(c, \phi) - E(c, \phi) - \cot\phi \sqrt{1-c^2\sin^2\phi} \right\}.$$

We may moreover write $c = \delta$, $b = \frac{C}{A}$, $\phi = 90^\circ - \lambda'$, $n = \frac{1}{\varepsilon^2}$, $M = \varepsilon$, and therefore $\frac{M}{n} = \varepsilon$, so that the formula is

$$\begin{aligned}\Lambda &= \varepsilon \left\{ \frac{1}{\varepsilon} \cdot \frac{1}{2}\pi + b^2 F(c, 90^\circ - \lambda') - E(c, 90^\circ - \lambda') - \tan\lambda' \sqrt{1-c^2\cos^2\lambda'} \right\} \\ &= \frac{1}{2}\pi - \varepsilon \left\{ \tan\lambda' \sqrt{1-c^2\cos^2\lambda'} + E(c, 90^\circ - \lambda') - b^2 F(c, 90^\circ - \lambda') \right\},\end{aligned}$$

where I retain c , b as standing for $\sqrt{1 - \frac{C^2}{A^2}}$, $\frac{C}{A}$ respectively.

Writing herein $\lambda' = 0$, we have

$$\Lambda = \frac{1}{2}\pi - \varepsilon (E, c - b^2 F, c),$$

where the coefficient $E, c - b^2 F, c$ is

$$= \int_0^{\frac{1}{2}\pi} d\theta \left(\sqrt{1-c^2\sin^2\theta} - \frac{1-c^2}{\sqrt{1-c^2\sin^2\theta}} \right) = c^2 \int_0^{\frac{1}{2}\pi} \frac{\cos^2\theta d\theta}{\sqrt{1-c^2\sin^2\theta}},$$

consequently positive; that is, Λ , the longitude of the node, is less than 90° , as it should be. Hence in order that Λ may be $= 90^\circ$, we must have λ' negative, say, $\lambda' = -\mu'$, where μ' is positive; and, observing that we may under the signs E, F write $90^\circ - \mu'$ instead of $90^\circ + \mu'$, we thus have

$$\frac{1}{2}\pi = \frac{1}{2}\pi + \varepsilon \left\{ \sqrt{1-c^2\cos^2\mu'} \tan\mu' - E(c, 90^\circ - \mu') + b^2 F(c, 90^\circ - \mu') \right\};$$

that is, we must have

$$\tan\mu' \sqrt{1-c^2\cos^2\mu'} = E(c, 90^\circ - \mu') - b^2 F(c, 90^\circ - \mu');$$

viz. μ' is here the parametric latitude (south) of the intersection of the meridian CB with the consecutive geodesic line—that is, of the point γ . As μ' increases from 0 to 90° , the left-hand side increases from 0 to ∞ ; and the right-hand side, beginning from a positive value and either attaining a maximum or not, ultimately decreases to 0; there is consequently a real root, which is easily found by trial.

Thus $\frac{C}{A} = \frac{1}{2}$, $C = \frac{1}{2}\sqrt{3}$ (angle of modulus $= 60^\circ$), $b = \frac{1}{2}$; or the equation is

$$\tan\mu' \sqrt{1 - \frac{3}{4}\cos^2\mu'} = E(90^\circ - \mu') - \frac{1}{4}F(90^\circ - \mu').$$

Using Legendre's Table IX., we have

μ'	$90^\circ - \mu'$	E	F	$E - \frac{1}{4}F$	$\tan \mu' \sqrt{1 - \frac{3}{4} \cos^2 \mu'}$
0°	90°	1.21105	2.15651	0.6719	0.0
10	80	1.12248	1.81252	0.6693	
20	70	1.02663	1.49441	0.6530	
30	60	0.91839	1.21253	0.6153	0.3819
40	50	0.79538	0.96465	0.5542	0.6278

so that we see the required value is between 30° and 40° ; and a rough interpolation gives the value $\mu' = 37^\circ 40'$. But repeating the calculation with the values 37° and 38° , we have

μ'	$90^\circ - \mu'$	E	F	$E - \frac{1}{4}F$	$\tan \mu' \sqrt{1 - \frac{3}{4} \cos^2 \mu'}$
37°	53°	0.833879	1.035870	0.57419	0.54425
38	52	0.821197	1.011849	0.56823	0.57108

whence, interpolating, $\mu' = 37^\circ 55'$.

The semiaxes of the geodesic evolute, measured according to their longitude and parametric latitude respectively, are thus $B\alpha$, long. of $\alpha = 45^\circ$; $B\gamma$, param. lat. = $37^\circ 55'$. But measuring them according to their geodesic distance, the equatorial radius A being taken = 1, we have

$$B\alpha = \frac{1}{4}\pi = 0.78540,$$

$$B\gamma = \left(\frac{C}{b} - 1\right) \{E, c - E(52^\circ 5')\} = 1.21106 - 0.82225 = 0.38881.$$

Reverting to the general formulæ for s , Λ , but writing therein $A = 1$, and therefore $C = \sqrt{1 - \delta^2}$; writing also $\phi = 90^\circ$ (that is, making the formulæ to refer to the node N of the geodesic line), we have

$$s = \frac{\sqrt{1 - \delta^2}}{\sqrt{1 - \delta^2 \sin^2 l}} E, c,$$

$$\Lambda = \frac{\sqrt{1 - \delta^2}}{n \cos l} \{(n + c^2) \Pi(n, c) - c^2 F, c\};$$

but for the calculation of the second of these formulæ by means of Legendre's Tables it is necessary to express $\Pi(n, c)$ in terms of the functions E, F .

The proper formula is given in *Fonct. Ellipt.* vol. i. p. 137; viz. this is

$$\frac{\Delta(b, \theta)}{\sin \theta \cos \theta} \Pi(n, c) = \frac{1}{2}\pi + \frac{\sin \theta}{\cos \theta} \Delta(b, \theta) F, c + F, c F(b, \theta) - F, c E(b, \theta) - E, c F(b, \theta),$$

where $\Delta(b, \theta) = \sqrt{1 - b^2 \sin^2 \theta}$. θ is an angle given by the equation $\cot \theta = \sqrt{n}$; we have $n = \tan^2 l'$; consequently $\theta = 90^\circ - l'$. Substituting this value, except that for shortness I retain $E(b, \theta), F(b, \theta)$ in place of $E(b, 90^\circ - l'), F(b, 90^\circ - l')$, we have

$$\Delta(b, \theta) = \sqrt{1 - b^2 \cos^2 l'}$$

$$= \sqrt{1 - (1 - \delta^2 \sin^2 l) \cos^2 l'} = \sin l,$$

and thence

$$\tan \theta \Delta(b, \theta) = \cot l' \sin l = \frac{\cos l}{\sqrt{1 - \delta^2}};$$

whence

$$\Pi(n, c) = \frac{\sin l' \cos l'}{\sin l} \left\{ \frac{1}{2}\pi + F, c \left[\frac{\cos l}{\sqrt{1 - \delta^2}} + F(b, \theta) - E(b, \theta) \right] - E, c F(b, \theta) \right\}.$$

But

$$n + c^2 = \tan^2 l' + \delta^2 \sin^2 l = \sin^2 l \sec^2 l'.$$

Hence

$$(n + c^2) \Pi(n, c) - c^2 F, c = \sin^2 l \{ \sec^2 l' \Pi(n, c) - \delta^2 F, c \},$$

and multiplying this by $\frac{\sqrt{1-\delta^2}}{n \cos l}$, $= \frac{\sqrt{1-\delta^2} \cos l}{\tan^2 l' \cos^2 l'}$, the exterior factor is

$$\frac{\sqrt{1-\delta^2} \cos l \tan^2 l'}{\tan^2 l' \cdot \sqrt{1-\delta^2}} = \frac{\cos l}{\sqrt{1-\delta^2}},$$

and we have

$$\Lambda = \frac{\cos l}{\sqrt{1-\delta^2}} \{ \sec^2 l' \Pi(n, c) - \delta^2 F, c \},$$

which is the formula I used in the calculations. It would, however, have been better to reduce a step further; viz. we have

$$\begin{aligned} \sec^2 l' \Pi(n, c) &= \frac{\sqrt{1-\delta^2}}{\cos l} \left\{ \frac{1}{2}\pi + F, c \left[\frac{\cos l}{\sqrt{1-\delta^2}} + F(b, \theta) - E(b, \theta) \right] - E, c F(b, \theta) \right\} \\ &= \frac{\sqrt{1-\delta^2}}{\cos l} \{ \frac{1}{2}\pi + F, c [F(b, \theta) - E(b, \theta)] - E, c F(b, \theta) \} + F, c, \end{aligned}$$

and thence

$$\sec^2 l' \Pi(n, c) - \delta^2 F, c = \frac{\sqrt{1-\delta^2}}{\cos l} \{ \frac{1}{2}\pi + F, c [F(b, \theta) - E(b, \theta)] - E, c F(b, \theta) \} + \sqrt{1-\delta^2} F, c;$$

or, finally,

$$\Lambda = \frac{1}{2}\pi + F, c F(b, \theta) - F, c E(b, \theta) - E, c F(b, \theta) + \sqrt{1-\delta^2} \cos l F, c.$$

It is easy with this expression of Λ to obtain the results already found for the extreme values $l' = 0^\circ$, $l' = 90^\circ$.

As Legendre's Tables have for argument, not the modulus c , but the angle of the modulus, say χ (that is, $\sin \chi = c = \delta \sin l$), it is convenient to replace $\sqrt{1-\delta^2} \sin^2 l$ by its value $\cos \chi$; and the formulæ thus are

$$\begin{aligned} s &= \frac{\sqrt{1-\delta^2}}{\cos \chi} E, c, \\ \Lambda &= \frac{1}{2}\pi + F, c \left[\sqrt{1-\delta^2} \cos l + F(b, \theta) - E(b, \theta) \right] - E, c F(b, \theta), \end{aligned}$$

where

$$c = \sin \chi = \delta \sin l, \quad \tan l' = \sqrt{1-\delta^2} \tan l, \quad \theta = 90^\circ - l';$$

and in the case intended to be numerically discussed, $\delta = \frac{1}{2}\sqrt{3}$, $\sqrt{1-\delta^2} = \frac{1}{2}$. I take l' as the argument, giving it the values $0^\circ, 10^\circ, \dots, 90^\circ$, and perform the calculation as shown in the Table.

[Table: The Table of numerical values of $l, \chi, 90^\circ - \chi, \frac{1}{2} \cos l, F, c, E, c, F(b, \theta), E(b, \theta)$, their combinations, Λ in radians and degrees, and the length s , for $l' = 0^\circ, 10^\circ, \dots, 90^\circ$, as displayed on page 23 of the original. Columns marked with an asterisk show respectively the longitude of the node and the length (or distance of node from vertex) for the geodesic lines belonging to the different values of the argument l' .]

The remarks which follow have reference to the stereographic projection of the figure on the plane of the equator, the centre of projection being the pole (say the South Pole) of the spheroid.

It is to be remarked that if a point P of the spheroid is projected as above, by means of an ordinate into the point Q of the sphere radius $OK (= OA)$, then projecting stereographically as to the spheroid and the sphere from the south poles thereof respectively, the points P and Q have the same projection. And it is hence easy to show that an azimuth α at a point of the meridian (parametric latitude λ' , normal latitude λ , and therefore $\tan \lambda' = \frac{C}{A} \tan \lambda$) is projected into an angle (α) such that

$$\tan(\alpha) = \frac{\sin \lambda'}{\sin \lambda} \tan \alpha.$$

In fact in fig. 3, if we take therein OK, OC for the axes of x, z respectively, and the axis of y at right angles to the plane of the paper, and if we have at P on the surface of the spheroid an element of length PR at the inclination α to the meridian PK , then if x, y, z are the coordinates of P , and $x + \delta x, y + \delta y, z + \delta z$ those of R , we have

$$\begin{aligned} \delta x &= \rho \cos \alpha \sin \lambda, \\ \delta z &= -\rho \cos \alpha \cos \lambda, \\ \delta y &= \rho \sin \alpha, \end{aligned}$$

and thence

$$\tan \alpha = \frac{\delta y}{\sqrt{\delta x^2 + \delta z^2}}.$$

Now, if the meridian and the points P, R are referred by lines parallel to Oz to the surface of the sphere radius OA , the only difference is that the ordinates z are increased in the ratio $C : A$; so that if the projected angle be (α) , we have

$$\tan(\alpha) = \frac{\delta y}{\sqrt{\delta x^2 + \frac{A^2}{C^2} \delta z^2}};$$

and then projecting the sphere stereographically from its south pole, the angle in the projection is $= (\alpha)$. And according to the foregoing remark, the angle (α) thus obtained is also the projection of α from the south pole of the spheroid. We have thus

$$\frac{\tan(\alpha)}{\tan \alpha} = \frac{\sqrt{\delta x^2 + \delta z^2}}{\sqrt{\delta x^2 + \frac{A^2}{C^2} \delta z^2}} = \frac{\sqrt{\sin^2 \lambda + \cos^2 \lambda}}{\sqrt{\sin^2 \lambda + \frac{A^2}{C^2} \cos^2 \lambda}} = \sqrt{\frac{1 + \cot^2 \lambda}{1 + \cot^2 \lambda'}} = \frac{\sin \lambda'}{\sin \lambda},$$

which is the required relation.

The foregoing equations,

$$\cos \lambda' \sin \alpha = \cos l', \quad \tan \lambda' = \frac{C}{A} \tan \lambda, \quad \tan(\alpha) = \frac{\sin \lambda'}{\sin \lambda} \tan \alpha,$$

determine in the stereographic projection the inclination (α) to the radius, or projection of the meridian, of the geodesic line (parametric latitude of vertex $= l'$) at the point the parametric latitude of which is $= \lambda'$; viz. they enable the construction (in the projection) of the direction of the successive elements of the geodesic line. There would be no difficulty in performing the construction geometrically; but it would, I think, be more convenient to calculate (α) numerically for a given value of l' and for the successive values of λ' . Observe that for $\lambda' = 0$ we have (as above) $90^\circ - \alpha = l'$, and then $\frac{\sin \lambda'}{\sin \lambda} = \frac{\tan \lambda'}{\tan \lambda} = \frac{C}{A}$, consequently $\tan(\alpha) = \frac{C}{A} \cot l'$: but we have also $\cot l' = \frac{A}{C} \cot l$, so that this equation becomes $\tan(\alpha) = \cot l$, or we have $90^\circ - (\alpha) = l$; viz. in the projection, the geodesic line cuts the equator at an angle $l =$ the normal latitude of the vertex of the geodesic line.

The preceding formulæ and results have enabled me to construct a drawing, on a large scale, of the stereographic projection of the geodesic lines for the spheroid, polar axis = $\frac{1}{2}$ equatorial axis.

[End of pages 1–25. Paper 422 continues to pages 26 ff. in subsequent files.]

[423]

On the Plane Representation of a Solid Figure.*[From the Philosophical Magazine, vol. XLI. (1871), pp. 286–290.]*

WE represent *in plano* the position of a point P whose coordinates in space are (x, y, z) by drawing these coordinates, on the same scale or on different scales, and in given directions from a fixed origin in the plane; $OM = x$, $MP' = y$, $P'P'' = z$. But observe that the point P'' *alone* does not completely represent the point P ; in fact P'' represents a whole series of points lying in a line; any one such point is the point whose coordinates are Om , mp' , $p'P''$. For the complete representation of P we require the *two points* P' , P'' : these might be distinguished as the projection P'' , and the foot-point P' . The two points P' , P'' are obviously such that the line joining them is in a given direction.

The preceding is, of course, the ordinary method of orthogonal projection, or geometrical delineation of a solid figure: it may be used under various forms; for example, the coordinates x, y, z may be taken on the same scale and in directions inclined to each other at angles of 120° (isometrical projection); or the coordinates x, y may be drawn on the same scale and at their actual inclination, 90° , to each other; and the coordinate z on the same or an altered scale in any given direction; the points P' then give a true ground-plan of the solid figure, and the lengths of the lines $P'P''$ give the altitudes of the several points P : this is also a method in ordinary use.

But it is to be observed that the points P' , P'' are both of them *projections*, and that the general theory is as follows: we represent the position of the point P by means of its projections P' , P'' , from two fixed points Ω' , Ω'' respectively; the line joining these points passes, it is clear, through a fixed point Ω which is the intersection of the plane of projection by the line which joins the two points Ω' , Ω'' .

Hence we say that a point P in space is represented *in plano* by any two points P' , P'' which are such that the line joining them passes through a fixed point Ω . And we have thus a *system of constructive geometry* which is the more simple on account of the generality of its basis, and which is at once applicable to any of the special projections above referred to. I establish the fundamental notions of such a geometry, and by way of illustration apply it to the solution of the well-known problem of finding the lines which meet four given lines in space.

A point P (as already mentioned) is given by its projections P' , P'' , which are points such that the line joining them passes through the fixed point Ω .

A line L is given by its projections L' , L'' , which are any two lines in the plane. We speak of the point (P', P'') , meaning the point P whose projections are P' , P'' ; and similarly of the line (L', L'') , meaning the line whose projections are L' , L'' .

If P' , P'' coincide, then the point P is in the plane of projection; and so if L' , L'' coincide, then the line L is in the plane of projection.

If through Ω we draw a line meeting L' , L'' in the points P' , P'' respectively, these are the projections of a point P on the line L . In particular the intersection of L' , L'' (considered as two coincident points) represents the intersection of the line L with the plane of projection.

The line through the points (P', P'') and (Q', Q'') has for its projections the lines $P'Q'$ and $P''Q''$.

Two lines (L', L'') and (M', M'') intersect each other if only the intersections $L'M'$ and $L''M''$ are the projections of a point P —that is, if the line through the points $L'M'$ and $L''M''$ passes through Ω . And then clearly P is the intersection of the two lines.

A plane Π is conveniently given by means of its trace Θ on the plane of projection, and of the projections (P', P'') of a point on the plane; or, say, by means of the trace Θ , and of a point P on the plane.

Suppose, however, that a plane is given by means of a line L and a point P on the plane. The trace Θ passes through the point of intersection of the line L with the plane of projection—that is, through the point of intersection of the projections L', L'' . To find another point on the trace, we have only to imagine on the line L a point Q , and, joining this with P , to suppose the line PQ produced to meet the plane of projection. The construction is obvious; but by way of illustration I give it in full. Through Ω draw a line meeting L', L'' in Q', Q'' respectively (then these are the projections of a point Q on the line L); the lines $P'Q'$ and $P''Q''$ are the projections of the line PQ , and the intersection of $P'Q'$ and $P''Q''$ is therefore the required point on the trace Θ .

The line of intersection of two planes passes through the point of intersection of their traces Θ_1, Θ_2 ; whence, if the planes have in common a point P , the line of intersection is the line joining P with the intersection of the traces Θ_1, Θ_2 .

In what precedes we have the solution of the following problem:—"Given a point P , and two lines L_1, L_2 , to find a line through P meeting the two lines L_1, L_2 ." The required line is in fact the line of intersection of the planes (P, L_1) and (P, L_2) ; we have seen how to construct the traces Θ_1 and Θ_2 of these planes respectively; and the required line is the line joining P with the intersection of Θ_1 and Θ_2 .

I proceed now to the problem to find the two lines, each of them meeting four given lines, L_1, L_2, L_3, L_4 (these being, of course, given by means of their projections (L'_1, L''_1) &c.). The question is in effect to find on the line L_1 a point P such that, drawing from it a line to meet L_2, L_3 , and also a line to meet L_3, L_4 , these shall be one and the same line.

Now, considering in the first instance P as an arbitrary point on the line L_1 , the line from P to meet L_2, L_3 is any line whatever meeting the lines L_1, L_2, L_3 : say it is a generating line of the hyperboloid whose directrices are L_1, L_2, L_3 , or of the hyperboloid $L_1L_2L_3$. Hence projecting from any point Ω' whatever, the generating lines and directrices are projected into tangents of one and the same conic. We know the projections L'_1, L'_2, L'_3 of the directrices; to find two other tangents of the conic, we take two arbitrary positions of P on the line L_1 , and construct as above the projections M', N' of the lines from these to meet the lines L_2, L_3 . The conic is then

given as the conic touching the five lines L'_1, L'_2, L'_3, M', N' : say this is the conic Σ' . Similarly, instead of Ω' , considering the point Ω'' , we have the lines L''_1, L''_2, L''_3 , and the lines M'', N'' , which are the other projections of the lines through the two positions of P ; and touching these five lines we have a conic Σ'' . Each tangent T' of Σ' , combined with the corresponding tangent T'' of Σ'' , represents a line T meeting L_1, L_2, L_3 ; to establish the correspondence, observe that, inasmuch as the line T meets L_1 , the intersections of T', L'_1 and of T'', L''_1 must lie in a line with Ω ; if T' be given, the point (T'', L''_1) is thus uniquely determined, and therefore also T'' (since L''_1 is a tangent of Σ''); and similarly if T'' be given, T' is uniquely determined; the correspondence T', T'' is thus, as it should be, a (1, 1) correspondence.

Considering in like manner the lines which meet L_1, L_3, L_4 , we have touching L'_1, L'_3, L'_4, M', N' a conic Σ'_1 ; and touching $L''_1, L''_3, L''_4, M'', N''$ a conic Σ''_1 ; each tangent T' of Σ'_1 , combined with the corresponding tangent T'' of Σ''_1 , represents a line meeting L_1, L_3, L_4 , the correspondence being a (1, 1) correspondence such as in the former case.

The conics Σ', Σ'_1 both touch L'_1, L'_3 ; hence they have in common two tangents. Say one of these is $T' = T'_1$; the corresponding tangents T'' and T''_1 will coincide with each other and be a common tangent of Σ'', Σ''_1 (these conics both touch L''_1, L''_3 , and have thus in common two tangents). We have thus $T' = T'_1$, and $T'' = T''_1$, as the projections of a line meeting L_1, L_2, L_3, L_4 ; and taking the other common tangents of Σ', Σ'_1 and of Σ'', Σ''_1 , we have the projections of the other line meeting L_1, L_2, L_3, L_4 .

The whole process is:—Construct M', M'' and N', N'' each of them the projections of a line through a point P of L_1 , which meets L_2, L_3 ; and M'_1, M''_1 and N'_1, N''_1 each of them the projections of a line through a point P of L_1 , which meets L_3, L_4 ; we have then the conics

$$\begin{aligned} \Sigma', \Sigma'' & \text{ touching } L'_1, L'_2, L'_3, M', N' \text{ and } L''_1, L''_2, L''_3, M'', N'' \text{ respectively,} \\ \Sigma'_1, \Sigma''_1 & \text{ touching } L'_1, L'_3, L'_4, M'_1, N'_1 \text{ and } L''_1, L''_3, L''_4, M''_1, N''_1 \text{ respectively;} \end{aligned}$$

and then the projections of each of the required lines are $T' = T'_1$, a common tangent of Σ', Σ'_1 , and $T'' = T''_1$, the corresponding common tangent of Σ'', Σ''_1 .

It is material to remark how the construction is simplified when there is given one of the lines, say M , which meets L_1, L_2, L_3, L_4 . Here if M is a common directrix of the two hyperboloids, we may for the hyperboloids Σ' and Σ'' consider, instead of L_1, L_2, L_3 and two new generating lines, the lines L_1, L_2, L_3, M , and a single new generating line N ; and similarly for the hyperboloids Σ'_1, Σ''_1 the lines L_1, L_3, L_4, M , and a single new generating line N_1 . Σ', Σ'_1 have thus in common the three tangents L'_1, L'_3, M' , and therefore only a single other common tangent, $T' = T'_1$; and similarly Σ'', Σ''_1 have in common the three tangents L''_1, L''_3, M'' , and therefore only a single other common tangent, $T'' = T''_1$; and we have thus the other line cutting the four given lines.

I take the opportunity of mentioning the following theorem:

“If in a given triangle we inscribe a variable triangle of given form, the envelope of each side of the variable triangle is a conic touching the two sides (of the given triangle) which contain the extremities of the variable side in question.”

We have thence a solution of the problem (*Principia*, Book I. Sect. V. Lemma XXVII.), in a given quadrilateral to inscribe a quadrangle of given form. The question in effect is: in the triangle ABC to inscribe a triangle $\alpha\beta\gamma$ of given form; and in the triangle ADE a triangle $\alpha'\beta'\gamma'$ of given form, in such wise that the sides $\alpha\gamma$, $\alpha'\gamma'$ may be coincident. The envelope of $\alpha\gamma$ is a conic touching AD , AE , and the envelope of $\alpha'\gamma'$ a conic also touching AD , AE : there are thus two other common tangents, either of which may be taken for the position of the side $\alpha\gamma = \alpha'\gamma'$; and the problem admits accordingly of two solutions.

[424]

On the Attraction of a Terminated Straight Line.

[From the *Philosophical Magazine*, vol. XLI. (1871), pp. 358–360.]

WRITE for shortness $(a, b, c; \varepsilon)$ to denote the shell included between the ellipsoids

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{and} \quad \frac{x^2}{a'^2} + \frac{y^2}{b'^2} + \frac{z^2}{c'^2} = (1 + \varepsilon)^2,$$

(where ε is indefinitely small); then, if the ellipsoids

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{and} \quad \frac{x^2}{a'^2} + \frac{y^2}{b'^2} + \frac{z^2}{c'^2} = 1$$

are confocal, the attractions of the shells $(a, b, c; \varepsilon)$ and $(a', b', c'; \varepsilon)$ upon any exterior point P are proportional to their masses. Hence, considering a prolate spheroid of revolution, $c = b$, the attractions of the shell $(a, b, b; \varepsilon)$ will be proportional to those of the shell $(\sqrt{a^2 - h}, \sqrt{b^2 - h}, \sqrt{b^2 - h}; \varepsilon)$; or if, as usual, $b^2 = a^2(1 - e^2)$, then, if h increases and becomes ultimately equal to b^2 , to those of the shell $(ae, 0, 0; \varepsilon)$; viz. this last is the portion of the axis of x included between the limits $x = -ae$, $x = +ae$; or say it is the terminated line $x = \pm ae$; and I say that the mass is distributed over this line uniformly.

To see that this is so, observe in general that, in the spheroid $\frac{x^2}{a^2} + \frac{y^2 + z^2}{b^2} = 1$, the volume included between the planes $x = \alpha$, $x = \alpha + d\alpha$, is $= \pi(y^2 + z^2) d\alpha = \pi b^2 \left(1 - \frac{\alpha^2}{a^2}\right) d\alpha$; and thence, writing $a(1 + \varepsilon)$, $b(1 + \varepsilon)$ for a , b , in the shell $(a, b, b; \varepsilon)$ the volume included between the planes $x = \alpha$, $x = \alpha + d\alpha$ is $= \pi b^2 \cdot 2\varepsilon d\alpha$; viz. this is independent of α , and simply proportional to $d\alpha$. Hence, writing $b = 0$, when the shell shrinks up into a line, the mass must be distributed uniformly over the line. It follows that for a line of uniform density the equipotential surfaces are each of them a prolate

spheroid of revolution having the extremities of the line for its foci, and that, if we have a shell bounded by any such surface and the consecutive similar surface, with its mass equal to that of the line, then such shell and the line will exert the same attractions upon any point P exterior to the shell. The attractions of the line are most easily obtained by means of its potential V ; viz. taking S, H for the extremities of the line, and, as above, the origin at the middle point, and the axis of x in the direction of the line, and writing $2ae$ for the length of the line, x, y, z for the coordinates of P , and r, s for the values of HP, SP (*that is*, $r = \sqrt{(x - ae)^2 + y^2 + z^2}$, $s = \sqrt{(x + ae)^2 + y^2 + z^2}$), *then the potential is at once found to be* $V = \log \frac{x+ae+s}{x-ae+r}$, and we can hereby verify that the equipotential surface is in fact a spheroid of revolution having the foci S, H ; for, taking the equation of such a spheroid to be

$$\frac{x^2}{\alpha^2} + \frac{y^2 + z^2}{\alpha^2 - a^2e^2} = 1$$

(α is an arbitrary parameter, since only the value of ae has been defined), we have

$$s = \alpha + ex, \quad r = \alpha - ex,$$

and thence

$$\begin{aligned} x + ae + s &= (1 + e)(x + \alpha), \\ x - ae + r &= (1 - e)(x + \alpha), \end{aligned}$$

and the quotient is $= \frac{1 + e}{1 - e}$, a constant value, as it should be. The equation $V = \text{const.}$ may in fact be written

$$\frac{1 + e}{1 - e} = \frac{x + ae + s}{x - ae + r},$$

viz. this equation, apparently of the fourth order, breaks up into the twofold plane $\frac{x^2}{a^2} + \frac{y^2 + z^2}{a^2(1 - e^2)} = 1$, and the spheroid $y^2 = 0$.

The foregoing results in regard to the attraction of a line are not new. See Green's *Essay on Electricity*, 1828, and *Collected Works*, Cambridge, 1871, p. 68; also

Joachimsthal, “On the Attraction of a Straight Line,” with Sir W. Thomson’s Note, *Camb. and Dubl. Math. Journ.*, vol. III. (1848), p. 93; but it does not appear to have been noticed that they are, in fact, included in the theory of the attraction of ellipsoids.

The like considerations show that the attractions of the ellipsoidal shell $(a, b, c; \varepsilon)$ upon an exterior point are equal to those of an elliptic disk $z = 0$, $\frac{x^2}{a^2 - c^2} + \frac{y^2}{b^2 - c^2} = 1$, the mass of which is equal to that of the shell, and which has the density at the point (x, y) proportional to $\left(1 - \frac{x^2}{a^2 - c^2} - \frac{y^2}{b^2 - c^2}\right)^{-\frac{1}{2}}$.

Sir W. Thomson informs me that the foregoing results have long been familiar to him.

[425]

Note on the Geodesic Lines on an Ellipsoid.*[From the Philosophical Magazine, vol. XLI. (1871), pp. 534, 535.]*

THE general configuration of the geodesic lines on an ellipsoid is established by means of the known theorem (an immediate consequence of Jacobi's fundamental formulæ, but which was first given by Mr Michael Roberts, *Comptes Rendus*, vol. XXI. p. 1470, Dec. 1845) that every geodesic line touches a curve of curvature; that is, attending to the two opposite ovals which constitute the curve of curvature, the geodesic line is in general an infinite curve undulating between these opposite ovals, and so touching each of them an infinite number of times (but possibly in particular cases it is a re-entrant curve touching each oval a finite number of times). The geodesic lines thus divide themselves into two kinds, according as they touch a curve of curvature of the one or the other kind; and there is besides a third limiting kind, the lines which pass through an umbilicus: any such geodesic line passes through the opposite umbilicus, and is in general an infinite curve passing an infinite number of times alternately through the two umbilici; but possibly it is in particular cases a re-entrant curve passing a finite number of times through the two umbilici. I annex a figure giving a general idea of the configuration of the geodesic lines drawn in different directions from a given point P on the surface of the ellipsoid: this is drawn (as it were) on the plane of the greatest and least axes; but it is not a perspective or geometrical representation of any kind, but a mere diagram for the purpose in question. We have A, A_1, B, C, C_1 the extremities of the axes; U_1, U_2, U_3, U_4 the umbilici; P the point on the surface; $1P2$ and $1P4$ the curves of curvature through P , viz. these are ovals containing the umbilici U_1, U_2 and U_1, U_4 respectively. Then U_1PU_3 and U_4PU_2 are the limiting geodesics passing through the umbilici; the line TPT' represents a geodesic line of the one kind, viz. this at T touches an oval (curve of curvature) U_1U_4 , and at T' the conjugate oval U_2U_3 . Similarly SPS' is a geodesic line of the other kind, viz. this at S touches an oval (curve of curvature) U_1U_2 , and at S' the conjugate oval U_3U_4 ; the dotted figure-of-eight curves are the loci of the points of contact T, T', S, S' .

[426]

On a supposed New Integration of Differential Equations of the Second Order.

[From the Philosophical Magazine, vol. XLII. (1871), pp. 197–199.]

THIS refers to a paper, Challis, “On the Application of a new Integration of Differential Equations of the Second Order to some unsolved Problems in the Calculus of Variations,” *Phil. Mag.* same volume, pp. 28–40.

[427]

On Gauss's Pentagramma Mirificum.*[From the Philosophical Magazine, vol. XLII. (1871), pp. 311, 312.]*

TAKE on a sphere (in the northern hemisphere) two points, A, B , whose longitudes differ by 90° , and refer them to the equator by the meridians AE and BG respectively; join A, B by an arc of great circle, and take in the southern hemisphere the pole D of this circle; and join D with E and C respectively by arcs of great circle. We have a spherical pentagon $ABCDE$, which is in fact the "Pentagramma mirificum," considered by Gauss, as appearing vol. III. pp. 481–490 of the *Collected Works*. Among its properties we have

$$\left. \begin{array}{l} \text{the distance of any two non-adjacent summits} \\ \text{the inclination of any two non-adjacent sides} \end{array} \right\} = 90^\circ;$$

so that each summit is the pole of the opposite side, or the pentagon is its own reciprocal.

Each angle is the supplement of the opposite side.

If the squared tangents of the sides (or angles) taken in order are $\alpha, \beta, \gamma, \delta, \varepsilon$, then

$$1 + \alpha = \gamma\delta, \quad 1 + \beta = \delta\varepsilon, \quad 1 + \gamma = \varepsilon\alpha, \quad 1 + \delta = \alpha\beta, \quad 1 + \varepsilon = \beta\gamma,$$

equivalent to three independent equations, so that any three of the quantities may be expressed in terms of the remaining two. (This agrees with the foregoing construction, where the arbitrary quantities are the latitudes of A, B respectively.)

Projecting from the centre of the sphere upon any plane, we have a plane pentagon which is such that the perpendiculars let fall from the summits upon the opposite sides respectively meet in a point. This (as easily seen) implies that the two portions into which each perpendicular is divided by the point in question have the same product.

Conversely, starting from the plane pentagon, and erecting from the point of intersection a perpendicular to the plane, the length of this perpendicular being equal to the square root of the product in question, we have the centre of a sphere such that the projection upon it of the plane polygon is the pentagramma mirificum.

I remark as to the analytical theory, that, taking the origin at the intersection of the perpendiculars, and for the coordinates of the summits $(\alpha_1, \beta_1), \dots, (\alpha_5, \beta_5)$ respectively, then we have

$$\alpha_1\alpha_3 + \beta_1\beta_3 = \alpha_2\alpha_4 + \beta_2\beta_4 = \alpha_3\alpha_5 + \beta_3\beta_5 = \alpha_4\alpha_1 + \beta_4\beta_1 = \alpha_5\alpha_2 + \beta_5\beta_2 = -\gamma^2,$$

where γ^2 is the above-mentioned product, or γ is the radius of the sphere.

Cambridge, September 14, 1871.

[428]

Note sur la Correspondance de Deux Points sur une Courbe.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, tom. LXII. (Janvier–Juin, 1866), pp. 586–590.]

THIS is to the same effect with the paper 385, “On the Correspondence of two Points on a Curve”; four examples of the theory are given, the first, second, and third of them the same as in this paper—the fourth example is as follows:

4°. *Recherche du nombre des points sextactiques*, c'est-à-dire des points qui sont tels que par chacun passe une conique qui a dans ce point un contact du cinquième ordre avec la courbe. Il faut prendre pour les points P les intersections avec la courbe de la conique qui a au point P' un contact du quatrième ordre: les points unis seront ceux dont il s'agit. La courbe $\theta = 0$ est la conique qui a au point P' un contact du quatrième ordre. On a ainsi parmi les intersections le point P' 5 fois; donc $k = 5$. À chaque point P' correspondent $2m - 5$ points P ; à chaque point P , $10m^2 - 20m - 5 - 20\delta$ points P' (j'emprunte le terme -20δ d'une formule que vient de donner M. Zeuthen); donc la formule donne pour le nombre des points unis

$$10m^2 - 18m - 10 - 20\delta + 10\delta,$$

c'est-à-dire

$$10m^2 - 18m - 10 - 20\delta.$$

Mais cette expression comprend le nombre $3m(m - 2) - 6\delta$ des inflexions; en effet pour un point d'inflexion la conique avec contact du quatrième ordre se réduit à la tangente prise deux fois, ce qui est une conique avec contact du cinquième ordre. Donc enfin le nombre des points sextactiques sera

$$m(12m - 27) - 24\delta,$$

ou, pour une courbe sans points doubles,

$$m(12m - 27),$$

ce qui s'accorde avec la valeur que j'ai trouvée par d'autres moyens.

[429]

Sur les Coniques déterminées par cinq Conditions de Contact avec une Courbe donnée.

*[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXIII.
(Juillet–Décembre, 1866), pp. 9–12.]*

THIS paper (dated Cambridge, 26 June 1866), contains the expressions for the numbers (5), (4, 1), (3, 2), (3, 1, 1), (2, 2, 1), (2, 1, 1, 1) and (1, 1, 1, 1, 1), of the conics which satisfy five conditions of contact with a given curve, as obtained in the paper 406 “On the Curves which satisfy given conditions,” see p. 214, and which expressions were found by the same process, viz. by consideration of functional equations obtained by supposing the given curve to break up into two curves of the orders m and m' respectively; there was in the expression for (1, 1, 1, 1, 1) a numerical error as mentioned in the footnote of the same page. The paper contains also the formula $\mu'' + \iota'' - \tau'' + \rho'' - \sigma'' = 0$, and the expression for $(2\nu, 3\rho)$ given, pp. 203, 204.

[430]

Note sur quelques Formules de M. E. de Jonquières, relatives aux Courbes qui satisfont à des Conditions données.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, tom. LXIII.
(Juillet–Décembre, 1866), pp. 666–670.]

LES formules dont il s'agit sont publiées dans les *Comptes Rendus*, séances du 3 et du 17 septembre 1866. En faisant une simple transformation algébrique pour y introduire la classe $M (= m^2 - m)$ de la courbe donnée U^m , et en changeant un peu la forme, les théorèmes de M. de Jonquières peuvent s'énoncer comme il suit:

1°. Le nombre des contacts des courbes C^r qui ont un contact de l'ordre n avec une courbe fixe U^m , et qui passent en outre par $\frac{1}{2}r(r+3) - n$ points donnés, est

$$= \frac{1}{2}(n+1)[nM + (2r - 2n)m].$$

Observation. Énoncé de cette manière, le théorème s'applique même au cas $n = 0$. En effet, pour $n = 0$, le nombre donné par le théorème est $= mr$, qui est le nombre des contacts de l'ordre 0 (intersections simples) de la courbe donnée U^m avec une courbe déterminée de l'ordre r .

2°. Le nombre des contacts de l'ordre n' ($=$ ou $< n$) des courbes C^r qui ont deux contacts des ordres n et n' respectivement avec une courbe fixe U^m , et qui passent en outre par $\frac{1}{2}r(r+3) - n - n'$ points donnés est

$$\begin{aligned} &= \frac{1}{4}(n+1)(n'+1) \left\{ [nM + (2r - 2n)m] [n'M + (2r - 2n')m] \right. \\ &\quad \left. - 2(n^2 + nn' + n'^2 + n + n') M \right. \\ &\quad \left. + [-4r(n + n' + 1) + 4(n^2 + nn' + n'^2 + n + n')] m \right\}. \end{aligned}$$

Observation. Énoncé de cette manière, le théorème s'applique même aux cas $n' = 0$, et $n' = n$. En effet, pour $n' = 0$, le nombre donné par le théorème est $= (rm - n - 1) \cdot \frac{1}{2}(n + 1)[nM + (2r - 2n)m]$, ce qui est égal au nombre des courbes qui ont avec la courbe donnée U^m un contact de l'ordre n , multiplié par $rm - n - 1$, nombre des contacts de l'ordre 0 (intersections simples) de chacune de ces courbes avec la courbe U^m . Et pour $n' = n$, le nombre des contacts est le double du nombre des courbes C^r .

Je remarque que les deux théorèmes peuvent se démontrer de la manière dont je me suis servi en cherchant le nombre des coniques qui satisfont à cinq conditions données; car, en remplaçant la courbe U^m par l'ensemble de deux courbes m et m' , on trouve que pour le théorème 1° le nombre cherché est

$$\frac{1}{2}(n + 1)[nM + (2r - 2n)m] + \alpha M + \beta m,$$

où les coefficients (α, β) ne dépendent que de (r, n) ; et puis, en supposant que ce théorème soit connu, on trouve que pour le théorème 2° le nombre cherché est

$$\frac{1}{4}(n + 1)(n' + 1)[nM + (2r - 2n)m][n'M + (2r - 2n')m] + \alpha M + \beta m,$$

où de même les coefficients (α, β) ne dépendent que de (r, n) .

Or voici comment on peut déterminer les coefficients dans les deux théorèmes:

Pour le théorème 1°, on démontre que pour U^m une droite, le nombre cherché est $= (n + 1)(r - n)$; et que pour U^m une conique, le nombre cherché se déduit de là en écrivant $2r$ au lieu de r ; c'est-à-dire, que pour la conique, le nombre est $= (n + 1)(2r - n)$. On a donc

$$\begin{aligned}\alpha + \beta &= (n + 1)(r - n), \\ 2\alpha + 2\beta &= (n + 1)(2r - n),\end{aligned}$$

et de là

$$\alpha = \frac{1}{2}(n + 1)n, \quad \beta = \frac{1}{2}(n + 1)(2r - 2n);$$

ce qui achève la démonstration.

Pour le théorème 2°, on démontre que pour U^m une droite, le nombre cherché est $= (n + 1)(n' + 1)(r - n - n')(r - n - n' - 1)$, et que pour U^m une conique, le nombre cherché se déduit de là en écrivant $2r$ au lieu de r ; c'est-à-dire, pour la conique, le nombre est

$$= (n + 1)(n' + 1)(2r - n - n')(2r - n - n' - 1).$$

On a donc

$$\begin{aligned}(n + 1)(n' + 1)(r - n - n')(r - n - n' - 1) &= (n + 1)(n' + 1)(r - n)(r - n') + \alpha + \beta, \\ (n + 1)(n' + 1)(2r - n - n')(2r - n - n' - 1) &= (n + 1)(n' + 1)(2r - n)(2r - n') + \alpha + 2\beta,\end{aligned}$$

cela donne pour α et β les valeurs

$$\begin{aligned}\alpha &= \frac{1}{4}(n + 1)(n' + 1)[-2(n^2 + nn' + n'^2 + n + n')], \\ \beta &= \frac{1}{4}(n + 1)(n' + 1)[-4r(n + n' + 1) + 4(n^2 + nn' + n'^2 + n + n')],\end{aligned}$$

et la démonstration est ainsi achevée.

Je remarque que sous les formes ici données les deux théorèmes s'appliquent à une courbe U^m avec des points doubles, mais sans point de rebroussement.

Le théorème dont je me suis servi pour la détermination des coefficients peut s'énoncer sous la forme plus générale que voici, savoir:

En dénotant par $\phi(r, n, n', \dots)$ le nombre des courbes C^r qui ont avec une droite donnée des contacts des ordres $n, n', \dots$ et qui passent en outre par $\frac{1}{2}r(r+3) - n - n' \dots$ points donnés, alors si, au lieu de la droite donnée, on a une conique donnée, le nombre des courbes C^r sera $= \phi(2r, n, n', \dots)$.

En effet, l'équation de la courbe cherchée C^r contient des coefficients indéterminés, lesquels, par les conditions de passer par les points donnés, se réduisent linéairement à $n + n' + \dots + 1$ coefficients; en dénotant par $(A, B, \dots)$ ces coefficients, l'équation de la courbe contiendra linéairement $(A, B, \dots)$ et sera ainsi de la forme $(A, B, \dots \tilde{a} x, y, z)^r = 0$. L'équation de la droite donnée est satisfaite en prenant pour (x, y, z) des fonctions linéaires déterminées d'un paramètre variable θ ; donc, en coupant la courbe C^r par la droite donnée, on obtient une équation $(A, B, \dots \tilde{a} \theta, 1)^r = 0$, et en exprimant que cette équation ait n racines égales, n' racines égales, etc., on obtient entre $(A, B, C, \dots)$ des équations, lesquelles, en éliminant tous les coefficients, exceptés deux quelconques (A, B) , conduisent à une équation finale $(A, B)^p = 0$, et le degré p de cette équation est ce qu'il s'agissait de trouver, le nombre des courbes C^r . Si au lieu d'une droite donnée on a une conique donnée, il n'y a rien à changer, sinon que les coordonnées (x, y, z) doivent être remplacées par des fonctions quadratiques de θ ; on a ainsi une équation $(A, B, \dots \tilde{a} \theta, 1)^{2r} = 0$, qui conduit à une équation finale $(A, B)^{p'} = 0$, où p' est la même fonction de $(2r, n, n', \dots)$ qu'est p de $(r, n, n', \dots)$; et le nombre des courbes C^r est $= p'$. Le théorème est donc démontré. Et, précisément de la même manière, on démontre le théorème encore plus général:

En dénotant par $\phi(r, n, n', \dots)$ le nombre des courbes C^r qui ont avec une droite donnée des contacts des ordres $n, n', \dots$, et qui passent en outre par $\frac{1}{2}r(r+3) - n - n' \dots$ points donnés, alors si, au lieu de la droite donnée, on a une courbe unicursale donnée de l'ordre m , le nombre des courbes C^r est $= \phi(mr, n, n', \dots)$.

On aurait pu se servir directement de cela pour démontrer les théorèmes 1° et 2°. Par exemple, pour le théorème 1°, la considération de la courbe unicursale U^m donne

$$\alpha M + \beta m = \alpha(2m - 2) + \beta m = (n + 1)(mr - n),$$

c'est-à-dire

$$\alpha = \frac{1}{2}(n + 1)n, \quad \beta = \frac{1}{2}(n + 1)(2r - 2n),$$

comme auparavant.

[431]

Sur la Transformation Cubique d'une Fonction Elliptique.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, tom. LXIV. (Janvier–Juin, 1867), pp. 560–563.]

SOIT $U = (a, b, c, d, e \text{ à } x, 1)^4$ une fonction quartique quelconque de x : I, J les deux invariants:

$$(I = ae - 4bd + 3c^2, \quad J = ace - ad^2 - b^2e + 2bcd - c^3),$$

et prenons $\Omega = \frac{I^3 - 27J^2}{I^3}$ pour l'invariant absolu de U . Soient de même $U' = (a', b', \dots \text{ à } x, 1)^4$ et $\Omega' = \frac{I'^3 - 27J'^2}{I'^3}$ l'invariant absolu de U' . En supposant que $\sqrt[4]{U}, \sqrt[4]{U'}$ soient les radicaux des deux fonctions elliptiques liées par la transformation du troisième ordre ou cubique, on peut se proposer la question quelle est la relation entre les deux invariants absolus Ω, Ω' ? J'ai trouvé cette relation d'abord par des considérations géométriques qui me furent suggérées par une lettre de M. Sylvestre; puis je l'ai déduite des formules pour la transformation cubique données par M. Hermite, *Crelle*, t. LX., 1862, p. 304), et enfin, à l'aide d'une considération tirée de ces formules, j'ai réussi à l'obtenir au moyen des formules des *Fundamenta Nova*. Je vais donner ici cette dernière investigation de la relation dont il s'agit.

En supposant que les fonctions U, U' soient transformées linéairement en $(1 - x^2)(1 - k^2x^2), (1 - y^2)(1 - \lambda^2y^2)$ respectivement, pour exprimer la liaison entre les modules k, λ , au lieu de l'équation explicite entre $\sqrt[4]{k}, \sqrt[4]{\lambda}$ (*Fund. Nova*, p. 23), je me sers des formules, p. 25, lesquelles en y écrivant

$$-\beta = \frac{\alpha + 2}{2\alpha + 1},$$

c'est-à-dire

$$2\alpha\beta + \alpha + \beta = -2,$$

deviennent

$$k^2 = -\alpha^2\beta, \quad \lambda^2 = \alpha\beta^2.$$

Les transformations linéaires donnent sans peine

$$\Omega = \frac{108k^2(k^2 - 1)^2}{(k^4 + 14k^2 + 1)^3}, \quad \Omega' = \frac{108\lambda^2(\lambda^2 - 1)^2}{(\lambda^4 + 14\lambda^2 + 1)^3},$$

et il s'agit entre ces équations d'éliminer $\alpha, \beta, k, \lambda$ de manière à obtenir une équation entre Ω, Ω' .

J'écris

$$\Omega = \frac{\frac{1}{4}(2\alpha + 1)(\alpha + 2)(\alpha - 1)^4}{(\alpha^2 + 4\alpha + 1)^3}, \quad \Omega' = \frac{\frac{1}{4}(2\beta + 1)(\beta + 2)(\beta - 1)^4}{(\beta^2 + 4\beta + 1)^3}.$$

L'équation entre α, β donne

$$2\beta + 1 = \frac{-3(\alpha + 1)}{2\alpha + 1}, \quad \beta + 2 = \frac{3(\alpha^2 + 4\alpha + 1)}{2\alpha + 1}, \quad \beta - 1 = \frac{-3(\alpha + 1)^2}{2\alpha + 1}, \quad \beta^2 + 4\beta + 1 = \frac{3(\alpha^2 + 4\alpha + 1)}{(2\alpha + 1)^2};$$

et on a de là

$$\beta' = \frac{\beta^2(\beta - 1)^2(\beta + 1)^2}{\beta^2(\beta + 2)^2}$$

puis, en faisant attention à l'identité

$$(2\alpha + 1)(\alpha + 2)(\alpha - 1)^4 + 27\alpha(\alpha + 1)^4 = 2(\alpha^2 + 4\alpha + 1)^3,$$

on obtient entre α', β' , la relation très simple $\alpha' + \beta' = 1$.

L'expression de k^2 donne

$$\begin{aligned} k^2 &= \frac{\alpha^2(\alpha + 2)}{2\alpha + 1}, \\ k^2 - 1 &= \frac{(\alpha - 1)(\alpha + 1)^2}{2\alpha + 1}, \\ k^4 + 14k^2 + 1 &= \frac{1}{(2\alpha + 1)^2} \{ \alpha^4(\alpha + 2)^2 + 14\alpha^2(\alpha + 2)(2\alpha + 1) + (2\alpha + 1)^2 \}, \\ &= \frac{1}{(2\alpha + 1)^2} \{ (\alpha^2 + 4\alpha + 1)(\alpha^4 + 3\alpha^3 + 16\alpha^2 + 3\alpha + 1) \}, \end{aligned}$$

et on a de là

$$\Omega = \frac{108\alpha^4(2\alpha + 1)(\alpha + 2)(\alpha - 1)^2(\alpha + 1)^4}{(\alpha^2 + 4\alpha + 1)^3 \cdot (\alpha^4 + 3\alpha^3 + 16\alpha^2 + 3\alpha + 1)^3}.$$

Or on a

$$\alpha' - 1 = \frac{\frac{1}{4}(2\alpha + 1)(\alpha + 2)(\alpha - 1)^4 - (\alpha^2 + 4\alpha + 1)^3}{(\alpha^2 + 4\alpha + 1)^3} = \frac{-\frac{27}{4}\alpha(\alpha + 1)^4}{(\alpha^2 + 4\alpha + 1)^3},$$

$$8\alpha' + 1 = \frac{4(2\alpha + 1)(\alpha + 2)(\alpha - 1)^4 + (\alpha^2 + 4\alpha + 1)^3}{(\alpha^2 + 4\alpha + 1)^3} = \frac{9(\alpha^4 + 3\alpha^3 + 16\alpha^2 + 3\alpha + 1)^2}{(\alpha^2 + 4\alpha + 1)^3},$$

et de là, en formant l'expression de la fonction $\frac{-64\alpha'(\alpha' - 1)^2}{(8\alpha' + 1)^3}$, on la trouve égale à la valeur qui vient d'être donnée pour Ω en termes de α : on a donc

$$\Omega = \frac{-64\alpha'(\alpha' - 1)^2}{(8\alpha' + 1)^3}$$

et de même

$$\Omega' = \frac{-64\beta'(\beta' - 1)^2}{(8\beta' + 1)^3}.$$

Avec la relation $\alpha' + \beta' = 1$, l'élimination de α' , β' entre ces équations ne présente pas de difficulté.

[432]

Théorème relatif à la Théorie des Substitutions.*Extrait d'une lettre adressée à M. J. A. Serret.**[From the Comptes Rendus de l'Académie des Sciences de Paris, tom. LXVII.
(Juillet-Décembre, 1868), pp. 784, 785.]*

ON peut énoncer par rapport aux substitutions un théorème qui comprend les trois théorèmes III., IV., V., t. II. pp. 260–263 de votre *Cours d'Algèbre Supérieure*.

Pour un nombre quelconque μ on peut former avec les $\phi(\mu)$ nombres inférieurs et premiers à μ plusieurs systèmes de nombres lesquels sont chacun un système conjugué (mod. μ); c'est-à-dire que le produit de deux nombres quelconques d'un tel système est congru suivant le module μ , à un nombre du système. Comme cas extrêmes, l'unité est un tel système, et les $\phi(\mu)$ nombres forment aussi un système conjugué.

Pour μ premier, en dénotant par a une racine primitive de μ et par f un diviseur quelconque de $\mu-1$, les nombres $\equiv a^{fx}$ (mod. μ), x étant un entier quelconque, forment un système conjugué. Et généralement pour un nombre μ quelconque ce nombre a des racines quasi-primitives $\alpha, \beta, \gamma, \dots$, aux exposants $A, B, C, \dots$, tels que $\alpha^A \equiv 1$ (mod. μ), $\beta^B \equiv 1$ (mod. μ), $\dots$ et $ABC\dots = \phi(\mu)$. En choisissant une combinaison quelconque, par exemple α, β de ces racines, soient f, g des diviseurs quelconques de A, B respectivement, les nombres $\equiv \alpha^{fx}\beta^{gy}$ (mod. μ) forment un système conjugué, l'ordre du système ou nombre des termes étant $= \frac{AB}{fg}$.

Cela étant, on a ce théorème:

Une substitution T quelconque de l'ordre μ étant formée avec n lettres, l'on forme toutes les substitutions S telles que le produit STS^{-1} se réduise à une puissance de T dont l'exposant soit un nombre quelconque appartenant à un système conjugué (mod. μ), les substitutions S constitueront un système conjugué de l'ordre θM , où θ dénote l'ordre du système conjugué (mod. μ) et M le nombre des substitutions échangeables avec T .

La démonstration est tout à fait la même que celle que vous donnez p. 62 pour votre théorème IV, en y ajoutant seulement que les nombres i, j qui appartiennent au système conjugué (mod. μ) auront leur produit ij congru à un nombre de ce même système conjugué.

[433]

Sur les Surfaces Tétraédrales.

[Notes to the work De la Gournerie, “Recherches sur les surfaces réglées tétraédrales symétriques,” 8vo. Paris, 1867.]

PREMIER MÉMOIRE. NOTES PP. 190–193.

Équations de certaines

1°. L'équation

$$\begin{aligned}
0 = & a^4 f^4 v^4 + b^4 g^4 u^4 + c^4 h^4 w^4 \\
& + 2c^2 ag (af - bg) y^2 z^2 - 2c^2 bf (af - bg) x^2 z^2 \\
& - 2b^2 ah (ch - af) z^2 x^2 + 2b^2 ch (ch - af) y^2 z^2 \\
& + 2a^2 gh (bg - ch) w^2 x^2 + 2a^2 bh (bg - ch) z^2 y^2 \\
& + 2g^2 hf (ch - af) w^2 y^2 \\
& + a^2 (b^2 y^2 + c^2 h^2 - 4bg ch) y^2 z^2 + b^2 (c^2 h^2 + a^2 f^2 - 4ch af) z^2 x^2 \\
& + g^2 (c^2 h^2 + a^2 f^2 - 4ch af) w^2 y^2 \\
& + f^2 (b^2 y^2 + c^2 h^2 - 4bg ch) w^2 x^2 \\
& + 2bf (afbg + c^2 h^2 - 2ch\chi) x^2 z^2 w^2 \\
& - 2ag (afbg + c^2 h^2 - 2ch\chi) y^2 w^2 z^2 \\
& + 2ah (chaf + b^2 y^2 - 2bg\chi) z^2 hu^2 - 2bh (bgch + a^2 f^2 - 2af\chi) z^2 \chi w^2 \\
& - 2gh (bcgh + af^2 - 2af\chi) w^2 y^2 z^2 - 2hf (cahf + by^2 - 2bg\chi) w^2 z^2 u^2 \\
& + 2\Omega h f^2 z^2 w^2.
\end{aligned}$$

SURFACES DU HUITIÈME ORDRE.

$$\begin{aligned}
& + 2b^2 cf (ch - af) x^2 z^2 - 2bc f^2 (bg - ch) x^2 w^2 \\
& - 2a^2 cg (bg - ch) y^2 z^2 - 2cag^2 (ch - af) y^2 w^2 \\
& - 2abh^2 (af - bg) z^2 w^2 + 2h^2 fg (af - bg) w^2 z^2 \\
& - c^2 (ch - af) z^2 x^2 + c^2 (a^2 f^2 + b^2 g^2 - 4afbg) x^4 y^4 \\
& - 4 (ch - af) w^2 y^2 + h^2 (a^2 f^2 + b^2 g^2 - 4afbg) w^2 z^2 \\
& - 2cf (chaf + b^2 f^2 - 2bg\chi) x^2 w^2 y^2 + 2bc (bcgh + a^2 f^2 - 2af\chi) x^2 y^2 z^2 \\
& + 2cg (bgch + a^2 f^2 - 2af\chi) y^2 x^2 w^2 + 2ca (cahf + by^2 - 2bg\chi) y^2 z^2 x^2 \\
& + 2ab (abfg + c^2 h^2 - 2ch\chi) z^2 x^2 y^2 \\
& - 2 (abfg + c^2 h^2 - 2ch\chi) w^2 x^2 y^2.
\end{aligned}$$

(où, pour abréger, on a écrit $\chi = af + bg + ch$, et Ω est une quantité quelconque) est celle d'une surface du huitième ordre qui a pour courbes doubles les quatre coniques

$$\begin{aligned} x = 0, & & +cy^2 - bz^2 + fw^2 &= 0; \\ y = 0, & & -cx^2 &+ az^2 + gw^2 = 0; \\ z = 0, & & +bx^2 - ay^2 &+ hw^2 = 0; \\ w = 0, & & -fx^2 - gy^2 - hz^2 &= 0. \end{aligned}$$

En déterminant Ω , savoir, en écrivant $\lambda + \mu + \nu = 0$, $af\lambda^2 + bg\mu^2 + ch\nu^2 = 0$ (ce qui donne deux systèmes de valeurs de $\lambda : \mu : \nu$), et puis

$$\Omega = 4 \sum \frac{\mu^2 \nu^2}{\lambda^2} (af + bg)^2 ch - 2 \sum \frac{\mu^2 \nu^2}{\lambda^2} (af + bg)^2 - 4 (af - bg) (bg - ch) (ch - af),$$

la surface devient une surface réglée, savoir, la quadrispinale de M. de la Gournerie; et, en particulier, en supposant $\frac{1}{af} + \frac{1}{bg} + \frac{1}{ch} = 0$, on obtient

$$\Omega = (-2 - 4) = -6 (af - bg) (bg - ch) (ch - af),$$

et la surface sera développable.

2°. L'équation

$$\begin{aligned} 0 = & a^2 y^2 z^4 + b^2 z^2 x^4 + c^2 x^2 y^4 + f^2 z^4 w^4 + g^2 x^4 w^4 + h^2 y^4 w^4 \\ & + 2bf x^2 z^2 w^2 - 2cf z x^2 y^2 w^2 + 2bc z x^2 y^2 z^2 \\ & - 2ag y^2 z^2 w^2 + \dots + 2cg x^2 y^2 w^2 + 2ca x^2 y^2 z^2 \\ & + 2ah z^2 y^2 w^2 - 2bh z^2 x^2 w^2 + \dots + 2ab x^2 y^2 z^2 \\ & - 2gh w^2 y^2 z^2 - 2hf w^2 z^2 x^2 - 2fg w^2 x^2 y^2 \\ & + 2\Omega x^2 y^2 z^2 w^2 \end{aligned}$$

(où Ω est une quantité quelconque) est celle d'une surface du huitième ordre ayant pour courbes doubles les quatre courbes du quatrième ordre

$$\begin{aligned} x = 0, & & -hz^2 w^2 + gw^2 y^2 - ay^2 z^2 &= 0; \\ y = 0, & & -hz^2 w^2 + fw^2 x^2 + bz^2 x^2 &= 0; \\ z = 0, & & +gw^2 x^2 - fw^2 y^2 + cx^2 y^2 &= 0; \\ w = 0, & & -ay^2 z^2 - bz^2 x^2 - cx^2 y^2 &= 0. \end{aligned}$$

En écrivant

$$\lambda + \mu + \nu = 0, \quad af\lambda^2 + bg\mu^2 + ch\nu^2 = 0,$$

...

(ce qui donne quatre systèmes de valeurs de $\lambda : \mu : \nu$), et puis

$$\Omega = af \frac{\nu - \mu}{\lambda} + bg \frac{\lambda - \nu}{\mu} + ch \frac{\mu - \lambda}{\nu} :$$

la surface devient une surface réglée, savoir, la *quadricuspidale* de M. de la Gournerie; et, en supposant

$$(af)^{\frac{1}{3}} + (bg)^{\frac{1}{3}} + (ch)^{\frac{1}{3}} = 0,$$

et

$$\lambda : \mu : \nu = (af)^{\frac{1}{3}} : (bg)^{\frac{1}{3}} : (ch)^{\frac{1}{3}},$$

ce qui donne

$$\Omega = [(af)^{\frac{1}{3}} - (bg)^{\frac{1}{3}}] [(bg)^{\frac{1}{3}} - (ch)^{\frac{1}{3}}] [(ch)^{\frac{1}{3}} - (af)^{\frac{1}{3}}] :$$

la surface devient développable.

Cambridge, 18 Octobre 1866.

DEUXIÈME MÉMOIRE. NOTES PP. 279–283.

NOTE I. SUR LA DÉCOMPOSITION DU LIEU DES GÉNÉRATRICES EN SURFACES TÉTRAÉDRALES DISTINCTES.

Il me semble qu'une de vos conclusions a besoin d'être modifiée. Ainsi la surface tétraédrale dérivée de deux courbes triangulaires à exposant $\frac{1}{m}$ (m étant un entier positif), laquelle, selon un de vos théorèmes, serait de l'ordre $2m^2$, paraît se décomposer en m surfaces chacune de l'ordre $2m$. Il y a pour cela une raison à *priori*; en effet,

[Figure : two planes “Plan $z = 0$ ” and “Plan $w = 0$ ”; the first carries the triangular curve through O with points $A, B, \dots$, the second through O' with points $A', B', \dots$; correspondence is established by lines OP and $O'P'$.]

pour deux triangulaires de la forme en question, en employant votre construction, on peut établir une correspondance non-seulement entre les deux systèmes de points $A, B, \dots$, et $A', B', \dots$, mais aussi entre chaque point A et un seul point correspondant A' , car l'équation de la première courbe étant de la forme

$$fx^{\frac{1}{m}} + gy^{\frac{1}{m}} + hz^{\frac{1}{m}} = 0,$$

on satisfait à cette équation en écrivant

$$x : y : z = a(\theta + \alpha)^m : b(\theta + \beta)^m : c(\theta + \gamma)^m,$$

où θ est un paramètre variable. De même, l'équation de la seconde courbe étant

$$f'x^{\frac{1}{m}} + g'y^{\frac{1}{m}} + k'w^{\frac{1}{m}} = 0,$$

on satisfait à cette équation en écrivant

$$x : y : w = a'(\theta' + \alpha')^m : b'(\theta' + \beta')^m : d'(\theta' + \delta')^m,$$

où θ' est aussi un paramètre variable.

Pour la droite OP , on a

$$\frac{x}{y} = \frac{a(\theta + \alpha)^m}{b(\theta + \beta)^m},$$

et pour la droite $O'P'$

$$\frac{x}{y} = \frac{a'(\theta' + \alpha')^m}{b'(\theta' + \beta')^m};$$

donc, la condition pour la correspondance des droites est

$$\frac{a(\theta + \alpha)^m}{b(\theta + \beta)^m} = \frac{a'(\theta' + \alpha')^m}{b'(\theta' + \beta')^m},$$

ce qui donne m valeurs différentes pour θ' en termes de θ . Mais chacune de ces valeurs est de la forme

$$\theta' = \frac{A\theta + B}{C\theta + D},$$

et, en ne faisant attention qu'à une seule valeur de θ' , on a le point

$$x : y : z = a(\theta + \alpha)^m : b(\theta + \beta)^m : c(\theta + \gamma)^m,$$

qui correspond à un point unique

$$x : y : w = a'(\theta' + \alpha')^m : b'(\theta' + \beta')^m : d'(\theta' + \delta')^m.$$

Pour le cas de l'exposant $\frac{1}{3}$, on a, de cette manière, une surface de l'ordre 6. J'ai vérifié cela dans le cas particulier de la surface développable. Il est très-singulier (c'est M. Salmon qui m'a fait cette remarque) qu'en écrivant dans cette équation (x^2, y^2, z^2, w^2) au lieu de (x, y, z, w) , on obtient l'équation d'une surface du douzième ordre, lieu des centres de courbure d'un ellipsoïde.

Cambridge, 15 Février 1866.

[433] p. 53

NOTE II. À L'OCCASION DE L'ORDRE DES SURFACES TÉTRAÉDRALES.

Je crois que j'ai négligé de vous faire connaître un théorème assez général au sujet de l'ordre de ces surfaces. En considérant dans l'espace deux courbes (planes ou à double courbure) des ordres m et m' respectivement, et en supposant qu'il y ait entre les points de ces deux courbes une correspondance (a, a') , (c'est-à-dire qu'à un point donné de la courbe m correspondent a' points sur la courbe m' , et à un point donné de la courbe m' correspondent a points sur la courbe m), alors la surface réglée que l'on obtient en unissant par des droites les points correspondants des courbes m et m' sera de l'ordre $ma' + m'a$.

Cambridge, 18 Octobre 1866.

NOTE III. SUR LA SURFACE COMPLÉMENTAIRE.

Je puis reconnaître, par mes propres formules, que, des pq^2 surfaces de l'ordre $2p^2q$, il n'y en a que pq qui passent par la troisième directrice. En effet, le rapport anharmonique k est donné en termes des paramètres de la troisième directrice, au moyen d'une équation qui contient la

quantité irrationnelle $k^{\frac{p}{q}}$. En rationalisant cette équation, on obtient pour k une équation de l'ordre pq ; à chaque racine k_1 correspondent q surfaces, savoir celles qui appartiennent aux q valeurs de $k_1^{\frac{p}{q}}$; mais l'équation irrationnelle n'est satisfaite que par une seule valeur de $k_1^{\frac{p}{q}}$, à savoir la valeur de $k_1^{\frac{p}{q}}$ donnée par l'équation irrationnelle, en y substituant pour k , en tant que k y entre rationnellement, la valeur $k = k_1$. Donc, à chaque racine k_1 correspond une seule surface qui passe par la troisième directrice. La question à laquelle donne lieu cette circonstance paraît très-intéressante. La surface déterminée par les trois directrices est composée de pq surfaces chacune de l'ordre $2p^2q$, et d'une surface résiduelle de l'ordre $2p^2(q^2 - q)$. Quelles sont la nature et les propriétés de cette surface résiduelle ? Je serais bien aise de savoir si vous avez fait des recherches à ce sujet.

Cambridge, 29 Mars 1866.

On certain Skew Surfaces, otherwise Scrolls.

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THE investigations contained in the present Memoir were suggested to me by the Memoirs of M. De la Gournerie, presented by him to the Academy of Sciences in 1865 and 1866, published in extract in the *Comptes Rendus*, and reproduced in his work “Recherches sur les surfaces réglées tétraédrales symétriques, par Jules De la Gournerie, avec des notes par Arthur Cayley,” 8vo. Paris, 1867. Although the results, or the greater part of them, agree with those in the work just referred to, the mode of treatment is different, and more general, the orders, &c. of the different scrolls being obtained by considerations founded on the theory of Correspondence; and I have thought it not improper to submit to geometers in this altered form the theory of the very interesting class of Scrolls for which they are indebted to M. De la Gournerie’s researches.

Article Nos. 1 to 10. Geometrical Construction of a Class of Scrolls.

1. Consider any two curves (plane or of double curvature) U, U' , of the orders m, m' respectively, and let the points of U have with those of U' an (α, α') correspondence; viz. let the points of the two curves be so related that to each point of U correspond α' points of U' , and to each point of U' correspond α points of U ; then the lines joining the corresponding points of U, U' form a scroll the order of which is $= m\alpha' + m'\alpha$.

2. In particular let U, U' be plane curves in the planes Π, Π' respectively; and let the correspondence between the points of the two curves be established as follows; viz. consider in the plane Π a curve Ω of the class μ , and in the plane Π' a curve Ω' of the class μ' ; and (to avoid useless generality) let the tangents of these two curves Ω, Ω' have to each other a $(1, 1)$ correspondence; that is, to each tangent of Ω there corresponds a single tangent of Ω' , and to each tangent of Ω' a single tangent of Ω (this assumes that the curves Ω, Ω' are rational transformations one of the other, and that they have consequently the same Deficiency). This being so, let the points of U which lie on any tangent of Ω and the points of U' which lie on the corresponding tangent of Ω' be taken to be corresponding points of U, U' . The correspondence is then $(\mu m', \mu' m)$: in fact, through a given point of U there pass μ tangents of Ω , and the corresponding μ tangents of Ω' meet U' in $\mu m'$ points, that is, to a given point of U correspond $\mu m'$ points of U' ; and similarly to a given point of U' correspond $\mu' m$ points of U . And hence the order of the scroll formed by the lines joining the corresponding points of U, U' is $= (\mu + \mu') mm'$.

3. This conclusion may be otherwise established as follows; let K, K' be any two corresponding points of U, U' , so that the scroll we are concerned with is that generated by the series of lines KK' ; and let I denote the line of intersection of the planes Π, Π' . The line I meets the curve U in m points, and taking one of these points for a point K we may from this point draw μ tangents to the curve Ω , that is, the point in question is a point K in respect of μ different tangents of the curve Ω ; to each of these tangents there corresponds a single tangent of Ω' , and such tangent of Ω' meets the curve U' in m' points, that is, to the point K in question there correspond $\mu m'$ points K' and consequently $\mu m'$ lines KK' in the plane Π' ; hence to each of the m points K on the line I there correspond $\mu m'$ lines KK' in the plane Π' , and we have thus $\mu mm'$ generating lines in the plane Π' ; there are in like manner $\mu' mm'$ generating lines in the plane Π .

Take K an arbitrary point on the curve U ; there are $\mu m'$ corresponding points K' , and consequently $\mu m'$ generating lines through K , that is, through each point of the curve U ; or the curve U (which is of the order m) is a $\mu m'$ -tuple line on the scroll; similarly the curve U' (which is of the order m') is a $\mu' m$ -tuple line on the scroll.

The complete section of the scroll by the plane Π' consists of the curve U taken $\mu m'$ times (order $\mu m m'$) and of the $\mu' m m'$ generating lines in the plane Π' ; that is, the order of the section is $= (\mu + \mu') m m'$; and we thus see that the order of the scroll is $= (\mu + \mu') m m'$. Of course in like manner the complete section of the scroll by the plane Π' consists of the curve U' taken $\mu' m$ times (order $\mu' m m'$) and of the $\mu m m'$ generating lines in the plane Π' , the order of the section being thus $= (\mu + \mu') m m'$.

4. There are on the scroll certain singular tangent planes; viz. if we have two corresponding tangents of Ω, Ω' meeting the line I in the same point, then we have m points K and m' points K' all lying in the plane of the two tangents; and of course the $m m'$ lines KK' will all lie in the plane of the two tangents, that is, the intersection of the scroll by the plane in question will be made up of the $m m'$ lines, and of a curve of the order $(\mu + \mu' - 1) m m'$; and the plane in question is thus a singular tangent plane.

5. The number of these singular tangent planes is $= \mu + \mu'$; in fact, considering as corresponding points on the line I , the intersection of this line by any tangent of Ω and the intersection by the corresponding tangent of Ω' , the correspondence is obviously (μ, μ') ; viz. through a given point P considered as belonging to the first system there pass μ tangents of Ω , and corresponding thereto we have μ tangents of Ω' each intersecting I in a single point P' ; that is, to a given point P correspond μ points P' ; and similarly to a given point P' correspond μ' points P . And this being so, the number of united points, that is, points of I through which pass corresponding tangents of Ω, Ω' , is $= \mu + \mu'$.

6. In particular the curves Ω, Ω' may reduce themselves each to a point: the tangents to the two curves are here the lines passing through the points Ω, Ω' respectively: and the condition for the $(1, 1)$ correspondence of the two tangents is that the pencils of lines shall be homographically related; or, what is the same thing, that these two pencils shall determine on the line I two ranges which are homographically related; the entire construction is then as follows:

Given in the plane Π a curve U and a point Ω , and in the plane Π' a curve U' and a point Ω' ; and taking in the plane Π a pencil of lines through Ω , and in the plane Π' a pencil of lines through Ω' , in such wise that the two pencils correspond homographically; then if a line of the first pencil meets the curve U in the m points K , and the corresponding line of the second pencil meets the curve U' in the m' points K' , the scroll in question is that generated by the $m m'$ lines KK' .

7. By what precedes, the scroll is of the order $2 m m'$; the curve U is a m' -tuple line, and the complete section by the plane Π is made up of this curve taken m' times and of $m m'$ generating lines; similarly the curve U' is a m -tuple line, and the complete section by the plane Π' is made up of this curve taken m times and of $m m'$ generating lines; there are two singular tangent planes such that the section by each of them is made up of $m m'$ generating lines and of a curve of the order $m m'$; the planes in question are obviously those through the line $\Omega \Omega'$ and the coincident points of the two ranges on the line I , say the points A, B respectively.

8. The foregoing results will be modified in special cases. Suppose, for instance, that the curve U passes ω times, α times, and β times through the points Ω, A, B respectively, and that the curve U' passes ω' times, α' times, and β' times through the points Ω', A, B respectively. Then to each point on the curve U there correspond the $m' - \omega'$ intersections (other than the point Ω') on a line through Ω' , so that U' is a $(m' - \omega')$ -tuple line on the surface. The curve U' meets the line I in m' points; and corresponding to each of them we have a line through Ω meeting the curve U in $(m - \omega)$ points, exclusive of the point Ω ; this would give $m'(m - \omega)$ generating lines in the plane Π ; but among the m' points are included the point A α' times, and the point B β' times; the $(m - \omega)$ points corresponding to A include the point A α times, and we have thus the point A corresponding to itself $\alpha \alpha'$ times, and giving a reduction $= \alpha \alpha'$ in the number $m'(m - \omega)$ of generating lines; similarly the $m - \omega$ points corresponding to B

include the point B β times, and we have thus the point B corresponding to itself $\beta\beta'$ times and giving a reduction $= \beta\beta'$ in the number $m'(m - \omega)$ of generating lines; the number of generating lines in the plane Π is thus $= m'(m - \omega) - \alpha\alpha' - \beta\beta'$. The complete section by the plane Π is made up of the curve U $(m - \omega')$ times (order $m(m' - \omega')$) and of the $m'(m - \omega) - \alpha\alpha' - \beta\beta'$ generating lines; the order of the section, and consequently also the order of the scroll, is thus $= 2mm' - m\omega' - m'\omega - \alpha\alpha' - \beta\beta'$. It is clear that in like manner the curve U' is a $(m - \omega)$ -tuple line on the surface, and that the complete section by the plane Π' is made up of this curve taken $(m - \omega)$ times, order $m'(m - \omega)$, and of $m(m' - \omega') - \alpha\alpha' - \beta\beta'$ generating lines.

9. The section by the tangent plane through A is made up of $(m - \omega)(m' - \omega') - \alpha\alpha'$ generating lines (viz. these are, the line $\Omega'A$ $\alpha'(m - \omega - \alpha)$ times, the line ΩA $\alpha(m' - \omega' - \alpha')$ times, and $(m - \omega - \alpha)(m' - \omega' - \alpha')$ other generating lines) and of a curve of the order $mm' - \omega\omega' - \beta\beta'$; similarly the section by the tangent plane through B is made up of $(m - \omega)(m' - \omega') - \beta\beta'$ generating lines (viz. these are, the line ΩB $\beta'(m - \omega - \beta)$ times, the line $\Omega'B$ $\beta(m' - \omega' - \beta')$ times, and $(m - \omega - \beta)(m' - \omega' - \beta')$ other generating lines), and of a curve of the order $mm' - \omega\omega' - \alpha\alpha'$.

10. A very interesting case is when $(m, m'$ being each even) we have

$$\omega = \alpha = \beta = \frac{1}{2}m, \quad \omega' = \alpha' = \beta' = \frac{1}{2}m'.$$

Here the curve U is a $\frac{1}{2}m'$ -tuple line on the scroll, and the complete section by the plane Π is this curve taken $\frac{1}{2}m'$ times; the order of the section, and therefore of the scroll, is thus $= \frac{1}{2}mm'$: of course in like manner the curve U' is a $\frac{1}{2}m$ -tuple line on the scroll, and the complete section by this plane is the curve U' taken $\frac{1}{2}m$ times; the section by each of the planes $\Omega\Omega'A, \Omega\Omega'B$ is a curve of the order $\frac{1}{2}mm'$, the planes in question being in the present case no longer singular tangent planes, or even tangent planes at all, of the scroll.

Article Nos. 11 to 14. Analytical Theory.

11. It will be convenient to denote by D, C respectively the points heretofore called Ω, Ω' respectively: this being so, we have a tetrahedron $ABCD$, of which the faces ABD, ABC are the planes heretofore called Π, Π' respectively, and the other two faces CDA, CDB are the singular tangent planes $\Omega\Omega'A, \Omega\Omega'B$ respectively. And then, taking

$$x = 0, \quad y = 0, \quad z = 0, \quad w = 0$$

for the equations of the faces BCD, CDA, DAB, ABC of the tetrahedron, we may write for the equations of the curve $U, z = 0, f_1(x, y, w) = 0$, for those of the curve $U', w = 0, f_2(x, y, z) = 0$; and take the homographic ranges on the line I ($z = 0, w = 0$) to be given as the intersections of this line with the pencils of planes $x - \theta y = 0, x - k\theta y = 0$ respectively (θ a variable parameter, k a constant). The points K are therefore given by

$$x - \theta y = 0, \quad z = 0, \quad f_1(x, y, w) = 0,$$

the points K' by

$$x - k\theta y = 0, \quad w = 0, \quad f_2(x, y, z) = 0;$$

and then the lines KK' belonging to the different values of the parameter θ generate the scroll.

12. Or, what is the same thing, taking (X, Y, Z, W) as the coordinates of $K, (X', Y', Z', W')$ as the coordinates of K' , we have

$$\begin{aligned} X - \theta Y &= 0, & Z &= 0, & f_1(X, Y, W) &= 0, \\ X' - k\theta Y' &= 0, & W' &= 0, & f_2(X', Y', Z') &= 0, \end{aligned}$$

and then the equations of the line KK' are

$$\begin{vmatrix} x & y & z & w \\ X & Y & 0 & W \\ X' & Y' & Z' & 0 \end{vmatrix} = 0;$$

or, as these may be written,

$$\begin{aligned} -WZ'y + W'Y'z + YZ'w &= 0, \\ WZ'x + W'X'z - XZ'w &= 0, \\ -W'Yx + WX'y + (XY' - X'Y)w &= 0, \\ -YZ'x + XZ'y - (XY' - X'Y)z &= 0, \end{aligned}$$

equivalent of course to two equations. The elimination of $X, Y, W, X', Y', Z', \theta$ from all the equations gives the equation of the scroll.

13. Substituting the values $X = \theta Y, X' = k\theta Y'$, we have

$$f_1(\theta Y, Y, W) = 0, \quad f_2(k\theta Y', Y', Z') = 0,$$

and

$$\begin{aligned} -WZ'y + WY'z + YZ'w &= 0, \\ WZ'x - k\theta WY'z - \theta YZ'w &= 0, \\ -WY'x + k\theta WY'y + \theta(1-k)YY'w &= 0, \\ -YZ'x + \theta YZ'y - \theta(1-k)YY'z &= 0; \end{aligned}$$

or, what is the same thing, writing $\frac{W}{Y} = \omega$ and $\frac{Z'}{Y'} = \zeta$, we have

$$f_1(\theta, 1, \omega) = 0, \quad f_2(k\theta, 1, \zeta) = 0,$$

$$\begin{aligned} \omega\zeta x - \omega\zeta y + \zeta\omega &= 0, \\ \omega\zeta x - k\theta\omega\zeta z - \theta\zeta\omega &= 0, \\ -\omega x + k\theta\omega y + \theta(1-k)\omega &= 0, \\ -\zeta x + \theta\zeta y - \theta(1-k)z &= 0. \end{aligned}$$

Recollecting that the last four equations are equivalent to two equations only, and substituting for ω, ζ their values in terms of θ , we have in effect two equations, which by the elimination of θ lead to a relation in (x, y, z, w) , the equation of the scroll.

14. We may find the sections of the scroll by the planes $x = 0, y = 0$ respectively. Writing first $x = 0$, we have

$$\omega y = \frac{k-1}{k} w, \quad \zeta y = -(k-1)z.$$

Hence taking the other two equations in the form

$$f_1\left(\theta y, y, \frac{k-1}{k} w\right) = 0, \quad f_2\left(u, \frac{y}{k}, \frac{-(k-1)z}{k}\right) = 0,$$

and putting $\theta y = u$, we have

$$f_1\left(u, y, \frac{k-1}{k} w\right) = 0, \quad f_2\left(u, \frac{y}{k}, \frac{-(k-1)z}{k}\right) = 0,$$

from which eliminating u we obtain an equation $f_3(y, z, w) = 0$, the equation of the section by the plane $x = 0$.

Similarly, writing $y = 0$, we have

$$\frac{\omega x}{\theta} = -(k-1)w, \quad \frac{\zeta x}{\theta} = (k-1)z,$$

whence taking the equations in the form

$$f_1\left(x, \frac{x}{\theta}, \frac{-(k-1)w}{\theta}\right) = 0, \quad f_2\left(kx, v, \frac{(k-1)z}{\theta}\right) = 0,$$

and writing $\frac{x}{\theta} = v$, we have

$$f_1(x, v, -(k-1)w) = 0, \quad f_2(kx, v, (k-1)z) = 0,$$

from which, eliminating v , we obtain an equation $f_4(x, z, w) = 0$, the equation of the section by the plane $y = 0$.

Article Nos. 15 to 29. The Curves U, U' are henceforward “triangular” curves.

15. Let $r = \pm \frac{p}{q}$, where p, q are positive integers prime to each other, and let the given sections be

$$\begin{aligned} z = 0, \quad Ax^r + By^r + Dw^r &= 0, \\ w = 0, \quad A'x^r + B'y^r + C'z^r &= 0, \end{aligned}$$

where it is to be observed that r being $= +\frac{p}{q}$, the two given sections are of the order pq , the order of the scroll is $= 2p^2q^2$; each of the given sections is a pq -tuple line on the scroll, and the plane thereof meets the scroll in the section taken pq times, and in the pq generating lines: but r being $= -\frac{p}{q}$, the two given sections are each of the order $2pq$, with three pq -tuple points ($\omega = \alpha = \beta = pq, \omega' = \alpha' = \beta' = pq$), and thence the order of the scroll is $\frac{1}{2}(2pq)^2 = 2p^2q^2$; each of the sections is a pq -tuple line on the scroll, and the plane meets the scroll only in the section taken pq times. But in either case, if q be > 1 , that is, if r be fractional, it will presently appear that the scroll of the order $2p^2q^2$ breaks up into q scrolls each of the order $2p^2q$.

16. To find the section by the plane $x = 0$, we have

$$\begin{aligned} Au^r + By^r + D\left(\frac{k-1}{k}w\right)^r &= 0, \\ A'(ku)^r + B'y^r + C'\left(\frac{-(k-1)z}{k}\right)^r &= 0, \end{aligned}$$

and eliminating u we obtain

$$(AB' - A'Bk^r)y^r + AC'\left(\frac{-(k-1)}{k}\right)^r z^r - A'D\left(\frac{k-1}{k}\right)^r w^r = 0;$$

writing $\left(\frac{k-1}{k}\right)^r = (-)^r \left(\frac{1-k}{k}\right)^r$, this is

$$\frac{AB' - A'Bk^r}{(1-k)^r} y^r - (-)^r AC' z^r + A'D w^r = 0;$$

or, what is the same thing, it is

$$\frac{AB' - A'Bk^r}{(1-k)^r} y^r - (-)^r AC' z^r + A'D w^r = 0.$$

And in regard to this and the other equations which contain $(-)^r$, it is to be observed that r being integral we have $(-)^r = (-)^r$, and that r being fractional, every value of $(-)^r$ is also a value of $(-)^{-r}$; so that we may in every case write $(-)^r$ in place of $(-)^{-r}$.

Similarly for the section by the plane $y = 0$, we have

$$\begin{aligned} Ax^r + Bv^r + D(-(k-1)w)^r &= 0, \\ A'(kx)^r + B'v^r + C'((k-1)z)^r &= 0, \end{aligned}$$

and eliminating v , we have

$$(AB' - A'Bk^r) x^r - BC' (k-1)^r z^r + BD(-(k-1))^r w^r = 0;$$

or, what is the same thing,

$$\frac{AB' - A'Bk^r}{(1-k)^r} x^r - (-)^r BC' z^r + B'D w^r = 0.$$

17. The four sections thus are

$$\begin{aligned} x = 0, & \quad \frac{AB' - A'Bk^r}{(1-k)^r} y^r - (-)^r AC' z^r + A'D w^r = 0, \\ y = 0, & \quad \frac{AB' - A'Bk^r}{(1-k)^r} x^r - (-)^r BC' z^r + B'D w^r = 0, \\ z = 0, & \quad Ax^r + By^r + Dw^r = 0, \\ w = 0, & \quad A'x^r + B'y^r + C'z^r = 0. \end{aligned}$$

It will be convenient to speak of these four curves as directrices of the scroll.

18. Suppose for a moment that r is integral; as either of the given equations may be multiplied by a constant, we may assume that $D = -(-)^r C'$; substituting this value and dividing the first and second equations each by C' , we have

$$\begin{aligned} x = 0, & \quad \frac{AB' - A'Bk^r}{(1-k)^r C'} y^r - (-)^r Az^r - A'w^r = 0, \\ y = 0, & \quad \frac{AB' - A'Bk^r}{(1-k)^r C'} x^r - (-)^r Bz^r - B'w^r = 0, \\ z = 0, & \quad Ax^r + By^r - (-)^r C'w^r = 0, \\ w = 0, & \quad A'x^r + B'y^r + C'z^r = 0, \end{aligned}$$

so that the diagonally opposite coefficients differ only by the factor $-(-)^r$; viz. the matrix is symmetrical or skew symmetrical according as r is odd or even.

19. If r be fractional, it is to be observed that, although the three symbols $(-)^r$ and the two symbols $(1-k)^r$ which enter into the first and second equations of No. 17, do not in the first instance represent of necessity the same values of $(-)^r$ and $(1-k)^r$ respectively, yet there is no loss of generality in assuming that they do so — the irrational equations are mere symbols for the rational equations to which they respectively give rise — and the irrationalities $(-)^r$ and $(1-k)^r$ will on the rationalisation of the equations disappear along with the irrationalities of x^r, y^r, z^r, w^r to which they are attached. But the case is otherwise with the irrationality k^r

involved in the expression $AB' - A'Bk^r$; writing as before $r = \pm \frac{p}{q}$ (p and q positive integers prime to each other), the symbol k^r has q different values; and there is not in the first instance any relation between the k^r of the first equation and the k^r of the second equation: for each of these equations the rationalised equation (that is, the equation rationalised in regard to the coordinates) will contain the irrationality k^r , and will thus for each of the q values of k^r represent a distinct curve. The given equations (viz. the first and second equations) represent each of them a single curve of the order pq or $2pq$, according as r is positive or negative; the first and second equations represent each of them q such curves.

20. Hence, starting from the two given curves in the planes $z = 0$ and $w = 0$, respectively, and with a given value of k , the section of the scroll by the plane $y = 0$ is made up of q curves, viz. the curves obtained from the second equation of No. 17, by assigning to the radical k^r each of its q different values; the scroll consequently breaks up into q different scrolls, viz. the lines passing through the two given curves, and any one of the q curves in the plane $y = 0$, constitute a distinct scroll. The lines in question meet the plane $x = 0$, not indifferently in any one of the q curves in that plane, but in a certain one of these curves, viz. in that curve for which the radical k^r has the same value as for the curve in question in the plane $y = 0$. Hence we may in the first and second equations regard the radicals k^r as having the same meaning, and the system of four equations in effect breaks up into q systems, viz. the systems obtained by giving to the radical k^r its q different values; each of these systems gives a scroll, and the scroll derived from the two given curves with a given value of k is made up of these q scrolls. And hence, attaching a unique value to each of the symbols $(-)^r$, $(1 - k)^r$ and k^r , we may, as before, write $D = -(-)^r C'$, and so reduce the original equations as in the case r integral, to the form No. 18, in which the diagonally opposite coefficients differ only by the factor $-(-)^r$.

21. Let the two given equations be taken to be

$$\begin{aligned} z = 0, \quad bx^r + yay^r + hw^r &= 0, \\ w = 0, \quad -(-)^r fx^r - (-)^r gy^r + (-)^r cz^r &= 0; \end{aligned}$$

we have then

$$\frac{AB' - A'Bk^r}{(1 - k)^r C'} = \frac{(-)^r bg + afk^r}{(-)^r c(1 - k)^r},$$

or, putting this $= -(-)^r c$, that is, writing

$$afk^r + bg(-)^r + ch(1 - k)^r = 0,$$

the four equations become

$$\begin{aligned} x = 0, & \quad + cy^r - (-)^r bz^r + fw^r = 0, \\ y = 0, & \quad -(-)^r cx^r + az^r + gw^r = 0, \\ z = 0, & \quad bx^r - (-)^r ay^r + hw^r = 0, \\ w = 0, & \quad -(-)^r fx^r - (-)^r gy^r - (-)^r hz^r = 0, \end{aligned}$$

where c being considered as given, k is determined as mentioned above; or, what is the same thing, $k : -1 : 1 - k = \lambda : \mu : \nu$, we have $\lambda : \mu : \nu$, and thence k , determined by the equations

$$\lambda + \mu + \nu = 0, \quad af\lambda^r + bg\mu^r + ch\nu^r = 0.$$

22. Consider for a moment λ, μ, ν , as the coordinates of a point in a plane, then ($r = \pm \frac{p}{q}$ as before), the equation $af\lambda^r + bg\mu^r + ch\nu^r = 0$, is that of a curve of the order pq or $2pq$, according as r is positive or negative: and this curve is met by the line $\lambda + \mu + \nu = 0$, in pq or $2pq$ points, that is, k has this number pq or $2pq$, of values: but to each of these values

of k there corresponds (not q values but) only a single value of k^r , viz. that value for which $afk^r + bg(-)^r + ch(1-k)^r = 0$; that is, starting from the two directrices in the planes $z = 0$, $w = 0$, respectively, and a given third directrix in the plane $y = 0$ (or in the plane $x = 0$), we may by means of each of the pq or $2pq$ values of k construct a scroll passing through the three directrices, and which will also pass through the fourth directrix in the plane $x = 0$ (or in the plane $y = 0$), but such scroll is only one (not each) of the q scrolls which can be constructed from the two given sections in the planes $z = 0$, $w = 0$, respectively, and from the assumed value of k . It has been mentioned that whether r is $+\frac{p}{q}$, or $-\frac{p}{q}$, the total scroll constructed from the two given directrices in the planes $z = 0$, $w = 0$, and from a given value of k is of the order $2p^2q^2$, and that such scroll breaks up into q distinct scrolls, hence the order of each of the distinct scrolls is $= 2p^2q$. Whence, starting with the given directrices in the planes $z = 0$, $w = 0$, and a given third directrix in the plane $y = 0$ (or in the plane $x = 0$), we have pq or $2pq$ scrolls each of the order $2p^2q$, and passing through these three directrices, and through the given fourth directrix in the plane $x = 0$ (or in the plane $y = 0$).

23. It is to be observed that when q is > 1 , then considering the three directrices as given, the pq or $2pq$ scrolls each of the order $2p^2q$, do not make up the total scroll generated by the lines which pass through the three given directrices. I call to mind that for three given directrices the orders of which are m, n, p , respectively, and which meet, the second and third, the third and first, and the first and second, in α points, β points, and γ points respectively, the order of the scroll generated by the lines which meet the three directrices is $= 2mnp - \alpha m - \beta n - \gamma p$. Suppose first, that $r = +\frac{p}{q}$, then the directrices are each of the order pq , and they do not any two of them meet; the order of the scroll is $= 2p^3q^3$. Suppose secondly, $r = -\frac{p}{q}$, then the directrices are each of the order $2pq$, but each two of them have in common two pq -tuple points counting as $2p^2q^2$ intersections; the order of the scroll is thus $(16 - 2 \cdot 4)p^3q^3 = 4p^3q^3$. In the first case the lines which meet the three directrices generate a residuary scroll of the order $2p^3(q^3 - q^2)$, and the pq scrolls each of the order $2p^2q$; in the second case they generate a residuary scroll of the order $4p^3(q^3 - q^2)$, and the $2pq$ scrolls each of the order $2p^2q$.

24. In the case $r = +\frac{p}{q}$, by way of illustration of the origin of the pq scrolls each of the order $2p^2q$, I consider the particular case $p = 1$, that is, $r = \frac{1}{q}$, the reciprocal of a positive integer q , and where it is to be shown that we have q scrolls each of the order $2q$. The given directrices are here

$$\begin{aligned} z = 0, \quad Ax^{\frac{1}{q}} + By^{\frac{1}{q}} + Dw^{\frac{1}{q}} &= 0, \\ w = 0, \quad A'x^{\frac{1}{q}} + B'y^{\frac{1}{q}} + C'z^{\frac{1}{q}} &= 0, \end{aligned}$$

each of them a unicursal curve; we may in fact satisfy the two equations respectively, by writing in the first of them

$$x : y : w = a(\phi + \alpha)^q : b(\phi + \beta)^q : d(\phi + \delta)^q;$$

and in the second

$$x : y : z = a'(\phi' + \alpha')^q : b'(\phi' + \beta')^q : c'(\phi' + \gamma')^q,$$

where $a, b, d, \alpha, \beta, \delta, a', b', c', \alpha', \beta', \gamma'$ are properly determined constants, ϕ, ϕ' are variable parameters. It follows that, considering the points K, K' which are the intersections of the first curve by the line $x - \theta y = 0$, and of the second curve by the corresponding line $x - k\theta y = 0$, we have not only a correspondence of q points K with q points K' , but we may establish, and that in q different manners, a correspondence between single points K and K' . For, substituting the

foregoing values of $x : y$ in the equations $x - \theta y = 0$ and $x - k\theta y = 0$ respectively, we have

$$\theta = \frac{a(\phi + \alpha)^q}{b(\phi + \beta)^q}, \quad k\theta = \frac{a'(\phi' + \alpha')^q}{b'(\phi' + \beta')^q},$$

and thence

$$\frac{a(\phi + \alpha)^q}{b(\phi + \beta)^q} = \frac{1}{k} \cdot \frac{a'(\phi' + \alpha')^q}{b'(\phi' + \beta')^q};$$

so that, extracting the q th root of each side, we have, in q different ways corresponding to the q values of the radical $\left(\frac{1}{k} \cdot \frac{ba'}{ab'}\right)^{\frac{1}{q}}$, a relation of the form $\phi' = \frac{\Lambda\phi + M}{N\phi + R}$; and considering ϕ' as having this value, the points K, K' as given by the equations

$$\begin{aligned} z = 0, \quad x : y : w &= a(\phi + \alpha)^q : b(\phi + \beta)^q : d(\phi + \delta)^q, \\ w = 0, \quad x : y : z &= a'(\phi' + \alpha')^q : b'(\phi' + \beta')^q : c'(\phi' + \gamma')^q, \end{aligned}$$

respectively, correspond as single points to each other. We have thus in q different ways a series of corresponding points K, K' , and consequently q series of lines KK' each of them generating a scroll which (as the order of the scroll generated by all the q series is $= 2q^2$), must be each of them of the order $2q$; and the decomposition in question is thus explained.

25. In the scroll of the order $2q^2$, each directrix is a q -tuple line, and the complete section by the plane of the directrix is made up of the directrix q times (order q^2), and of q q -fold generating lines, in fact, of q q -fold generating lines: to show that this is so, consider the directrix in the plane $z = 0$, viz. the equation of this is $Ax^{\frac{1}{q}} + By^{\frac{1}{q}} + Dw^{\frac{1}{q}} = 0$. Writing herein $w = 0$, we have $Ax^{\frac{1}{q}} + By^{\frac{1}{q}} = 0$, that is, $A^q x - (-)^q B^q y = 0$; it is clear that the rationalised equation must reduce itself to $[A^q x - (-)^q B^q y]^q = 0$, and that the line $w = 0$, is thus a tangent of q -pointic intersection at the point $w = 0, A^q x - (-)^q B^q y = 0$. Taking K at this point we have, in each of the scrolls of the order $2q$, q coincident positions of K' , that is, a q -fold line KK' in the plane $w = 0$; and the like for the plane $z = 0$, so that the total section by the plane $z = 0$ is made up of the directrix q times and of q q -fold generating lines; and it follows that for each of the scrolls of the order $2q$, the section by the plane $z = 0$ is made up of the directrix once, and of a q -fold generating line.

26. It is easy to see that in the general case $r = +\frac{p}{q}$, the like conclusion holds for the scroll of the order $2p^2q^2$, the section by the plane of the directrix consists of the directrix pq times (order p^2q^2), and of p^2q q -fold generating lines; whence for each of the q component scrolls of the order $2p^2q$, the section is made up of the directrix p times (that is, the directrix is a p -tuple line on the scroll) and of p^2 q -fold generating lines.

27. In the case $r = -\frac{p}{q}$, where the order of the directrix is $= 2pq$, then in the scroll of the order $2p^2q^2$, the directrix is a pq -tuple line on the scroll, and taken pq times it constitutes the complete section by the plane of the directrix; whence in each of the q component scrolls of the order $2p^2q$, the directrix is a p -tuple line; and taken p times it constitutes the complete section by the plane of the directrix.

28. It is convenient to exhibit the foregoing results in a tabular form as follows:

$r = +\frac{p}{q}$.	$r = -\frac{p}{q}$.
Each directrix is of order pq .	Each directrix is of order $2pq$, with three pq -tuple points.
<i>Scroll belonging to two directrices, and a given value of k, is of the order</i>	
$2p^2q^2$, breaking up into q scrolls each of order $2p^2q$, each which scroll of the order $2p^2q$ has each directrix for a p -tuple line and has besides p^2 q -fold generating lines in the plane of the directrix.	$2p^2q^2$, breaking up into q scrolls each of the order $2p^2q$, each which scroll of the order $2p^2q$ has each directrix for a p -tuple line, and conse- quently no generating line in the plane of the directrix.
<i>Considering two directrices and a given third directrix,</i>	
k has pq values.	k has $2pq$ values.
<i>Total scroll for the three directrices is made up of</i>	
pq scrolls each of order $2p^2q$ (viz. one for each value of k), and residuary scroll of order $2p^3(q^3 - q^2)$.	$2pq$ scrolls each of order $2p^2q$ (viz. one for each value of k), and residuary scroll of order $4p^3(q^3 - q^2)$.

29. The following are noticeable cases; $r = 1$ gives the hyperboloid as derived from three directrix lines; $r = -1$ the hyperboloid as derived from three plane sections thereof; $r = 2$, an octic surface, M. De la Gournerie's Quadrispinal; $r = -2$, an octic surface, his Quadricuspidal; $r = \frac{1}{3}$, a sextic surface which (as remarked by Dr Salmon), on writing therein (x^2, y^2, z^2, w^2) , in place of (x, y, z, w) , is converted into a surface of the twelfth order, locus of the centres of curvature of an ellipsoid.

[435]

On the Six Coordinates of a Line.

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THE notion of the six coordinates of a line was, so far as I am aware, first established in my paper “On a new analytical representation of Curves in Space,” *Quart. Math. Jour.* t. III. (1860), pp. 225–236, [284]; see p. 226, where writing p, q, r, s, t, u for the six determinants of the matrix $\begin{pmatrix} x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \end{pmatrix}$, I remark that these values give identically $ps + qt + ru = 0$; and I consider a cone as represented by a homogeneous equation $V = 0$ between the six coordinates (p, q, r, s, t, u) ; and many of the investigations of the present memoir, in which these coordinates are employed, have been in my possession for some years past. But these coordinates presented themselves independently to Prof. Plücker, and the theory of them is set forth in his most interesting and valuable memoir, “On a new Geometry of Space,” *Phil. Trans.* t. CLV. (1865), pp. 725–791; the course of development there given to the theory is however altogether different from that in the present memoir. They have also more recently been made use of in a paper by Herr Lüroth, “Zur Theorie der windschiefen Flächen,” *Crelle*, t. LXII. (1867), pp. 130–152.

I have in the present memoir applied these coordinates to the question of the Involution of six lines; the notion of this relation of six lines is due to Prof. Sylvester, to whom it presented itself in the year 1861, in connexion with a theorem in the *Lehrbuch der Statik*, by Möbius (Leipzig, 1837), that if four forces acting on a solid body are in equilibrium the lines along which the forces act are the generating lines of a hyperboloid. Prof. Sylvester was thereby led to consider six lines such that (regarding them as lines in a solid body) there exist along them forces which are in equilibrium; and he thence obtained, by the statical considerations reproduced in the present memoir, the construction (when five of the lines are given) of a sixth line to pass through a given point or to be situate in a given plane.

Article, Nos. 1 to 8. The Six Coordinates of a Line; definition and general notions.

1. Using any quadriplanar coordinates (x, y, z, w) whatever, consider a line; on the line two points the coordinates of which are $(\alpha, \beta, \gamma, \delta)$ and $(\alpha', \beta', \gamma', \delta')$ respectively; and through the line two planes, the equations whereof are $(A, B, C, D \parallel x, y, z, w) = 0$, and $(A', B', C', D' \parallel x, y, z, w) = 0$ respectively; we have

$$\begin{aligned} (A, B, C, D \parallel \alpha, \beta, \gamma, \delta) &= 0, \\ (A, B, C, D \parallel \alpha', \beta', \gamma', \delta') &= 0, \\ (A', B', C', D' \parallel \alpha, \beta, \gamma, \delta) &= 0, \\ (A', B', C', D' \parallel \alpha', \beta', \gamma', \delta') &= 0. \end{aligned}$$

2. From the first and second equations, eliminating successively A, B, C, D , we find

$$\begin{vmatrix} 0 & \alpha\beta' - \alpha'\beta & -(\gamma\alpha' - \gamma'\alpha) & \alpha\delta' - \alpha'\delta \\ -(\alpha\beta' - \alpha'\beta) & 0 & \beta\gamma' - \beta'\gamma & \beta\delta' - \beta'\delta \\ \gamma\alpha' - \gamma'\alpha & -(\beta\gamma' - \beta'\gamma) & 0 & \gamma\delta' - \gamma'\delta \\ -(\alpha\delta' - \alpha'\delta) & -(\beta\delta' - \beta'\delta) & -(\gamma\delta' - \gamma'\delta) & 0 \end{vmatrix} (A, B, C, D) = 0,$$

and from the third and fourth equations we find the like system with (A', B', C', D') in place of (A, B, C, D) . Comparing the corresponding equations of the two systems, we find an equality of ratios, as will presently be mentioned.

3. From the first and third equations, eliminating successively $\alpha, \beta, \gamma, \delta$, we find

$$\begin{vmatrix} 0 & AB' - A'B & -(CA' - C'A) & AD' - A'D \\ -(AB' - A'B) & 0 & BC' - B'C & BD' - B'D \\ CA' - C'A & -(BC' - B'C) & 0 & CD' - C'D \\ -(AD' - A'D) & -(BD' - B'D) & -(CD' - C'D) & 0 \end{vmatrix} (\alpha, \beta, \gamma, \delta) = 0,$$

and from the second and fourth equations we find the like system with $(\alpha', \beta', \gamma', \delta')$ in place of $(\alpha, \beta, \gamma, \delta)$; comparing the corresponding equations of the two systems, we find the same equality of ratios as before, viz.

4. This is

$$\begin{aligned} \beta\gamma' - \beta'\gamma : \gamma\alpha' - \gamma'\alpha : \alpha\beta' - \alpha'\beta : \alpha\delta' - \alpha'\delta : \beta\delta' - \beta'\delta : \gamma\delta' - \gamma'\delta \\ = AD' - A'D : BD' - B'D : CD' - C'D : BC' - B'C : CA' - C'A : AB' - A'B, \end{aligned}$$

and putting each of these two equal sets of ratios

$$= a : b : c : f : g : h,$$

then the quantities (a, b, c, f, g, h) , which it is easy to see satisfy the condition

$$af + bg + ch = 0,$$

are said to be the ‘six coordinates’ of the line: as only the ratios of the six quantities are material, and as the last-mentioned equation establishes a single relation between these ratios, the system of the six coordinates contain four arbitrary ratios or parameters, for the determination of the particular line.

5. A line is thus determined by its six coordinates (a, b, c, f, g, h) , which are such that $af + bg + ch = 0$; and conversely any six quantities (a, b, c, f, g, h) satisfying this relation may be taken to be the six coordinates of a line.

6. It is proper to show that the ratios $a : b : c : f : g : h$ are independent of the particular two points on the line, or two planes through the line, used for their determination. In fact, if instead of the points

$$\alpha, \beta, \gamma, \delta, \quad \alpha', \beta', \gamma', \delta',$$

we have any other two points on the line, say the points

$$\lambda\alpha + \mu\alpha', \quad \lambda\beta + \mu\beta', \quad \lambda\gamma + \mu\gamma', \quad \lambda\delta + \mu\delta',$$

$$\nu\alpha + \rho\alpha', \quad \nu\beta + \rho\beta', \quad \nu\gamma + \rho\gamma', \quad \nu\delta + \rho\delta',$$

then the six determinants have their original values each multiplied by $\lambda\rho - \mu\nu$; and the ratios are unaltered.

And the like is the case, if instead of the planes

$$A, B, C, D, \quad A', B', C', D',$$

we have any other two planes through the line, say the planes

$$\lambda A + \mu A', \quad \lambda B + \mu B', \quad \lambda C + \mu C', \quad \lambda D + \mu D',$$

$$\nu A + \rho A', \quad \nu B + \rho B', \quad \nu C + \rho C', \quad \nu D + \rho D';$$

the determinants have their original values each multiplied by $\lambda\rho - \mu\nu$; and the ratios are unaltered.

7. It may be remarked, that the theory of the six coordinates considered as derived from the two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, and as derived from the two planes (A, B, C, D) , (A', B', C', D') is precisely the same in each case; and we may confine ourselves to the first point of view, regarding therefore the six coordinates as derived from the two points $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$. I further remark, that I do not at present in anywise fix the absolute magnitudes of the coordinates (a, b, c, f, g, h) : it is only the ratios that we are concerned with.

8. The values of the ratios $a : b : c : f : g : h$ of the six coordinates do however depend on the particular coordinate planes $x = 0, y = 0, z = 0, w = 0$, made use of for their determination; and in the sequel it will be necessary to investigate the formulae of transformation to a new set of coordinate planes $x_0 = 0, y_0 = 0, z_0 = 0, w_0 = 0$. And I shall also show in what manner the absolute magnitudes of the coordinates may be fixed. But deferring the consideration of these questions, I consider the planes $x = 0, y = 0, z = 0, w = 0$ as given planes, and take the six coordinates (a, b, c, f, g, h) of a line to be determined as above in reference to these given planes, the absolute values of these coordinates remaining indeterminate, and their ratios only being attended to. And I proceed to consider the various questions which present themselves in the geometry of the line, considered as thus determined by means of its six coordinates (a, b, c, f, g, h) .

Article, Nos. 9 to 18. (Various Sub-headings.) Elementary Theorems.

Condition that a line may be in a given plane.

9. Taking the line to be (a, b, c, f, g, h) , the equation of the given plane to be

$$(A, B, C, D \parallel x, y, z, w) = 0;$$

then if $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$ are the coordinates of any two points on the line, we have the system of equations *ante*, No. 2, and substituting therein for $\beta\gamma' - \beta'\gamma$, &c. the values (a, b, c, f, g, h) , we find

$$\begin{vmatrix} 0 & c & -b & f \\ -c & 0 & a & g \\ b & -a & 0 & h \\ -f & -g & -h & 0 \end{vmatrix} (A, B, C, D) = 0;$$

which equations, equivalent to a twofold relation, are the required condition. It may be remarked that, treating (A, B, C, D) as current plane coordinates, each equation of the system is that of a point lying in the line.

Condition that a line may pass through a given point.

10. The coordinates of the given point are taken to be $(\alpha, \beta, \gamma, \delta)$. If

$$(A, B, C, D \parallel x, y, z, w) = 0, \quad (A', B', C', D' \parallel x, y, z, w) = 0,$$

are the equations of any two planes through the line, then we have the system of equations *ante* No. 3, and substituting therein for $AB' - A'B$, &c. their values in terms of the coordinates (a, b, c, f, g, h) of the line, we have

$$\begin{vmatrix} 0 & h & -g & a \\ -h & 0 & f & b \\ g & -f & 0 & c \\ -a & -b & -c & 0 \end{vmatrix} (\alpha, \beta, \gamma, \delta) = 0;$$

which equations, equivalent to a twofold relation, are the required condition. It is obvious that, treating $(\alpha, \beta, \gamma, \delta)$ as current point coordinates, each equation of the system is the equation of a plane through the given line.

Condition for the intersection of two lines.

11. The coordinates of the lines are taken to be (a, b, c, f, g, h) , and $(a_1, b_1, c_1, f_1, g_1, h_1)$, respectively. If $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, are the coordinates of any two points in the first line, and $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, $(\alpha'_1, \beta'_1, \gamma'_1, \delta'_1)$, are the coordinates of any two points on the second line, then the four points are in a plane, that is, we have

$$\begin{vmatrix} \alpha & \beta & \gamma & \delta \\ \alpha' & \beta' & \gamma' & \delta' \\ \alpha_1 & \beta_1 & \gamma_1 & \delta_1 \\ \alpha'_1 & \beta'_1 & \gamma'_1 & \delta'_1 \end{vmatrix} = 0,$$

that is, expanding the determinant and substituting for $\beta\gamma' - \beta'\gamma$, &c. and $\beta_1\gamma'_1 - \beta'_1\gamma_1$, &c. their values in terms of the coordinates of the two lines respectively, we have

$$af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1 = 0,$$

or, as this may also be written,

$$(f_1, g_1, h_1, a_1, b_1, c_1 \parallel a, b, c, f, g, h) = 0,$$

for the condition that the two lines may intersect.

12. The same result will be obtained if we take

$$(A, B, C, D \parallel x, y, z, w) = 0, \quad (A', B', C', D' \parallel x, y, z, w) = 0,$$

for the equations of any two planes through the first line, and

$$(A_1, B_1, C_1, D_1 \parallel x, y, z, w) = 0, \quad (A'_1, B'_1, C'_1, D'_1 \parallel x, y, z, w) = 0,$$

for the equations of any two planes through the second line. The four planes will meet in a point, that is, we have

$$\begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A_1 & B_1 & C_1 & D_1 \\ A'_1 & B'_1 & C'_1 & D'_1 \end{vmatrix} = 0,$$

or, expanding and substituting, we have the same condition as before.

13. In the case of any two lines (a, b, c, f, g, h) , and $(a_1, b_1, c_1, f_1, g_1, h_1)$, we may define the ‘moment’ of the two lines to be the function

$$af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1,$$

it being understood that we have not as yet any complete quantitative definition of the moment; this being so, we have, in what precedes, the theorem that the moment of two intersecting lines is = 0.

Plane through two intersecting lines.

14. Let $(A, B, C, D \parallel x, y, z, w) = 0$ be the equation of the plane through the two intersecting lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$. We have two systems of equations, as in No. 9, and comparing the corresponding equations of the two systems, we find in the first instance

$$\begin{aligned} A : B : C : D &= \lambda : bf_1 - b_1f : cf_1 - c_1f : -(bc_1 - b_1c) \\ &= ag_1 - a_1g : \mu : cg_1 - c_1g : -(ca_1 - c_1a) \\ &= ah_1 - a_1h : bh_1 - b_1h : \nu : -(ab_1 - a_1b) \\ &= gh_1 - g_1h : hf_1 - h_1f : fg_1 - f_1g : \rho, \end{aligned}$$

where λ, μ, ν, ρ , are in the first instance unknown; the different sets of ratios are of course identical in virtue of the relation

$$(f_1, g_1, h_1, a_1, b_1, c_1 \parallel a, b, c, f, g, h) = 0,$$

and comparing them we have equations which lead to the values of λ, μ, ν, ρ ; and we thus obtain more completely.

$$\begin{aligned} A : B : C : D &= f_1a + b_1g + c_1h : bf_1 - b_1f : cf_1 - c_1f : -(bc_1 - b_1c) \\ &= ag_1 - a_1g : a_1f + g_1b + c_1h : cg_1 - c_1g : -(ca_1 - c_1a) \\ &= ah_1 - a_1h : bh_1 - b_1h : a_1f + b_1g + h_1c : -(ab_1 - a_1b) \\ &= gh_1 - g_1h : hf_1 - h_1f : fg_1 - f_1g : f_1a + g_1b + c_1h. \end{aligned}$$

15. It is in these equations easy to verify the identity of the different sets of values: we ought, for instance, to have

$$\frac{a_1f + bg + h_1c}{fg_1 - f_1g} = \frac{ab_1 - a_1b}{a_1f + bg_1 + ch_1};$$

that is,

$$(h_1c + a_1f + b_1g)(h_1c + af_1 + bg_1) + (ab_1 - a_1b)(fg_1 - f_1g) = 0,$$

and, observing that

$$\begin{aligned} (a_1f + b_1g)(af_1 + bg_1) + (ab_1 - a_1b)(fg_1 - f_1g) &= (af + bg)(a_1f_1 + b_1g_1) \\ &= ch \cdot c_1h_1, \end{aligned}$$

the left-hand side is

$$\begin{aligned} &= ch_1(ch_1 + af_1 + bg_1 + a_1f + b_1g + c_1h), \\ &= ch_1(af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1) = 0. \end{aligned}$$

Point on two intersecting lines.

16. Let $(\alpha, \beta, \gamma, \delta)$ be the coordinates of the point of intersection of the two intersecting lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$. We have two systems of equations such as in No. 10, and comparing the corresponding equations of the two systems, we find

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= L : ag_1 - a_1g : ah_1 - a_1h : gh_1 - g_1h \\ &= bf_1 - b_1f : M : bh_1 - b_1h : hf_1 - h_1f \\ &= cf_1 - c_1f : cg_1 - c_1g : N : fg_1 - f_1g \\ &= -(bc_1 - b_1c) : -(ca_1 - c_1a) : -(ab_1 - a_1b) : P, \end{aligned}$$

where L, M, N, P , are in the first instance unknown; but, comparing the different sets of values, we have equations for finding the values of these quantities, and we thus obtain the more complete system

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= f_1a + b_1g + c_1h : ag_1 - a_1g : ah_1 - a_1h : gh_1 - g_1h \\ &= bf_1 - b_1f : a_1f + g_1b + c_1h : bh_1 - b_1h : hf_1 - h_1f \\ &= cf_1 - c_1f : cg_1 - c_1g : a_1f + b_1g + h_1c : fg_1 - f_1g \\ &= -(bc_1 - b_1c) : -(ca_1 - c_1a) : -(ab_1 - a_1b) : f_1a + g_1b + h_1c, \end{aligned}$$

where it is to be observed that the right-hand side considered as a matrix is the transposed matrix of that which occurs in No. 13, in the formula for $A : B : C : D$. The verification of the identity of the different sets of values can of course be effected as in No. 15.

Expression for an arbitrary plane through a line.

17. The condition in order that the plane $(A, B, C, D \parallel x, y, z, w) = 0$, may pass through the line (a, b, c, f, g, h) , is the twofold relation given, No. 9; it is satisfied by any one of the four systems

$$\begin{aligned} A : B : C : D &= 0 : h : -g : a, \\ &\text{or} = -h : 0 : f : b, \\ &\text{or} = g : -f : 0 : c, \\ &\text{or} = -a : -b : -c : 0; \end{aligned}$$

and consequently also by

$$\begin{aligned} A : B : C : D &= (0, -h, g, -a \parallel \xi, \eta, \zeta, \omega) \\ &: (h, 0, -f, -b \parallel \xi, \eta, \zeta, \omega) \\ &: (-g, f, 0, -c \parallel \xi, \eta, \zeta, \omega) \\ &: (a, b, c, 0 \parallel \xi, \eta, \zeta, \omega); \end{aligned}$$

or, what is the same thing, by

$$\begin{aligned} A : B : C : D &= (0, h, -g, a \parallel \xi, \eta, \zeta, \omega) \\ &: (-h, 0, f, b \parallel \xi, \eta, \zeta, \omega) \\ &: (g, -f, 0, c \parallel \xi, \eta, \zeta, \omega) \\ &: (-a, -b, -c, 0 \parallel \xi, \eta, \zeta, \omega), \end{aligned}$$

where $(\xi, \eta, \zeta, \omega)$ are arbitrary: there is, however, no loss of generality in putting any two of these quantities = 0.

Expression for an arbitrary point in a line.

18. The condition in order that the point $(\alpha, \beta, \gamma, \delta)$, may lie in the line (a, b, c, f, g, h) , is the twofold relation given, No. 10; it is satisfied by any one of the four systems

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= 0 : c : -b : f, \\ &\text{or} = -c : 0 : a : g, \\ &\text{or} = b : -a : 0 : h, \\ &\text{or} = -f : -g : -h : 0; \end{aligned}$$

and consequently, also by

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= (0, -c, b, -f \parallel x, y, z, w) \\ &: (c, 0, -a, -g \parallel x, y, z, w) \\ &: (-b, a, 0, -h \parallel x, y, z, w) \\ &: (f, g, h, 0 \parallel x, y, z, w); \end{aligned}$$

or, what is the same thing, by

$$\begin{aligned} \alpha : \beta : \gamma : \delta &= (0, c, -b, f \parallel x, y, z, w) \\ &: (-c, 0, a, g \parallel x, y, z, w) \\ &: (b, -a, 0, h \parallel x, y, z, w) \\ &: (-f, -g, -h, 0 \parallel x, y, z, w), \end{aligned}$$

where (x, y, z, w) are arbitrary: there is, however, no loss of generality in putting two of these quantities $= 0$.

Article Nos. 19 to 25. Geometrical considerations in regard to three, four, five, and six lines.

Before proceeding further, I will establish certain geometrical notions in regard to three, four, five, and six lines. I use the term ‘tractor’ to denote a line which meets any given lines.

19. Three given lines have an infinity of tractors; viz. these are the generating lines of a hyperboloid having the three given lines for directrices.

[435] ON THE SIX COORDINATES OF A LINE (CONTINUED).

29. If the two sets of coefficients are each of them the coordinates of a line, then the two equations express that the line (a, b, c, f, g, h) is any line whatever cutting each of the two given lines. And the general case is in fact reducible to this particular one; for suppose that neither set of coefficients belongs to a line, then we may from the two given linear relations form the relation

$$(\lambda F + \lambda_1 F_1, \lambda G + \lambda_1 G_1, \lambda H + \lambda_1 H_1, \lambda A + \lambda_1 A_1, \lambda B + \lambda_1 B_1, \lambda C + \lambda_1 C_1) \cdot (a, b, c, f, g, h) = 0,$$

and if the ratio $\lambda : \lambda_1$ be properly determined, then $(\lambda A + \lambda_1 A_1, \dots)$ will be the coordinates of a line. This will in fact be the case if

$$(\lambda A + \lambda_1 A_1)(\lambda F + \lambda_1 F_1) + (\lambda B + \lambda_1 B_1)(\lambda G + \lambda_1 G_1) + (\lambda C + \lambda_1 C_1)(\lambda H + \lambda_1 H_1) = 0,$$

that is, if

$$(AF + BG + CH, AF_1 + BG_1 + CH_1 + FA_1 + GB_1 + HC_1, A_1F_1 + B_1G_1 + C_1H_1)(\lambda, \lambda_1)^2 = 0,$$

a quadric equation giving two values of the ratio $\lambda : \lambda_1$, that is, two linear relations in each of which the coefficients are the coordinates of a line: we have thus two derived lines, and the line (a, b, c, f, g, h) meets each of these derived lines.

There is no real difference if one or the other of the given systems of coefficients, say the system (A, B, C, F, G, H) , are the coordinates of a line. We have then $AF + BG + CH = 0$; the quadric equation in $\lambda : \lambda_1$ has a root $\lambda_1 : \lambda = 0$, and rejecting it, the other root is determined by a simple equation: this only means that the line (A, B, C, F, G, H) is itself one of the two derived lines.

But there is a real difference in the case where the equation in $\lambda : \lambda_1$ has equal roots; to explain this special case, observe that if in the general case we consider the two derived lines as a pair of tractors of any four lines, then the linear relations express that the line (a, b, c, f, g, h) has with these four lines a pair of tractors; and in the special case under consideration the linear relations express that the line (a, b, c, f, g, h) has with the four lines, or (what is the same thing) with any three of them, that is with some three lines, a twofold tractor. According to what precedes (No. 21), the construction of the line (a, b, c, f, g, h) is in fact as follows, viz. if on the twofold tractor considered as given, we take a series of points p , and through the tractor, homographic with the range, a pencil of planes P , then the sought-for line will be any line through a point p , in the corresponding plane P . But it is proper to give an analytical proof of the construction.

30. I observe that we may without loss of generality assume $A_1F_1 + B_1G_1 + C_1H_1 = 0$, and this being so, the condition for the equality of the roots of the quadric equation is

$$AF_1 + BG_1 + CH_1 + FA_1 + GB_1 + HC_1 = 0,$$

that is, writing $(a_1, b_1, c_1, f_1, g_1, h_1)$ in place of $(A_1, B_1, C_1, F_1, G_1, H_1)$, the case in question may be taken to be that of

Two linear relations

$$\begin{aligned} (f_1, g_1, h_1, a_1, b_1, c_1) (a, b, c, f, g, h) &= 0, \\ (F, G, H, A, B, C) (a, b, c, f, g, h) &= 0, \end{aligned}$$

where $(a_1, b_1, c_1, f_1, g_1, h_1)$ are, (A, B, C, F, G, H) are not, the coordinates of a line, and where

$$(f_1, g_1, h_1, a_1, b_1, c_1) (A, B, C, F, G, H) = 0;$$

that is, where the twofold derived line is in fact the original line

$$(a_1, b_1, c_1, f_1, g_1, h_1).$$

31. To simplify, we may take $x = 0, y = 0$ for the equations of the line; the coordinates of the line then are $(a_1, b_1, c_1, f_1, g_1, h_1) = (0, 0, 0, 0, 0, 1)$. Taking moreover $x = 0, y = 0, \frac{z}{\gamma} = \frac{w}{\delta}$ for the coordinates of the point p , and $\frac{x}{\alpha} = \frac{y}{\beta}$ for the equation of the plane P , the homographic relation of the point and plane is given by an equation of the form

$$-F\beta\gamma + G\alpha\gamma - A\alpha\delta - B\beta\delta = 0,$$

or, as this may be written,

$$(F, G, H, A, B, C)(-\beta\gamma, \alpha\gamma, 0, -\alpha\delta, -\beta\delta, \omega) = 0,$$

where H and ω , being each multiplied by 0, do not really enter into the equation.

The equations of any line whatever through the point p and in the plane P may be written $\beta x - \alpha y = 0, A'x + B'y + \delta z - \gamma w = 0$, where A', B' are arbitrary: hence arranging the coefficients in the order

$$\begin{array}{cccc} \beta, & -\alpha, & 0, & 0, \\ A', & B', & \delta, & -\gamma, \end{array}$$

the coordinates (a, b, c, f, g, h) of the line in question are

$$(-\alpha\beta\gamma, \alpha\gamma\beta', 0, -\alpha\delta, -\beta\delta, A'\alpha + B'\beta);$$

correctly written:

$$(a, b, c, f, g, h) = (-\beta\gamma, \alpha\gamma, 0, -\alpha\delta, -\beta\delta, A'\alpha + B'\beta);$$

so that we have

$$(f_1, g_1, h_1, a_1, b_1, c_1)(a, b, c, f, g, h) = (0, 0, 1, 0, 0, 0)(a, b, c, f, g, h) = c = 0;$$

and moreover the homographic relation, replacing therein the arbitrary quantity ω by $A'\alpha + B'\beta$, becomes

$$(F, G, H, A, B, C)(a, b, c, f, g, h) = 0.$$

Hence the linear relations satisfied by the coordinates (a, b, c, f, g, h) of the line in question are

$$\begin{aligned} (f_1, g_1, h_1, a_1, b_1, c_1)(a, b, c, f, g, h) &= 0, \\ (F, G, H, A, B, C)(a, b, c, f, g, h) &= 0, \end{aligned}$$

with the coefficients

$$\begin{aligned} (f_1, g_1, h_1, a_1, b_1, c_1) &= (0, 0, 1, 0, 0, 0), \\ (A, B, C, F, G, H) &= (A, B, 0, F, G, H), \end{aligned}$$

values which satisfy the condition

$$(f_1, g_1, h_1, a_1, b_1, c_1)(A, B, C, F, G, H) = 0.$$

Hence the line (a, b, c, f, g, h) through the point p and in the plane P is a line the coordinates of which satisfy two linear relations as mentioned in the heading; and the theorem is thus proved. The demonstration would be simplified by taking, as is allowable, the homographic relation to be $\frac{\alpha}{\beta} = k\frac{\gamma}{\delta}$.

32. It appears from the foregoing examination of the case of two linear relations that in the following cases of three or more linear relations there is no real loss of generality in assuming that the coefficients of each set are the coordinates of a line; for if originally this be not so, we have only to replace the given relations by linear functions of these relations, and to assign such values to the multipliers $\lambda, \lambda_1, \lambda_2 \dots$ as in each case to make the new coefficients to be the coordinates of a line; and as there are two or more arbitrary ratios $\lambda : \lambda_1 : \lambda_2 : \dots$ to be assigned at pleasure and only a single condition to be satisfied, no cases of failure can arise. The remaining cases may consequently be stated in a more simple form.

Three linear relations, the coefficients of each set being the coordinates of a line.

33. The three relations express that the line (a, b, c, f, g, h) meets each of the three given lines; that is, that the line is any generating line of a hyperboloid having the three given lines for directrices.

Four linear relations, the coefficients of each set being the coordinates of a line.

34. The four relations express that the line (a, b, c, f, g, h) meets each of four given lines; or what is the same thing, that the line is a tractor of four given lines. It is to be noticed that the four linear relations serve to express the ratios $a : b : c : f : g : h$ linearly in terms of any one of these ratios, or what is the same thing, to express the several ratios in terms of an arbitrary ratio $u : v$. Substituting the resulting values in the equation

$$af + bg + ch = 0,$$

we have a quadric equation for the determination of the remaining ratio, or of the ratio $u : v$; and then each of the ratios of the coordinates can be expressed rationally in terms of either root of the quadric equation; we thus obtain the coordinates of each of the two tractors of the four given lines; or we have a complete analytical solution of the problem, to find the tractors of four given lines. The quadric equation may have equal roots; that is, the four given lines may have a twofold tractor, which is then determined linearly.

35. The theory of the linear relations of the coordinates (a, b, c, f, g, h) of a line may be considered in a different manner. It will be convenient to take the different cases in a reverse order, beginning with the extreme case (not before mentioned) of a fivefold relation and ascending to the case of a onefold or single relation.

Case of the fivefold relation.

36. The fivefold relation

$$\begin{vmatrix} a & b & c & f & g & h \\ a_1 & b_1 & c_1 & f_1 & g_1 & h_1 \end{vmatrix} = 0,$$

expresses that the quantities (a, b, c, f, g, h) are proportional to $(a_1, b_1, c_1, f_1, g_1, h_1)$. As the former set are by hypothesis the coordinates of a line, the given set $(a_1, b_1, c_1, f_1, g_1, h_1)$ must, it is clear, also be the coordinates of a line, and the relation then expresses that the line (a, b, c, f, g, h) coincides with the given line.

Case of the fourfold relation.

37. The fourfold relation is

$$\begin{vmatrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \end{vmatrix} = 0,$$

or what is the same thing, we have the six equations $\lambda a + \lambda_1 a_1 + \lambda_2 a_2 = 0$, &c., involving the indeterminate quantities $\lambda, \lambda_1, \lambda_2$. If the coefficients $(a_1, b_1, c_1, f_1, g_1, h_1), (a_2, b_2, c_2, f_2, g_2, h_2)$ are not either set the coordinates of a line; then substituting the foregoing values $-\lambda a = \lambda_1 a_1 + \lambda_2 a_2$, &c. in the equation $af + bg + ch = 0$, we have a quadric equation in $(\lambda_1 : \lambda_2)$: and for each root of this equation, the coefficients $\lambda_1 a_1 + \lambda_2 a_2$, &c. will be the coordinates of a line. There are thus in general two derived lines; and the fourfold relation expresses that the line (a, b, c, f, g, h) coincides with one or other of these derived lines. There is no real difference if one or the other of the two sets $(a_1, b_1, c_1, f_1, g_1, h_1), (a_2, b_2, c_2, f_2, g_2, h_2)$, or if each set, are the coordinates of a line; one of the derived lines or both of them will in these cases coincide with one or both of the given lines. And if the quadric equation has equal roots, then instead of two derived lines there is a twofold derived line, and the line (a, b, c, f, g, h) must coincide with this twofold line.

38. A case presenting peculiarity is however that in which the coefficients of the quadric equation vanish identically; this is only so when the coefficients $(a_1, b_1, c_1, f_1, g_1, h_1)$ and $(a_2, b_2, c_2, f_2, g_2, h_2)$ are the coordinates of two intersecting lines. The equations $-\lambda a = \lambda_1 a_1 + \lambda_2 a_2$, &c. here show that every line whatever which meets each of the two lines $(a_1, b_1, c_1, f_1, g_1, h_1)$ and $(a_2, b_2, c_2, f_2, g_2, h_2)$ meets also the line (a, b, c, f, g, h) ; that is, the line (a, b, c, f, g, h) is any line whatever in the plane and through the point of intersection of the two intersecting lines. We see moreover that not only

$$a_1 f_2 + b_1 g_2 + c_1 h_2 + a_2 f_1 + b_2 g_1 + c_2 h_1 = 0,$$

but also that $af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1 = 0$ and $af_2 + bg_2 + ch_2 + fa_2 + gb_2 + hc_2 = 0$; that is, the moment of each pair of lines is $= 0$. It may be remarked that the ratios $\lambda : \lambda_1 : \lambda_2$ may be determined from any two of the six equations

$$\lambda a + \lambda_1 a_1 + \lambda_2 a_2 = 0, \dots \lambda h + \lambda_1 h_1 + \lambda_2 h_2 = 0;$$

but that in consequence of the moments being each $= 0$, there is not for the determination of these ratios any such set of equations as occur in the cases subsequently considered of a threefold relation, &c.

39. In what follows we have three or more sets $(a_1, b_1, c_1, f_1, g_1, h_1)$, &c.; and we may without loss of generality assume that each of these are the coordinates of a line: for replacing the several coefficients $a_1, \dots$ by linear functions $\mu_1 a_1 + \mu_2 a_2 + \mu_3 a_3 + \dots$, &c., the multipliers may be determined so that these are the coordinates of a line: and since for each set there is only a single condition to be satisfied by the two or more ratios $\mu_1 : \mu_2 : \mu_3 \dots$, it is easy to see that no cases of failure will arise.

Case of the threefold relation.

40. The threefold relation is

$$\begin{vmatrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \end{vmatrix} = 0,$$

where $(a_1, \dots)$, $(a_2, \dots)$, $(a_3, \dots)$ are each the coordinates of a line. Here writing

$$\lambda a + \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 = 0, \dots$$

it is clear that every line which meets each of the lines $(a_1, \dots)$, $(a_2, \dots)$, $(a_3, \dots)$ will also meet the line (a, b, c, f, g, h) ; the lines which meet the first-mentioned three lines are the generating lines of a hyperboloid having these three lines for directrices, and it hence appears that the line (a, b, c, f, g, h) is any directrix line whatever of the hyperboloid in question.

41. Using the notations 01, 02, 12, &c. to denote the moments of the several pairs of lines, viz.

$$\begin{aligned} 01 &= af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1, \\ 12 &= a_1f_2 + b_1g_2 + c_1h_2 + f_1a_2 + g_1b_2 + h_1c_2, \\ &\text{\&c.}, \end{aligned}$$

then from the equations $\lambda a + \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 = 0$, &c., we deduce

$$\begin{aligned} \lambda_1 01 + \lambda_2 02 + \lambda_3 03 &= 0, \\ \lambda 10 + \lambda_1 21 + \lambda_2 12 + \lambda_3 13 &= 0, \\ \lambda 20 + \lambda_1 21 + \lambda_2 32 + \lambda_3 23 &= 0, \\ \lambda 30 + \lambda_1 31 + \lambda_2 32 + \lambda_3 33 &= 0, \end{aligned}$$

and hence eliminating λ , λ_1 , λ_2 , λ_3 , we find

$$\begin{vmatrix} \cdot & 01 & 02 & 03 \\ 10 & \cdot & 12 & 13 \\ 20 & 21 & \cdot & 23 \\ 30 & 31 & 32 & \cdot \end{vmatrix} = 0,$$

a relation between the moments satisfied in virtue of the given threefold relation; but which as a mere onefold relation is of course not equivalent to the threefold relation. It will subsequently appear that the equation expresses that any one of the four lines, say the line (a, b, c, f, g, h) touches the hyperboloid having the other three lines for generatrices; this condition is satisfied in virtue of the threefold relation which, as we have seen, expresses that the line (a, b, c, f, g, h) lies wholly in the hyperboloid in question.

42. The last mentioned determinant is the Norm of

$$\sqrt{01 \cdot 23} + \sqrt{02 \cdot 31} + \sqrt{03 \cdot 12},$$

so that the equation may be written

$$\sqrt{01 \cdot 23} + \sqrt{02 \cdot 31} + \sqrt{03 \cdot 12} = 0,$$

or, what is the same thing,

$$\sqrt{01} \sqrt{23} + \sqrt{02} \sqrt{31} + \sqrt{03} \sqrt{12} = 0,$$

it being of course understood that the signs of the radicals must be determined in accordance with this equation; we then find

$$\lambda : \lambda_1 : \lambda_2 : \lambda_3 = \sqrt{23 \cdot 31 \cdot 12} : \sqrt{02 \cdot 03 \cdot 23} : \sqrt{03 \cdot 01 \cdot 31} : \sqrt{01 \cdot 02 \cdot 12},$$

or say

$$= \sqrt{23} \sqrt{31} \sqrt{12} : \sqrt{02} \sqrt{03} \sqrt{23} : \sqrt{03} \sqrt{01} \sqrt{31} : \sqrt{01} \sqrt{02} \sqrt{12};$$

in fact, substituting these last values in the linear equations for $\lambda, \lambda_1, \lambda_2, \lambda_3$, we find that the equations are all satisfied in virtue of the single equation

$$\sqrt{01} \sqrt{23} + \sqrt{02} \sqrt{31} + \sqrt{03} \sqrt{12} = 0.$$

Case of the twofold relation.

43. We have here

$$\begin{vmatrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \\ a_4, & b_4, & c_4, & f_4, & g_4, & h_4 \end{vmatrix} = 0,$$

where $(a_1, \dots), (a_2, \dots), (a_3, \dots), (a_4, \dots)$, are each the coordinates of a line. Here, writing

$$\lambda a + \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 + \lambda_4 a_4 = 0,$$

it is clear that every line which meets each of the four given lines, will also meet the line (a, b, c, f, g, h) ; but the only lines meeting the four given lines are two determinate lines, the tractors of the four given lines; and the conclusion is, that the line (a, b, c, f, g, h) is any line whatever which meets the two tractors.

44. If, however, the four given lines have a twofold tractor, then the line (a, b, c, f, g, h) is still a line having two conditions imposed upon it; it is in fact a line determined as in No. 21, viz. if on the tractor we take a series of points p , and through the tractor a series of planes P , corresponding homographically to the points, then the line (a, b, c, f, g, h) is any line through a point p , in the corresponding plane P .

45. Using as before 01, 02, ... 12, &c. to denote the moments of the several pairs of lines, we have

$$\begin{aligned} \lambda_1 01 + \lambda_2 02 + \lambda_3 03 + \lambda_4 04 &= 0, \\ \lambda 10 + \lambda_2 12 + \lambda_3 13 + \lambda_4 14 &= 0, \\ \lambda 20 + \lambda_1 21 + \lambda_3 23 + \lambda_4 24 &= 0, \\ \lambda 30 + \lambda_1 31 + \lambda_2 32 + \lambda_4 34 &= 0, \\ \lambda 40 + \lambda_1 41 + \lambda_2 42 + \lambda_3 43 &= 0, \end{aligned}$$

and thence also

$$\begin{vmatrix} \cdot & 01 & 02 & 03 & 04 \\ 10 & \cdot & 12 & 13 & 14 \\ 20 & 21 & \cdot & 23 & 24 \\ 30 & 31 & 32 & \cdot & 34 \\ 40 & 41 & 42 & 43 & \cdot \end{vmatrix} = 0,$$

a relation between the moments satisfied in virtue of the original twofold relation; but which, as a single equation, is of course not equivalent to the twofold relation. It is in fact easy to see that this equation expresses that the five lines have a common tractor; this is true, since in virtue of the twofold relation there are really two common tractors.

I have not obtained from the linear equations any symmetrical expressions for the ratios $\lambda : \lambda_1 : \lambda_2 : \lambda_3 : \lambda_4$.

Case of a onefold relation.

46. The onefold relation is

$$\begin{vmatrix} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \\ a_4, & b_4, & c_4, & f_4, & g_4, & h_4 \\ a_5, & b_5, & c_5, & f_5, & g_5, & h_5 \end{vmatrix} = 0,$$

where $(a_1, \dots), (a_2, \dots), (a_3, \dots), (a_4, \dots), (a_5, \dots)$, are each the coordinates of points in a line. The preceding mode of dealing with the question is inapplicable, since there is not in general any line which meets the five given lines; in the particular case, however, where the five given lines are met by a single line, say when they have a common tractor, then the line (a, b, c, f, g, h) is any line meeting this common tractor. The general case is that of the involution of six lines, mentioned No. 25, and the consideration of which was deferred.

47. The onefold relation implies that we can find multipliers $\lambda, \mu, \nu, \rho, \sigma, \tau$, such that

$$\begin{aligned} \lambda a + \mu b + \nu c + \rho f + \sigma g + \tau h &= 0, \\ \lambda a_1 + \mu b_1 + \nu c_1 + \rho f_1 + \sigma g_1 + \tau h_1 &= 0, \\ &\vdots \\ \lambda a_5 + \mu b_5 + \nu c_5 + \rho f_5 + \sigma g_5 + \tau h_5 &= 0; \end{aligned}$$

we may by means of the last five equations determine the ratios of $\lambda, \mu, \nu, \rho, \sigma, \tau$, viz. these quantities will be proportional to the determinants formed out of the matrix

$$\begin{pmatrix} a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \\ a_4, & b_4, & c_4, & f_4, & g_4, & h_4 \\ a_5, & b_5, & c_5, & f_5, & g_5, & h_5 \end{pmatrix}$$

and the first equation is then a linear relation in (a, b, c, f, g, h) , expressing the relation that exists between these coordinates.

48. Consider an arbitrary point O on the line (a, b, c, f, g, h) ; taking this point as origin, the coordinates of O are $0, 0, 0, 1$; and if x, y, z, w , are the coordinates of any other point on the line, then writing

$$\begin{array}{cccc} x, & y, & z, & w, \\ 0, & 0, & 0, & 1, \end{array}$$

we find $a : b : c : f : g : h = 0 : 0 : 0 : x : y : z$ and the equation

$$\lambda a + \mu b + \nu c + \rho f + \sigma g + \tau h = 0$$

becomes simply $\rho x + \sigma y + \tau z = 0$; viz. this equation expresses that the line (a, b, c, f, g, h) , assumed to pass through a given point O , lies in a determinate plane Ω through this point.

49. To construct this plane Ω , I consider any four of the five given lines, say the lines 2, 3, 4, 5, and I endeavour to find the line OQ_1 through O , which has with these lines a pair of tractors. OQ_1 being a line through O , the coordinates of the line in question may be taken to

be $0, 0, 0, F_1, G_1, H_1$, (where F_1, G_1, H_1 , are in fact the coordinates x, y, z , of any point on the line OQ_1); and then the condition for the pair of tractors may be written

$$\begin{aligned} p_2a_2 + p_3a_3 + p_4a_4 + p_5a_5 &= 0, \\ p_2b_2 + p_3b_3 + p_4b_4 + p_5b_5 &= 0, \\ p_2c_2 + p_3c_3 + p_4c_4 + p_5c_5 &= 0, \\ p_2f_2 + p_3f_3 + p_4f_4 + p_5f_5 &= F_1, \\ p_2g_2 + p_3g_3 + p_4g_4 + p_5g_5 &= G_1, \\ p_2h_2 + p_3h_3 + p_4h_4 + p_5h_5 &= H_1, \end{aligned}$$

where $p_2, p_3, \dots$ are arbitrary coefficients; and we hence deduce

$$\rho F_1 + \sigma G_1 + \tau H_1 = 0;$$

but in precisely the same way, if the line OQ_2 have with the lines 1, 3, 4, 5, a pair of tractors, and if F_2, G_2, H_2 , be the coordinates of a point on the line OQ_2 , and similarly for the lines OQ_3, OQ_4, OQ_5 , and the coordinates (F_3, G_3, H_3) , (F_4, G_4, H_4) , (F_5, G_5, H_5) , we have

$$\begin{aligned} \rho F_2 + \sigma G_2 + \tau H_2 &= 0, \\ \rho F_3 + \sigma G_3 + \tau H_3 &= 0, \\ \rho F_4 + \sigma G_4 + \tau H_4 &= 0, \\ \rho F_5 + \sigma G_5 + \tau H_5 &= 0, \end{aligned}$$

and these equations show that the five lines $OQ_1, OQ_2, OQ_3, OQ_4, OQ_5$, lie in the plane

$$\rho x + \sigma y + \tau z = 0;$$

so that this plane is given as the plane through the lines $OQ_1, OQ_2, OQ_3, OQ_4, OQ_5$; and we have thus (given the lines 1, 2, 3, 4, 5, and the arbitrary point O) the construction of the line (a, b, c, f, g, h) through O in involution with the given lines.

50. The original onefold relation may be replaced by the six equations

$$\begin{aligned} \lambda a + \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 + \lambda_4 a_4 + \lambda_5 a_5 &= 0, \\ \lambda b + \lambda_1 b_1 + \lambda_2 b_2 + \lambda_3 b_3 + \lambda_4 b_4 + \lambda_5 b_5 &= 0, \\ &\vdots \\ \lambda h + \lambda_1 h_1 + \lambda_2 h_2 + \lambda_3 h_3 + \lambda_4 h_4 + \lambda_5 h_5 &= 0, \end{aligned}$$

and hence denoting as before the moments by 01, 02, 12, &c. we have

$$\begin{aligned} \lambda_1 01 + \lambda_2 02 + \lambda_3 03 + \lambda_4 04 + \lambda_5 05 &= 0, \\ \lambda 10 + \lambda_2 12 + \lambda_3 13 + \lambda_4 14 + \lambda_5 15 &= 0, \\ \lambda 20 + \lambda_1 21 + \lambda_3 23 + \lambda_4 24 + \lambda_5 25 &= 0, \\ \lambda 30 + \lambda_1 31 + \lambda_2 32 + \lambda_4 34 + \lambda_5 35 &= 0, \\ \lambda 40 + \lambda_1 41 + \lambda_2 42 + \lambda_3 43 + \lambda_5 45 &= 0, \\ \lambda 50 + \lambda_1 51 + \lambda_2 52 + \lambda_3 53 + \lambda_4 54 &= 0, \end{aligned}$$

which lead to

$$\begin{vmatrix} \cdot & 01 & 02 & 03 & 04 & 05 \\ 10 & \cdot & 12 & 13 & 14 & 15 \\ 20 & 21 & \cdot & 23 & 24 & 25 \\ 30 & 31 & 32 & \cdot & 34 & 35 \\ 40 & 41 & 42 & 43 & \cdot & 45 \\ 50 & 51 & 52 & 53 & 54 & \cdot \end{vmatrix} = 0,$$

a relation between the moments equivalent to the original onefold relation, and consequently expressing that the six lines are in involution. I have not obtained a symmetrical system of values for the ratios $\lambda : \lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 : \lambda_5$.

51. Reverting to the relation which exists between the point O and the plane Ω , it is proper to remark that, since to any given point O there corresponds a single plane Ω , and to any given plane Ω a single point O , it follows that the point O and plane Ω are reciprocal figures; viz. they are reciprocals of the particular kind treated of by Möbius, wherein the reciprocal of a point is a plane through the point, and the reciprocal of a plane a point in the plane; and of which the analytical character is that the reciprocal of the point $(\alpha, \beta, \gamma, \delta)$ is the plane

$$\begin{aligned} & (\cdot + h\beta - g\gamma + l\delta) x \\ & + (-h\alpha + \cdot + f\gamma + m\delta) y \\ & + (g\alpha - f\beta + \cdot + n\delta) z \\ & + (-l\alpha - m\beta - n\gamma + \cdot) w = 0. \end{aligned}$$

Article No. 52. A geometrical property of an involution of six lines.

52. The figure of six lines in involution is connected in various ways with the theory of cubic curves in space, for instance, considering a point A of the curve, this determines with any given line l a plane meeting the curve in two other points, and the line λ which joins these two points may be called the projection of the line l . This being so, if in any osculating plane of the cubic we have six lines $l, l_1, l_2, l_3, l_4, l_5$, tangents of a conic in that plane, the six projections $\lambda, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$ of these tangents will be a set of lines in involution. I do not stop to prove this theorem or to develop any of its consequences.

Article No. 53. To find the condition that four given lines may have a twofold tractor.

53. Taking the coordinates of the given lines to be

$$(a, b, c, f, g, h), (a_1, b_1, c_1, f_1, g_1, h_1), (a_2, b_2, c_2, f_2, g_2, h_2), (a_3, b_3, c_3, f_3, g_3, h_3),$$

then if (A, B, C, F, G, H) be the coordinates of a tractor of these lines, we have

$$\begin{aligned} & (F, G, H, A, B, C) (a, b, c, f, g, h) = 0, \\ & (F, G, H, A, B, C) (a_1, b_1, c_1, f_1, g_1, h_1) = 0, \\ & (F, G, H, A, B, C) (a_2, b_2, c_2, f_2, g_2, h_2) = 0, \\ & (F, G, H, A, B, C) (a_3, b_3, c_3, f_3, g_3, h_3) = 0. \end{aligned}$$

In virtue of these relations the ratios $A : B : C : F : G : H$ are given linear functions of any one of these ratios or of an arbitrary ratio $u : v$; and we then have $AF + BG + CH = 0$, a quadric equation for determining the unknown ratio. In the case of a twofold tractor, this equation must have equal roots; whence employing as usual the method of indeterminate multipliers, we find

$$\begin{aligned} A + \lambda a + \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3 &= 0, \\ B + \lambda b + \lambda_1 b_1 + \lambda_2 b_2 + \lambda_3 b_3 &= 0, \\ C + \lambda c + \lambda_1 c_1 + \lambda_2 c_2 + \lambda_3 c_3 &= 0, \\ F + \lambda f + \lambda_1 f_1 + \lambda_2 f_2 + \lambda_3 f_3 &= 0, \\ G + \lambda g + \lambda_1 g_1 + \lambda_2 g_2 + \lambda_3 g_3 &= 0, \\ H + \lambda h + \lambda_1 h_1 + \lambda_2 h_2 + \lambda_3 h_3 &= 0. \end{aligned}$$

Hence representing as before the moments of the pairs of lines by 01, 02, &c., we deduce

$$\begin{aligned}\lambda_1 01 + \lambda_2 02 + \lambda_3 03 &= 0, \\ \lambda 10 + \lambda_2 12 + \lambda_3 13 &= 0, \\ \lambda 20 + \lambda_1 21 + \lambda_3 23 &= 0, \\ \lambda 30 + \lambda_1 31 + \lambda_2 32 &= 0,\end{aligned}$$

so that, as already mentioned, we have

$$\begin{vmatrix} \cdot & 01 & 02 & 03 \\ 10 & \cdot & 12 & 13 \\ 20 & 21 & \cdot & 23 \\ 30 & 31 & 32 & \cdot \end{vmatrix} = 0,$$

as the condition that the four given lines may have a twofold tractor.

Article Nos. 54 to 56. Hyperboloid passing through three given lines.

54. The direct investigation is somewhat tedious; but I write down, and will afterwards verify, the equation of the hyperboloid passing through the three given lines

$$(a_1, b_1, c_1, f_1, g_1, h_1), (a_2, b_2, c_2, f_2, g_2, h_2), (a_3, b_3, c_3, f_3, g_3, h_3).$$

Writing for shortness (agh), &c. to denote the determinants

$$\begin{vmatrix} a_1 & g_1 & h_1 \\ a_2 & g_2 & h_2 \\ a_3 & g_3 & h_3 \end{vmatrix}, \text{ \&c.},$$

the equation of the hyperboloid is

$$\begin{aligned}(agh) x^2 + (bhf) y^2 + (cfg) z^2 + (abc) w^2 \\ + [(afb) + (cah)] xw + [(bfg) + (chf)] yz \\ + [(bch) - (ahf)] yw + [(cgh) + (afg)] zx \\ + [(caf) - (bcg)] zw + [(ahf) + (bgh)] xy = 0.\end{aligned}$$

In fact, we have

$$(agh) = a_1 (g_2 h_3 - g_3 h_2) - g_1 (h_2 a_3 - h_3 a_2) + h_1 (a_2 g_3 - a_3 g_2) = a \cdot gh + g \cdot ha + h \cdot ag,$$

where a , &c. stand for a_1 , &c. and gh , &c. for $g_2 h_3 - g_3 h_2$, &c. Hence the foregoing equation may be written

$$\begin{aligned}x^2 (a \cdot gh + g \cdot ha + h \cdot ag) \\ + y^2 (b \cdot hf + h \cdot fb + f \cdot bh) \\ + z^2 (c \cdot fg + f \cdot gc + g \cdot cf) \\ + w^2 (a \cdot bc + b \cdot ca + c \cdot ab) \\ + xw \begin{pmatrix} a \cdot fb + b \cdot ga + g \cdot ab \\ - c \cdot ah - a \cdot hc - h \cdot ca \end{pmatrix} + yz \begin{pmatrix} b \cdot fg + f \cdot gb + g \cdot bf \\ + c \cdot hf + h \cdot fc + f \cdot ch \end{pmatrix} \\ + yw \begin{pmatrix} b \cdot ch + c \cdot hb - h \cdot bc \\ - a \cdot bf - b \cdot fa - f \cdot ab \end{pmatrix} + zx \begin{pmatrix} c \cdot gh + g \cdot hc + h \cdot cg \\ + a \cdot fg + f \cdot ga + g \cdot af \end{pmatrix} \\ + zw \begin{pmatrix} c \cdot af + a \cdot fc + f \cdot ca \\ - b \cdot cg - c \cdot gb - g \cdot bc \end{pmatrix} + xy \begin{pmatrix} a \cdot hf + h \cdot fa + f \cdot ah \\ + b \cdot gh + g \cdot hb + h \cdot bg \end{pmatrix} = 0.\end{aligned}$$

55. This is

$$\begin{aligned}
 & bc \cdot w(hy - gz + aw) \\
 & + ca \cdot w(-hx + fz + bw) \\
 & + ab \cdot w(gx - fy + cw) \\
 & + gh \cdot x(ax + by + cz) \\
 & + hf \cdot y(ax + by + cz) \\
 & + fg \cdot z(ax + by + cz) \\
 & + af\{w(ax + by + cz) - x(hy - gz + aw)\} \\
 & + bg\{w(ax + by + cz) - y(-hx + fz + bw)\} \\
 & + ch\{w(ax + by + cz) - z(gx - fy + cw)\} \\
 & - bf \cdot y(hy - gz + aw) \\
 & - cf \cdot z(hy - gz + aw) \\
 & - cg \cdot z(-hx + fz + bw) \\
 & - ag \cdot x(-hx + fz + bw) \\
 & - ah \cdot x(gx - fy + cw) \\
 & - bh \cdot y(gx - fy + cw) = 0.
 \end{aligned}$$

56. Hence writing

$$(X, Y, Z, W) = \begin{pmatrix} \cdot & -h & g & a \\ h & \cdot & -f & b \\ -g & f & \cdot & c \\ -a & -b & -c & \cdot \end{pmatrix} (x, y, z, w),$$

the foregoing equation is

$$\begin{aligned}
 & bc \cdot wX + ca \cdot wY + ab \cdot wZ - gh \cdot xW - hf \cdot yW - fg \cdot zW \\
 & - af(wW + xX) - bg(wW + yY) - ch(wW + zZ) \\
 & - bf \cdot yX - cg \cdot zY - ah \cdot xZ - cf \cdot zX - ag \cdot xY - bh \cdot yZ = 0;
 \end{aligned}$$

or, collecting and arranging, this is

$$\begin{aligned}
 & X\{-af \cdot x - bf \cdot y - cf \cdot z + bc \cdot w\} \\
 & + Y\{-ag \cdot x - bg \cdot y - cg \cdot z + ca \cdot w\} \\
 & + Z\{-ah \cdot x - bh \cdot y - ch \cdot z + ab \cdot w\} \\
 & + W\{-gh \cdot x - hf \cdot y - fg \cdot z + (af + bg + ch)w\} = 0,
 \end{aligned}$$

which is satisfied by $X = 0, Y = 0, Z = 0, W = 0$; that is, since (a, b, c, f, g, h) have been written in place of $(a_1, b_1, c_1, f_1, g_1, h_1)$, by $X_1 = 0, Y_1 = 0, Z_1 = 0, W_1 = 0$ (if we thus denote the corresponding functions of $(a_1, b_1, c_1, f_1, g_1, h_1)$), that is, the hyperboloid passes through the line $(a_1, b_1, c_1, f_1, g_1, h_1)$; and similarly it passes through the other two lines.

Article Nos. 57 and 58. The six coordinates defined as to their absolute magnitudes.

57. In all that precedes, the absolute magnitudes of the coordinates have been left indeterminate, only the ratios being attended to. But the magnitudes of the six coordinates may be fixed in a very simple manner as follows; viz. using ordinary rectangular coordinates, then for any

line, if x_0, y_0, z_0 are the coordinates of a particular point on this line, and α, β, γ the inclinations of the line to the axes, the coordinates of another point on the line are

$$x_0 + r \cos \alpha, \quad y_0 + r \cos \beta, \quad z_0 + r \cos \gamma,$$

and hence writing

$$\begin{array}{cccc} x_0 + r \cos \alpha, & y_0 + r \cos \beta, & z_0 + r \cos \gamma, & 1, \\ x_0, & y_0, & z_0, & 1, \end{array}$$

we have

$$a : b : c : f : g : h = z_0 \cos \beta - y_0 \cos \gamma : x_0 \cos \gamma - z_0 \cos \alpha : y_0 \cos \alpha - x_0 \cos \beta : \cos \alpha : \cos \beta : \cos \gamma.$$

Or we may take

$$\begin{array}{ll} a = z_0 \cos \beta - y_0 \cos \gamma, & f = \cos \alpha, \\ b = x_0 \cos \gamma - z_0 \cos \alpha, & g = \cos \beta, \\ c = y_0 \cos \alpha - x_0 \cos \beta, & h = \cos \gamma, \end{array}$$

values which of course satisfy, as they should do, the relation $af + bg + ch = 0$. It is hardly necessary to remark, that the values of a, b, c are not altered on substituting for x_0, y_0, z_0 the coordinates $x_0 + s \cos \alpha, y_0 + s \cos \beta, z_0 + s \cos \gamma$ of any other point on the line.

58. Considering any two lines $(a, b, c, f, g, h), (a_1, b_1, c_1, f_1, g_1, h_1)$, if we define the moment of the two lines to be the product of the perpendicular distance into the sine of the inclination of the two lines, then we have, Moment

$$= af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1,$$

viz. we have now a quantitative definition of the function of the coordinates previously called the moment of the two lines.

For the demonstration of this formula it is to be remarked, that taking on the first line a segment of the length r , the coordinates of its extremities being (x_0, y_0, z_0) and $(x_0 + r \cos \alpha, y_0 + r \cos \beta, z_0 + r \cos \gamma)$, and on the second line a segment of the length r_1 , the coordinates of its extremities being (x'_0, y'_0, z'_0) and $(x'_0 + r_1 \cos \alpha_1, y'_0 + r_1 \cos \beta_1, z'_0 + r_1 \cos \gamma_1)$ and joining the extremities of these segments so as to form a tetrahedron, the volume of the tetrahedron is

$$\frac{1}{6} rr_1 (af_1 + bg_1 + ch_1 + fa_1 + gb_1 + hc_1).$$

But the volume of the tetrahedron is also equal to $\frac{1}{6}$ of the product of the opposite edges into their perpendicular distance into the sine of the inclination of the two edges¹; that is, it is $\frac{1}{6} rr_1$ into the moment of the two lines, and we have thus the formula in question.

Article Nos. 59 to 75. Statical and Kinematical Applications.

The coordinates (a, b, c, f, g, h) , as last defined, are peculiarly convenient in kinematical and mechanical questions, as will appear from the following investigations.

¹I take the opportunity of mentioning a very simple demonstration of this formula: taking the opposite edges to be r, r_1 , their inclination $= \theta$, and perpendicular distance $= h$; the section of the tetrahedron by a plane parallel to the two edges at the distances $z, h - z$ from the two edges respectively is a parallelogram, the sides of which are $\frac{rz}{h}$ and $\frac{r_1(h-z)}{h}$ respectively, and their inclination is $= \theta$; the area of the section is therefore $\frac{rr_1}{h^2} \sin \theta \cdot z(h-z)$, and the volume of the tetrahedron is $= \frac{rr_1}{h^2} \sin \theta \int_0^h z(h-z) dz = \frac{1}{6} rr_1 h \sin \theta$. The same result is however obtained still more simply by drawing a plane through one of the two edges perpendicular to the other edge; the volume is then equal to the sum or the difference of the volumes of two tetrahedra standing on a common triangular base; and the required result at once follows.

59. Using the term rotation to denote an infinitesimal rotation, I say first that a rotation λ round the line (a, b, c, f, g, h) produces in the point (x, y, z) rigidly connected with this line the displacements

$$\begin{aligned}\delta x &= \lambda(\cdot - hy + gz - a), \\ \delta y &= \lambda(hx + \cdot - fz - b), \\ \delta z &= \lambda(-gx + fy + \cdot - c).\end{aligned}$$

In fact assuming for a moment that the axis of rotation passes through the origin, then for the point P coordinates x, y, z , the square of the perpendicular distance from the axis is

$$\begin{aligned}(\cdot - y \cos \gamma + z \cos \beta)^2 \\ + (x \cos \gamma + \cdot - z \cos \alpha)^2 \\ + (-x \cos \beta + y \cos \alpha + \cdot)^2,\end{aligned}$$

and the expressions which enter into this formula denote as follows; viz. if through the point P at right angles to the plane through P and the axis of rotation we draw a line PQ , = perpendicular distance of P from the axis of rotation, then the coordinates of Q referred to P as origin are

$$\begin{aligned}\cdot - y \cos \gamma + z \cos \beta, \\ x \cos \gamma + \cdot - z \cos \alpha, \\ - x \cos \beta + y \cos \alpha + \cdot,\end{aligned}$$

respectively. Hence the foregoing quantities each multiplied by λ are the displacements of the point P in the directions of the axes, produced by the rotation λ .

60. Suppose that the axis of rotation (instead of passing through the origin) pass through the point (x_0, y_0, z_0) ; the only difference is that we must in the formula write $(x - x_0, y - y_0, z - z_0)$ in place of (x, y, z) : and attending to the significations of the six coordinates, it thus appears that the displacements produced by the rotation are equal to λ into the expressions

$$\begin{aligned}\cdot - hy + gz - a, \\ hx + \cdot - fz - b, \\ - gx + fy + \cdot - c,\end{aligned}$$

respectively; which is the theorem in question.

61. I say secondly that considering in a solid body the point (x, y, z) situate in the line (a, b, c, f, g, h) , and writing

$$a, b, c, f, g, h = z \cos \beta - y \cos \gamma, \quad x \cos \gamma - z \cos \alpha, \quad y \cos \alpha - x \cos \beta, \quad \cos \alpha, \quad \cos \beta, \quad \cos \gamma,$$

then for any infinitesimal motion of the solid body the displacement of the point in the direction of the line is

$$= ap + bq + cr + fl + gm + hn,$$

where p, q, r, l, m, n are constants depending on the infinitesimal motion.

In fact for any infinitesimal motion of a solid body the displacements of the point (x, y, z) are

$$\begin{aligned}\delta x &= l + \cdot + ry - qz, \\ \delta y &= m - rx + \cdot + pz, \\ \delta z &= n + qx - py + \cdot,\end{aligned}$$

and hence the displacement in the direction of the line is

$$= \cos \alpha \delta x + \cos \beta \delta y + \cos \gamma \delta z,$$

which attending to the significations of (a, b, c, f, g, h) is

$$= ap + bq + cr + fl + gm + hn,$$

and we have thus the theorem in question.

62. It thus appears that for a system of rotations

$$\begin{aligned} \lambda_1 & \text{ about the line } (a_1, b_1, c_1, f_1, g_1, h_1), \\ \lambda_2 & \quad , , \quad (a_2, b_2, c_2, f_2, g_2, h_2), \\ \&c. & \quad , , \quad \&c. \end{aligned}$$

the displacements of the point (x, y, z) rigidly connected with the several lines are

$$\begin{aligned} \delta x &= \quad \cdot -y \Sigma h \lambda + z \Sigma g \lambda - \Sigma a \lambda, \\ \delta y &= x \Sigma h \lambda + \quad \cdot -z \Sigma f \lambda - \Sigma b \lambda, \\ \delta z &= -x \Sigma g \lambda + y \Sigma f \lambda + \quad \cdot - \Sigma c \lambda, \end{aligned}$$

and when the rotations are in equilibrium then the displacements $(\delta x, \delta y, \delta z)$ of any point (x, y, z) whatever must each of them vanish; that is, we must have

$$\Sigma \lambda a = 0, \quad \Sigma \lambda b = 0, \quad \Sigma \lambda c = 0, \quad \Sigma \lambda f = 0, \quad \Sigma \lambda g = 0, \quad \Sigma \lambda h = 0,$$

which are therefore the conditions for the equilibrium of the system of rotations.

63. And it further appears that for a system of forces acting on a rigid body,

$$\begin{aligned} \lambda_1 & \text{ along the line } (a_1, b_1, c_1, f_1, g_1, h_1), \\ \lambda_2 & \quad , , \quad (a_2, b_2, c_2, f_2, g_2, h_2), \\ \&c. & \end{aligned}$$

the conditions of equilibrium as given by the Principle of Virtual Velocities is

$$\Sigma \lambda (ap + bq + cr + fl + gm + hn) = 0,$$

or what is the same thing, that we have

$$\Sigma \lambda a = 0, \quad \Sigma \lambda b = 0, \quad \Sigma \lambda c = 0, \quad \Sigma \lambda f = 0, \quad \Sigma \lambda g = 0, \quad \Sigma \lambda h = 0,$$

for the conditions of equilibrium of the system of forces $\lambda_1, \lambda_2, \&c.$ The conditions of equilibrium are thus precisely the same in the case of a system of rotations (infinitesimal rotations) and in that of a system of forces.

64. It now appears that the greater portion of the investigations in the first part of the present paper are applicable, and may be considered as relating, to the equilibrium of forces (or of rotations; but as the two theories are identical, it is sufficient to attend to one of them), and that we have in effect solved the following question, “Given any system of two, three, four, five or six lines considered as belonging to a solid body, to determine the relations between these lines in order that there may exist along them forces which are in equilibrium;” but for greater clearness I will consider the several cases in order; it is hardly necessary to remark that when the forces exist the equilibrium will depend on the ratios only, and that the absolute magnitude of any one of the forces may be assumed at pleasure.

65. The condition in the case of two lines is of course that these shall coincide together, or form one and the same line; and the forces are then equal and opposite forces.

66. In the case of three lines, these must meet in a point and lie in a plane; and the force along each line must then be as the sine of the angle between the other two lines.

67. Supposing that the forces are λ along the line (a, b, c, f, g, h) , λ_1 along the line $(a_1, b_1, c_1, f_1, g_1, h_1)$, and λ_2 along the line $(a_2, b_2, c_2, f_2, g_2, h_2)$, the conditions of equilibrium are $\lambda a + \lambda_1 a_1 + \lambda_2 a_2 = 0, \dots \lambda h + \lambda_1 h_1 + \lambda_2 h_2 = 0$, any two of which determine the ratios $\lambda : \lambda_1 : \lambda_2$; these ratios were not worked out *ante* No. 38 for the reason that with the coordinates there made use of, a symmetrical solution was not obtainable; but in the present case, selecting the last three equations, these are

$$\begin{aligned}\lambda \cos \alpha + \lambda_1 \cos \alpha_1 + \lambda_2 \cos \alpha_2 &= 0, \\ \lambda \cos \beta + \lambda_1 \cos \beta_1 + \lambda_2 \cos \beta_2 &= 0, \\ \lambda \cos \gamma + \lambda_1 \cos \gamma_1 + \lambda_2 \cos \gamma_2 &= 0,\end{aligned}$$

giving in the first instance an equation which expresses that the three lines (assumed to meet in a point) lie in the same plane: and then if 01, 02, 12 be the angles between the pairs of lines respectively, giving by an easy transformation

$$\begin{aligned}\lambda + \lambda_1 \cos 01 + \lambda_2 \cos 02 &= 0, \\ \lambda \cos 10 + \lambda_1 + \lambda_2 \cos 12 &= 0, \\ \lambda \cos 20 + \lambda_1 \cos 21 + \lambda_2 &= 0.\end{aligned}$$

68. Putting for shortness A, B, C in the place of 12, 20, 01 respectively, we thence find

$$\begin{vmatrix} 1 & \cos C & \cos B \\ \cos C & 1 & \cos A \\ \cos B & \cos A & 1 \end{vmatrix} = 0,$$

that is

$$1 - \cos^2 A - \cos^2 B - \cos^2 C + 2 \cos A \cos B \cos C = 0,$$

equivalent to $A + B + C = 2\pi$; and then from the first and second equations

$$\begin{aligned}\lambda : \lambda_1 : \lambda_2 &= \cos A - \cos B \cos C : \cos B - \cos C \cos A : 1 - \cos^2 C, \\ &= \sin A \sin C : \sin B \sin C : \sin^2 C, \\ &= \sin A : \sin B : \sin C,\end{aligned}$$

which is the required formula.

69. In the case of four given lines the condition (as noticed by Mobius) is that the four lines shall be generating lines of the same hyperboloid. In fact every line which meets three of the four lines must also meet the fourth line; for otherwise the moment of the system about such line would not be $= 0$. Calling the lines 0, 1, 2, 3 and writing as before 01, 02, &c. for the moments of the several pairs of lines, then taking the moments of the system about the four lines respectively, we obtain directly the before-mentioned system of equations

$$\begin{aligned}\lambda_1 01 + \lambda_2 02 + \lambda_3 03 &= 0, \\ \lambda 10 + \lambda_2 12 + \lambda_3 13 &= 0, \\ \lambda 20 + \lambda_1 21 + \lambda_3 23 &= 0, \\ \lambda 30 + \lambda_1 31 + \lambda_2 32 + &= 0,\end{aligned}$$

leading as before to the relation

$$\sqrt{01} \sqrt{23} + \sqrt{02} \sqrt{31} + \sqrt{03} \sqrt{12} = 0,$$

and to the values

$$\lambda : \lambda_1 : \lambda_2 : \lambda_3 = \sqrt{12} \sqrt{23} \sqrt{31} : \sqrt{23} \sqrt{30} \sqrt{02} : \sqrt{30} \sqrt{01} \sqrt{13} : \sqrt{01} \sqrt{12} \sqrt{20}$$

for the proportional magnitudes of the forces. These last equations give

$$\lambda \lambda_1 01 = \lambda_2 \lambda_3 23,$$

which, representing each force by a segment on the line along which the force acts, denotes that the tetrahedron of any two of the forces is equal to the tetrahedron of the other two forces; this is in fact equivalent to the theorem of M. Chasles, that if a system of forces be in any manner whatever reduced to two forces, the tetrahedron formed by these two forces has a constant volume.

70. In the case of five given lines, the lines must have a pair of tractors. Any four of the lines have in fact two tractors; and each of these tractors must meet the fifth line, for otherwise the moment of the system about the tractor would not be $= 0$. In the case where the four lines have a twofold tractor, the foregoing consideration shows only that the fifth line meets the twofold tractor, but it fails to show that the twofold tractor is a twofold tractor in regard to the fifth line.

71. I stop to consider this particular case under the present statical point of view. Taking the twofold tractor for the axis of z ; let the line meet this line in the point $(0, 0, c)$, the coordinates (a, b, c, f, g, h) of this line being consequently

$$(-c \cos \beta, c \cos \alpha, 0, \cos \alpha, \cos \beta, \cos \gamma)$$

and the like for the other four lines 1, 2, 3, 4. Using the sign Σ to refer to the last-mentioned four lines the equations of equilibrium become

$$\begin{aligned} -\lambda c \cos \beta + \Sigma \lambda_i c_i \cos \beta_i &= 0, \\ \lambda c \cos \alpha + \Sigma \lambda_i c_i \cos \alpha_i &= 0, \\ \lambda \cos \alpha + \Sigma \lambda_i \cos \alpha_i &= 0, \\ \lambda \cos \beta + \Sigma \lambda_i \cos \beta_i &= 0, \\ \lambda \cos \gamma + \Sigma \lambda_i \cos \gamma_i &= 0. \end{aligned}$$

These equations give

$$\frac{\Sigma \lambda_i c_i \cos \beta_i}{\Sigma \lambda_i \cos \alpha_i} = \frac{c \cos \beta}{\cos \alpha},$$

we may without loss of generality take the homographic conditions which express that the axis of z is a twofold tractor of the four lines to be

$$\frac{c_1 \cos \beta_1}{\cos \alpha_1} = \frac{c_2 \cos \beta_2}{\cos \alpha_2} = \frac{c_3 \cos \beta_3}{\cos \alpha_3} = \frac{c_4 \cos \beta_4}{\cos \alpha_4},$$

and this being so, the last-mentioned equation becomes

$$\frac{c \cos \beta}{\cos \alpha},$$

and it thus appears that the axis of z is a twofold tractor in regard also to the line 0.

72. In the case of six lines such that there exist along them forces which are in equilibrium, taking this as a definition of the involution of six lines, we may very readily obtain from statical considerations the before-mentioned construction of the sixth line; viz. it may be shown that given any five of the lines, say the lines 1, 2, 3, 4, 5, and a point O , we can through the point O determine a plane Ω , such that any line whatever through the point O and in the plane Ω is in involution with the five given lines. Consider the tractors of any four of the lines, say the lines 2, 3, 4, 5; we may through the point O draw a line OA meeting the two tractors; that is, the lines 2, 3, 4, 5 and the line OA will have a pair of common tractors. There consequently exist along these lines forces which are in equilibrium; and since only the ratios are material, the absolute magnitude of the force along the line OA may be anything whatever. Similarly, considering the tractors of the lines 1, 3, 4, 5, and through O a line OB meeting these tractors, then there exist along the lines 1, 3, 4, 5 and the line OB forces which are in equilibrium, and the absolute magnitude of the force along the line OB may be anything whatever. Hence, combining the two sets of forces, we have, along a line through O in the plane OA, OB , but otherwise indeterminate in its direction, a force in equilibrium with forces along the lines 1, 2, 3, 4, 5; that is, the line found as above is a line in involution with the lines 1, 2, 3, 4, 5.

73. It is to be added, that through O we cannot, out of the plane OA, OB , draw a line in involution with the lines 1, 2, 3, 4, 5; for if any such line OK existed, then we should have along each of the lines OA, OB, OK forces in equilibrium with forces along the lines 1, 2, 3, 4, 5; and the magnitudes of the three forces being each of them anything whatever, it would follow that along any line whatever through the point O there would exist a force in equilibrium with forces along the lines 1, 2, 3, 4, 5; that is, any line whatever through the point O would be a line in involution with these lines.

74. It hence appears that drawing OA to meet the tractors of 2, 3, 4, 5; OB to meet those of 3, 4, 5, 1; OC to meet those of 4, 5, 1, 2; OD to meet those of 5, 1, 2, 3; and OE to meet those of 1, 2, 3, 4; the lines OA, OB, OC, OD, OE will be in one plane, say the plane Ω : and that any line through O in the plane Ω will be a line in involution with the lines 1, 2, 3, 4, 5.

75. There is another statical representation of the involution of six lines. If a system of forces act on a solid body, then taking six lines at random, the system will be in equilibrium if the sum of the moments be $= 0$ in regard to each of the six lines. But if the six lines be in involution; then, for the very reason that a rotation about one of these lines is resolvable into rotations about the other five lines, if the sum of the moments be $= 0$ for each of the five lines, it will also be $= 0$ for the sixth line: that is, it is not sufficient for the equilibrium of the forces that the sum of the moments shall be $= 0$ for each of the six lines. And we thus see that six lines in involution are lines such that the equilibrium of a system of forces about each of the six lines as axes does not insure the equilibrium of the system.

Article Nos. 76 and 77. Transformation of Coordinates.

76. Reverting to the general definition of the six coordinates (a, b, c, f, g, h) of a line by means of the points $(\alpha, \beta, \gamma, \delta)$ and $(\alpha', \beta', \gamma', \delta')$ on the line; suppose that instead of the original coordinate planes $x = 0, y = 0, z = 0, w = 0$ (forming a tetrahedron $ABCD$) we have new coordinate planes $x_0 = 0, y_0 = 0, z_0 = 0, w_0 = 0$ (forming a tetrahedron $A_0B_0C_0D_0$); and that

the relations between the two sets of current coordinates are given by the equations

$$\begin{aligned} x &= (\lambda_1, \mu_1, \nu_1, \rho_1) (x_0, y_0, z_0, w_0) \\ y &= (\lambda_2, \mu_2, \nu_2, \rho_2) (x_0, y_0, z_0, w_0) \\ z &= (\lambda_3, \mu_3, \nu_3, \rho_3) (x_0, y_0, z_0, w_0) \\ w &= (\lambda_4, \mu_4, \nu_4, \rho_4) (x_0, y_0, z_0, w_0), \end{aligned}$$

with, of course, the like relations between the original coordinates $(\alpha, \beta, \gamma, \delta)$ and new coordinates $(\alpha_0, \beta_0, \gamma_0, \delta_0)$, and between the original coordinates $(\alpha', \beta', \gamma', \delta')$ and the new coordinates $(\alpha'_0, \beta'_0, \gamma'_0, \delta'_0)$, of the two points on the line (a, b, c, f, g, h) ; then taking $(a_0, b_0, c_0, f_0, g_0, h_0)$ as the new values of the six coordinates of the line, viz. writing

$$\begin{aligned} & a_0, & b_0, & c_0, & f_0, & g_0, & h_0 \\ = & \beta_0 \gamma'_0 - \beta'_0 \gamma_0, & \gamma_0 \alpha'_0 - \gamma'_0 \alpha_0, & \alpha_0 \beta'_0 - \alpha'_0 \beta_0, & \alpha_0 \delta'_0 - \alpha'_0 \delta_0, & \beta_0 \delta'_0 - \beta'_0 \delta_0, & \gamma_0 \delta'_0 - \gamma'_0 \delta_0, \end{aligned}$$

we obtain a system of formulae which may be conveniently written as follows:

$$\begin{aligned} a : b : c : f : g : h = \\ & \frac{\mu\nu}{23} a_0 + \frac{\nu\lambda}{23} b_0 + \frac{\lambda\mu}{23} c_0 + \frac{\lambda\rho}{23} f_0 + \frac{\mu\rho}{23} g_0 + \frac{\nu\rho}{23} h_0 \\ & \text{(suffix 31)} \\ & \text{(suffix 12)} \\ & \text{(suffix 14)} \\ & \text{(suffix 24)} \\ & \text{(suffix 34)} \end{aligned}$$

viz. the top line stands for $(\mu_2\nu_3 - \mu_3\nu_2) a_0 + (\nu_2\lambda_3 - \nu_3\lambda_2) b_0 + \&c.$, and the other lines are obtained from this by mere alterations of the suffixes.

77. As to the interpretation of these formulae, taking $ABCD$ as the fundamental tetrahedron for (x, y, z, w) , then

$$\begin{aligned} (\lambda_1, \mu_1, \nu_1, \rho_1) (x_0, y_0, z_0, w_0) &= 0 \text{ is the equation of plane } BCD, \\ (\lambda_2, \mu_2, \nu_2, \rho_2) (\quad, \quad, \quad) &= 0 \quad, \quad CDA, \\ (\lambda_3, \mu_3, \nu_3, \rho_3) (\quad, \quad, \quad) &= 0 \quad, \quad DAB, \end{aligned}$$

whence, observing that the second and third equations belong to two planes each passing through the line DA , it appears that the coefficients

$$\frac{\mu\nu}{23}, \frac{\nu\lambda}{23}, \frac{\lambda\mu}{23}, \frac{\lambda\rho}{23}, \frac{\mu\rho}{23}, \frac{\nu\rho}{23}$$

are the six coordinates of the line DA , expressed in regard to the tetrahedron $(A_0B_0C_0D_0)$; and similarly that the coefficients in the six expressions of the transformation formula are the six coordinates of the lines AD, BD, CD, BC, CA, AB respectively in regard to the tetrahedron $(A_0B_0C_0D_0)$.

In the preceding formulae for the transformation of coordinates the ratios only have been attended to, no determinate absolute magnitudes have been assigned to the coordinates (a, b, c, f, g, h) . But I will nevertheless show how we may attribute absolute magnitudes to these coordinates.

Article Nos. 78 to 80. New definition of the six coordinates as to their absolute magnitudes.

78. I assume (x, y, z, w) to be “volume” coordinates; viz. taking as before $ABCD$ for the fundamental tetrahedron, and denoting the point (x, y, z, w) by P , I assume that we have

$$x : y : z : w : 1 = PBCD : APCD : ABPD : ABCP : ABCD,$$

where $PBCD$, &c. denote the volumes of the several tetrahedra $PBCD$, &c. It is to be noticed that the volume is in every case taken with a determinate sign: analytically the sign may be fixed by taking (x_a, y_a, z_a) , &c., as the Cartesian coordinates of the points A , &c. and writing

$$PBCD = \begin{vmatrix} x_p & x_b & x_c & x_d \\ y_p & y_b & y_c & y_d \\ z_p & z_b & z_c & z_d \\ 1 & 1 & 1 & 1 \end{vmatrix}, \text{ \&c.}$$

(whence of course $PBCD = PCDA = -PCBD$, &c. according to the rule of signs): or we may in an equivalent manner, but less easily, determine the sign, by considering the sense of the rotation about CD (considered as an axis drawn from C to D) which would be produced by a force along PB (from P to B).

79. It is to be observed that the foregoing values give identically $x + y + z + w = 1$, so that the equation of the plane infinity is $x + y + z + w = 0$. The values of the coordinates (x, y, z, w) may be written

$$x : y : z : w : 1 = PBCD : PCAD : PABD : PCBA : ABCD;$$

or in the original form

$$x : y : z : w : 1 = PBCD : APCD : ABPD : ABCP : ABCD,$$

as may be most convenient.

80. Denoting the points $(\alpha, \beta, \gamma, \delta)$ and $(\alpha', \beta', \gamma', \delta')$ by Q, Q' respectively, we have

$$\alpha : \beta : \gamma : \delta : 1 = QBCD : AQCD : ABQD : ABCQ : ABCD,$$

and

$$\alpha' : \beta' : \gamma' : \delta' : 1 = Q'BCD : AQ'CD : ABQ'D : ABCQ' : ABCD,$$

and writing

$$(a, b, c, f, g, h) = (\beta\gamma' - \beta'\gamma, \gamma\alpha' - \gamma'\alpha, \alpha\beta' - \alpha'\beta, \alpha\delta' - \alpha'\delta, \beta\delta' - \beta'\delta, \gamma\delta' - \gamma'\delta),$$

viz. the two sets being taken to be equal, $a = \beta\gamma' - \beta'\gamma$, &c. instead of merely proportional, then it is easily seen that we obtain

$$a : b : c : f : g : h : 1 = AQQ'D : Q'BQD : QQ'CD : QBCQ' : AQCQ' : ABQQ' : ABCD,$$

that is, in order to form the first six combinations we successively replace

$$(B, C), (C, A), (A, B), (A, D), (B, D), (C, D)$$

in $ABCD$ by (Q, Q') .

Article No. 81. Resulting formulae of Transformation.

81. For the transformation of coordinates if we assume

$$\begin{aligned} x &= (\lambda_1, \mu_1, \nu_1, \rho_1) (x_0, y_0, z_0, w_0), \\ y &= (\lambda_2, \mu_2, \nu_2, \rho_2) (\quad, \quad, \quad), \\ z &= (\lambda_3, \mu_3, \nu_3, \rho_3) (\quad, \quad, \quad), \\ w &= (\lambda_4, \mu_4, \nu_4, \rho_4) (\quad, \quad, \quad), \end{aligned}$$

and take also (a, b, c, f, g, h) , $(a_0, b_0, c_0, f_0, g_0, h_0)$ respectively equal, instead of merely proportional, to the foregoing values, then, observing that for the point A_0 we have $(x_0, y_0, z_0, w_0) = (1, 0, 0, 0)$ we see that $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are the $A_0(BCD)$ coordinates of A_0 ; and the like as to the other sets of coefficients; viz. we have

$$\begin{aligned} \lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 : 1 &= A_0BCD : AA_0CD : ABA_0D : ABCA_0 : ABCD, \\ \mu_1 : \mu_2 : \mu_3 : \mu_4 : 1 &= B_0\cdot, \cdot, \cdot, \cdot, \cdot B_0\cdot, \cdot, \cdot, \cdot, \cdot B_0\cdot, \cdot, \cdot, \cdot, \cdot B_0 : \cdot, \cdot, \cdot, \\ \nu_1 : \nu_2 : \nu_3 : \nu_4 : 1 &= C_0\cdot, \cdot, \cdot, \cdot, \cdot C_0\cdot, \cdot, \cdot, \cdot, \cdot C_0\cdot, \cdot, \cdot, \cdot, \cdot C_0 : \cdot, \cdot, \cdot, \\ \rho_1 : \rho_2 : \rho_3 : \rho_4 : 1 &= D_0\cdot, \cdot, \cdot, \cdot, \cdot D_0\cdot, \cdot, \cdot, \cdot, \cdot D_0\cdot, \cdot, \cdot, \cdot, \cdot D_0 : \cdot, \cdot, \cdot, \end{aligned}$$

and we hence find

$$\begin{aligned} &\frac{\mu\nu}{23} : \frac{\nu\lambda}{23} : \frac{\lambda\mu}{23} : \frac{\lambda\rho}{23} : \frac{\mu\rho}{23} : \frac{\nu\rho}{23} : 1 \\ &= AB_0C_0D : AC_0A_0D : AA_0B_0D : AA_0D_0D : AB_0D_0D : AC_0D_0D : ABCD \end{aligned}$$

viz. multiplying the last-mentioned set of terms by $A_0B_0C_0D_0 \div ABCD$, in order to make the last term equal to unity, we see that the coefficients $\frac{\mu\nu}{23}$, &c. are equal to $\frac{A_0B_0C_0D_0}{ABCD}$ into the six $(A_0B_0C_0D_0)$ -coordinates respectively of the line AD by means of the points A, D thereof. And similarly in the six expressions which enter into the formula of transformation, the coefficients are $\frac{A_0B_0C_0D_0}{ABCD}$ into the six $(A_0B_0C_0D_0)$ -coordinates of the

line AD in regard to points A, D thereof

$$\begin{array}{ll} \cdot, BD & \cdot, B, D \\ \cdot, CD & \cdot, C, D \\ \cdot, BC & \cdot, B, C \\ \cdot, CA & \cdot, C, A \\ \cdot, AB & \cdot, A, B \end{array}$$

The foregoing theory of the transformation of coordinates seemed to me interesting for its own sake, and I have developed it in preference to the more simple theory which might easily be established of the case in which the coordinates are quantitatively defined as being equal to

$$(z_0 \cos \beta - y_0 \cos \gamma, x_0 \cos \gamma - z_0 \cos \alpha, y_0 \cos \alpha - x_0 \cos \beta, \cos \alpha, \cos \beta, \cos \gamma)$$

respectively.

[436] ON A CERTAIN SEXTIC TORSE.

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The torse (developable surface) intended to be considered is that which has for its edge of regression an excubo-quartic curve, or say a unicursal quartic curve. I call to mind that (excluding the plane quartic) a quartic curve is either a quadriquadric, viz. it is the complete intersection of two quadric surfaces; or else it is an excubo-quartic, viz. there is through the curve only one quadric surface, and the curve is the partial intersection of this quadric surface with a cubic surface through two generating lines (of the same kind) of the quadric surface. Returning to the quadriquadric curve, this may be general, nodal, or cuspidal; viz. if the two quadric surfaces have an ordinary contact, the curve of intersection is a nodal quadriquadric; if they have a stationary contact, the curve is a cuspidal quadriquadric.

The unicursal quartic is a curve such that the coordinates (x, y, z, w) of any point thereof are proportional to rational and integral quartic functions $(\theta, 1)^4$ of a variable parameter θ ; and the general unicursal quartic is in fact the excubo-quartic; but included as particular cases of the unicursal curve (although not as cases of the excubo-quartic as above defined) we have the nodal quadriquadric and the cuspidal quadriquadric. The torse having for its edge of regression a unicursal curve is a sextic torse; and this is in fact the order of the torse derived from the excubo-quartic, and from the nodal quadriquadric; but for the cuspidal quadriquadric, there is a depression of one, and the torse becomes a quintic torse. The equations have been obtained of (1) the sextic torse derived from the nodal quadriquadric, (2) the quintic torse derived from the cuspidal quadriquadric, (3) the sextic torse derived from a certain special excubo-quartic; but the equation of the torse derived from the general unicursal quartic has not yet been found. To show at the outset what the analytical problem is, I anticipate the remark that the coordinates (x, y, z, w) of a point on the curve may by an obvious reduction be rendered proportional to the fourth powers $(\theta + \alpha)^4, (\theta + \beta)^4, (\theta + \gamma)^4, (\theta + \delta)^4$ in the parameter θ ; this leads to an equation

$$\frac{x}{(\theta + \alpha)^4} + \frac{y}{(\theta + \beta)^4} + \frac{z}{(\theta + \gamma)^4} + \frac{w}{(\theta + \delta)^4} = 0,$$

for the osculating plane at the point (x, y, z, w) ; or observing that this equation, when integralised, is of the form $(x, y, z, w)(\theta, 1)^{12} = 0$, we see that the equation is obtained by equating to zero the discriminant of a certain sextic function in θ ; the discriminant is of the order 10 in the coordinates (x, y, z, w) , but it obviously contains the factor $xyzw$, or throwing this out we have an equation of the order 6, so that the torse is (as above stated) a sextic torse.

Theorem relating to Four Binary Quartics.

1. Consider the four quartics:

$$\begin{aligned} &(a_1, b_1, c_1, d_1, e_1)(x, y)^4, \\ &(a_2, b_2, c_2, d_2, e_2)(x, y)^4, \\ &(a_3, b_3, c_3, d_3, e_3)(x, y)^4, \\ &(a_4, b_4, c_4, d_4, e_4)(x, y)^4, \end{aligned}$$

then if $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are any four quantities, these may be determined, and that in four different ways, so that

$$\lambda_1 (a_1, \dots)(x, y)^4 + \lambda_2 (a_2, \dots)(x, y)^4 + \lambda_3 (a_3, \dots)(x, y)^4 + \lambda_4 (a_4, \dots)(x, y)^4 = (\beta x + \alpha y)^4,$$

a perfect fourth power; in fact, equating the coefficients of the different powers of $(x, y)^4$, we have five equations, which determine the ratios of the unknown quantities $\lambda_1, \lambda_2, \lambda_3, \lambda_4; \alpha, \beta$: eliminating $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, we find the equation

$$\begin{vmatrix} \beta^4 & 4\beta^3\alpha & 6\beta^2\alpha^2 & 4\beta\alpha^3 & \alpha^4 \\ a_1, & b_1, & c_1, & d_1, & e_1 \\ a_2, & b_2, & c_2, & d_2, & e_2 \\ a_3, & b_3, & c_3, & d_3, & e_3 \\ a_4, & b_4, & c_4, & d_4, & e_4 \end{vmatrix} = 0,$$

giving four different values of the ratio $\alpha : \beta$; or, assigning at pleasure a value to α or β (say $\beta = 1$), then to each of the four sets of values of (α, β) there correspond a determinate set of values of $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$; that is, we have as stated four sets of values of $\lambda_1, \lambda_2, \lambda_3, \lambda_4; \alpha, \beta$.

[436] ON A CERTAIN SEXTIC TORSE (CONTINUED).

Standard Equation of the Unicursal Quartic.

2. The coordinates (x, y, z, w) being originally taken to be proportional to any four given quartic functions $(\ast \theta, 1)^4$ of the parameter θ , then forming a linear function of the coordinates, we have four sets of values of the multipliers, each reducing the function of θ to a perfect fourth power; that is, writing (X, Y, Z, W) for the linear functions of the original coordinates, and taking (X, Y, Z, W) as coordinates, it appears that the unicursal quartic may be represented by the equations

$$X : Y : Z : W = (\theta + \alpha)^4 : (\theta + \beta)^4 : (\theta + \gamma)^4 : (\theta + \delta)^4.$$

Tangent Line, and Osculating Plane of the Unicursal Quartic.

3. The equations of the tangent line at the point (θ) (that is, the point the coordinates whereof are as $(\theta + \alpha)^4 : (\theta + \beta)^4 : (\theta + \gamma)^4 : (\theta + \delta)^4$) are at once seen to be

$$\begin{vmatrix} X, & Y, & Z, & W \\ (\theta + \alpha)^4, & (\theta + \beta)^4, & (\theta + \gamma)^4, & (\theta + \delta)^4 \\ (\theta + \alpha)^3, & (\theta + \beta)^3, & (\theta + \gamma)^3, & (\theta + \delta)^3 \end{vmatrix} = 0,$$

and that of the osculating plane to be

$$\begin{vmatrix} X, & Y, & Z, & W \\ (\theta + \alpha)^4, & (\theta + \beta)^4, & (\theta + \gamma)^4, & (\theta + \delta)^4 \\ (\theta + \alpha)^3, & (\theta + \beta)^3, & (\theta + \gamma)^3, & (\theta + \delta)^3 \\ (\theta + \alpha)^2, & (\theta + \beta)^2, & (\theta + \gamma)^2, & (\theta + \delta)^2 \end{vmatrix} = 0.$$

Writing as in the sequel

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, \\ b &= \gamma - \alpha, & g &= \beta - \delta, \\ c &= \alpha - \beta, & h &= \gamma - \delta, \end{aligned}$$

the equations of the tangent line become

$$\begin{aligned} \frac{hY}{(\theta + \beta)^4} - \frac{gZ}{(\theta + \gamma)^4} + \frac{aW}{(\theta + \delta)^4} &= 0, \\ -\frac{hX}{(\theta + \alpha)^4} + \frac{fZ}{(\theta + \gamma)^4} + \frac{bW}{(\theta + \delta)^4} &= 0, \\ \frac{gX}{(\theta + \alpha)^4} - \frac{fY}{(\theta + \beta)^4} + \frac{cW}{(\theta + \delta)^4} &= 0, \\ -\frac{aX}{(\theta + \alpha)^4} - \frac{bY}{(\theta + \beta)^4} - \frac{cZ}{(\theta + \gamma)^4} &= 0 \end{aligned}$$

(equivalent of course to two equations), and the equation of the osculating plane becomes

$$\frac{ahgX}{(\theta + \alpha)^2} + \frac{hbfY}{(\theta + \beta)^2} + \frac{gfcZ}{(\theta + \gamma)^2} + \frac{abcW}{(\theta + \delta)^2} = 0.$$

Modification of the foregoing notation, and final form for the Unicursal Quartic.

4. If instead of the coordinates (X, Y, Z, W) we introduce the coordinates (x, y, z, w) connected therewith by the relations

$$ahgX : hbfY : gfcZ : abcW = x : y : z : w,$$

or, what is the same thing,

$$X : Y : Z : W = bcfx : cagy : abhz : fghw,$$

then the curve is given by the equations

$$x : y : z : w = ahg(\theta + \alpha)^4 : hbf(\theta + \beta)^4 : cfg(\theta + \gamma)^4 : abc(\theta + \delta)^4.$$

The equations of the tangent line are

$$\begin{aligned} \frac{cy}{(\theta + \beta)^2} - \frac{bz}{(\theta + \gamma)^2} + \frac{fw}{(\theta + \delta)^2} &= 0, \\ -\frac{cx}{(\theta + \alpha)^2} + \frac{az}{(\theta + \gamma)^2} + \frac{gw}{(\theta + \delta)^2} &= 0, \\ \frac{bx}{(\theta + \alpha)^2} - \frac{ay}{(\theta + \beta)^2} + \frac{hw}{(\theta + \delta)^2} &= 0, \\ -\frac{fx}{(\theta + \alpha)^2} - \frac{gy}{(\theta + \beta)^2} - \frac{hz}{(\theta + \gamma)^2} &= 0, \end{aligned}$$

and the equation of the osculating plane is

$$\frac{x}{(\theta + \alpha)^2} + \frac{y}{(\theta + \beta)^2} + \frac{z}{(\theta + \gamma)^2} + \frac{w}{(\theta + \delta)^2} = 0.$$

Determination of the Sextic Torse.

5. Starting from the equation of the osculating plane written under the form

$$\begin{aligned} &x(\theta + \beta)^2(\theta + \gamma)^2(\theta + \delta)^2 \\ &+ y(\theta + \gamma)^2(\theta + \delta)^2(\theta + \alpha)^2 \\ &+ z(\theta + \delta)^2(\theta + \alpha)^2(\theta + \beta)^2 \\ &+ w(\theta + \alpha)^2(\theta + \beta)^2(\theta + \gamma)^2 = 0, \end{aligned}$$

the equation of the torse is obtained by equating to zero the discriminant of the sextic function. Writing as before

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, \\ b &= \gamma - \alpha, & g &= \beta - \delta, \\ c &= \alpha - \beta, & h &= \gamma - \delta, \end{aligned}$$

equations which give

$$\begin{aligned} h \cdot -g \cdot +a \cdot &= 0, \\ -h \cdot +f \cdot +b \cdot &= 0, \\ g \cdot -f \cdot +c \cdot &= 0, \\ -a \cdot -b \cdot -c \cdot &= 0, \end{aligned}$$

$$\begin{aligned} h\beta - g\gamma + a\delta &= 0, \\ -h\alpha + f\gamma + b\delta &= 0, \\ g\alpha - f\beta + c\delta &= 0, \\ -a\alpha - b\beta - c\gamma &= 0, \end{aligned}$$

and also

$$af + bg + ch = 0,$$

then the discriminant is a function of (x, y, z, w) , (a, b, c, f, g, h) of the degree 10 in (x, y, z, w) and the degree 30 in (a, b, c, f, g, h) . But the equation in θ has two equal roots, or the discriminant vanishes, if any one of the quantities (x, y, z, w) is $= 0$; and again, if any one of the differences $\alpha - \beta$, &c. (that is any one of the quantities a, b, c, f, g, h) is $= 0$: the discriminant thus contains the factors $xyzw$ and $(abcfgh)^2$, and throwing these out, we have an equation of the form

$$\Delta = (a, b, c, f, g, h)^{18} (x, y, z, w)^6 = 0,$$

which is the equation of the sextic torse.

Principal Sections of the Torse.

6. Consider for instance the section by the plane $w = 0$. Writing $w = 0$, the equation of the osculating plane is

$$(\theta + \delta)^2 [x(\theta + \beta)^2(\theta + \gamma)^2 + y(\theta + \gamma)^2(\theta + \alpha)^2 + z(\theta + \alpha)^2(\theta + \beta)^2] = 0.$$

The discriminant of the sextic function vanishes identically in virtue of the double factor $(\theta + \delta)^2$. But omitting this factor, the equation becomes

$$x(\theta + \beta)^2(\theta + \gamma)^2 + y(\theta + \gamma)^2(\theta + \alpha)^2 + z(\theta + \alpha)^2(\theta + \beta)^2 = 0.$$

The discriminant of this quartic function of θ is a function of x, y, z, a, b, c of the degree 6 in (x, y, z) and 12 in (a, b, c) ; it contains however the factors $xyz, a^2b^2c^2$, and the remaining factor is of the degree 3 in (x, y, z) and 6 in (a, b, c) ; this remaining factor is as will presently be seen

$$= (a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2xyz.$$

The last mentioned sextic equation in θ will have a triple root $\theta = -\delta$, if only the value $\theta = -\delta$ makes to vanish the factor in [], that is if we have

$$0 = g^2h^2x + h^2f^2y + f^2g^2z.$$

The foregoing results lead to the conclusion that for $w = 0$, we have

$$\Delta = (g^2h^2x + h^2f^2y + f^2g^2z)^3 [(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2xyz];$$

but this will appear more distinctly as follows.

7. First, as to the factor $(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2xyz$: writing in the equation of the osculating plane $w = 0$, the equation becomes

$$\frac{x}{(\theta + \alpha)^2} + \frac{y}{(\theta + \beta)^2} + \frac{z}{(\theta + \gamma)^2} = 0,$$

which equation is therefore that of the trace of the osculating plane on the plane $w = 0$; the envelope of the trace in question is a part of the section of the torse by the plane $w = 0$. To find the equation of this envelope we must eliminate θ from the foregoing, and its derived equation

$$\frac{x}{(\theta + \alpha)^3} + \frac{y}{(\theta + \beta)^3} + \frac{z}{(\theta + \gamma)^3} = 0;$$

the two equations give

$$x : y : z = a(\theta + \alpha)^3 : b(\theta + \beta)^3 : c(\theta + \gamma)^3,$$

and thence

$$(a^2x)^{1/3} + (b^2y)^{1/3} + (c^2z)^{1/3} = a(\theta + \alpha) + b(\theta + \beta) + c(\theta + \gamma) = 0,$$

that is, we have

$$(a^2x)^{1/3} + (b^2y)^{1/3} + (c^2z)^{1/3} = 0,$$

or, what is the same thing,

$$(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2xyz = 0$$

for a part of the section in question.

8. I have said that the foregoing cubic is a part of the section; the equations

$$x : y : z : w = ahg(\theta + \alpha)^4 : bhf(\theta + \beta)^4 : cfg(\theta + \gamma)^4 : abc(\theta + \delta)^4,$$

which for $w = 0$ give $\theta = -\delta$, and thence $x : y : z = af^4 : bg^4 : ch^4$, show that the last-mentioned point is a four-pointic intersection of the curve with the plane $w = 0$. But the curve, having four consecutive points, will have three consecutive tangents in the plane $w = 0$; that is, the tangent at the point in question will present itself as a threefold factor in the equation of the torse. Writing in the equations of the tangent $w = 0$, $\theta = -\delta$, we find for the equation of the tangent in question

$$\frac{x}{f^2} : \frac{y}{g^2} : \frac{z}{h^2} = \frac{1}{a} : \frac{1}{b} : \frac{1}{c},$$

or, what is the same thing,

$$g^2h^2x + h^2f^2y + f^2g^2z = 0.$$

Hence the section by the plane $w = 0$ is made up of this line taken three times, and of the last mentioned cubic curve.

9. By symmetry, we conclude that the sections by the principal planes $x = 0$, $y = 0$, $z = 0$, $w = 0$, are each made up of a line taken three times, and of a cubic curve: viz. these are

$$\begin{array}{lll} x = 0, & f^2h^2y + c^2f^2z + b^2c^2w = 0, & (b^2y)^{1/3} + (g^2z)^{1/3} + (a^2w)^{1/3} = 0, \\ y = 0, & a^2f^2x + c^2g^2z + c^2a^2w = 0, & (h^2x)^{1/3} + (f^2z)^{1/3} + (b^2w)^{1/3} = 0, \\ z = 0, & a^2h^2x + b^2h^2y + a^2b^2w = 0, & (g^2x)^{1/3} + (f^2y)^{1/3} + (c^2w)^{1/3} = 0, \\ w = 0, & g^2h^2x + h^2f^2y + f^2g^2z = 0, & (a^2x)^{1/3} + (b^2y)^{1/3} + (c^2z)^{1/3} = 0, \end{array}$$

where for shortness I have written the equations of the four cubics in their irrational forms respectively.

Partial Determination of the Equation.

10. As the value of Δ is known when any one of the coordinates x, y, z, w is put $= 0$, we in fact know all the terms of Δ , except those which contain the factor $xyzw$, which unknown terms, as Δ is of the degree 6, are of the form $(*)(x, y, z, w)^2$.

I remark that if $(xyzw)$ is any homogeneous function $(*)(x, y, z, w)^n$, and $(xyz), (xy), (x)$ are what $(xyzw)$ become on putting therein $(w = 0), (z = 0, w = 0), (y = 0, z = 0, w = 0)$ respectively, and the like for the other similar symbols, then that

$$\begin{aligned} (xyzw) &= (x) + (y) + (z) + (w) \\ &\quad - (xy) - (xz) - (xw) - (yz) - (yw) - (zw) \\ &\quad + (xyz) + (xyw) + (xzw) + (yzw) \\ &\quad + \text{terms multiplied by } xyzw \end{aligned}$$

in fact, omitting the last line, this equation on writing therein $x = 0$ or $y = 0$ or $z = 0$ or $w = 0$, becomes an identity, that is, the difference of the two sides vanishes when any one of these equations is satisfied, and such difference contains therefore the factor $xyzw$; which proves the theorem. It hence appears that the equation $\Delta = 0$ of the torse is

$$\begin{aligned} \Delta &= a^2 g^2 h^2 x^6 + b^2 h^2 f^2 y^6 + c^2 f^2 g^2 z^6 + a^2 b^2 c^2 w^6 \\ &\quad - (g^2 h^2 x + h^2 f^2 y)^3 (a^2 x + b^2 y)^3 \\ &\quad - (h^2 f^2 y + f^2 g^2 z)^3 (b^2 y + c^2 z)^3 \\ &\quad - (g^2 h^2 x + f^2 g^2 z)^3 (a^2 x + c^2 z)^3 \\ &\quad - (a^2 h^2 x + a^2 b^2 w)^3 (g^2 x + b^2 w)^3 \\ &\quad - (b^2 h^2 y + a^2 b^2 w)^3 (f^2 y + c^2 w)^3 \\ &\quad - (c^2 f^2 z + c^2 a^2 w)^3 (g^2 z + b^2 w)^3 \\ &\quad + (b^2 h^2 y + c^2 f^2 z + b^2 c^2 w)^3 [(h^2 y + g^2 z + a^2 w)^3 - 27a^2 g^2 h^2 yzw] \\ &\quad + (a^2 f^2 x + c^2 g^2 z + c^2 a^2 w)^3 [(h^2 x + f^2 z + b^2 w)^3 - 27b^2 h^2 f^2 xzw] \\ &\quad + (a^2 h^2 x + b^2 h^2 y + a^2 b^2 w)^3 [(g^2 x + f^2 y + c^2 w)^3 - 27c^2 f^2 g^2 xyw] \\ &\quad + (g^2 h^2 x + h^2 f^2 y + f^2 g^2 z)^3 [(a^2 x + b^2 y + c^2 z)^3 - 27a^2 b^2 c^2 xyz] \\ &\quad + xyzw \cdot (*)(x, y, z, w)^2, \end{aligned}$$

where the ten coefficients of $(*)(x, y, z, w)^2$ remain to be found.

Process for the Determination of the Unknown Coefficients.

11. At a point of the cubic curve in the plane $w = 0$, we have

$$x : y : z = a(\theta + \alpha)^3 : b(\theta + \beta)^3 : c(\theta + \gamma)^3;$$

and the tangent plane at this point is the osculating plane of the curve; that is, it is the plane

$$\frac{x'}{(\theta + \alpha)^2} + \frac{y'}{(\theta + \beta)^2} + \frac{z'}{(\theta + \gamma)^2} + \frac{w'}{(\theta + \delta)^2} = 0,$$

if for a moment (x', y', z', w') are the current coordinates of a point in the tangent plane. But the equation of the tangent plane as deduced from the equation $\Delta = 0$ is

$$x' \frac{d\Delta}{dx} + y' \frac{d\Delta}{dy} + z' \frac{d\Delta}{dz} + w' \frac{d\Delta}{dw} = 0,$$

where in the differential coefficients of Δ , the coordinates (x, y, z, w) are considered as having the values

$$x : y : z : w = a(\theta + \alpha)^3 : b(\theta + \beta)^3 : c(\theta + \gamma)^3 : 0.$$

Hence, with these values of (x, y, z, w) , we have

$$\frac{d\Delta}{dx} : \frac{d\Delta}{dy} : \frac{d\Delta}{dz} : \frac{d\Delta}{dw} = \frac{1}{(\theta + \alpha)^2} : \frac{1}{(\theta + \beta)^2} : \frac{1}{(\theta + \gamma)^2} : \frac{1}{(\theta + \delta)^2},$$

conditions which determine the values of certain of the coefficients of $(*)(x, y, z, w)^2$, viz. the six coefficients of the terms independent of w ; and when these are known the values of the remaining four coefficients are at once obtained by symmetry.

12. To develop this process, disregarding the higher powers of w , we may write

$$\Delta = \Theta + 3\Phi w + xyzw(*) (x, y, z)^2,$$

where Θ denotes the terms independent of w , $3\Phi w$ the known terms which contain the factor w , and $xyzw(*) (x, y, z)^2$ the unknown terms which contain this same factor; the value of $(*)(x, y, z)^2$ being clearly $= (*)(x, y, z, 0)^2$.

We have, moreover,

$$\Theta = (g^2 h^2 x + h^2 f^2 y + f^2 g^2 z)^3 [(a^2 x + b^2 y + c^2 z)^3 - 27a^2 b^2 c^2 xyz],$$

and

$$\begin{aligned} \Phi = & f^2 (h^2 y + g^2 z)^2 [a^2 f^2 (h^4 y^2 - 7h^2 g^2 yz + g^4 z^2) (b^2 y + c^2 z) + b^2 c^2 (h^2 y + g^2 z)^2] \\ & + g^4 (a^2 x + c^2 z)^2 [b^2 g^2 (h^4 x^2 - 7h^2 f^2 xz + f^4 z^2) (a^2 x + c^2 z) + c^2 a^2 (h^2 x + f^2 z)^2] \\ & + h^4 (a^2 x + b^2 y)^2 [c^2 h^2 (g^4 x^2 - 7g^2 f^2 xy + f^4 y^2) (a^2 x + b^2 y) + a^2 b^2 (g^2 x + f^2 y)^2] \\ & - a^2 h^4 g^4 (b^2 y + c^2 z)^2 x^2 \\ & - b^2 h^4 f^4 (c^2 z + a^2 x)^2 y^2 \\ & - c^2 f^4 g^4 (a^2 x + b^2 y)^2 z^2. \end{aligned}$$

13. The equations, putting after the differentiations $w = 0$, and writing for shortness $(*)$ in place of $(*)(x, y, z)^2$, become

$$\frac{d\Theta}{dx} : \frac{d\Theta}{dy} : \frac{d\Theta}{dz} : 3\Phi + xyz(*) = \frac{1}{(\theta + \alpha)^2} : \frac{1}{(\theta + \beta)^2} : \frac{1}{(\theta + \gamma)^2} : \frac{1}{(\theta + \delta)^2}.$$

Now, observing that the second factor of Θ vanishes for the values

$$a(\theta + \alpha)^3, \quad b(\theta + \beta)^3, \quad c(\theta + \gamma)^3 \text{ of } (x, y, z),$$

we have simply

$$\frac{d\Theta}{dx} = (g^2 h^2 x + h^2 f^2 y + f^2 g^2 z)^3 \cdot 3a^2 [(a^2 x + b^2 y + c^2 z)^2 - 9b^2 c^2 yz].$$

But

$$a^2 x + b^2 y + c^2 z = a^2 (\theta + \alpha)^3 + b^2 (\theta + \beta)^3 + c^2 (\theta + \gamma)^3,$$

in virtue of the relation $a(\theta + \alpha) + b(\theta + \beta) + c(\theta + \gamma) = 0$, and hence

$$\begin{aligned} [(a^2 x + b^2 y + c^2 z)^2 - 9b^2 c^2 yz] &= 9b^2 c^2 (\theta + \beta)^2 (\theta + \gamma)^2 [a^2 (\theta + \alpha)^2 - bc(\theta + \beta)(\theta + \gamma)], \\ &= 9b^2 c^2 (\theta + \beta)^2 (\theta + \gamma)^2 Q, \end{aligned}$$

where

$$\begin{aligned} Q &= a^2 (\theta + \alpha)^2 - bc(\theta + \beta)(\theta + \gamma), \\ &= b^2 (\theta + \beta)^2 - ca(\theta + \gamma)(\theta + \alpha), \\ &= c^2 (\theta + \gamma)^2 - ab(\theta + \alpha)(\theta + \beta). \end{aligned}$$

Hence

$$\frac{d\Theta}{dx} = 27a^2 b^2 c^2 (g^2 h^2 x + h^2 f^2 y + f^2 g^2 z)^3 (\theta + \beta)^2 (\theta + \gamma)^2 Q,$$

and similarly

$$\frac{d\Theta}{dy} = 27a^2b^2c^2 (g^2h^2x + h^2f^2y + f^2g^2z)^3(\theta + \gamma)^2(\theta + \alpha)^2 Q,$$

$$\frac{d\Theta}{dz} = 27a^2b^2c^2 (g^2h^2x + h^2f^2y + f^2g^2z)^3(\theta + \alpha)^2(\theta + \beta)^2 Q;$$

whence the above-mentioned conditions reduce themselves to the single condition

$$(\theta + \delta)^2 \{3\Phi + xyz(*)\} = 27a^2b^2c^2 (g^2h^2x + h^2f^2y + f^2g^2z)^3(\theta + \alpha)^2(\theta + \beta)^2(\theta + \gamma)^2 Q.$$

14. But we have

$$\begin{aligned} g^2h^2x + h^2f^2y + f^2g^2z &= g^2h^2a(\theta + \delta + f)^3 + h^2f^2b(\theta + \delta + g)^3 + f^2g^2c(\theta + \delta + h)^3, \\ &= (\theta + \delta)^3 [(g^2h^2a + h^2f^2b + f^2g^2c)(\theta + \delta) + 3(gha + hfb + fgc)fgh], \\ &= -abc(\theta + \delta)^2 [(gh + hf + fg)(\theta + \delta) + 3fgh], \\ &= -abc(\theta + \delta)^2 [gh(\theta + \alpha) + hf(\theta + \beta) + fg(\theta + \gamma)], \\ &= -abc(\theta + \delta)^2 P, \end{aligned}$$

if for shortness

$$P = gh(\theta + \alpha) + hf(\theta + \beta) + fg(\theta + \gamma).$$

Hence, substituting,

$$3\Phi + abc(\theta + \alpha)^3(\theta + \beta)^3(\theta + \gamma)^3(*) = -27(abc)^5(\theta + \delta)^4(\theta + \alpha)^2(\theta + \beta)^2(\theta + \gamma)^2 P^3Q$$

which when the values $a(\theta + \alpha)^3$, $b(\theta + \beta)^3$, $c(\theta + \gamma)^3$ for (x, y, z) are substituted in the functions Φ and $(*)$, will be an identical equation in θ .

15. It is right to remark that what we require is the expression of $(*)$, $= (*) (x, y, z)^2$; the foregoing equation leads to the value of $(*)$ expressed in terms of θ ; and it is necessary to show that this leads back to the expression for $(*)$ as a function of (x, y, z) ; in fact, that the function of θ is transformable in a definite manner into a function of (x, y, z) . Suppose that the function of θ could be expressed in two different manners as a function of (x, y, z) ; then we should have two different functions $(x, y, z)^2$ each equivalent to the same function of θ , and the difference of these functions would be identically $= 0$; that is, we should have a function $(x, y, z)^2$ vanishing identically by the substitution

$$x : y : z = a(\theta + \alpha)^3 : b(\theta + \beta)^3 : c(\theta + \gamma)^3;$$

but these relations are equivalent to the single relation

$$(a^2x + b^2y + c^2z)^3 - 27a^2b^2c^2 xyz = 0,$$

which, *quâ* cubic equation, is not equivalent to any equation whatever of the form

$$(x, y, z)^2 = 0;$$

that is, the function of θ is equivalent to a definite function $(x, y, z)^2$.

16. To proceed with the reduction, I remark that we have

$$\begin{aligned}
 \Phi &= (a^2x + b^2y + c^2z) \\
 &\times \left\{ f^4 [a^2 f^2 (h^4 y^2 - 7h^2 g^2 yz + g^4 z^2)(b^2 y + c^2 z) + b^2 c^2 (h^2 y + g^2 z)^2] \right. \\
 &\quad + g^4 [b^2 g^2 (h^4 x^2 - 7h^2 f^2 xz + f^4 z^2)(a^2 x + c^2 z) + c^2 a^2 (h^2 x + f^2 z)^2] \\
 &\quad + h^4 [c^2 h^2 (g^4 x^2 - 7g^2 f^2 xy + f^4 y^2)(a^2 x + b^2 y) + a^2 b^2 (g^2 x + f^2 y)^2] \\
 &\quad - a^2 h^4 g^4 (b^2 y + c^2 z)^2 x^2 \\
 &\quad - b^2 h^4 f^4 (c^2 z + a^2 x)^2 y^2 \\
 &\quad \left. - c^2 f^4 g^4 (a^2 x + b^2 y)^2 z^2 \right\} \\
 &+ xyz \Omega,
 \end{aligned}$$

where

$$\begin{aligned}
 -\Omega &= 2(b^2 c^2 A x^2 + c^2 a^2 B y^2 + a^2 b^2 C z^2) \\
 &\quad + (My + Nz)(a^4 x + 2a^2 b^2 y + 2a^2 c^2 z) \\
 &\quad + (Oz + Px)(2a^2 b^2 x + b^4 y + 2b^2 c^2 z) \\
 &\quad + (Qx + Ry)(2a^2 c^2 x + 2b^2 c^2 y + c^4 z),
 \end{aligned}$$

if for shortness

$$\begin{aligned}
 A &= a^2 f^4 h^4 (b^2 g^2 + c^2 h^2), \\
 B &= b^2 g^4 f^4 (c^2 h^2 + a^2 f^2), \\
 C &= c^2 h^4 g^4 (a^2 f^2 + b^2 g^2),
 \end{aligned}$$

$$\begin{aligned}
 M &= f^2 h^2 (a^2 f^2 b^2 h^2 - 7a^2 f^4 g^2 + 3b^2 g^4 c^2 h^2), & N &= f^2 g^2 (-7a^2 f^2 c^2 h^2 + a^2 f^2 b^2 h^2 + 3b^2 c^4 g^2), \\
 O &= g^2 f^2 (b^2 g^2 c^2 f^2 - 7b^2 c^2 h^2 g^2 + 3c^2 h^4 a^2 f^2), & P &= g^2 h^2 (-7b^2 c^2 a^2 f^2 + b^2 c^4 h^2 + 3c^2 h^4 a^2 f^2), \\
 Q &= h^2 g^2 (c^2 h^2 b^2 g^2 - 7c^2 h^2 a^2 b^2 + 3a^2 f^4 b^2 g^2), & R &= h^2 f^2 (-7c^2 h^2 b^2 g^2 + c^2 h^2 a^2 f^2 + 3a^2 f^4 b^2 g^2),
 \end{aligned}$$

and I represent the foregoing equation by

$$\Phi = (a^2x + b^2y + c^2z)U + xyz \Omega.$$

Hence, writing for x, y, z the foregoing values, we have

$$\Phi = 9a^2 b^2 c^2 (\theta + \alpha)^2 (\theta + \beta)^2 (\theta + \gamma)^2 U + abc (\theta + \alpha)^3 (\theta + \beta)^3 (\theta + \gamma)^3 \Omega,$$

and thence

$$27U + \frac{\Omega}{(\theta + \alpha)(\theta + \beta)(\theta + \gamma)} (3\Omega + (*)) = -27(abc)^3 (\theta + \delta)^4 P^3 Q,$$

that is

$$27[U + (abc)^3 (\theta + \delta)^4 P^3 Q] + \frac{\Omega}{(\theta + \alpha)(\theta + \beta)(\theta + \gamma)} (3\Omega + (*)) = 0.$$

In order that this may be the case, it is clear that we must have

$$U + (abc)^3 (\theta + \delta)^4 P^3 Q = (\theta + \alpha)(\theta + \beta)(\theta + \gamma) M,$$

viz. the left-hand side expressed as a function of θ must be divisible by the product $(\theta + \alpha)(\theta + \beta)(\theta + \gamma)$. Assuming for a moment that this is so, the quotient M will be a function $(\theta, 1)^6$ expressible in a unique manner in the form $(x, y, z)^2$, and assuming it to be so expressed, we have

$$27Mabc + 3\Omega + (*) = 0;$$

which equation, without any further substitution of the θ -values of (x, y, z) , gives $(*)$ in its proper form as a function of (x, y, z) .

Reduction of the Equation, $U + (abc)^3(\theta + \delta)^4 P^3 Q = (\theta + \alpha)(\theta + \beta)(\theta + \gamma)M$.

17. We have by an easy transformation

$$\begin{aligned} U = (a^2x + b^2y + c^2z) & \left\{ a^2f^2(h^4y^2 - 7h^2g^2yz + g^4z^2) \right. \\ & + b^2g^2(f^4z^2 - 7f^2h^2zx + h^4x^2) \\ & \left. + c^2h^2(g^4x^2 - 7g^2f^2xy + f^4y^2) \right\} \\ & + 7(a^4f^4 + b^4g^4 + c^4h^4)f^2g^2h^2xyz \\ & + U', \end{aligned}$$

if for shortness

$$\begin{aligned} U' = y^2z \cdot a^2f^4h^4(3b^2g^2 - c^2h^2) \\ + yz^2 \cdot b^2g^4f^4(3c^2h^2 - b^2g^2) \\ + z^2x \cdot c^2f^4g^4(3a^2h^2 - a^2f^2) \\ + zx^2 \cdot c^2h^4g^4(3a^2f^2 - b^2h^2) \\ + x^2y \cdot b^2g^4h^4(3a^2f^2 - b^2g^2) \\ + xy^2 \cdot a^2f^4h^4(3b^2g^2 - a^2f^2). \end{aligned}$$

Substituting the θ -values, the terms of U , other than U' , are at once seen to contain the factor $(\theta + \alpha)(\theta + \beta)(\theta + \gamma)$, and we have

$$\begin{aligned} M = 3abc & \left\{ a^4(h^4y^2 - 7h^2g^2yz + g^4z^2) \right. \\ & + b^4(f^4z^2 - 7f^2h^2zx + h^4x^2) \\ & \left. + c^4(g^4x^2 - 7g^2f^2xy + f^4y^2) \right\} \\ & + 7(a^4f^4 + b^4g^4 + c^4h^4)f^2g^2h^2abc(\theta + \alpha)^2(\theta + \beta)^2(\theta + \gamma)^2 \\ & + M', \end{aligned}$$

where

$$U' + (abc)^3(\theta + \delta)^4 P^3 Q = (\theta + \alpha)(\theta + \beta)(\theta + \gamma) M'.$$

18. Write for shortness $p, q, r = (af, bg, ch)$; after a complicated reduction, I obtain

$$\begin{aligned} 3abc M' = a^2f^2h^2(r - p)(p - q)(-2p^4 + 5p^2qr - 6q^2r^2)x^2 \\ + b^2h^2f^2(p - q)(q - r)(-2q^4 + 5q^2rp - 6r^2p^2)y^2 \\ + c^2f^2g^2(q - r)(r - p)(-2r^4 + 5r^2pq - 6p^2q^2)z^2 \\ + 2f^2g^2h^2b^2c^2(7p^4 - 20p^2qr + 4q^2r^2)yz \\ + 2f^2g^2h^2c^2a^2(7q^4 - 20pq^2r + 4r^2p^2)zx \\ + 2f^2g^2h^2a^2b^2(7r^4 - 20pqr^2 + 4p^2q^2)xy \\ - 2f^2g^2h^2(p^4 + q^4 + r^4)(a^2x + b^2y + c^2z)^2. \end{aligned}$$

We then have

$$9abc M = \text{terms } (x, y, z)^2 + 9abc M', \quad \Omega = \text{terms } (x, y, z)^2$$

as above; and

$$27Mabc + 3\Omega + (*) = 0,$$

which gives (*).

19. After all reductions we find:

$$\begin{aligned}
 -\frac{1}{3}(\ast) = & a^2 f^2 h^2 (28p^6 - 84p^4 qr + 62p^2 q^2 r^2 - 28q^3 r^3) x^2 \\
 & + b^2 h^2 f^2 (28q^6 - 84pq^4 r + 62p^2 q^2 r^2 - 28r^3 p^3) y^2 \\
 & + c^2 f^2 g^2 (28r^6 - 84pqr^4 + 62p^2 q^2 r^2 - 28p^3 q^3) z^2 \\
 & + f^2 (-3p^6 + 14p^4 qr - 130p^4 q^2 r^2 - 136p^2 q^3 r^3 + 42q^4 r^4) yz \\
 & + g^2 (-3q^6 + 14pq^4 r - 130p^2 q^4 r^2 - 136p^3 q^2 r^3 + 42r^4 p^4) zx \\
 & + h^2 (-3r^6 + 14pqr^4 - 130p^2 q^2 r^4 - 136p^3 q^3 r^2 + 42p^4 q^4) xy
 \end{aligned}$$

or observing that the coefficients of $a^2 f^2 h^2 x^2$, $b^2 h^2 f^2 y^2$ and $c^2 f^2 g^2 z^2$ are equal to each other and to

$$62p^2 q^2 r^2 - 28(q^3 r^3 + r^3 p^3 + p^3 q^3),$$

the equation becomes

$$\begin{aligned}
 (\ast) = & -3[62p^2 q^2 r^2 - 28(q^3 r^3 + r^3 p^3 + p^3 q^3)](a^2 f^2 h^2 x^2 + b^2 h^2 f^2 y^2 + c^2 f^2 g^2 z^2) \\
 & + 3(3p^6 - 14p^4 qr + 130p^4 q^2 r^2 + 136p^2 q^3 r^3 - 42q^4 r^4) f^2 yz \\
 & + 3(3q^6 - 14q^4 pr + 130q^4 p^2 r^2 + 136q^2 p^3 r^3 - 42r^4 p^4) g^2 zx \\
 & + 3(3r^6 - 14r^4 pq + 130r^4 p^2 q^2 + 136r^2 p^3 q^3 - 42p^4 q^4) h^2 xy;
 \end{aligned}$$

and we thence obtain by symmetry the complete value of $(\ast)(x, y, z, w)^2$; viz. we have only to complete the literal parts of the foregoing expression into the forms

$$\begin{aligned}
 & f^2 yz + a^2 xw, \\
 & g^2 zx + b^2 yw, \\
 & h^2 xy + c^2 zw,
 \end{aligned}$$

respectively.

20. The equation of the torse thus is

$$\begin{aligned}
 \Delta = & a^2 g^2 h^2 x^6 + b^2 h^2 f^2 y^6 + c^2 f^2 g^2 z^6 + a^2 b^2 c^2 w^6 \\
 & - g^6 (f^2 z + h^2 x)^3 (b^2 z + a^2 x)^3 \\
 & - h^6 (g^2 x + f^2 y)^3 (c^2 x + b^2 y)^3 \\
 & - a^6 (g^2 x + b^2 w)^3 (h^2 x + b^2 w)^3 \\
 & - b^6 (h^2 y + a^2 w)^3 (f^2 y + c^2 w)^3 \\
 & - c^6 (f^2 z + b^2 y)^3 (g^2 z + a^2 y)^3 \\
 & + (a^2 f^2 x + c^2 g^2 z + c^2 a^2 w)^3 [(h^2 x + f^2 z + b^2 w)^3 - 27b^2 h^2 f^2 xzw] \\
 & + (a^2 h^2 x + b^2 h^2 y + a^2 b^2 w)^3 [(g^2 x + f^2 y + c^2 w)^3 - 27c^2 f^2 g^2 xyw] \\
 & + (g^2 h^2 x + h^2 f^2 y + f^2 g^2 z)^3 [(a^2 x + b^2 y + c^2 z)^3 - 27a^2 b^2 c^2 xyz] \\
 & + xyzw (\ast)(x, y, z, w)^2 = 0.
 \end{aligned}$$

I recall that

$$\begin{aligned}
 a &= \beta - \gamma, \quad f = \alpha - \delta, \quad p = af = (\alpha - \delta)(\beta - \gamma), \\
 b &= \gamma - \alpha, \quad g = \beta - \delta, \quad q = bg = (\beta - \delta)(\gamma - \alpha), \\
 c &= \alpha - \beta, \quad h = \gamma - \delta, \quad r = ch = (\gamma - \delta)(\alpha - \beta).
 \end{aligned}$$

Developing, we have finally the equation of the torse in the form following.

Equation of the Sextic Torse.

21. The equation is

$$\begin{aligned}
 0 = & (a^2g^2h^2, b^2h^2f^2, c^2f^2g^2, a^2b^2c^2)(x^6, y^6, z^6, w^6) \\
 & + 3(q^2 + r^2)(b^4h^4f^2, c^4g^4f^2, g^4h^4a^2, b^4c^4a^2)(y^4z^2, y^2z^4, x^4w^2, x^2w^4) \\
 & + 3(r^2 + p^2)(c^4f^4g^2, a^4h^4f^2, h^4f^4b^2, c^4a^4b^2)(z^4x^2, z^2x^4, y^4w^2, y^2w^4) \\
 & + 3(p^2 + q^2)(a^4g^4h^2, b^4f^4h^2, f^4g^4c^2, a^4b^4c^2)(x^4y^2, x^2y^4, z^4w^2, z^2w^4) \\
 & + 3(q^4 + 3q^2r^2 + r^4)(b^2h^2f^2, c^2g^2f^2, g^2h^2a^2, b^2c^2a^2)(y^4z^2, y^2z^4, x^4w^2, x^2w^4) \\
 & + 3(r^4 + 3r^2p^2 + p^4)(c^2f^2g^2, a^2h^2f^2, h^2f^2b^2, c^2a^2b^2)(z^4x^2, z^2x^4, y^4w^2, y^2w^4) \\
 & + 3(p^4 + 3p^2q^2 + q^4)(a^2g^2h^2, b^2f^2h^2, f^2g^2c^2, a^2b^2c^2)(x^4y^2, x^2y^4, z^4w^2, z^2w^4) \\
 & + (6p^4 + 9p^2q^2 + 9p^2r^2 - 21q^2r^2)(a^2g^2h^2, a^2b^2c^2, f^2h^2b^2, f^2g^2c^2)(x^4yz, w^4yz, y^4wx, z^4wy) \\
 & + (6q^4 + 9q^2r^2 + 9q^2p^2 - 21r^2p^2)(b^2h^2f^2, b^2c^2a^2, g^2f^2c^2, g^2h^2a^2)(y^4zx, w^4zx, z^4wy, x^4wz) \\
 & + (6r^4 + 9r^2p^2 + 9r^2q^2 - 21p^2q^2)(c^2f^2g^2, c^2a^2b^2, h^2g^2a^2, h^2f^2b^2)(z^4xy, w^4xy, x^4wz, y^4wx) \\
 & + (q^6 + 9q^4r^2 + 9q^2r^4 + r^6)(f^6, a^6)(y^3z^3, x^3w^3) \\
 & + (r^6 + 9r^4p^2 + 9r^2p^4 + p^6)(g^6, b^6)(z^3x^3, y^3w^3) \\
 & + (p^6 + 9p^4q^2 + 9p^2q^4 + q^6)(h^6, c^6)(x^3y^3, z^3w^3) \\
 & + 9(q^2r^4 + q^4r^2 + r^2p^4 + r^4p^2 + p^2q^4 + p^4q^2 - 14p^2q^2r^2) \\
 & \quad \times (f^2g^2h^2, f^2b^2c^2, g^2c^2a^2, h^2a^2b^2)(x^2y^2z^2, y^2z^2w^2, z^2x^2w^2, x^2y^2w^2) \\
 & + 3\{p^6 + 3p^4(2q^2 + r^2) + 3p^2(q^4 - 7q^2r^2) + q^6\} \\
 & \quad \times (g^4h^2, h^4g^2, g^4c^2, c^4b^2)(x^4y^2z, y^4wx^2, z^4w^2x, w^4yz^2) \\
 & + 3\{q^6 + 3q^4(2r^2 + p^2) + 3q^2(r^4 - 7r^2p^2) + r^6\} \\
 & \quad \times (h^4f^2, f^4c^2, h^4a^2, a^4c^2)(y^4z^2x, z^4wy^2, x^4w^2y, w^4zx^2) \\
 & + 3\{r^6 + 3r^4(2p^2 + q^2) + 3r^2(p^4 - 7p^2q^2) + p^6\} \\
 & \quad \times (f^4g^2, g^4a^2, f^4b^2, b^4a^2)(x^2yz^4, x^4wz^2, y^4w^2z, w^4xy^2) \\
 & + 3\{p^6 + 3p^4(2r^2 + q^2) + 3p^2(r^4 - 7r^2q^2) + q^6\} \\
 & \quad \times (g^4h^2, h^4b^2, g^4c^2, c^4b^2)(x^4yz^2, y^2w^2x, z^4wx^2, w^4y^2z) \\
 & + 3\{q^6 + 3q^4(2p^2 + r^2) + 3q^2(p^4 - 7p^2r^2) + r^6\} \\
 & \quad \times (h^4f^2, f^4a^2, h^4a^2, a^4c^2)(y^4zx^2, z^2wy^2, x^4wy^2, w^4z^2x) \\
 & + 3\{r^6 + 3r^4(2q^2 + p^2) + 3r^2(q^4 - 7q^2p^2) + p^6\} \\
 & \quad \times (f^4g^2, g^4c^2, f^4b^2, b^4a^2)(x^2yz^4, z^4w^2z, y^4wz^2, w^4xy^2) \\
 & + xyzw [\\
 & \quad - 3\{62p^2q^2r^2 - 28(q^3r^3 + r^3p^3 + p^3q^3)\}(a^2f^2h^2, b^2h^2f^2, c^2f^2g^2, a^2b^2c^2)(x^2, y^2, z^2, w^2) \\
 & \quad + 3(3p^6 - 14p^4qr + 130p^4q^2r^2 + 136p^2q^3r^3 - 42q^4r^4)(f^2, a^2)(yz, xw) \\
 & \quad + 3(3q^6 - 14q^4rp + 130q^4r^2p^2 + 136q^2r^3p^3 - 42r^4p^4)(g^2, b^2)(zx, yw) \\
 & \quad + 3(3r^6 - 14r^4pq + 130r^4p^2q^2 + 136r^2p^3q^3 - 42p^4q^4)(h^2, c^2)(xy, zw) \\
 & \quad].
 \end{aligned}$$

Comparison with the Equation of the Centro-surface of an Ellipsoid.

22. In the Equation

$$\frac{x}{(\theta + \alpha)^2} + \frac{y}{(\theta + \beta)^2} + \frac{z}{(\theta + \gamma)^2} + \frac{w}{(\theta + \delta)^2} = 0,$$

for x, y, z, w write $\xi^2, \eta^2, \zeta^2, \omega^2$, and then $\delta = \infty$, the equation is converted into

$$\frac{\xi^2}{(\theta + \alpha)^2} + \frac{\eta^2}{(\theta + \beta)^2} + \frac{\zeta^2}{(\theta + \gamma)^2} + \omega^2 = 0;$$

or writing a^2, b^2, c^2 for α, β, γ , and understanding $\xi^2, \eta^2, \zeta^2, \omega^2$ to mean $a^2x^2, b^2y^2, c^2z^2, -1$, this is

$$\frac{a^2x^2}{(\theta + a^2)^2} + \frac{b^2y^2}{(\theta + b^2)^2} + \frac{c^2z^2}{(\theta + c^2)^2} - 1 = 0.$$

This is an equation, the envelope of which in regard to the variable parameter θ , gives the surface which is the locus of the centres of curvature of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, or say the *Centro-surface of the Ellipsoid*. (Salmon's *Solid Geometry*, Ed. 2, p. 400, [Ed. 4, p. 465].)

Making the same substitution in the foregoing equation $(*)(x, y, z, w)^2 = 0$, the quantities f, g, h become equal to $-\delta$, and p, q, r to $-a\delta, -b\delta, -c\delta$ respectively, and the whole equation divides by δ^n ; throwing out this factor, we have a result which is obtained more simply by changing

$$x, y, z, w, \quad a, b, c, f, g, h, \quad p, q, r,$$

into

$$\xi^2, \eta^2, \zeta^2, \omega^2, \quad \alpha, \beta, \gamma, 1, 1, 1, \quad \alpha, \beta, \gamma,$$

where α, β, γ now signify $b^2 - c^2, c^2 - a^2, a^2 - b^2$ respectively, and $\xi^2, \eta^2, \zeta^2, \omega^2$ are retained as standing for $a^2x^2, b^2y^2, c^2z^2, -1$ respectively; viz. the equation of the centro-surface is found to

be

$$\begin{aligned}
 0 = & (\alpha^2, \beta^2, \gamma^2, \alpha^2\beta^2\gamma^2)(\xi^6, \eta^6, \zeta^6, \omega^6) \\
 & + 3(\beta^2 + \gamma^2)(\beta^2, \gamma^2, \alpha^2, \alpha^2\beta^2\gamma^2)(\eta^4\zeta^2, \eta^2\zeta^4, \xi^4\omega^2, \xi^2\omega^4) \\
 & + 3(\gamma^2 + \alpha^2)(\gamma^2, \alpha^2, \beta^2, \beta^2\alpha^2\gamma^2)(\zeta^4\xi^2, \zeta^2\xi^4, \eta^4\omega^2, \eta^2\omega^4) \\
 & + 3(\alpha^2 + \beta^2)(\alpha^2, \beta^2, \gamma^2, \gamma^2\alpha^2\beta^2)(\xi^4\eta^2, \xi^2\eta^4, \zeta^4\omega^2, \zeta^2\omega^4) \\
 & + 3(\beta^4 + 3\beta^2\gamma^2 + \gamma^4)(\beta^2, \gamma^2, \alpha^2, \alpha^2\beta^2\gamma^2)(\eta^4\zeta^2, \eta^2\zeta^4, \xi^4\omega^2, \xi^2\omega^4) \\
 & + 3(\gamma^4 + 3\gamma^2\alpha^2 + \alpha^4)(\gamma^2, \alpha^2, \beta^2, \beta^2\gamma^2\alpha^2)(\zeta^4\xi^2, \zeta^2\xi^4, \eta^4\omega^2, \eta^2\omega^4) \\
 & + 3(\alpha^4 + 3\alpha^2\beta^2 + \beta^4)(\alpha^2, \beta^2, \gamma^2, \gamma^2\alpha^2\beta^2)(\xi^4\eta^2, \xi^2\eta^4, \zeta^4\omega^2, \zeta^2\omega^4) \\
 & + 3(2\alpha^4 + 3\alpha^2\beta^2 + 3\alpha^2\gamma^2 - 7\beta^2\gamma^2)(\alpha^2, \alpha^2\beta^2\gamma^2, \beta^2, \gamma^2)(\xi^4\eta\zeta, \omega^4\eta\zeta, \eta^4\omega\xi, \zeta^4\omega\xi) \\
 & + 3(2\beta^4 + 3\beta^2\gamma^2 + 3\beta^2\alpha^2 - 7\gamma^2\alpha^2)(\beta^2, \beta^2\gamma^2\alpha^2, \gamma^2, \alpha^2)(\eta^4\zeta\xi, \omega^4\zeta\xi, \zeta^4\omega\eta, \xi^4\omega\zeta) \\
 & + 3(2\gamma^4 + 3\gamma^2\alpha^2 + 3\gamma^2\beta^2 - 7\alpha^2\beta^2)(\gamma^2, \gamma^2\alpha^2\beta^2, \alpha^2, \beta^2)(\zeta^4\xi\eta, \omega^4\xi\eta, \xi^4\omega\zeta, \eta^4\omega\xi) \\
 & + (\beta^6 + 9\beta^4\gamma^2 + 9\beta^2\gamma^4 + \gamma^6)(1, \alpha^6)(\eta^3\zeta^3, \xi^3\omega^3) \\
 & + (\gamma^6 + 9\gamma^4\alpha^2 + 9\gamma^2\alpha^4 + \alpha^6)(1, \beta^6)(\zeta^3\xi^3, \eta^3\omega^3) \\
 & + (\alpha^6 + 9\alpha^4\beta^2 + 9\alpha^2\beta^4 + \beta^6)(1, \gamma^6)(\xi^3\eta^3, \zeta^3\omega^3) \\
 & + 3\{\alpha^6 + 3\alpha^4(2\beta^2 + \gamma^2) + 3\alpha^2(\beta^4 - 7\beta^2\gamma^2) + \beta^6\}(1, \beta^2, \gamma^2, \beta^2\gamma^2)(\xi^4\eta\zeta, \eta^4\omega\xi^2, \zeta^4\omega^2\xi, \omega^4\eta\zeta^2) \\
 & + 3\{\beta^6 + 3\beta^4(2\gamma^2 + \alpha^2) + 3\beta^2(\gamma^4 - 7\gamma^2\alpha^2) + \gamma^6\}(1, \gamma^2, \alpha^2, \gamma^2\alpha^2)(\eta^4\zeta\xi, \zeta^4\omega\eta^2, \xi^4\omega^2\eta, \omega^4\zeta\xi^2) \\
 & + 3\{\gamma^6 + 3\gamma^4(2\alpha^2 + \beta^2) + 3\gamma^2(\alpha^4 - 7\alpha^2\beta^2) + \alpha^6\}(1, \alpha^2, \beta^2, \alpha^2\beta^2)(\zeta^4\xi^2\eta, \xi^4\omega\zeta^2, \eta^4\omega^2\zeta, \omega^4\xi\eta^2) \\
 & + 3\{\alpha^6 + 3\alpha^4(2\gamma^2 + \beta^2) + 3\alpha^2(\gamma^4 - 7\gamma^2\beta^2) + \beta^6\}(1, \gamma^2, \beta^2, \beta^2\gamma^2)(\xi^4\eta\zeta^2, \zeta^2\omega^2\xi, \eta^4\omega\xi^2, \omega^4\eta^2\zeta) \\
 & + 3\{\beta^6 + 3\beta^4(2\alpha^2 + \gamma^2) + 3\beta^2(\alpha^4 - 7\alpha^2\gamma^2) + \gamma^4\alpha^2\}(1, \alpha^2, \gamma^2, \gamma^2\alpha^2)(\eta^4\zeta\xi^2, \xi^2\omega^2\eta, \zeta^4\omega\eta^2, \omega^4\zeta^2\xi) \\
 & + 3\{\gamma^6 + 3\gamma^4(2\beta^2 + \alpha^2) + 3\gamma^2(\beta^4 - 7\beta^2\alpha^2) + \alpha^4\beta^2\}(1, \beta^2, \alpha^2, \alpha^2\beta^2)(\zeta^4\xi\eta^2, \eta^2\omega^2\zeta, \xi^4\omega\zeta^2, \omega^4\xi^2\eta) \\
 & + 9(\beta^2\gamma^4 + \beta^4\gamma^2 + \gamma^2\alpha^4 + \gamma^4\alpha^2 + \alpha^2\beta^4 + \alpha^4\beta^2 - 14\alpha^2\beta^2\gamma^2)(1, \beta^2\gamma^2, \gamma^2\alpha^2, \alpha^2\beta^2)(\xi^2\eta^2\zeta^2, \eta^2\zeta^2\omega^2, \zeta^2\xi^2\omega^2, \xi^2\omega^2\zeta^2) \\
 & + \xi^2\eta^2\zeta^2\omega^2 [\\
 & \quad - 3\{62\alpha^2\beta^2\gamma^2 - 28(\beta^3\gamma^3 + \gamma^3\alpha^3 + \alpha^3\beta^3)\}(\alpha^2, \beta^2, \gamma^2, \alpha^2\beta^2\gamma^2)(\xi^2, \eta^2, \zeta^2, \omega^2) \\
 & \quad + 3(3\alpha^6 - 14\alpha^4\beta\gamma + 130\alpha^4\beta^2\gamma^2 + 136\alpha^2\beta^3\gamma^3 - 42\beta^4\gamma^4)(1, \alpha^2)(\eta^2\zeta^2, \xi^2\omega^2) \\
 & \quad + 3(3\beta^6 - 14\beta^4\gamma\alpha + 130\beta^4\gamma^2\alpha^2 + 136\beta^2\gamma^3\alpha^3 - 42\gamma^4\alpha^4)(1, \beta^2)(\zeta^2\xi^2, \eta^2\omega^2) \\
 & \quad + 3(3\gamma^6 - 14\gamma^4\alpha\beta + 130\gamma^4\alpha^2\beta^2 + 136\gamma^2\alpha^3\beta^3 - 42\alpha^4\beta^4)(1, \gamma^2)(\xi^2\eta^2, \zeta^2\omega^2) \\
 & \quad] .
 \end{aligned}$$

This agrees with the result given in Salmon's *Solid Geometry*, Ed. 2, p. 151, [Ed. 4, p. 178], and *Quarterly Mathematical Journal*, vol. II. p. 220 (1858); in the latter place, however, the term

$$\beta^2\gamma^2\xi^4 + \gamma^2\alpha^2\eta^4 + \alpha^2\beta^2\omega^4 + \alpha^2\beta^2\gamma^2\xi^4\omega^4$$

is by mistake written

$$\beta^2\gamma^2\xi^4 + \gamma^2\alpha^2\eta^4 + \alpha^2\beta^2\omega^4 + \beta^2\gamma^2\xi^4\omega^4$$

viz. a factor α^2 is omitted in one of the coefficients.

Some of the coefficients are presented under slightly different forms; viz. instead of

$$62\alpha^2\beta^2\gamma^2 - 28(\beta^3\gamma^3 + \gamma^3\alpha^3 + \alpha^3\beta^3)$$

Salmon has

$$14(\beta^4\gamma^2 + \beta^2\gamma^4 + \gamma^4\alpha^2 + \gamma^2\alpha^4 + \alpha^4\beta^2 + \alpha^2\beta^4) + 20\alpha^2\beta^2\gamma^2;$$

and instead of

$$3\alpha^6 - 14\alpha^4\beta\gamma + 130\alpha^4\beta^2\gamma^2 + 136\alpha^2\beta^3\gamma^3 - 42\beta^4\gamma^4,$$

he has

$$-4\alpha^8 + 7\alpha^6(\beta^2 + \gamma^2) + 196\alpha^4\beta^2\gamma^2 - 53\alpha^2\beta^2\gamma^2(\beta^2 + \gamma^2) - 42\beta^4\gamma^4,$$

but these different forms are respectively equivalent in virtue of the relation

$$\alpha + \beta + \gamma = 0.$$

[437] DÉMONSTRATION NOUVELLE DU THÉORÈME DE M. CASEY PAR RAPPORT AUX
CERCLES QUI TOUCHENT À TROIS CERCLES DONNÉS.

[From the Annali di Matematica pura ed applicata, tom. 1. (1867), pp. 132–134.]

This is in fact the investigation contained in the paper 414, “On Polyzomal Curves otherwise the curves $\sqrt{U} + \sqrt{V} + \&c. = 0$,” Annex II. pp. 568–573, “On Casey’s theorem for the circle which touches three given circles,” viz. it is based on the identity of the two problems 1° to find a circle touching three given circles, 2° to find a cone-sphere (sphere of radius zero) passing through three given points in space.

[438] NOTE SUR QUELQUES TORSES SEXTIQUES.

[From the *Annali di Matematica pura ed applicata*, tom. II. (1868), pp. 99, 100.]

Je désire d'appeler attention aux surfaces développables, ou torses, données par l'équation

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0.$$

Dans cette équation (a, b, c, d, e) sont des fonctions linéaires quelconques des quatre coordonnées (x, y, z, t) ; ces quantités sont donc liées par une équation linéaire

$$Aa + 4Bb + 6Cc + 4Dd + Ee = 0,$$

et je remarque que la classification des torses comprises sous l'équation mentionnée dépend des propriétés invariantives de la fonction $(A, B, C, D, E)(t, 1)^4$.

En effet la torse a une courbe cuspidale, ou arête de rebroussement, donnée par les équations

$$ae - 4bd + 3c^2 = 0, \quad ace - ad^2 - b^2e + 2bcd - c^3 = 0,$$

et une courbe nodale donnée par les équations

$$\frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

ces deux courbes se rencontrent dans les points donnés par les équations

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e},$$

lesquels sont des points stationnaires de la courbe cuspidale. Pour trouver ces points, en écrivant $a : b : c : d : e = \tau^4 : \tau^3 : \tau^2 : \tau : 1$, on obtient pour le paramètre τ l'équation

$$(A, B, C, D, E)(\tau, 1)^4 = 0$$

et l'on voit ainsi qu'il y a un rapport entre la théorie de la surface et cette équation; à toute particularité invariante de l'équation, il y correspondra quelque particularité de la torse.

Les cas à considérer sont:

1°. Racines inégales, sans aucune relation invariante. C'est le cas général; je l'ai considéré dans le Mémoire, "On a certain Sextic Developable," *Quart. Math. Journ.*, t. IX. (1868), pp. 129–142. [398].

2°. Deux racines égales. Ce cas n'a pas été considéré; je remarque que la courbe cuspidale est du cinquième ordre. En effet on peut supposer que les racines égales soient $= 0$, ce qui revient à prendre $D = 0, E = 0$. On a donc $Aa + 4Bb + 6Cc = 0$, c'est-à-dire les équations $a = 0, b = 0$ impliquent l'équation $c = 0$; et on voit de là que les surfaces $ae - 4bd + 3c^2 = 0, ace - ad^2 - b^2e + 2bcd - c^3 = 0$ se coupent selon la droite $a = 0, b = 0$; il reste ainsi une courbe du cinquième ordre pour la courbe cuspidale.

3°. Trois racines égales. On peut supposer que ces racines sont $= 0$, ce qui donne $C = 0, D = 0, E = 0$; et l'on a ainsi $Aa + 4Bb = 0$, c'est-à-dire les plans $a = 0, b = 0$ sont ici un seul plan. L'équation de la torse contient le facteur a , et en l'écartant elle se réduit au cinquième ordre; on obtient ainsi la torse générale du cinquième ordre.

4°. Deux paires de racines égales. On peut supposer que ces racines sont $= \infty, 0$; cela donne $A = 0, B = 0, D = 0, E = 0$, et l'on a ainsi identiquement $c = 0$. L'équation de la torse est $(ae - 4bd)^3 - 27(-ad^2 - b^2e)^2 = 0$. J'ai considéré ce cas dans le Mémoire, "On a Special Sextic Developable," *Quart. Math. Journ.*, t. VII. (1866), pp. 105–113, [373]; la courbe cuspidale est du quatrième ordre, une courbe excubo-quartique d'une forme particulière.

5°. Quatre racines égales; en prenant ces racines $= 0$, on a $B = C = D = E = 0$, donc identiquement $a = 0$; l'équation de la torse contient le facteur b^2 , et en l'écartant elle se réduit au quatrième ordre: on a dans ce cas la torse générale du quatrième ordre.

Il y a encore deux cas à considérer.

6°. L'invariant I de la fonction $(A, B, C, D, E)(t, 1)^4$ est $= 0$;

7°. L'invariant J de cette fonction est $= 0$;

Mais je n'ai pas encore examiné ce que cela veut dire¹. Il n'y a pas le cas à considérer où l'on a à la fois $I = 0, J = 0$; car cela revient au cas 3° de trois racines égales.

Cambridge, le 16 mai 1868.

¹La courbe cuspidale étant du genre 0 (unicursale), on peut considérer la série des points de la courbe comme correspondant anharmoniquement aux points d'une droite. Si le système des quatre points stationnaires est harmonique on a $J = 0$; si ce système est équi-anharmonique, on a $I = 0$.

[439] ADDITION À LA NOTE SUR QUELQUES TORSES SEXTIQUES.

[From the *Annali di Matematica pura ed applicata*, tom. II. (1868), pp. 219–221.]

Je viens de trouver ce que signifie la condition $J = 0$. Considérons deux surfaces quadriques qui se touchent (d'un contact ordinaire). Les équations tangentielles peuvent s'écrire sous la forme

$$\begin{aligned} a\xi^2 + b\eta^2 + c\zeta^2 + 2n\zeta\omega &= 0, \\ a'\xi^2 + b'\eta^2 + c'\zeta^2 + 2n'\zeta\omega &= 0, \end{aligned}$$

et l'on satisfait à ces équations par des valeurs de ξ, η, ζ, ω qui contiennent un paramètre arbitraire θ , en écrivant

$$\xi : \eta : \zeta : \omega = \alpha \left(\theta + \frac{1}{\theta} \right) : \beta \left(\theta - \frac{1}{\theta} \right) : \epsilon : \gamma \left(\theta^2 + \frac{1}{\theta^2} \right) + \delta;$$

en effet cela donne

$$\begin{aligned} a\alpha^2 + b\beta^2 + 2n\gamma\epsilon &= 0, & 2a\alpha^2 - 2b\beta^2 + c\epsilon^2 + 2n\delta\epsilon &= 0, \\ a'\alpha^2 + b'\beta^2 + 2n'\gamma\epsilon &= 0, & 2a'\alpha^2 - 2b'\beta^2 + c'\epsilon^2 + 2n'\delta\epsilon &= 0, \end{aligned}$$

ce qui détermine les valeurs de $\alpha : \beta : \gamma : \delta : \epsilon$. L'équation du plan tangent commun sera donc

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

ou, en multipliant par $12\theta^2$, cette équation sera

$$(a, b, c, d, e)(\theta, 1)^4 = 0,$$

les valeurs des coefficients étant

$$\begin{aligned} a &= 12\xi w, \\ b &= 3(\alpha x + \beta y), \\ c &= 2(\delta w + \epsilon z), \\ d &= 3(\alpha x - \beta y), \\ e &= 12\gamma w. \end{aligned}$$

En représentant par $Aa + 4Bb + 6Cc + 4Dd + Ee = 0$ l'équation qui lie les fonctions linéaires a, b, c, d, e , cette équation sera $a - e = 0$; on a donc $A = -E = 1$, $B = C = D = 0$; l'invariant J de la fonction $(A, B, C, D, E)(\tau, 1)^4$ est donc $= 0$.

Nous arrivons ainsi à la conclusion que la torse sextique

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e - c^3 + 2bcd)^2 = 0,$$

où les fonctions linéaires a, b, c, d, e sont liées par une équation

$$Aa + 4Bb + 6Cc + 4Dd + Ee = 0,$$

telle que l'invariant

$$J = ACE - AD^2 - EB^2 - C^3 - 2BCD$$

de la fonction $(A, B, C, D, E)(t, 1)^4$ est $= 0$ (cas 7° de la Note), est la torse enveloppée par le plan tangent commun de deux surfaces quadriques qui se touchent d'un contact ordinaire. J'ai trouvé l'équation de cette torse dans le Mémoire, "On the Developable Surfaces which arise from two Surfaces of the Second Order," *Camb. and Dubl. Math. Jour.*, t. v. (1850), pp. 46–57, voir

p. 56, [85]; où, en supposant que $a, b, c, n, a', b', c', n'$ dénotent les mêmes coefficients comme à présent, et en écrivant

$$\begin{aligned} bc' - b'c &= f, & an' - a'n &= p, \\ ca' - c'a &= g, & bn' - b'n &= q, \\ ab' - a'b &= h, & cn' - c'n &= r, \end{aligned}$$

(et de là $pf + qg + rh = 0$), l'équation trouvée est

$$f^2g^2h^2w^4 + \dots + 4pqr(qx^2 + py^2)^2 = 0.$$

Je vais vérifier ces termes. Partant de l'équation $(a, b, c, d, e)(\theta, 1)^4 = 1$, l'équation de la torse, en y introduisant pour commodité le facteur $-\frac{1}{27}pqr$, sera

$$\begin{aligned} & -\frac{1}{27}pqr \{[(\delta w + \epsilon z)^2 + 12\gamma^2 - 3(\alpha^2x^2 - \beta^2y^2)]^3 \\ & - [2(\delta w + \epsilon z)^3 - 9(\delta w + \epsilon z)(\alpha^2x^2 - \beta^2y^2) - 27(\delta w + \epsilon z)\gamma w^2 + 54(\alpha^2x^2 + \beta^2y^2)\gamma w]^2\} = 0. \end{aligned}$$

En prenant $\gamma\epsilon = ab' - a'b = h$, on obtient pour $\alpha, \beta, \gamma, \delta, \epsilon$ les valeurs

$$\alpha^2 = 2q, \quad \beta^2 = -2p, \quad \gamma\epsilon = h, \quad \epsilon^2 = -\frac{9pq}{r}, \quad \delta\epsilon = \frac{2(fp - gq)}{r},$$

et de là

$$\gamma^2 = -\frac{h^2r}{9pq}, \quad \delta^2 = -\frac{(fp - gq)^2}{2pqr} = -\frac{h^2r + 4fgpq}{2pqr}, \quad \gamma^2\delta^2 = \frac{h^2(h^2r + 4fgpq)}{18p^2q^2r}, \quad \delta^2 - 4\gamma^2 = \frac{2fg}{r}.$$

Les termes en w^4 et $(x^2, y^2)^2$ sont

$$\begin{aligned} & -\frac{1}{27}pqr \{[4(\delta^2 + 12\gamma^2)^3 - (2\delta^3 - 72\gamma^2\delta)^2]w^4 - 108(\alpha^2x^2 - \beta^2y^2)^3\}, \\ & = -\frac{1}{27}pqr \{432\gamma^2(\delta^2 - 4\gamma^2)^2w^4 - 108(\alpha^2x^2 - \beta^2y^2)^3\}; \end{aligned}$$

ces termes sont donc

$$\begin{aligned} & = -\frac{1}{27}pqr \left\{ -432 \cdot \frac{h^2r}{9pq} \cdot \frac{4f^2g^2}{r^2} w^4 + 864(qx^2 + py^2)^3 \right\} \\ & = f^2g^2h^2w^4 + 4pqr(qx^2 + py^2)^3, \end{aligned}$$

comme cela doit être.

Cambridge, le 22 septembre 1868.

[440] NOTE SUR UNE TRANSFORMATION GÉOMÉTRIQUE.

[From the *Journal für die reine und angewandte Mathematik (Crelle)*, tom. LXVII. (1867), pp. 95, 96.]

La lecture de la Note de M. Hesse, “Ein Uebertragungsprincip” (t. LXVI. p. 15 de ce Journal) m’a suggéré les remarques suivantes:

Soient (a_1, b_1, c_1, d_1) , (a_2, b_2, c_2, d_2) , (a_3, b_3, c_3, d_3) des constantes données, on peut supposer que les coordonnées (x, y) d’un point quelconque dans un plan soient exprimées en fonctions des paramètres variables (u, v) par les équations

$$x = \frac{a_1 + b_1u + c_1v + d_1uv}{a_3 + b_3u + c_3v + d_3uv}, \quad y = \frac{a_2 + b_2u + c_2v + d_2uv}{a_3 + b_3u + c_3v + d_3uv}.$$

En introduisant une nouvelle indéterminée s , ces équations peuvent être écrites dans la forme

$$sx = a_1 + b_1u + c_1v + d_1uv,$$

$$sy = a_2 + b_2u + c_2v + d_2uv,$$

$$s = a_3 + b_3u + c_3v + d_3uv;$$

pour des valeurs données des coordonnées (x, y) la quantité s est en général déterminée par une équation quadratique, et les paramètres u et v sont des fonctions linéaires données de s ; il y a cependant deux cas particuliers qu’il convient de distinguer.

1°. L’équation quadratique en s peut avoir la racine $s = 0$, et débarrassée de ce facteur, se réduire par conséquent à une équation linéaire; ce cas particulier a lieu si la condition $(abc)(bcd) = (abd)(acd)$ est remplie, où la notation (abc) désigne le déterminant

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

Dans ce cas u et v sont des fonctions rationnelles de (x, y) et la transformation a la signification géométrique suivante:

En considérant deux droites quelconques L, M dans l’espace et en menant par le point donné (x, y) la droite unique G qui rencontre ces deux droites, on peut supposer que u et v soient des paramètres qui déterminent les positions des points de rencontre sur les deux droites respectivement; c. à. d. que u soit la distance d’un point fixe sur la droite L au point de rencontre avec la droite G , et de même que v soit la distance d’un point fixe sur la droite M au point de rencontre avec la droite G .

2°. Supposons $b_1 : b_3 = c_1 : c_3$ ou ce qui est au fond la même chose $b_2 : b_3 = c_2 : c_3$, $b_1 - c_1 = 0$, $b_2 - c_2 = 0$, $b_3 - c_3 = 0$; alors s est déterminée par une équation simple, mais u et v ne sont plus des fonctions rationnelles de s ; on voit que dans ce cas $u + v$ et uv sont des fonctions rationnelles de (x, y) , et que par conséquent u et v sont les racines d’une équation quadratique qui contient (x, y) linéairement. On peut supposer que u et v soient les paramètres de deux points sur une droite donnée, c. à. d. que u et v soient les distances de ces deux points respectivement à un point fixe situé sur la droite donnée; on a ainsi la transformation de M. Hesse.

Je n’ai pas cherché la signification géométrique des formules générales.

Cambridge, 10 octobre 1866.

[441] NOTE SUR L'ALGORITHME DES TANGENTES DOUBLES D'UNE COURBE DU QUATRIÈME ORDRE.

[From the *Journal für die reine und angewandte Mathematik (Crelle)*, tom. LXVIII. (1868), pp. 176–179.]

On n'a pas, je crois, assez fait attention à l'algorithme (tiré de la considération d'une figure dans l'espace) qu'a trouvé M. Hesse (dans le mémoire "Ueber die Doppeltangenten der Curven vierter Ordnung," t. XLIX de ce Journal, 1855) pour dénoter les tangentes doubles (ou bitangentes) d'une courbe du quatrième ordre. Voici en quoi cet algorithme consiste. En employant les huit symboles 1, 2, 3, ..., 8, les 28 bitangentes sont représentées par les combinaisons binaires 12, 13, 14, ..., 78. Cela posé, considérons une expression quelconque 12.13.14, ou 12.34, ... ou disons un "terme" qui représente un système d'une seule ou de plusieurs des bitangentes. On peut opérer sur ce terme avec deux espèces de substitutions; la substitution ordinaire qui consiste à changer l'arrangement 12345678 des huit symboles en un autre arrangement quelconque; et la substitution "bifide" représentée par un symbole tel que 1234.5678, lequel dénote qu'il faut entrechanger les combinaisons 12 et 34, 13 et 24, 14 et 23, 56 et 78, 57 et 68, 58 et 67, en ne changeant pas les autres combinaisons. Par exemple en opérant avec 1234.5678 sur 34.45.56.17 on obtient 12.45.78.17. Le nombre de ces substitutions bifides est 35, ou en comptant la substitution unité, qui ne change aucune des combinaisons, ce nombre est 36.

Appelons "homotypiques" deux termes qui se dérivent l'un de l'autre par une substitution ordinaire; "syntypiques" qui se dérivent l'un de l'autre par une substitution ordinaire ou bifide; "sous-groupe" le système entier des termes homotypiques à un terme donné; "groupe" le système entier des termes syntypiques à un terme donné. Un groupe peut contenir un seul sous-groupe, ou plusieurs sous-groupes; mais il importe de remarquer que la notion du sous-groupe n'a pas de signification géométrique, et ne sert que comme moyen de former les termes du groupe. Cela étant, le théorème géométrique est celui-ci: "les systèmes de bitangentes représentées par des termes syntypiques (ou autrement dit, par des termes qui appartiennent au même groupe) ont les mêmes propriétés géométriques."

Par exemple, en considérant les bitangentes deux à deux, on a deux sous-groupes, l'un composé de termes homotypiques à 12.13; l'autre, de termes homotypiques à 12.34—ou disons, le sous-groupe 12.13 de 168 termes et le sous-groupe 12.34 de 210 termes; mais ces deux sous-groupes ne forment qu'un seul groupe: pour montrer cela il suffit d'opérer sur 12.13, par exemple avec la substitution 1245.3678, ce qui donne 45.13, terme homotypique à 12.34. Cela veut dire qu'il n'y a pas de combinaison de deux bitangentes qui se distingue d'une manière quelconque de toute autre combinaison de deux bitangentes.

Mais en combinant les bitangentes trois à trois, on a les deux sous-groupes 12.34.56 (420 termes) et 12.23.34 (840 termes) qui forment un groupe de 1260 termes; les trois bitangentes représentées par un quelconque des 1260 termes ont leurs six points de contact sur une même conique. Les trois autres sous-groupes 12.23.31 (56 termes), 12.23.45 (1680 termes) et 12.13.14 (280 termes) forment un groupe de 2016 termes, et pour trois bitangentes représentées par un terme quelconque de ce groupe, les six points de contact ne sont pas situés sur une même conique.

Comme un autre exemple j'explique la constitution des 63 "groupes" de Steiner (voir le mémoire de Steiner, "Eigenschaften der Curven vierten Grades rücksichtlich ihrer Doppeltangenten," t. XLIX. de ce journal, 1855) ou (pour éviter l'emploi de ce mot groupe dans une nouvelle signification) disons les 63 termes G de Steiner, chaque terme composé de 6 paires de bitangentes. On a ici un sous-groupe de 35 termes G , de la forme

$$12.34; 13.42; 14.23; 56.78; 57.86; 58.67$$

(pour abrégé on peut dénoter ce terme par 1234.5678), et un sous-groupe de 28 termes G , de la forme

$$13.32; 14.42; 15.52; 16.62; 17.72; 18.82$$

(pour abrégé on peut de même dénoter ce terme par 12.345678), les deux sous-groupes forment le groupe des 63 termes G .

Steiner a de plus considéré les “systèmes” ou disons les termes S_1, S_2 , composés chacun de trois termes G ; savoir 315 termes S_1 et 336 termes S_2 . Les 315 termes S_1 sont ici un groupe composé d'un sous-groupe de 105 termes $3G_1$, de la forme

$$1234.5678; 1256.3478; 1278.3456$$

et un sous-groupe de 210 termes $2G_1 + G_2$ de la forme

$$12.345678; 34.125678 \text{ et } 1234.5678.$$

Et de même les 336 termes S_2 sont un groupe composé d'un sous-groupe de 280 termes $2G_1 + G_2$ de la forme

$$1234.5678; 5234.1678 \text{ et } 15.234678$$

et un sous-groupe de 56 termes $3G_2$ de la forme

$$12.34.5678; 13.24.5678; 31.245678.$$

Il va sans dire que je me suis servi de l'abréviation 1234.5678 pour dénoter le terme 12.34; 13.42; 14.23; 56.78; 57.86; 58.67; et de même pour les autres termes G_1 ou G_2 .

M. Aronhold (dans le mémoire “Ueber den gegenseitigen Zusammenhang der 28 Doppeltangenten einer allgemeinen Curve vierten Grades,” *Berl. Monatsber.* Juli 1864), partant de 7 bitangentes données, a trouvé une construction pour les autres 21 bitangentes. Les bitangentes données doivent être indépendantes; savoir pour trois quelconques de ces 7 bitangentes, les six points de contact ne sont pas situés sur une même conique. Les bitangentes représentées par les termes 12, 13, 14, 15, 16, 17, 18 sont un tel système de bitangentes indépendantes; et en dénotant de cette manière les 7 bitangentes données, la bitangente construite par le moyen de la conique qui touche cinq de ces droites, par exemple les droites 38, 48, 58, 68, 78, (ou conique 34567) peut être dénotée par 12, et de même pour les autres bitangentes cherchées; on a ainsi le système entier des bitangentes dénotées comme auparavant par 12, 13, 14, ... 78.

J'ajoute que le groupe qui contient 18, 28, 38, 48, 58, 68, 78 est composé d'un sous-groupe 18, 28, 38, 48, 58, 68, 78 de 8 termes, et d'un sous-groupe 12, 23, 31, 48, 58, 68, 78 de 280 termes; le groupe contient donc 288 termes; savoir il y a ce nombre 288 de systèmes de sept bitangentes indépendantes qui peuvent chacun servir à trouver par la construction d'Aronhold les autres 21 bitangentes.

P.S. J'ai trouvé à propos de la méthode de M. Aronhold une forme commode pour l'équation de la conique qui touche cinq droites données; en supposant que l'on ait identiquement $x + y + z + w = 0$, et que les droites données soient $x = 0$, $y = 0$, $z = 0$, $w = 0$, et $ax + by + cz + dw = 0$, l'équation de la conique est

$$(a - d)^2(b - c)^2(xw + yz) + (b - d)^2(c - a)^2(yw + zx) + (c - d)^2(a - b)^2(zw + xy) = 0.$$

J'ajoute qu'en écrivant pour abrégé

$$\alpha : \beta : \gamma = (a - d)(b - c) : (b - d)(c - a) : (c - d)(a - b)$$

(d'où $\alpha + \beta + \gamma = 0$) les coordonnées (x, y, z, w) des points de contact avec les droites

$$x = 0, y = 0, z = 0, w = 0 \text{ sont } (0, \gamma, \beta, \alpha), (\gamma, 0, \alpha, \beta), (\beta, \alpha, 0, \gamma), (\alpha, \beta, \gamma, 0)$$

respectivement; et que les coordonnées du point de contact avec la droite $ax + by + cz + dw = 0$ sont

$$x : y : z : w = (bcd) : -(cda) : (dab) : -(abc)$$

où, pour abrégé, (bcd) dénote $(b - c)(c - d)(d - b)$, et de même pour (cda) , (dab) , (abc) .

Cambridge, le 23 septembre 1867.

[442] NOTE SUR LA SURFACE DU QUATRIÈME ORDRE DOUÉE DE SEIZE POINTS SINGULIERS ET DE SEIZE PLANS SINGULIERS.

[From the *Journal für die reine und angewandte Mathematik (Crelle)*, tom. LXXIII. (1871), pp. 292–293.]

L'équation de M. Kummer se transforme sans difficulté en celle-ci

$$\sqrt{\alpha x} \left(\gamma' \gamma'' y - \beta' \beta'' z - \frac{w}{\alpha} \right) + \sqrt{\beta y} \left(\alpha' \alpha'' z - \gamma' \gamma'' x - \frac{w}{\beta} \right) + \sqrt{\gamma z} \left(\beta' \beta'' x - \alpha' \alpha'' y - \frac{w}{\gamma} \right) = 0,$$

où

$$\alpha + \beta + \gamma = 0, \quad \alpha' + \beta' + \gamma' = 0, \quad \alpha'' + \beta'' + \gamma'' = 0.$$

Or cette équation rendue rationnelle prend, après toutes les réductions nécessaires, la forme suivante:

$$\begin{aligned} & w^2(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) \\ & + 2w(\alpha\alpha'(y^2z - yz^2) + \beta\beta'(z^2x - zx^2) + \gamma\gamma'(x^2y - xy^2) + \theta xyz) \\ & + (\alpha\alpha'y z + \beta\beta'z x + \gamma\gamma'x y)^2 = 0, \end{aligned}$$

où, pour abréger, l'on a écrit

$$\begin{aligned} \theta &= (\beta - \gamma)\alpha'\alpha'' + (\gamma - \alpha)\beta'\beta'' + (\alpha - \beta)\gamma'\gamma'' \\ &= (\beta' - \gamma')\alpha''\alpha + (\gamma' - \alpha')\beta''\beta + (\alpha' - \beta')\gamma''\gamma \\ &= (\beta'' - \gamma'')\alpha\alpha' + (\gamma'' - \alpha'')\beta\beta' + (\alpha'' - \beta'')\gamma\gamma' \\ &= -\frac{1}{2}\{(\beta - \gamma)(\beta' - \gamma')(\beta'' - \gamma'') + (\gamma - \alpha)(\gamma' - \alpha')(\gamma'' - \alpha'') + (\alpha - \beta)(\alpha' - \beta')(\alpha'' - \beta'')\}. \end{aligned}$$

L'identité de ces différentes valeurs de θ étant facile à vérifier.

En représentant par $Aw^2 + 2Bw + C = 0$ la forme rationnelle de l'équation de la surface, on trouve pour le discriminant $AC - B^2$ de cette équation du second degré en w la valeur

$$AC - B^2 = 4\alpha\alpha''\beta\beta'\gamma\gamma'\gamma'' - 2\theta xyz \left(\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} \right) \left(\frac{x}{\alpha'} + \frac{y}{\beta'} + \frac{z}{\gamma'} \right) \left(\frac{x}{\alpha''} + \frac{y}{\beta''} + \frac{z}{\gamma''} \right).$$

L'équation de la surface rendue rationnelle est symétrique par rapport aux trois systèmes de quantités (α, β, γ) , $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$; la forme irrationnelle de la même équation peut donc être présentée de trois manières différentes, savoir:

$$\begin{aligned} & \sqrt{\alpha x} \left(\gamma' \gamma'' y - \beta' \beta'' z - \frac{w}{\alpha} \right) + \sqrt{\beta y} \left(\alpha' \alpha'' z - \gamma' \gamma'' x - \frac{w}{\beta} \right) + \sqrt{\gamma z} \left(\beta' \beta'' x - \alpha' \alpha'' y - \frac{w}{\gamma} \right) = 0, \\ & \sqrt{\alpha' x} \left(\gamma'' \gamma y - \beta'' \beta z - \frac{w}{\alpha'} \right) + \sqrt{\beta' y} \left(\alpha'' \alpha z - \gamma'' \gamma x - \frac{w}{\beta'} \right) + \sqrt{\gamma' z} \left(\beta'' \beta x - \alpha'' \alpha y - \frac{w}{\gamma'} \right) = 0, \\ & \sqrt{\alpha'' x} \left(\gamma \gamma' y - \beta \beta' z - \frac{w}{\alpha''} \right) + \sqrt{\beta'' y} \left(\alpha \alpha' z - \gamma \gamma' x - \frac{w}{\beta''} \right) + \sqrt{\gamma'' z} \left(\beta \beta' x - \alpha \alpha' y - \frac{w}{\gamma''} \right) = 0; \end{aligned}$$

et l'on voit de plus que les équations des seize plans singuliers sont

$$\begin{aligned} & x = 0, \quad y = 0, \quad z = 0, \quad w = 0, \\ & \frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 0, \quad \frac{x}{\alpha'} + \frac{y}{\beta'} + \frac{z}{\gamma'} = 0, \quad \frac{x}{\alpha''} + \frac{y}{\beta''} + \frac{z}{\gamma''} = 0, \\ & \gamma' \gamma'' y - \beta' \beta'' z - \frac{w}{\alpha} = 0, \quad \alpha' \alpha'' z - \gamma' \gamma'' x - \frac{w}{\beta} = 0, \quad \beta' \beta'' x - \alpha' \alpha'' y - \frac{w}{\gamma} = 0, \\ & \gamma'' \gamma y - \beta'' \beta z - \frac{w}{\alpha'} = 0, \quad \alpha'' \alpha z - \gamma'' \gamma x - \frac{w}{\beta'} = 0, \quad \beta'' \beta x - \alpha'' \alpha y - \frac{w}{\gamma'} = 0, \\ & \gamma \gamma' y - \beta \beta' z - \frac{w}{\alpha''} = 0, \quad \alpha \alpha' z - \gamma \gamma' x - \frac{w}{\beta''} = 0, \quad \beta \beta' x - \alpha \alpha' y - \frac{w}{\gamma''} = 0, \end{aligned}$$

les quantités α, β, γ , etc. étant liées entre elles par les trois équations

$$\alpha + \beta + \gamma = 0, \quad \alpha' + \beta' + \gamma' = 0, \quad \alpha'' + \beta'' + \gamma'' = 0.$$

Voilà ce me semble la forme la plus simple pour l'équation de cette surface.

Cambridge, le 23 février 1871.

[443] NOTE ON THE SOLUTION OF THE QUARTIC EQUATION $\alpha U + 6\beta H = 0$.

[From the *Mathematische Annalen*, vol. I. (1869), pp. 54, 55.]

If U denote the quartic function $(a, b, c, d, e \mid x, y)^4$, H its Hessian

$$H = (ac - b^2, 2(ad - bc), ae + 2bd - 3c^2, 2(be - cd), ce - d^2 \mid x, y)^4,$$

α and β constants, then we may find the linear factors of the function $\alpha U + 6\beta H$ (or what is the same thing solve the equation $\alpha U + 6\beta H = 0$) by a formula almost identical with that given by me (Fifth Memoir on Quantics, *Phil. Trans.* vol. CXLVIII. (1858), see p. 446, [156]) in regard to the original quartic function U .

In fact (reproducing the investigation) if I, J are the two invariants, $M = \frac{I^3}{4J^2}$, Φ the cubicovariant

$$\Phi = (-a^2d + 3abc - 2b^3, \dots \mid x, y)^6,$$

then the identical equation $JU^3 - IU^2H + 4H^3 = -\Phi^2$, may be written $(1, 0, -M, M \mid IH, JU)^3 = -\frac{1}{4}I^3\Phi^2$, whence if $\omega_1, \omega_2, \omega_3$ are the roots of the equation $(1, 0, -M, M \mid \omega, 1)^3 = 0$, or what is the same thing $\omega^3 - M(6\omega - 1) = 0$, then the functions

$$IH - \omega_1JU, \quad IH - \omega_2JU, \quad IH - \omega_3JU$$

are each of them a square: writing

$$(\omega_2 - \omega_3)(IH - \omega_1JU) = X^2,$$

$$(\omega_3 - \omega_1)(IH - \omega_2JU) = Y^2,$$

$$(\omega_1 - \omega_2)(IH - \omega_3JU) = Z^2,$$

so that identically $X^2 + Y^2 + Z^2 = 0$, the expression $\alpha X + \beta Y + \gamma Z$ will be a square if only $\alpha^2 + \beta^2 + \gamma^2 = 0$. (To see this observe that in virtue of the equation $X^2 + Y^2 + Z^2 = 0$, we have $X + iY, X - iY$ each of them a square, and thence

$$\alpha X + \beta Y + \gamma Z = \frac{1}{2}(\alpha + i\beta)(X - iY) + \frac{1}{2}(\alpha - i\beta)(X + iY) - \frac{1}{2}i\gamma\sqrt{X^2 + Y^2},$$

is a square if the condition in question be satisfied.)

Hence in particular writing

$$\sqrt{\omega_2 - \omega_3} \sqrt{\alpha I + 6\beta\omega_1J}, \dots \sqrt{\omega_1 - \omega_2} \sqrt{\alpha I + 6\beta\omega_3J},$$

for α, β, γ , we have

$$(\omega_2 - \omega_3)\sqrt{\alpha I + 6\beta\omega_1J} \sqrt{IH + \omega_1JU} + \dots + (\omega_1 - \omega_2)\sqrt{\alpha I + 6\beta\omega_3J} \sqrt{IH + \omega_3JU}$$

a perfect square, and since the product of the four different values is a multiple of $(\alpha U + 6\beta H)^2$ (this is most readily seen by observing that for $\alpha U + 6\beta H = 0$, the irrational expression omitting a factor is $(\omega_2 - \omega_3)(\alpha I + 6\beta\omega_1J) + \dots + (\omega_1 - \omega_2)(\alpha I + 6\beta\omega_3J)$, which vanishes identically) it follows that the expression in question is the square of a linear factor of $\alpha U + 6\beta H$.

It thus appears that the radicals (other than those arising from the solution of $U = 0$) contained in the solution of the equation $\alpha U + 6\beta H = 0$ are the three roots

$$\sqrt{\alpha I + 6\beta\omega_1J}; \quad \sqrt{\alpha I + 6\beta\omega_2J}; \quad \sqrt{\alpha I + 6\beta\omega_3J}.$$

Cambridge, September 2, 1868.

[444] ON THE CENTRO-SURFACE OF AN ELLIPSOID.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 16–18.]

The President [Prof. Cayley] gave an account of his investigations on the centro-surface of an ellipsoid (locus of the centres of curvature of the ellipsoid). The surface has been studied by Dr Salmon, and also by Prof. Clebsch, but in particular the theory of the nodal curve on the surface admits of further development. The position of a point on the ellipsoid is determined by means of the parameters, or elliptic coordinates, h, k ; viz., if as usual a, b, c are the semi-axes, and if X, Y, Z are the coordinates of the point in question, then

$$\frac{X^2}{a^2 + h} + \frac{Y^2}{b^2 + h} + \frac{Z^2}{c^2 + h} = 1,$$

$$\frac{X^2}{a^2 + k} + \frac{Y^2}{b^2 + k} + \frac{Z^2}{c^2 + k} = 1;$$

and hence

$$-\beta\gamma X^2 = a^2(a^2 + h)(a^2 + k),$$

$$-\gamma\alpha Y^2 = b^2(b^2 + h)(b^2 + k),$$

$$-\alpha\beta Z^2 = c^2(c^2 + h)(c^2 + k),$$

if for shortness

$$\alpha = b^2 - c^2, \quad \beta = c^2 - a^2, \quad \gamma = a^2 - b^2 \quad (\alpha + \beta + \gamma = 0).$$

This being so, the coordinates of the point of intersection of the normal at (X, Y, Z) by the normal at the consecutive point of the curve of curvature

$$\frac{X^2}{a^2 + k} + \frac{Y^2}{b^2 + k} + \frac{Z^2}{c^2 + k} = 1$$

are given by the formulae

$$-\beta\gamma a^2 x^2 = (a^2 + h)^2(a^2 + k),$$

$$-\gamma\alpha b^2 y^2 = (b^2 + h)^2(b^2 + k),$$

$$-\alpha\beta c^2 z^2 = (c^2 + h)^2(c^2 + k);$$

viz., these equations, considering therein (h, k) as arbitrary parameters, determine the coordinates (x, y, z) of a point on the centre-surface. The principal sections (as is known) consist each of them of an ellipse counting three times, and of an evolute of an ellipse; the evolute and ellipse have four contacts (two-fold intersections) and four simple intersections, but the contacts and intersections respectively are in the different sections real and imaginary; and if (as we may without loss of generality assume) $a^2 + c^2 > 2b^2$, then the form of the principal sections is as shown in the figure (which represents only an octant of the surface); viz., there is a real contact at P in the plane of xz , and a real intersection at Q in the plane of xy . The surface has thus an exterior and an interior sheet, but (instead of meeting in a conical point, as in the wave surface) these intersect in a nodal curve QP . The curve has a cusp at Q , and a node at P ; viz., the curve extends beyond P , but from that point is acnodal, or without any real sheet of the surface passing through it. For the nodal curve there must be two values (h, k) , (h_1, k_1) , giving the same values of (x, y, z) ; viz., there must exist the relations

$$(a^2 + h)^2(a^2 + k) = (a^2 + h_1)^2(a^2 + k_1),$$

$$(b^2 + h)^2(b^2 + k) = (b^2 + h_1)^2(b^2 + k_1),$$

$$(c^2 + h)^2(c^2 + k) = (c^2 + h_1)^2(c^2 + k_1);$$

from which equations eliminating k and k_1 we should have between h, k a relation which, combined with the expressions of x, y, z in terms of (h, k) , determines the nodal curve. But the better course is to eliminate k, k_1 , thus obtaining a relation between h and h_1 , in virtue whereof h_1 may be regarded as a known function of h ; k and k_1 can then be readily expressed in terms of h, h_1 ; that is, we have k as a function of h, h_1 , or in effect as a function of h . The relation between h, h_1 (after a reduction of some complexity) assumes ultimately a form which is very simple and remarkable; viz., writing

$$P = a^2 + b^2 + c^2, \quad Q = b^2c^2 + c^2a^2 + a^2b^2, \quad R = a^2b^2c^2,$$

the relation is

$$(QR + 3Qh^2 + Ph^4) + h_1(3Qh + 4Ph^2 + 3h^3) + h_1^2(P + 3h^2) = 0;$$

this is a $(2, 2)$ correspondence between the two parameters h, h_1 ; the united values $h_1 = h$, are given by the equation $6(R + Qh + Ph^2 + h^3) = 0$, that is

$$(a^2 + h)(b^2 + h)(c^2 + h) = 0;$$

viz., the two points on the ellipsoid which have their common centre of curvature on the nodal curve are only situate on the same curve of curvature when this curve is a principal section of the ellipsoid.

[Since the date of the foregoing communication, Prof. Cayley has found that the squared coordinates x^2, y^2, z^2 of a point on the nodal curve can be expressed as rational functions of a single variable parameter σ .]

[445] A MEMOIR ON QUARTIC SURFACES.

*[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 19–69.
Read February 10, 1870.]*

The present Memoir is intended as a commencement of the theory of the quartic surfaces which have nodes (conical points). A quartic surface may be without nodes, or it may have any number of nodes up to 16. I show that this is so, and I consider how many of the nodes may be given points. Although it would at first sight appear that the number is 8, it is in fact 7; viz., we can, with 7 given points as nodes (but not in a proper sense with 8 or more given points), find a quartic surface; such surface contains in its equation 6 constants, which may be such that the surface has an additional node or nodes. Suppose that the surface has an 8th node: there are two distinct cases; viz., (1) the 8 nodes are the points of intersection of 3 quadric surfaces, or say they are an octad, and the surface is said to be octadic; (2) the 8th node is any point whatever on a certain sextic surface determined by means of the 7 given nodes, and called the dianodal surface of these 7 points; the quartic surface is said to be a dianome. The two cases are in general exclusive of each other; viz., the 7 given points being any points whatever, the dianodal surface does not pass through the 8th point of the octad; and thus the quartic surface with the 8 nodes is either octadic or else a dianome. Assuming it to be a dianome, the constants may be further determined so that there shall be a 9th node; it is necessary to examine whether this forms with 7 of the 8 nodes an octad. Supposing that it does not (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 9th node is then any point whatever on a certain curve of the order 18, determined by means of the 8 nodes, and called the dianodal curve of these 8 points. And, finally, the constants may be further determined so that there shall be a

10th node; supposing, as before, that this does not form an octad with any 7 of the 9 nodes (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 10th node is then any one of a system of 22 [should be 13] points determined by means of the 9 nodes, and called the dianodal system of these 9 points. But the quartic surface is now completely determined; viz., starting with any 7 given points as nodes, we have a dianome with 8 nodes, 9 nodes, or 10 nodes, say, an octodianome, enneadianome, or decadianome, but not with any greater number of nodes; these can only present themselves when particular conditions are satisfied in regard to the 7 given nodes, and to the 8th and 9th node; and the consideration of the quartic surfaces with more than 10 nodes would thus form a separate branch of the subject.

The case of the decadianome (or quartic surface with 10 nodes formed as above with 7 given points as nodes) is peculiarly interesting. I identify this with the surface which I call a symmetroid; viz., the surface represented by an equation $\Delta = 0$, where Δ is a symmetrical determinant of the 4th order the several terms whereof are linear functions of the coordinates (x, y, z, w) ; this surface is related to the Jacobian surface of 4 quadric surfaces (itself a very remarkable surface), and this theory of the symmetroid and the Jacobian, and of questions connected therewith, forms a large portion of the present Memoir.

The theory of the Jacobian is connected also with the researches in regard to nodal quartic surfaces in general; and, for greater clearness, it has seemed to me proper to commence the Memoir with certain definitions, &c., in regard to this theory. It will be seen in what manner I extend the notion of the Jacobian.

I remark that the present researches on Quartic Surfaces were suggested to me by Professor Kummer's most interesting Memoir "Ueber die algebraischen Strahlensysteme u.s.w.," *Berl. Abh.* 1866, in which, without entering upon the general theory, he is led to consider the quartic surfaces, or certain quartic surfaces, with 16, 15, 14, 13, 12, or 11 nodes; the last of these, or surface with 11 nodes, being in fact a particular case of the symmetroid.

Considerations in regard to the Jacobian of four, or more or less than four, Surfaces.

1. In the case of any four surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, the differential coefficients of P , Q , R , S in regard to the coordinates (x, y, z, w) may be arranged as a square matrix in either of the ways

$$\begin{array}{c|cccc} & P & Q & R & S \\ \hline \partial_x & & & & \\ \partial_y & & & & \\ \partial_z & & & & \\ \partial_w & & & & \end{array} ; \begin{array}{c|cccc} & \partial_x & \partial_y & \partial_z & \partial_w \\ \hline P & & & & \\ Q & & & & \\ R & & & & \\ S & & & & \end{array}$$

and with either arrangement we may form one and the same determinant, the Jacobian determinant $J(P, Q, R, S)$, or, equating it to zero, the Jacobian surface $J(P, Q, R, S) = 0$ of the four surfaces.

2. In the case of more than four surfaces, adopting the arrangement

$$\begin{array}{c|cccccc} & P & Q & R & S & T & \dots \\ \hline \partial_x & & & & & & \\ \partial_y & & & & & & \\ \partial_z & & & & & & \\ \partial_w & & & & & & \end{array}$$

and considering the several determinants which can be formed with any four columns of the matrix, these equated to zero establish a more than one-fold relation between the coordinates;

viz., in the case of five surfaces, we have $J(P, Q, R, S, T) = 0$ a twofold relation representing a curve; and in the case of six surfaces, $J(P, Q, R, S, T, U) = 0$ a threefold relation representing a point-system; and (since with four coordinates a relation is at most threefold) these are the only cases to be considered.

3. In the case of fewer than four surfaces, adopting the arrangement

$$\begin{array}{c|cccc} & \partial_x & \partial_y & \partial_z & \partial_w \\ \hline P & & & & \\ Q & & & & \end{array}$$

and considering the several determinants which can be formed with any 3 or 2 columns of the matrix and equating these to zero, we have in like manner a more than onefold relation between the coordinates; viz., in the case of three surfaces, we have $J(P, Q, R) = 0$ a twofold relation representing a curve; and in the case of two surfaces $J(P, Q) = 0$ a threefold relation representing a point-system; and (since with four coordinates a relation is at most threefold) these are the only cases to be considered (viz., this denotes the points common to the surfaces $\partial_x P = 0$, $\partial_y P = 0$, $\partial_z P = 0$, $\partial_w P = 0$, that is, the points which are at once a node of the surface $P = 0$ and a node of the surface $Q = 0$).

If the notation were used, $J^4(P) = 0$ would denote the system of four equations $\partial_x P = 0$, $\partial_y P = 0$, $\partial_z P = 0$, $\partial_w P = 0$, which are satisfied simultaneously by the coordinates (x, y, z, w) of any node of the surface $P = 0$. Although in what precedes I have used the sign $=$, it should be understood that, while $J(P, Q, R, S) = 0$ denotes a single equation, $J(P, Q, R, S, T) = 0$ or $J(P, Q) = 0$ are each of them a twofold relation, and $J(P, Q, R, S, T, U) = 0$ or $J(P, Q, R) = 0$ each of them a threefold relation. We should have a fourfold relation $J^4(P) = 0$.

4. It is not asserted that $\dots J(P, Q, R) = 0$, $J(P, Q, R, S) = 0$, $J(P, Q, R, S, T) = 0$ form a continuous series of analogous relations; and there might even be a hiatus in using, in regard to four or more surfaces, J , and in regard to four or fewer surfaces an inverted J (viz., in regard to four surfaces, either symbol used indifferently); but there is no ambiguity in, and I have preferred to adopt, the use of the single symbol J .

5. Suppose that the orders of the surfaces $P = 0$, $Q = 0$, $\dots$ are $a + 1$, $b + 1$, $\dots$ (so that the orders of the differential coefficients of P , Q , $\dots$ are a , b , $\dots$), then we have for the orders of the several loci,

$$\begin{array}{ll} J(P, Q) = 0, \text{ point-system, order} & a^3 + a^2b + ab^2 + b^3; \\ J(P, Q, R) = 0, \text{ curve,} & a^2 + b^2 + c^2 + bc + ca + ab; \\ J(P, Q, R, S) = 0, \text{ surface,} & a + b + c + d; \\ J(P, Q, R, S, T) = 0, \text{ curve,} & ab + ac + \dots + de; \\ J(P, Q, R, S, T, U) = 0, \text{ point-system,} & abc + abd + \dots + def; \end{array}$$

see, as to this, Salmon's *Solid Geometry*, Ed. 2, (1865), Appendix IV., "On the Order of Systems of Equations" [not reproduced in the later editions]. In particular, if $a = b = c \dots = 1$, then the orders are 4, 6, 4, 10, 20.

As to the Surface obtained by equating to zero a Symmetrical Determinant.

6. It is also shown (Salmon, Ed. 2, p. 495) that the surface obtained by equating to zero any symmetrical determinant has a determinate number of nodes; viz., if the orders of the terms in

the diagonal be a, b, c , &c., then the number of nodes is $= \frac{1}{6}(\Sigma a \cdot \Sigma ab - 2\Sigma abc)$, or, as this may also be written, $\frac{1}{2}(\Sigma a^2b + 2\Sigma abc)$. In particular, the formula applies to the case of the surface

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0,$$

(a, b, c, d) being here the orders of A, B, C, D respectively, and the orders of F, G , &c., being $\frac{1}{2}(b + c)$, $\frac{1}{2}(a + c)$, &c. If the terms are all of them linear functions of the coordinates, or $a = b = c = d = 1$, then the number of nodes is $= 10$.

7. That the surface has nodes is, in fact, clear from the consideration that any point for which the minors of the determinant all vanish will be a node; and that (for the symmetrical determinant), by making the minors all of them vanish, we establish only a threefold relation between the coordinates. The expression for the number of the nodes is, I think, obtained most readily as follows.

The nodes will be points of intersection of the curve and surface

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0, \quad \begin{vmatrix} B, & F, & M \\ F, & C, & N \\ M, & N, & D \end{vmatrix} = 0,$$

the points

$$\begin{vmatrix} H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0;$$

and not only so, but they touch at the points in question; so that, multiplying together the orders of the curve and surface, and subtracting twice the order of the point-system, we obtain the expression for the number of nodes. In the particular case where the functions are all linear, we have a sextic curve and cubic surface intersecting in 18 points; but the curve and surface touch in 4 points, and the number of nodes is $(18 - 2 \cdot 4) = 10$. And in the same way the formula may be established for the general case.

8. The subsidiary theorem of the contact of the curve and surface requires, however, to be proved. Seeking for the equation of the tangent plane of the surface at any one of the points in question, we have first

$$\begin{vmatrix} \delta B, & \delta F, & \delta M \\ F, & C, & N \\ M, & N, & D \end{vmatrix} + \begin{vmatrix} B, & F, & M \\ \delta F, & \delta C, & \delta N \\ M, & N, & D \end{vmatrix} + \begin{vmatrix} B, & F, & M \\ F, & C, & N \\ \delta M, & \delta N, & \delta D \end{vmatrix} = 0,$$

where, in virtue of the equations

$$\begin{vmatrix} H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0,$$

the last term vanishes. Expanding the other two terms, the equation becomes

$$D(C\delta B + B\delta C - 2F\delta F) - (N^2\delta B - 2MN\delta F + M^2\delta C) + \delta M(FN - CM) + \delta N(BN - MF) = 0;$$

in virtue of the same equations, the coefficients of δM and δN each of them vanish, and we have also

$$N^2\delta B + M^2\delta C - 2MN\delta F = \frac{M^2}{D}(C\delta B + B\delta C - 2F\delta F),$$

so that the equation becomes finally $C\delta B + B\delta C - 2F\delta F = 0$. Investigating by a like process the equation of the tangent of the curve

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \end{vmatrix} = 0,$$

we find between the differentials δA , δB , &c., a twofold linear relation, expressible by means of the foregoing equation $C\delta B + B\delta C - 2F\delta F = 0$, and one other equation; that is, at each of the points in question the tangent of the curve lies in the tangent plane of the surface, or, what is the same thing, the curve and surface touch at these points.

Surfaces represented by an equation $F(P, Q) = 0$, &c.

9. In the remarks which follow as to the surfaces $F(P, Q) = 0$, $F(P, Q, R) = 0$, &c., the function F is a rational and integral function of (P, Q) , (P, Q, R) , &c., not in general homogeneous in regard to $P, Q, R, \dots$ but of such degrees in regard to these functions respectively as to be homogeneous in regard to the coordinates (x, y, z, w) .

The surface $F(P, Q) = 0$ has in general a nodal curve $\partial_P F = 0$, $\partial_Q F = 0$; if it has besides any nodes, these are points of the point-system $J(P, Q) = 0$.

The surface $F(P, Q, R) = 0$ has in general nodes $\partial_P F = 0$, $\partial_Q F = 0$, $\partial_R F = 0$; if it has besides any nodes, these are points on the curve $J(P, Q, R) = 0$.

The surface $F(P, Q, R, S) = 0$ has not in general, but it may have, nodes $\partial_P F = 0$, $\partial_Q F = 0$, $\partial_R F = 0$, $\partial_S F = 0$; if it has any other nodes, these are points on the surface $J(P, Q, R, S) = 0$.

Nodes of a Quartic Surface; Circumscribed Cone having its vertex at a Node.

10. A quartic surface may be without nodes; or it may have any number of nodes up to 16. Consider a quartic surface having a node; and take the single node, or (if more nodes than one) any one of the nodes, as the vertex of a circumscribed cone; then, considering any plane through the vertex, the section will be a quartic curve having a node at the vertex, and the generating lines in the plane will be the tangents from the node to the quartic curve; the number of them is therefore 6, and the order of the circumscribed cone is thus = 6. Each tangent intersects the quartic curve in the node counting as two intersections, and in the point of contact counting as two intersections; there are consequently no singular tangents; and therefore in the circumscribed cone no singular lines arising from a singular tangency of the generating line. Hence, in the case of a single node on the surface, the circumscribed cone is a cone of the order 6 without nodal or stationary lines; and the class is = 30; but in the case of more than one node, say k nodes, the circumscribed cone passes through the remaining $k - 1$ nodes, and the generating line through each of these nodes is a nodal line of the cone; that is, the cone has $k - 1$ nodal lines, and its class is = $30 - 2k + 2$. The cone is not of necessity a proper cone; the maximum number of nodal lines is when it breaks up into 6 planes, and we have then $k - 1 = 15$; that is, the number of nodes of the surface is at most = 16.

11. It is easy to form a table of the different *primâ facie* possible forms of the sextic according to the number of nodes of the surface; viz., writing 6 for a proper sextic cone without nodal lines, $6_1, 6_2, \dots, 6_{10}$ for the proper sextic cone with 1, 2, $\dots$ or 10 nodal lines; and so $5, 5_1, \dots, 5_6$ for the proper quintic cones, $4, 4_1, 4_2, 4_3, 3, 3_1, 2$ for the quartic, cubic, and quadric cones, and 1 for the plane, the table is

Nodes of Surface.	
1	6
2	6_1
3	6_2
4	6_3
5	6_4
6	$6_5; 5, 1$
7	$6_6; 5_1, 1$
8	$6_7; 5_2, 1$
9	$6_8; 5_3, 1; 4, 2$
10	$6_9; 5_4, 1; 4_1, 2; 4, 1, 1; 3, 3$
11	$6_{10}; 5_5, 1; 4_2, 2; 4_1, 1, 1; 3_1, 3$
12	$5_6, 1; 4_3, 2; 4_2, 1, 1; 3_2, 3; 3, 2, 1$
13	$\dots; \dots; 4_3, 1, 1; \dots; 3, 2, 1; 3, 1, 1, 1; 2, 2, 2$
14	$\dots; \dots; \dots; \dots; 3_1, 1, 1, 1; 2, 2, 1, 1$
15	$\dots; \dots; \dots; \dots; \dots; 2_1, 1, 1, 1, 1$
16	$\dots; \dots; \dots; \dots; \dots; 1, 1, 1, 1, 1, 1$

and moreover, in the cases where there are two or more forms of the sextic cone, then the k sextic cones may be of the different forms in various combinations. The total number of cases *primâ facie* possible is thus very great; but only a comparatively small number of them actually exist.

12. In the case where there is a plane 1, the sextic cone breaks up into this plane, and into a (proper or improper) quintic cone intersecting the plane in 5 lines; that is, there will be in the plane 6 nodes; the plane is, in fact, a singular tangent plane meeting the surface in a conic twice repeated; and the 6 nodes lie on this conic. Taking any one of these nodes as vertex, the corresponding sextic cone breaks up into the plane, and into a (proper or improper) quintic cone.

13. In the cases $k = 1, 2, 3, 4, 5$, and $k = 15, 16$, there is only one form of sextic cone; so that each node (at least so far as appears) stands in the same relation to the surface. Considering the last mentioned two cases; $k = 16$, each of the 16 nodes gives 6 singular tangent planes, but each of these passes through 6 nodes; therefore the number of planes is $= 16$: similarly, $k = 15$, the number of singular tangent planes is $15 \times 4 \div 6 = 10$.

For $k = 14$, the cones are $3_1, 1, 1, 1$, or $2, 2, 1, 1$: it is easy to see that we have only the three cases

Cones $3_1, 1, 1, 1$	:	$2, 2, 1, 1$		Singular tangent planes
may be 14,	0		gives	$(14 \cdot 3 + 0 \cdot 6) \div 6 = 7$
8,	6		,	$(8 \cdot 3 + 6 \cdot 2) \div 6 = 6$
2,	12		,	$(2 \cdot 3 + 12 \cdot 2) \div 6 = 5$

and we may in the like manner limit the number of possible cases, for other values of k . But I do not at present further pursue the inquiry.

As to the Number of Constants contained in a Surface.

14. We say that a surface $P = 0$ contains or depends upon a certain number of constants; viz., this is the number of constants contained in the equation $P = 0$ of the surface, taking the coefficient of any one term to be equal to unity; thus the general quadric surface contains 9 constants; the surface can in fact be determined so as to satisfy 9 conditions; or, as we might express it, the Postulation of the surface is $= 9$. [I have elsewhere said *Postulandum* and *Capacity*: I prefer this last expression.] And if, in the general equation so containing 9 constants,

k of these are given, or, what is the same thing, if the quadric surface be made to satisfy any k conditions, then the number of constants, or postulation of the surface, is $= 9 - k$.

15. But a different form of expression is sometimes convenient; the conditions to be satisfied are frequently such that, being satisfied by the surfaces $P = 0, Q = 0, \dots$ they will be satisfied by the surface $\alpha P + \beta Q + \dots = 0$, where $\alpha, \beta, \dots$ are any constant multipliers whatever. When this is so, there will be a certain number of solutions $P = 0, Q = 0, \dots$ not connected by any such relation, or say of aszygetic solutions, such that the general surface satisfying the conditions in question is $\alpha P + \beta Q + \dots = 0$; and hence, taking one of these coefficients as unity, the number of constants, or postulation of the surface, is equal to the number of the remaining coefficients, or, what is the same thing, it is less by unity than the number of the aszygetic solutions $P = 0, Q = 0, \dots$. Instead of considering the number of constants, or postulation, we may consider the number of solutions (that is, aszygetic solutions) or surfaces $P = 0, Q = 0, \dots$ which satisfy the conditions in question.

16. Thus, for the quadric not subjected to any conditions, there are 10 surfaces (for example, these may be taken to be the surfaces $x^2 = 0, y^2 = 0, z^2 = 0, w^2 = 0, yz = 0, zx = 0, xy = 0, xw = 0, yw = 0, zw = 0$); and the general quadric surface is by means of these expressed linearly in the form $(a, \dots | x, y, z, w)^2 = 0$. So for the quadric surfaces through k given points, the number of these is $= 10 - k$; thus for the surfaces through 4 given points, say the points $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)$, the 6 given surfaces may be taken to be $yz = 0, zx = 0, xy = 0, xw = 0, yw = 0, zw = 0$, and every other quadric surface through the 4 points is by means of these expressed linearly in the form $(a, \dots | yz, zx, xy, xw, yw, zw) = 0$; for the quadric surfaces through 8 points there are two surfaces $P = 0, Q = 0$; and every quadric surface through the 8 points is by means of these expressed linearly in the form $\alpha P + \beta Q = 0$; and (as the extreme case) if the quadric surface passes through 9 given points, then there is the single quadric surface $P = 0$.

17. In the questions in which such quadric surfaces present themselves, it is in general quite immaterial what particular surfaces are selected as the surfaces $P = 0, Q = 0, \dots$; the selection may be made at pleasure; and, being so made, the surfaces are to be regarded as completely determinate; viz., there would be no gain of generality if these were replaced by any other surfaces $\alpha P + \beta Q + \dots = 0$. For instance, in the theory of the quartic surfaces with 6 given points as nodes, we have through the 6 given points the 4 quartic surfaces $P = 0, Q = 0, R = 0, S = 0$, and we consider the quartic functions $(a, \dots | P, Q, R, S)^2$ and $J(P, Q, R, S)$: each of these is unaltered as to its form when P, Q, R, S are replaced each of them by any linear function of these quantities; viz., $(a, \dots | P, Q, R, S)^2$ is changed into a new quadric function $(a', \dots | P, Q, R, S)^2$, and $J(P, Q, R, S)$ into a mere constant multiple of its original value. We have herein a justification of the expressions in question, through 6 given points there are 4 quadric surfaces, &c.

General theory of the Quartic Surface with a given Node or Nodes.

18. A quartic surface contains 34 constants; and the number of conditions to be satisfied in order that a given point may be a node is $= 4$. Hence, if the surface has k given points as nodes, the number of constants is $= 34 - 4k$; and it would at first sight appear that k might be $= 8$, and that with the 8 given points as nodes we should have a quartic surface containing 2 constants. But this is not so in a proper sense; through the 8 given points we have 2 quadric surfaces $P = 0, Q = 0$; and we can by means of these form a quartic surface $(a, b, c | P, Q)^2 = 0$, containing 2 constants, and having in a sense the 8 points as nodes. This, however, is no proper quartic surface, but is a system of 2 quadric surfaces, each of them passing through the 8 points, and

the two quadric surfaces therefore intersecting in a quadri-quadric curve through the 8 points; which curve is therefore a nodal curve on the compound surface; and it is only as points on this nodal curve, and not in a proper sense, that the 8 given points are nodes of the quartic surface. The greatest value of k is thus $k = 7$.

19. Of course, if $k = 0$, we have the general quartic surface $U = 0$, containing 34 constants. The cases $k = 1$, $k = 2$, $k = 3$ (viz., a single given node, 2 given nodes, 3 given nodes), may be at once disposed of; taking for instance the 1st node to be the point $(1, 0, 0, 0)$, the 2nd node the point $(0, 1, 0, 0)$, the 3rd node the point $(0, 0, 1, 0)$, we find at once an equation $U = 0$, with 30, 26, or 22 constants, having the given node or nodes.

Four given Nodes.

20. The case of 4 given nodes is just as easy; but in reference to what follows, it is proper to consider it more in detail. The equation should contain 18 constants; we have through the 4 given points 6 quadric surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, $T = 0$, $U = 0$, and we can by means of them form a quartic equation $(a, \dots | P, Q, R, S, T, U)^2 = 0$, having the 4 given points as nodes; this contains, however, $(21 - 1 =) 20$ constants; the reduction to the right number 18 occurs by reason that the functions (P, Q, R, S, T, U) , although linearly independent, are connected by two quadric equations

$$(* | P, Q, R, S, T, U)^2 = 0, \quad (*' | P, Q, R, S, T, U)^2 = 0;$$

hence writing the equation of the quartic surface in the form

$$(a, \dots | P_*)^2 - \lambda (* | P_*)^2 - \mu (*' | P_*)^2 = 0,$$

the coefficients λ, μ may be so determined as to reduce to zero the coefficients of any two terms of the equation, and the number of constants really is $20 - 2 = 18$, as it should be.

21. In proof, observe that, taking the 4 given nodes to be the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, the quadric surfaces may be taken to be $yz = 0$, $zx = 0$, $xy = 0$, $xw = 0$, $yw = 0$, $zw = 0$; the equation of the quartic surface will thus be

$$(a, \dots | yz, zx, xy, xw, yw, zw)^2 = 0;$$

but we have between the functions xy , &c., the two identical relations

$$xy \cdot zw - xz \cdot yw = 0, \quad xy \cdot zw - xw \cdot yz = 0;$$

and the number of constants is thus = 18.

Five given Nodes.

22. In the case of 5 given nodes, the number of constants should be = 14. We have through the 5 given points, 5 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$, $T = 0$, and we form herewith the quartic equation $(a, \dots | P, Q, R, S, T)^2 = 0$, containing the right number 14 of arbitrary constants. The functions P, Q , &c. are in this case not connected by any quadric relation, and the equation just written down is in fact the general equation of the quartic surface with the 5 given nodes.

23. In verification, take the first 4 nodes to be as above, and the 5th node to be the point $(1, 1, 1, 1)$; we may write

$$(P, Q, R, S, T) = \{x(y - z), x(y - w), y(x - z), y(x - w), xy - zw\};$$

and if from the 5 equations $P = x(y - z)$, &c., we eliminate (x, y, z, w) , we obtain one, and only one, relation between the functions P, Q, R, S, T ; this is found to be

$$PS(Q + R - T) - QR(P + S - T) = 0,$$

or, what is the same thing,

$$R(P - Q)(S - T) - P(R - S)(Q - T) = 0;$$

viz., it is a cubic relation, and there is consequently no quadric relation between the 5 functions.

Six given Nodes.

24. In the case of 6 given nodes, the quartic surface should contain 10 constants. We have through the 6 given points 4 quadric surfaces $P = 0$, $Q = 0$, $R = 0$, $S = 0$; but if we form herewith the quartic surface $(a, \dots | P, Q, R, S)^2 = 0$, this contains only 9 constants. It is to be shown that the Jacobian surface $J(P, Q, R, S) = 0$ of the 4 quadric surfaces (or say of the 6 points) is a quartic surface having the 6 given points as nodes, and not included in the foregoing form $(a, \dots | P, Q, R, S)^2 = 0$; this being so, we have the quartic surface

$$(a, \dots | P, Q, R, S)^2 + \theta J(P, Q, R, S) = 0,$$

having the 6 given points as nodes, and containing the complete number of constants, viz., 10.

25. The 6 given nodes being any points whatever, their coordinates may be taken to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, $(1, 1, 1, 1)$, and $(\alpha, \beta, \gamma, \delta)$. I proceed to find the Jacobian of these 6 points. For this purpose, let (a, b, c, f, g, h) be the 6 coordinates of the line through the points $(1, 1, 1, 1)$ and $(\alpha, \beta, \gamma, \delta)$, viz.,

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, \\ b &= \gamma - \alpha, & g &= \beta - \delta, \\ c &= \alpha - \beta, & h &= \gamma - \delta, \end{aligned}$$

whence $af + bg + ch = 0$, and also

$$\begin{aligned} 0 \cdot -h \cdot -g \cdot + a &= 0, \\ -h \cdot + 0 \cdot + f \cdot + b &= 0, \\ g \cdot -f \cdot + 0 \cdot + c &= 0, \\ -a \cdot -b \cdot -c \cdot + 0 &= 0; \end{aligned}$$

we have through the 6 points the plane pairs

$$\begin{aligned} x(\dots - hy - gz + aw) &= 0, \\ y(-hx \dots + fz + bw) &= 0, \\ z(gx - fy \dots + cw) &= 0, \\ w(-ax - by - cz \dots) &= 0, \end{aligned}$$

where, adding the four equations, we have identically $= 0$. For this reason, we cannot take these to be the equations of the 4 quadric surfaces, but we may take the first 3 of them for the surfaces

$P = 0, Q = 0, R = 0$; and for the 4th surface $S = 0$, I take the quadric cone having its vertex at the point $(0, 0, 0, 1)$; viz., the equation is

$$a\alpha yz + b\beta zx + c\gamma xy = 0;$$

that is, I write

$$(P, Q, R, S) = \{x(hy - gz + aw), y(-hx + fz + bw), z(gx - fy + cw), (a\alpha yz + b\beta zx + c\gamma xy)\}.$$

26. The Jacobian is then easily found to be

$$\begin{aligned} & (b\beta zx + c\gamma xy)(-agh, bhf, cfg, abc, -af^2, -gB, hC, aA, bfg, -c^2h \mid x, y, z, w)^2 \\ & + (c\gamma xy + a\alpha yz)(agh, -bhf, cfg, abc, fA, -bg^2, -hC, -a^2f, bB, c^2h \mid x, y, z, w)^2 \\ & + (a\alpha yz + b\beta zx)(agh, bhf, -cfg, abc, -fA, gB, -ch^2, a^2f, -b^2g, cC \mid x, y, z, w)^2 = 0; \end{aligned}$$

where for the moment A, B, C denote $bg - ch, ch - af, af - bg$ respectively. Collecting and reducing, the whole divides by $2abc$; and if finally we replace a, b, c, f, g, h by their values, the result is

$$J = \left\{ \begin{aligned} & (\beta - \gamma)yz(\alpha w^2 - \delta x^2) + (\alpha - \delta)xw(\beta z^2 - \gamma y^2) \\ & + (\gamma - \alpha)zx(\beta w^2 - \delta y^2) + (\beta - \delta)yw(\gamma x^2 - \alpha z^2) \\ & + (\alpha - \beta)xy(\gamma w^2 - \delta z^2) + (\gamma - \delta)zw(\alpha y^2 - \beta x^2) \end{aligned} \right\} = 0.$$

27. It may be shown *à posteriori* that J is not a quadric function of P, Q, R, S . For, attempting to express it in this form, J does not contain the terms x^2w^2, y^2w^2, z^2w^2 , and it thence at once appears that the coefficients of P^2, Q^2, R^2 each of them vanish. Hence, introducing for convenience the factor 2, I assume $(0, 0, 0, D, F, G, H, L, M, N \mid P, Q, R, S)^2 = 2J$. Comparing the terms in $w^2(yz, zx, xy)$, we obtain

$$bcF = a\alpha, \quad caG = b\beta, \quad abH = c\gamma;$$

and comparing the coefficients of $w(y^2z, z^2x, x^2y, yz^2, zx^2, xy^2)$, we obtain

$$\begin{aligned} -Ff + a\alpha M &= \frac{a\alpha}{c}\beta, & Ff + a\alpha N &= -\frac{a\alpha}{b}\gamma, \\ -Gg + b\beta N &= \frac{b\beta}{a}\gamma, & Gg + b\beta L &= -\frac{b\beta}{c}\alpha, \\ -Hh + c\gamma L &= -\frac{c\gamma}{b}\alpha, & Hh + c\gamma M &= -\frac{c\gamma}{a}\beta; \end{aligned}$$

substituting for F, G, H their values, we obtain from the first 3 equations $L, M, N = \frac{f}{bc}, \frac{-g}{ca}, \frac{-h}{ab}$, and from the second 3 equations, $L, M, N = \frac{f}{bc}, \frac{g}{ca}, \frac{h}{ab}$; that is, the equations are inconsistent, and the function J is not expressible in the form in question.

Jacobian Surface of Six given Points.

28. The equation $J = 0$ is the locus of the vertices of the quadric cones which pass through the given 6 points; calling these 1, 2, 3, 4, 5, 6, we see at once that the surface passes through the 15 lines 12, 13, ... 56, and also through the ten lines 123.456 (viz., line of intersection of the planes through 1, 2, 3, and through 4, 5, 6), &c. In fact, taking the vertex at any point 0 in the line 1, 2, the lines drawn to the six points are 01 = 02, 03, 04, 05, 06; viz., there are only five lines, so that these lie in a quadric cone. And taking the vertex at any point in the line 123.456, the lines to the 6 points lie in these planes 123 and 456 respectively, and the quadric cone is in fact this plane-pair. Moreover, the surface containing the lines 12, 13, 14, 15, 16, must have the point 1 for a node; and similarly, the points 2, 3, 4, 5, 6 are each of them a node on the surface. It is to be added that the surface contains the skew cubic through the 6 points, or say the skew cubic 123456. See, as to this, *post* No. 108.

29. The surface in question (the Jacobian of the 6 points) is a particular case of the Jacobian of any 4 quadric surfaces. This more general surface will be considered in the sequel; I only remark here that it contains 10 lines, corresponding to the 10 lines 123.456, &c., but it has not any other lines, or any nodes.

Jacobian Curve of Seven given Points, or of an Octad of Points.

30. In connexion with what precedes, we may here consider a curve which presents itself in the sequel; viz., the curve which is the locus of the vertices of the quadric cones which pass through seven given points. The general case is when no one of the points is the vertex of a quadric cone through the other 6 points. We have through the 7 points the three quadric surfaces $P = 0$, $Q = 0$, $R = 0$; hence, forming the equation $\alpha P + \beta Q + \gamma R = 0$ of the general quadric surface through the 7 points, and making this a cone, we find as the locus of the vertex $J(P, Q, R) = 0$; the analytical form shows that this is a sextic curve. It appears, moreover, that the curve is symmetrically related to all the 8 points $P = 0$, $Q = 0$, $R = 0$; and instead of calling it the Jacobian of the 7 points, we may call it the Jacobian of the octad. But in further explanation, take the points to be 1, 2, 3, 4, 5, 6, 7; the vertex will lie on each of the Jacobian surfaces 123456 and 123457; and it is at present assumed that 7 is not a point on the first surface, nor 6 a point on the second surface. The two surfaces have in common the lines 12, 13, ... 45, and they consequently besides intersect in a curve of the 6th order, or sextic curve, which is the locus in question. At the point 1 there is on the first surface a tangent cone through the lines 12, 13, 14, 15, 16, and on the second surface a tangent cone through the lines 12, 13, 14, 15, 17; these two cones have for their complete intersection the lines 12, 13, 14, 15, which lines belong to the complete intersection of the two surfaces, but not to the sextic curve. It thus appears, *à posteriori*, that the sextic curve does not pass through the point 1; and similarly, that it does not pass through any of the points 2, 3, 4, or 5. As to the points 6 and 7, each of these is on only one of the quartic surfaces, and therefore the curve of intersection does not pass through either of these points.

31. Suppose, however, that one of the seven points is the vertex of a cone through the other six; it is of course the same thing whether we take this to be one of the points 1, 2, 3, 4, 5, or one of the points 6 and 7, but the result comes out more easily in the latter case; viz., in the former case, taking 1 to be the point in question, the two tangent cones at 1 are one and the same cone, and all that appears is that there is nothing to hinder a branch or branches of the sextic curve from passing through the point 1. But in the latter case, taking 7 for the point in question, then 7 lies on the surface 123456, being a simple point on this surface, but a node on the surface 123457; and it thus appears that there are through 7 two branches of the sextic curve; so that any one of the seven points, being the vertex of a cone through the other six, is an actual double point on the sextic curve.

32. In the case where two of the points are each of them the vertex of a cone through the other six points, then the seven points lie on a skew cubic; and the sextic curve of the general case becomes this skew cubic twice repeated.

Seven given Nodes.

33. In the case of 7 given nodes, the number of constants should be $= 6$; the 7 given points determine 3 quadric surfaces $P = 0$, $Q = 0$, $R = 0$: and we have hence the quartic surface $(a, \dots | P, Q, R)^2 = 0$, containing 5 constants only. That this is not the general quartic surface with the 7 given nodes, is also clear from the consideration that the surface in question has 8 nodes; viz., the 8 points of intersection of the three quadric surfaces. Suppose that a particular

quartic surface, having the 7 given nodes, but not of the last mentioned form, is $\Delta = 0$; then a quartic surface having the 7 given nodes is

$$(a, \dots | P, Q, R)^2 + \theta \Delta = 0;$$

and this, as containing 6 constants, will be the general quartic surface with the 7 given nodes.

34. It follows that, if $\Delta' = 0$ be another quartic surface having the 7 given nodes, we must have identically $\Delta' - \rho\Delta = (* | P, Q, R)^2$, where ρ is a determinate constant and $(* | P, Q, R)^2$ a determinate quadric function of (P, Q, R) . The formula extends to the case where $\Delta' = 0$ has the 8 nodes ($P = 0, Q = 0, R = 0$), but we have then $\rho = 0$, and the meaning is simply that the general quartic surface having the 8 nodes is $(* | P, Q, R)^2 = 0$.

35. A particular quartic surface having (in an improper sense) the 7 given nodes, but not having the 8th node, is $M\Omega = 0$, where $M = 0$ in the plane through any 3 of the 7 points and $\Omega = 0$ is the cubic surface through these same 3 points, and having the remaining 4 points as nodes. The equation of the cubic surface, if the 4 points are taken to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, is obviously of the form

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} + \frac{d}{w} = 0, \quad (\text{that is, } ayzw + bzwx + cxyw + dxyz = 0),$$

and by making the surface pass through the 3 points we determine linearly the coefficients (a, b, c, d) , that is, their ratios. The equation of the quartic surface thus is

$$(a, \dots | P, Q, R)^2 + \theta M\Omega = 0,$$

the 7 given points being here proper nodes; and the formula being precisely equivalent to the preceding one containing Δ .

36. We can with the 7 given points form 35 such combinations $M\Omega = 0$ of a plane and a cubic surface, and so present the equation of the quartic surface under 35 different forms; these are of course equivalent in virtue of the before mentioned formula for $\Delta' - \rho\Delta$; viz., we must have identically $M\Omega - \rho'M'\Omega' = (* | P, Q, R)^2$: a theorem of some interest, which it might be difficult to verify *à posteriori*.

Investigation of the cases of 8 Nodes.

37. It has already been shown that a quartic surface cannot in a proper sense have 8 given nodes. In regard to the quartic surfaces with 8 nodes, we start from the surface with 7 given nodes; viz.,

$$(a, \dots | P, Q, R)^2 + \theta \nabla = 0,$$

or, what is the same thing,

$$(a, \dots | P, Q, R)^2 + \theta M\Omega = 0;$$

and we inquire in what cases this surface has an 8th node. Obviously if $\theta = 0$, that is, if the surface is $(a, \dots | P, Q, R)^2 = 0$, the surface will have an 8th node, the remaining intersection of the quadric surfaces $P = 0, Q = 0, R = 0$ (observe that this is a point in no wise depending on the particular quadric surfaces, but uniquely determined by means of the 7 given points); and we have thus one kind, say the “octadic” surface, of the quartic surfaces with 8 nodes; viz., the nodes are the 8 points of intersection of any 3 quadric surfaces, or they are an octad of points. By what precedes, 7 of the nodes may be given points, and the remaining node is then a uniquely determinate point, the 8th point of the octad.

38. But if θ be not $= 0$, there may still be an 8th node; viz., this must then be a point on the Jacobian surface $J(P, Q, R, \nabla) = 0$, which is of the order 6. It is clear *à priori* that this must be a surface depending only on the 7 points, but independent of the particular surfaces $P = 0, Q = 0, R = 0, \nabla = 0$; to verify this, observe that, substituting for ∇ the function $\nabla_1 = \nabla + (* | P, Q, R)^2$, we in fact leave the Jacobian unaltered; I call it the dianodal surface of the 7 points.

39. I say that the 8th node may be any point whatever on the dianodal surface; in fact, regarding for a moment the coordinates of the node as given, and expressing that the point is a node on the quartic surface, we have 4 equations containing

$$aP_0 + hQ_0 + gR_0, \quad hP_0 + bQ_0 + fR_0, \quad gP_0 + fQ_0 + cR_0,$$

(P_0, Q_0, R_0 the values of P, Q, R at the node,) but which, if only the point be on the dianodal surface, reduce themselves to three equations; viz., we have between the coefficients (a, b, c, f, g, h) three equations which being satisfied, the point in question will be a node. And it thus appears that, taking the 8th node to be a given point on the dianodal surface, the equation $(a, \dots | P, Q, R)^2 + \theta \nabla = 0$ of the quartic surface will contain 3 constants. Observe that we may through the 8 nodes draw 2 quadric surfaces $P = 0, Q = 0$; and this being so if $\Delta = 0$ be a particular quartic surface with the 8 nodes, then the general quartic surface will be

$$(a, b, c | P, Q)^2 + \theta \Delta = 0,$$

containing the right number 3 of constants. But there is not here any simple form of the surface $\Delta = 0$, such as the form $M\Omega = 0$ for the surface through 7 given points.

40. It is clear *à priori* that the relation between the 8 nodes is a symmetrical one; so that the 8th point being situate anywhere on the dianodal surface of the remaining 7 points, each of the 7 points will be situate on the dianodal surface of the remaining 7 points. This is a remarkable property of the dianodal surface, which will have to be again considered.

41. In what precedes, we have the second kind of quartic surfaces with 8 nodes, say the “dianome”; viz., each node is a point on the dianodal surface of the remaining nodes; any 7 of the nodes may be taken to be given points, and the remaining node to be any point whatever on the dianodal surface of the 7 points.

The Dianodal Surface.

42. Consider the seven points 1, 2, 3, 4, 5, 6, 7. As already mentioned, through three of these say 1, 2, 3, we may draw a plane $M = 0$; and through the same three points and the remaining points 4, 5, 6, 7 as nodes ($3 + 4 \cdot 4 = 19$ conditions), a cubic surface $\Omega = 0$; this surface passes through the six lines 45, 46, ... 67. Hence we have $\Delta = M\Omega = 0$, a quartic surface with the seven points as nodes. And using this form of Δ it may be shown that the dianodal $J(P, Q, R, \Delta) = 0$ passes through the 21 lines 12, 13, ... 67, and through 35 plane cubics such as $M = 0, \Omega = 0$; viz., this is a cubic in the plane 123 passing through the points 1, 2, 3, and through the intersection of the plane with each of the six lines 45, 46, ... 67 (nine points determining the cubic); the complete intersection by the plane 123 being therefore composed of this cubic and of the three lines 12, 13, 23. For the passage through the cubic, we have only to observe that

$$J(P, Q, R, M\Omega) = J(P, Q, R, \Omega) M + J(P, Q, R, M) \Omega = 0$$

is satisfied by $M = 0, \Omega = 0$; and for the passage through the lines, taking $x = 0, y = 0, z = 0, w = 0$ for the equations of the planes 567, 674, 745, and 456 respectively, each of the functions

P, Q, R is of the form $ayz + bzx + cxy + fxw + gyw + hzw$, and the function Ω is of the form $Ayzw + Bzwx + Cwxy + Dxyz$. Hence, writing in the derived functions $z = 0, w = 0$, the first and second lines of the determinant $J(P, Q, R, \Omega)$ will be of the form

$$\begin{pmatrix} cy, & c'y, & c''y, & * \\ cx, & c'x, & c''x, & * \end{pmatrix},$$

or the determinant vanishes for $z = 0, w = 0$; that is for any point of the line 45 we have $\Omega = 0$ and also $J(P, Q, R, \Omega) = 0$; consequently $J(P, Q, R, M\Omega) = 0$, and the like for the other lines. The theorem is thus proved.

43. I say that the dianodal surface passes through each of the 7 skew cubics, such as 123456. To prove this, it is only necessary to show that the skew cubic 123456 lies on the dianodal surface. For this purpose it will be enough to show that the skew cubic meets the plane 712 in a point of the surface; for then it will, in like manner, meet each of the 15 planes 712, 713, ... 756 in a point of the surface; that is, we shall have 15 intersections of the curve and surface, and there are, besides, the intersections 1, 2, 3, 4, 5, 6, in all 21 intersections; that is, the skew cubic must lie on the surface.

44. The plane 712 meets the surface in three lines and in a plane cubic determined by the points 7, 1, 2 and the six intersections of the plane with the lines 34, 35, ... 56. We have therefore to show that this plane cubic meets the skew cubic 123456. Consider for a moment the points 1, 2, 3, 4, 5, 6 and another point $7'$. As seen above, we have in general, through the points 1, 2, $7'$ and with the points 3, 4, 5, 6 as nodes, a determinate cubic surface, which surface passes through the lines 34, 35, ... 56. But the cubic surface becomes indeterminate if the points 1, 2, $7'$, 3, 4, 5, 6 are on the same skew cubic; that is, if $7'$ is any point whatever on the skew cubic 123456 (the proof presently). Taking, then, $7'$ as the intersection of the skew cubic by the plane 712, we have in this plane the points $7', 1, 2$, and the intersections of the plane by the lines 34, 35, ... 56, nine points through which there pass an infinity of plane cubics; that is, the plane cubic determined by the points 7, 1, 2 and the six intersections will pass through the point $7'$; viz., it meets the skew cubic 123456.

45. For the subsidiary theorem, taking X, Y, Z, W as current coordinates, viz., $X = 0, Y = 0, Z = 0, W = 0$ as the equations of the planes 456, 563, 634, 345 respectively, (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2) as the coordinates of the points 1 and 2 respectively, and (x, y, z, w) for those of $7'$; the equation of the cubic surface passing through $7', 1, 2$, and having the nodes 3, 4, 5, 6, is

$$\begin{vmatrix} \frac{1}{X}, & \frac{1}{Y}, & \frac{1}{Z}, & \frac{1}{W} \\ \frac{x}{1}, & \frac{y}{1}, & \frac{z}{1}, & \frac{w}{1} \\ \frac{x_1}{1}, & \frac{y_1}{1}, & \frac{z_1}{1}, & \frac{w_1}{1} \\ \frac{x_2}{1}, & \frac{y_2}{1}, & \frac{z_2}{1}, & \frac{w_2}{1} \end{vmatrix} = 0;$$

and this ceases to be a determinate function if only

$$\begin{vmatrix} \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{x_1}{1}, & \frac{y_1}{1}, & \frac{z_1}{1}, & \frac{w_1}{1} \\ \frac{x_2}{1}, & \frac{y_2}{1}, & \frac{z_2}{1}, & \frac{w_2}{1} \end{vmatrix} = 0;$$

viz., considering (x_1, y_1, z_1, w_1) , (x_2, y_2, z_2, w_2) as given, this is a twofold relation between the coordinates (x, y, z, w) of the point 7'. The relation may be represented by the four equations $(yzw) = 0$, $(zwx) = 0$, $(wxy) = 0$, $(xyz) = 0$, if for shortness

$$(yzw) = \begin{vmatrix} yz, & zw, & wy \\ y_1z_1, & z_1w_1, & w_1y_1 \\ y_2z_2, & z_2w_2, & w_2y_2 \end{vmatrix}$$

and the like as to the other symbols. The four equations represent quadric surfaces, each two intersecting in a line [e.g., $(yzw) = 0$, $(zwx) = 0$ in the line $z = 0$, $w = 0$], and the four surfaces besides intersecting in a skew cubic, which is the required locus of the point 7', and which, as is seen at once, passes through the points 1, 2, 3, 4, 5, 6.

46. By what precedes, we have on the dianodal surface through the point 1 the lines 12, 13, 14, 15, 16, 17, and the skew cubics 123456, &c. The six lines are not on the same quadric cone, and it thus appears that the point 1 must be a cubic-node (point where, instead of the tangent plane, we have a cubic cone) on the surface. It is to be remarked that the lines 12, 13, 14, 15, 16, and the tangent at 1 to the skew cubic 123456, lie in a quadric cone; viz., this tangent is given as the sixth intersection of the cubic cone with the quadric cone through the lines 12, 13, 14, 15, 16.

47. I revert to the equation of the dianodal surface as given in the form $J = J(P, Q, R, M\Omega) = 0$, where $M = 0$ is the plane through the points 1, 2, 3, and $\Omega = 0$ the cubic surface through these points, and having the points 4, 5, 6, 7, as nodes. We can find the orders of the several functions P, Q, R, M, Ω in the coordinates (x_1, y_1, z_1, w_1) , &c., of the several points; viz., writing for shortness x_1^2 to denote the order 2 in regard to (x_1, y_1, z_1, w_1) , and so in other cases, we have

$$P = Q = R = x^2(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^2,$$

$$M = x(x_1, x_2, x_3),$$

$$\Omega = x^3(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^3,$$

(where, of course, the x 's show in like manner the orders in regard to the current coordinates (x, y, z, w) ; the proof in regard to Ω is easily supplied.) The order of J is equal that of $PQRM\Omega$, less 4 as regards the current coordinates, by reason of the differentiations; that is, we have $J = x^6(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$; and we thus see that the equation of the dianodal surface as above obtained is encumbered with a constant factor of the form $(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$. In fact, the relation between the 7 points and the current point (x, y, z, w) , or say the point 8, as expressing that the 8 points are the nodes of a dianome, should be a symmetrical one in regard to the coordinates of the several points; and being of the order 6 in regard to the coordinates (x, y, z, w) , it should be of the same order in regard to the other coordinates; that is, the true form would be $J = (xx_1x_2x_3x_4x_5x_6x_7)^6 = 0$.

48. It is possible that taking the 4 points, say 1, 2, 3, 4, to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, and the 3 points, say 5, 6, 7, to be $(1, 1, 1, 1)$, $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, the extraneous factor might exhibit itself, and that the [paper 445 continues on book p. 151].

[443] NOTE ON THE SOLUTION OF THE QUARTIC EQUATION $aU + 6\beta H = 0$
(CONCLUDED).

so that identically $X^2 + Y^2 + Z^2 = 0$, the expression $\alpha X + \beta Y + \gamma Z$ will be a square if only $\alpha^2 + \beta^2 + \gamma^2 = 0$. (To see this observe that in virtue of the equation $X^2 + Y^2 + Z^2 = 0$, we have $X + iY$, $X - iY$ each of them a square, and thence

$$\alpha X + \beta Y + \gamma Z = \frac{1}{2}(\alpha + i\beta)(X - iY) + \frac{1}{2}(\alpha - i\beta)(X + iY) - \gamma i \sqrt{X^2 + Y^2}$$

is a square if the condition in question be satisfied.)

Hence in particular writing

$$\sqrt{\omega_2 - \omega_3} \sqrt{aJ + 6\beta\omega_1 J}, \dots, \sqrt{\omega_1 - \omega_2} \sqrt{aJ + 6\beta\omega_3 J},$$

for α, β, γ , we have

$$(\omega_2 - \omega_3) \sqrt{aJ + 6\beta\omega_1 J} \sqrt{H} + (\omega_3 - \omega_1) \sqrt{aJ + 6\beta\omega_2 J} \sqrt{H} + (\omega_1 - \omega_2) \sqrt{aJ + 6\beta\omega_3 J} \sqrt{H}$$

a perfect square, and since the product of the four different values is a multiple of $(aU + 6\beta H)^2$ (this is most readily seen by observing that for $aU + 6\beta H = 0$, the irrational expression omitting a factor is

$$(\omega_2 - \omega_3)(aJ + 6\beta\omega_1 J) + (\omega_3 - \omega_1)(aJ + 6\beta\omega_2 J) + (\omega_1 - \omega_2)(aJ + 6\beta\omega_3 J),$$

which vanishes identically) it follows that the expression in question is the square of a linear factor of $aU + 6\beta H$.

It thus appears that the radicals (other than those arising from the solution of $U = 0$) contained in the solution of the equation $aU + 6\beta H = 0$ are the three roots

$$\sqrt{aJ + 6\beta\omega_1 J}, \quad \sqrt{aJ + 6\beta\omega_2 J}, \quad \sqrt{aJ + 6\beta\omega_3 J}.$$

Cambridge, September 2, 1868.

[444] ON THE CENTRO-SURFACE OF AN ELLIPSOID.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 16–18.]

The President [Prof. Cayley] gave an account of his investigations on the centro-surface of an ellipsoid (locus of the centres of curvature of the ellipsoid). The surface has been studied by Dr Salmon, and also by Prof. Clebsch, but in particular the theory of the nodal curve on the surface admits of further development. The position of a point on the ellipsoid is determined by means of the parameters, or elliptic coordinates, h, k ; viz., if as usual a, b, c are the semi-axes, and if X, Y, Z are the coordinates of the point in question, then

$$\frac{X^2}{a^2 + h} + \frac{Y^2}{b^2 + h} + \frac{Z^2}{c^2 + h} = 1,$$

$$\frac{X^2}{a^2 + k} + \frac{Y^2}{b^2 + k} + \frac{Z^2}{c^2 + k} = 1;$$

and hence

$$-\beta\gamma X^2 = a^2(a^2 + h)(a^2 + k),$$

$$\begin{aligned} -\gamma\alpha Y^2 &= b^2(b^2 + h)(b^2 + k), \\ -\alpha\beta Z^2 &= c^2(c^2 + h)(c^2 + k), \end{aligned}$$

if for shortness

$$\alpha = b^2 - c^2, \quad \beta = c^2 - a^2, \quad \gamma = a^2 - b^2 \quad (\alpha + \beta + \gamma = 0).$$

This being so, the coordinates of the point of intersection of the normal at (X, Y, Z) by the normal at the consecutive point of the curve of curvature

$$\frac{X^2}{a^2 + k} + \frac{Y^2}{b^2 + k} + \frac{Z^2}{c^2 + k} = 1$$

are given by the formulae

$$\begin{aligned} -\beta\gamma a^2 x^2 &= (a^2 + h)^2(a^2 + k), \\ -\gamma\alpha b^2 y^2 &= (b^2 + h)^2(b^2 + k), \\ -\alpha\beta c^2 z^2 &= (c^2 + h)^2(c^2 + k); \end{aligned}$$

viz., these equations, considering therein (h, k) as arbitrary parameters, determine the coordinates (x, y, z) of a point on the centre-surface. The principal sections (as is known) consist each of them of an ellipse counting three times, and of an evolute of an ellipse; the evolute and ellipse have four contacts (two-fold intersections) and four simple intersections, but the contacts and intersections respectively are in the different sections real and imaginary; and if (as we may without loss of generality assume) $a^2 + c^2 > 2b^2$, then the form of the principal sections is as shown in the figure (which represents only an octant of the surface); viz., there is a real contact at P in the plane of xz , and a real intersection at Q in the plane of xy . The surface has thus an exterior and an interior sheet, but (instead of meeting in a conical point, as in the wave surface) these intersect in a nodal curve QP . The curve has a cusp at Q , and a node at P ; viz., the curve extends beyond P , but from that point is acnodal, or without any real sheet of the surface passing through it. For the nodal curve there must be two values (h, k) , (h_1, k_1) , giving the same values of (x, y, z) ; viz., there must exist the relations

$$\begin{aligned} (a^2 + h)^2(a^2 + k) &= (a^2 + h_1)^2(a^2 + k_1), \\ (b^2 + h)^2(b^2 + k) &= (b^2 + h_1)^2(b^2 + k_1), \\ (c^2 + h)^2(c^2 + k) &= (c^2 + h_1)^2(c^2 + k_1); \end{aligned}$$

from which equations eliminating k and k_1 we should have between h, k a relation which, combined with the expressions of x, y, z in terms of (h, k) , determines the nodal curve. But the better course is to eliminate k, k_1 , thus obtaining a relation between h and h_1 , in virtue whereof h_1 may be regarded as a known function of h ; k and k_1 can then be readily expressed in terms of h, h_1 ; that is, we have k as a function of h, h_1 , or in effect as a function of h . The relation between h, h_1 (after a reduction of some complexity) assumes ultimately a form which is very simple and remarkable; viz., writing

$$P = a^2 + b^2 + c^2, \quad Q = b^2c^2 + c^2a^2 + a^2b^2, \quad R = a^2b^2c^2,$$

the relation is

$$\begin{aligned} &(QR + 3Qh^2 + Ph^4) \\ &+ h_1(3Qh + 4Ph^2 + 3h^3) \\ &+ h_1^2(P + 3h^2) = 0; \end{aligned}$$

this is a (2, 2) correspondence between the two parameters h, h_1 ; the united values $h_1 = h$, are given by the equation $6(R + Qh + Ph^2 + h^3) = 0$, that is

$$(a^2 + h)(b^2 + h)(c^2 + h) = 0;$$

viz., the two points on the ellipsoid which have their common centre of curvature on the nodal curve are only situate on the same curve of curvature when this curve is a principal section of the ellipsoid.

[Since the date of the foregoing communication, Prof. Cayley has found that the squared coordinates x^2, y^2, z^2 of a point on the nodal curve can be expressed as rational functions of a single variable parameter σ .]

[445] A MEMOIR ON QUARTIC SURFACES.

*[From the Proceedings of the London Mathematical Society, vol. III. (1869–1871), pp. 19–69.
Read February 10, 1870.]*

The present Memoir is intended as a commencement of the theory of the quartic surfaces which have nodes (conical points). A quartic surface may be without nodes, or it may have any number of nodes up to 16. I show that this is so, and I consider how many of the nodes may be given points. Although it would at first sight appear that the number is 8, it is in fact 7; viz., we can, with 7 given points as nodes (but not in a proper sense with 8 or more given points), find a quartic surface; such surface contains in its equation 6 constants, which may be such that the surface has an additional node or nodes. Suppose that the surface has an 8th node: there are two distinct cases; viz., (1) the 8 nodes are the points of intersection of 3 quadric surfaces, or say they are an octad, and the surface is said to be octadic; (2) the 8th node is any point whatever on a certain sextic surface determined by means of the 7 given nodes, and called the dianodal surface of these 7 points; the quartic surface is said to be a dianome. The two cases are in general exclusive of each other; viz., the 7 given points being any points whatever, the dianodal surface does not pass through the 8th point of the octad; and thus the quartic surface with the 8 nodes is either octadic or else a dianome. Assuming it to be a dianome, the constants may be further determined so that there shall be a 9th node; it is necessary to examine whether this forms with 7 of the 8 nodes an octad. Supposing that it does not (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 9th node is then any point whatever on a certain curve of the order 18, determined by means of the 8 nodes, and called the dianodal curve of these 8 points. And, finally, the constants may be further determined so that there shall be a 10th node; supposing, as before, that this does not form an octad with any 7 of the 9 nodes (viz., that there are not any 8 nodes in regard to which the surface is octadic), the 10th node is then any one of a system of 22 [should be 13] points determined by means of the 9 nodes, and called the dianodal system of these 9 points. But the quartic surface is now completely determined; viz., starting with any 7 given points as nodes, we have a dianome with 8 nodes, 9 nodes, or 10 nodes, say, an octodianome, enneadianome, or decadianome, but not with any greater number of nodes; these can only present themselves when particular conditions are satisfied in regard to the 7 given nodes, and to the 8th and 9th node; and the consideration of the quartic surfaces with more than 10 nodes would thus form a separate branch of the subject.

The case of the decadianome (or quartic surface with 10 nodes formed as above with 7 given points as nodes) is peculiarly interesting. I identify this with the surface which I call a symmetroid; viz., the surface represented by an equation $\Delta = 0$, where Δ is a symmetrical determinant of the 4th order the several terms whereof are linear functions of the coordinates (x, y, z, w) ; this surface is related to the Jacobian surface of 4 quadric surfaces (itself a very

remarkable surface), and this theory of the symmetroid and the Jacobian, and of questions connected therewith, forms a large portion of the present Memoir.

The theory of the Jacobian is connected also with the researches in regard to nodal quartic surfaces in general; and, for greater clearness, it has seemed to me proper to commence the Memoir with certain definitions, &c., in regard to this theory. It will be seen in what manner I extend the notion of the Jacobian.

I remark that the present researches on Quartic Surfaces were suggested to me by Professor Kummer's most interesting Memoir "Ueber die algebraischen Strahlensysteme u.s.w.," *Berl. Abh.* 1866, in which, without entering upon the general theory, he is led to consider the quartic surfaces, or certain quartic surfaces, with 16, 15, 14, 13, 12, or 11 nodes; the last of these, or surface with 11 nodes, being in fact a particular case of the symmetroid.

Considerations in regard to the Jacobian of four, or more or less than four, Surfaces.

1. In the case of any four surfaces, $P = 0$, $Q = 0$, $R = 0$, $S = 0$, the differential coefficients of P , Q , R , S in regard to the coordinates (x, y, z, w) may be arranged as a square matrix in either of the ways

$$\begin{array}{c|cccc} & P, & Q, & R, & S \\ \hline \partial_x & & & & \\ \partial_y & & & & \\ \partial_z & & & & \\ \partial_w & & & & \end{array} \quad \text{or} \quad \begin{array}{c|cccc} & \partial_x, & \partial_y, & \partial_z, & \partial_w \\ \hline P & & & & \\ Q & & & & \\ R & & & & \\ S & & & & \end{array}$$

according as we make P, Q, R, S the columns, or as we make them the lines, of the matrix; and we then have one and the same determinant, the Jacobian $J(P, Q, R, S) = 0$, regarding as zero the term in question.

2. In the case of more than four surfaces, adopting the arrangement

$$\begin{array}{c|cccccc} & P, & Q, & R, & S, & T, & \dots \\ \hline \partial_x & & & & & & \\ \partial_y & & & & & & \\ \partial_z & & & & & & \\ \partial_w & & & & & & \end{array}$$

and considering the several determinants which can be formed with any four columns of the matrix, these equated to zero establish a more than one-fold relation between the coordinates, viz., in the case of five surfaces, we have a twofold relation, the intersection of the 5 surfaces in lines, instead of in points; and so in other cases. Or, what is the same thing, the Jacobians $J(P, Q, R, S)$, etc., equated to zero, are not independent equations.

3. In the case of fewer than four surfaces, adopting the arrangement

$$\begin{array}{c|ccc} & P, & Q, & R \\ \hline \partial_x & & & \\ \partial_y & & & \\ \partial_z & & & \\ \partial_w & & & \end{array}$$

and considering the several determinants which can be formed with any 3 of 4 columns of the matrix, we establish a more than ordinary relation, the simultaneous vanishing of J_1, J_2, J_3, J_4 formed by omitting in succession each of the rows; this gives a curve of intersection of higher order than the proper intersection. So in other cases. It will be seen in what manner I extend the notion of the Jacobian.

4. We say that a surface $P = 0$ contains an arbitrary point, or that the equation $P = 0$ is satisfied by the coordinates of an arbitrary point, when there is one or more relations between the coefficients of P , viz., P is regarded as a function of (x, y, z, w) and of a series of coefficients which enter linearly therein. To say that the surface passes through 1 given point is to assert one such relation; to say that it passes through 2 given points is to assert two such relations; and so on.

5. Suppose that the orders of the surfaces $P = 0, Q = 0, \dots$ are $a + 1, b + 1, \dots$ (so that the orders of the differential coefficients are $a, b, \dots$); then the Jacobian $J(P, Q, R, S)$ is of the order $a + b + c + d$, and the corresponding surface has $(a + b + c + d)$ -tic contact with the surfaces $P = 0, Q = 0, R = 0, S = 0$ at the points of common intersection of the four surfaces.

6. It is also shown (Salmon, Ed. 2, p. 495) that the surface obtained by equating the discriminant of a quadric surface to zero is the Jacobian of the four polar quadric surfaces with respect to four arbitrary points, viz., the Jacobian determinant $J(P, Q, R, S)$, regarded as a function of (x, y, z, w) , expresses the condition that the four planes of polarity should pass through one and the same point. So in other analogous statements about polars and Jacobians.

7. The subsidiary theorem of the contact of the curve and surface above mentioned is, that if for shortness the differential coefficients of P, Q, R, S in regard to (x, y, z, w) be denoted by $P_x, \dots, Q_x, \dots$, etc., the Jacobian $J(P, Q, R, S) = 0$ may be written, for x, y, z, w replaced by the coordinates of a point on the curve of intersection,

$$\begin{vmatrix} P_x & Q_x & R_x & S_x \\ P_y & Q_y & R_y & S_y \\ P_z & Q_z & R_z & S_z \\ P_w & Q_w & R_w & S_w \end{vmatrix} = 0,$$

and this determinant vanishes identically by virtue of the relation between the partial derivatives at the points common to the surfaces $P = 0, Q = 0, R = 0, S = 0$. The contact is therefore of the order $a + b + c + d$ at the points of common intersection.

8. The subsidiary theorem of the contact of the curve and surface just mentioned gives rise to the Jacobian as the locus of points the polar planes of which, with respect to P, Q, R, S , pass through one and the same point. We shall in the sequel make considerable use of this theorem in various forms.

9. The Jacobian is then easily reduced to its proper form by a process of elimination, viz., introducing as many auxiliary equations as may be required to render it determinate.

10. We have, in particular, the theorem that the Jacobian of four quadric surfaces is a quartic surface, since each of the differential coefficients is of order 1 and there are four of them.

11. The Jacobian of four quadric surfaces is in fact a quartic surface, and the corresponding surface has 4-tic contact with the four quadric surfaces $P = 0, Q = 0, R = 0, S = 0$ at the points of common intersection.

12. In the case where there is a plane 1, the sextic cone breaks up into this plane and into a quintic cone, etc.

13. In the cases $k = 1, 2, 3, 4, 5$, and $k = 15, 16$, there is only one figure, and the number of arrangements may, on referring to the table, be at once seen.

Circumscribed Sextic Cone.

Order of Surface	
2	6
3	6, 1
4	6, 2, 1
5	6, 3, 1, 1
6	6, 4, 2, 1, 1
7	6, 5, 2, 1, 1, 1
8	6, 5, 3, 2, 1, 1, 1
9	6, 5, 4, 2, 1, 1, 1, 1
10	6, 5, 4, 3, 2, 1, 1, 1, 1
11	6, 5, 4, 3, 2, 1, 1, 1, 1, 1
12	... 6, 4, 2, 1, 1, 1, 1, 1, 1, 1, 1
13	... 5, 3, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
14	... 3, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
15	... 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
16	... 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

and moreover, in the cases where there are two or more lines of the sextic cone, this the k sextic cones may be of the different forms in various combinations. The total number of cases, proper *facies* (*i.e.* this very gens), but not only a comparatively small number of them actually exist.

14. In the case where there is a plane 1, the sextic cone breaks up into this plane and into a (quintic or improper quintic cone intersecting this plane in 5 lines, etc.). There is, in fact, in the plane 1 a cubic, the plane is, in fact, a singular tangent plane meeting the surface in a conic twice repeated; and the 5 residual lines on the quintic cone touching the surface in a single point of the same kind (the line through the plane and the surface meeting this plane in three points each), and the like for the conic.

15. But a different form of expression is sometimes convenient: the conditions to be satisfied are frequently such that, being satisfied by the surfaces $P = 0, Q = 0$, they will be satisfied by the surface $\lambda P + \mu Q = 0$ where, for a given value of $\lambda : \mu$, there is a 1-tuple infinity of surfaces; or, in the same way, satisfied by the surfaces $P = 0, Q = 0, R = 0$ they will be satisfied by the surface $\lambda P + \mu Q + \nu R = 0$, where there is a 2-tuple infinity of surfaces; and so on.

16. Thus, for the quadric not subjected to any conditions, there are 9 constants (this is, 1 to fix the surface itself, and 8 in the equation thereof; the surface contains 9 constants); and we have therefore one and the same thing in the various forms in which the quartic surface in 9 nodes presents itself.

17. In the questions in which such quadric surfaces present themselves, we have to fix our ideas in regard to the number of constants. If, in such a question, there occur quadric surfaces $P = 0$, $Q = 0$, etc., we are to consider that we may through this or that function of P , Q , etc., have, for any given relations between the surfaces, so many constants; and the calculations are to be conducted accordingly.

18. A quartic surface contains 34 constants. We have, in fact, in the equation of a quartic surface

$$(a, b, c, d, e, f, g, h, i, j, k, l, m, n, p, q, r, s, t, u, v, w, x, y, z, \dots)$$

35 coefficients, and so on. In a way which it is needless to recapitulate here, we ultimately arrive at the result that the quartic surface contains 34 constants.

19. In particular if we are concerned with a quartic surface having a node, then we may take the node as the point $(0, 0, 0, 1)$, viz., as the point of intersection of three of the four reference planes. The quartic surface then has the term in w^4 absent, etc.

20. The case of 4 given nodes is just as easy. In particular if we are concerned with a quartic surface having 4 nodes, then we may take these as the four points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, viz., as the four vertices of the tetrahedron of reference.

21. In proof, observe that, taking the 4 given nodes to be the points $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$ as before, the equation of a quartic surface having these four nodes contains a definite number of constants.

22. In the case of 5 given nodes, the number of constants should be reduced still further. We have through the 5 given points $\binom{5}{2} = 10$ lines, the common nodes; the remaining number of constants is $34 - 5 \cdot \text{constraints} = \text{a definite count}$.

23. In verification, take the first 4 nodes to be as above, and let the 5th node be the point $(\alpha, \beta, \gamma, \delta)$.

24. In the case of 6 given nodes, the quartic surface should contain 10 constants. We have through the 6 given points $\binom{6}{2} = 15$ lines, and the calculation proceeds in the manner just indicated.

25. The 6 given nodes being any points whatever, their coordinates may be taken to be $(\alpha_1, \beta_1, \gamma_1, \delta_1), \dots, (\alpha_6, \beta_6, \gamma_6, \delta_6)$; we may by a linear transformation take the first four of them to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$; and then the remaining two as $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$. The conditions that the quartic surface should have nodes at these 6 points reduce the number of constants to 10.

As to the Surface obtained by equating to zero a Symmetrical Determinant.

(continued). In the above (Salmon, Ed. 2, p. 493) that the surface obtained by equating to zero any symmetrical determinant has a determinate number of nodes; viz., if the order of the determinant is n , the number is $\binom{n+1}{3}$. In particular, the symmetrical determinant of the 4th order (Jacobian) has $\binom{5}{3} = 10$ nodes, as in the text of this Memoir.

26. The Jacobian is then easily found to be the symmetrical determinant

$$\left| \partial_i \partial_j K \right|_{i,j=1,\dots,4} = 0,$$

that is, the corresponding surface is a quartic surface having a determinate number of nodes, viz., 10.

27. It may be shown *a posteriori* that J is not a quadric function of P, Q, R, S ; and consequently the equation $J = 0$ represents a quartic surface, the Jacobian of the 4 quadric surfaces $P = 0, Q = 0, R = 0, S = 0$.

28. The equation $J = 0$ is the locus of the vertices of the quadric cones which can be drawn through the curve of intersection of the surfaces $P = 0, Q = 0, R = 0, S = 0$; that is, the locus of the vertices of the quadric cones which can be drawn through the 8 points of intersection; in particular, this Jacobian surface passes through the 6 lines 12, 34, 56 (viz., line of intersection of the planes through 1, 2, 3, and through 4, 5, 6); through 13, 24, 56; etc.

29. The surface in question (the Jacobian of the four quadric surfaces, or of the six points) is then a quartic surface having the six points as nodes; the order of the surface is 4 and the number of nodes is 10; we have, in fact, in the equation of the Jacobian surface $34 - 10 \cdot \text{some count} =$ the right number of constants.

30. In connexion with what precedes, we may here consider a curve which presents itself in connexion with the Jacobian surface above mentioned. We have, in fact, through any 7 given points $\binom{7}{2} = 21$ lines, through any 3 of which $4 \cdot \binom{7}{3}/3$ planes are determined; the Jacobian surface contains these lines as part of itself, together with a curve of certain order containing the points.

31. Suppose, however, that one of the seven points, instead of being a node of the dianodal surface, is itself a point on the dianodal surface; the question then arises whether the dianodal surface still passes through this point and whether the remaining structure is preserved.

32. In the case where two of the points are each of them the vertex of a cone passing through the other 5 points, the dianodal surface degenerates further: one or more of the conic relations comes to a duplication.

33. *Particular cases of the dianodal surface.* There are many particular cases of the dianodal surface, but it is unnecessary to enumerate them; in fact, throughout the general theory, the form of the dianodal surface remains the same.

34. It follows that, if $\Delta = 0$ be another quartic surface having the 7 given nodes, then the equation of the dianodal surface (in regard to the symmetrical determinant) must contain the function of Δ , etc.

35. A particular quartic surface having (in an improper sense) the 7 given nodes, viz., in many ways, but in special cases as follows: take any plane $M = 0$ through three of the nodes, say through 1, 2, 3; and any cubic surface $\Omega = 0$ having the other 4 nodes, viz., 4, 5, 6, 7; then the product $M \cdot \Omega$ vanishes at each of the 7 given points.

36. We can with the 7 given points form 35 such combinations $M \cdot \Omega$; these are of course equivalent in virtue of the dianodal relations among them.

37. It has already been shown that a quartic surface cannot in a proper sense have 8 given nodes. In regard to the quartic surfaces with 8 nodes, we start from the dianodal surface as described, and add an 8th node.

38. But if θ be not $= 0$, there may still be an 8th node; viz., this must then lie on the dianodal surface, etc.

39. I say that the 8th node may be any point whatever on the dianodal surface, that is, the dianodal surface is the locus of the 8th node.

40. [*Article 40 develops further the parametric form of the dianodal surface and its relation to the 7 given nodes.*]

41. [*Article 41 contains supplementary remarks bearing on the same construction.*]

42. Consider the seven points 1, 2, 3, 4, 5, 6, 7. As already mentioned, through any 3 of them, say 1, 2, 3, we have a plane $M = 0$; and through the same 7 points, with the 4 points 4, 5, 6, 7 as nodes, a cubic surface $\Omega = 0$. We may take the planes through 123, 145, and 456 respectively each as M , and the cubic surfaces having Ω as nodes the remaining 4 points; and similarly, taking other combinations, we have $\Delta = M\Omega = 0$, a quartic surface with the seven points as nodes. And using this form of Δ it may be shown that the dianodal $J(P, Q, R, \Delta) = 0$ passes through the 21 lines 12, 13, ... 67, and through 35 plane cubics such as $M = 0$, $\Omega = 0$; viz., this is a cubic in the plane 123 passing through the points 1, 2, 3, and through the intersection of the plane with each of the six lines 45, 46, ... 67 (nine points determining the cubic); the complete intersection by the plane 123 being therefore composed of this cubic and of the three lines 12, 13, 23. For the passage through the cubic, we have only to observe that

$$J(P, Q, R, M\Omega) = J(P, Q, R, \Omega) M + J(P, Q, R, M) \Omega = 0$$

is satisfied by $M = 0$, $\Omega = 0$; and for the passage through the lines, taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the planes 567, 674, 745, and 456 respectively, each of the functions P , Q , R is of the form $ayz + bzx + cxy + fxw + gyw + hzw$, and the function Ω is of the form $Ayzw + Bzwx + Cwxy + Dxyz$. Hence, writing in the derived functions $z = 0$, $w = 0$, the first and second lines of the determinant $J(P, Q, R, \Omega)$ will be of the form

$$\begin{pmatrix} cy, & c'y, & c''y, & * \\ cx, & c'x, & c''x, & * \end{pmatrix},$$

or the determinant vanishes for $z = 0$, $w = 0$; that is for any point of the line 45 we have $\Omega = 0$ and also $J(P, Q, R, \Omega) = 0$; consequently $J(P, Q, R, M\Omega) = 0$, and the like for the other lines. The theorem is thus proved.

43. I say that the dianodal surface passes through each of the 7 skew cubics, such as 123456. To prove this, it is only necessary to show that the skew cubic 123456 lies on the dianodal surface. For this purpose it will be enough to show that the skew cubic meets the plane 712 in a point of the surface; for then it will, in like manner, meet each of the 15 planes 712, 713, ... 756 in a point of the surface; that is, we shall have 15 intersections of the curve and surface, and there are, besides, the intersections 1, 2, 3, 4, 5, 6, in all 21 intersections; that is, the skew cubic must lie on the surface.

44. The plane 712 meets the surface in three lines and in a plane cubic determined by the points 7, 1, 2 and the six intersections of the plane with the lines 34, 35, . . . 56. We have therefore to show that this plane cubic meets the skew cubic 123456. Consider for a moment the points 1, 2, 3, 4, 5, 6 and another point 7'. As seen above, we have in general, through the points 1, 2, 7' and with the points 3, 4, 5, 6 as nodes, a determinate cubic surface, which surface passes through the lines 34, 35, . . . 56. But the cubic surface becomes indeterminate if the points 1, 2, 7', 3, 4, 5, 6 are on the same skew cubic; that is, if 7' is any point whatever on the skew cubic 123456 (the proof presently). Taking, then, 7' as the intersection of the skew cubic by the plane 712, we have in this plane the points 7', 1, 2, and the intersections of the plane by the lines 34, 35, . . . 56, nine points through which there pass an infinity of plane cubics; that is, the plane cubic determined by the points 7, 1, 2 and the six intersections will pass through the point 7'; viz., it meets the skew cubic 123456.

45. For the subsidiary theorem, taking X, Y, Z, W as current coordinates, viz., $X = 0, Y = 0, Z = 0, W = 0$ as the equations of the planes 456, 563, 634, 345 respectively, (x_1, y_1, z_1, w_1) and (x_2, y_2, z_2, w_2) as the coordinates of the points 1 and 2 respectively, and (x, y, z, w) for those of 7'; the equation of the cubic surface passing through 7', 1, 2, and having the nodes 3, 4, 5, 6, is

$$\begin{vmatrix} \frac{1}{X}, & \frac{1}{Y}, & \frac{1}{Z}, & \frac{1}{W} \\ \frac{x}{1}, & \frac{y}{1}, & \frac{z}{1}, & \frac{w}{1} \\ \frac{x_1}{1}, & \frac{y_1}{1}, & \frac{z_1}{1}, & \frac{w_1}{1} \\ \frac{x_2}{1}, & \frac{y_2}{1}, & \frac{z_2}{1}, & \frac{w_2}{1} \end{vmatrix} = 0;$$

and this ceases to be a determinate function if only

$$\begin{vmatrix} \frac{1}{x}, & \frac{1}{y}, & \frac{1}{z}, & \frac{1}{w} \\ \frac{x_1}{1}, & \frac{y_1}{1}, & \frac{z_1}{1}, & \frac{w_1}{1} \\ \frac{x_2}{1}, & \frac{y_2}{1}, & \frac{z_2}{1}, & \frac{w_2}{1} \end{vmatrix} = 0;$$

viz., considering $(x_1, y_1, z_1, w_1), (x_2, y_2, z_2, w_2)$ as given, this is a twofold relation between the coordinates (x, y, z, w) of the point 7'. The relation may be represented by the four equations $(yzw) = 0, (zwx) = 0, (wxy) = 0, (xyz) = 0$, if for shortness

$$(yzw) = \begin{vmatrix} yz, & zw, & wy \\ y_1z_1, & z_1w_1, & w_1y_1 \\ y_2z_2, & z_2w_2, & w_2y_2 \end{vmatrix}$$

and the like as to the other symbols. The four equations represent quadric surfaces, each two intersecting in a line [e.g., $(yzw) = 0, (zwx) = 0$ in the line $z = 0, w = 0$], and the four surfaces besides intersecting in a skew cubic, which is the required locus of the point 7', and which, as is seen at once, passes through the points 1, 2, 3, 4, 5, 6.

46. By what precedes, we have on the dianodal surface through the point 1 the lines 12, 13, 14, 15, 16, 17, and the skew cubics 123456, &c. The six lines are not on the same quadric cone, and it thus appears that the point 1 must be a cubic-node (point where, instead of the tangent plane, we have a cubic cone) on the surface. It is to be remarked that the lines 12, 13, 14, 15, 16, and the tangent at 1 to the skew cubic 123456, lie in a quadric cone; viz., this tangent is given as the sixth intersection of the cubic cone with the quadric cone through the lines 12, 13, 14, 15, 16.

47. I revert to the equation of the dianodal surface as given in the form $J = J(P, Q, R, M\Omega) = 0$, where $M = 0$ is the plane through the points 1, 2, 3, and $\Omega = 0$ the cubic surface through these points, and having the points 4, 5, 6, 7, as nodes. We can find the orders of the several functions P, Q, R, M, Ω in the coordinates (x_1, y_1, z_1, w_1) , &c., of the several points; viz., writing for shortness x_1^2 to denote the order 2 in regard to (x_1, y_1, z_1, w_1) , and so in other cases, we have

$$P = Q = R = x^2(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^2,$$

$$M = x(x_1, x_2, x_3),$$

$$\Omega = x^3(x_5, x_6, x_7)(x_1, x_2, x_3, x_4)^3,$$

(where, of course, the x 's show in like manner the orders in regard to the current coordinates (x, y, z, w) ; the proof in regard to Ω is easily supplied.) The order of J is equal that of $PQRM\Omega$, less 4 as regards the current coordinates, by reason of the differentiations; that is, we have $J = x^6(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$; and we thus see that the equation of the dianodal surface as above obtained is encumbered with a constant factor of the form $(x_1x_2x_3)^4(x_4x_5x_6x_7)^4$. In fact, the relation between the 7 points and the current point (x, y, z, w) , or say the point 8, as expressing that the 8 points are the nodes of a dianome, should be a symmetrical one in regard to the coordinates of the several points; and being of the order 6 in regard to the coordinates (x, y, z, w) , it should be of the same order in regard to the other coordinates; that is, the true form would be $J = (x x_1 x_2 x_3 x_4 x_5 x_6 x_7)^6 = 0$.

48. It is possible that taking the 4 points, say 1, 2, 3, 4, to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, and the 3 points, say 5, 6, 7, to be $(1, 1, 1, 1)$, $(\alpha, \beta, \gamma, \delta)$, $(\alpha', \beta', \gamma', \delta')$, the extraneous factor might exhibit itself, and that the calculation thereby be simplified. But I do not at present further pursue the inquiry.

Dianodal Curve of 8 given Points, or say a Nodo or Section.

49. The dianodal surface, qua surface having the last-mentioned 8 points for nodes, has on it various special curves, and the discussion of these is here in some sort necessary; viz., we have on it a curve, the section by an arbitrary plane, of the order 6; the curve of intersection of the dianodal surface with another quartic surface through the 7 given nodes and called the dianodal curve of these 8 points. There exists between $P, Q, R, S = M\Omega$ a syzygetic relation

$$\alpha P + \beta Q + \gamma R + \delta M\Omega = 0,$$

where in $\alpha, \beta, \gamma, \delta$ are linear functions of the current coordinates, α for instance being a function which vanishes at the points 1, 2, 3, and which contains a number of arbitrary coefficients; whence the syzygetic relation is one satisfied at every one of the 7 given nodes.

50. This being so, the cubic curve through the last-mentioned six points has its equation of the form $J(P, Q, R, \alpha M\Omega) = 0$; and the dianodal curve of these 8 points is the intersection of the dianodal surface with this cubic curve.

51. Considering the orders in regard to $(a, b, c, d, f, g, h, l, m, n, P, Q, R, S, T, U, V)$, the dianodal curve through the points 1, 2, ..., 7, and an 8th point lies on the dianodal surface, and is of the order 18. It contains in fact one set of intersections (the 21 lines 12, 13, ..., 67 counted as one with multiplicity 2) and the curve proper; this last is the dianodal curve of the 8 points.

52. We have $5 \cdot 6 = 30$ as the order of intersection of the dianodal surface (sextic) and the auxiliary quartic surface; the auxiliary quartic passing through the 7 given points counts as $\binom{7}{2} = 21$, and the dianodal curve is of the order 18. There would be a reduction of 25 if the line through the 6 given points 1, 2, 3, 4, 5, 6 were in involution with the 6 given quadric surfaces; whence the dianodal curve order would be $30 - 25 = 5$. But this is not the case.

53. In order to find it, taking as above 9 a given point on the Jacobian, we have a tenth point on the surface only if there be a relation between the coordinates of the points 1 through 9. The condition is in fact a determinate condition which fixes the order of the dianodal curve at 18.

54. If there is a 10th node, say 10, this is also a point on the Jacobian curve of the 8 points 1, 2, ..., 8, and similarly a point on the dianodal curves through suitable octads of the 9 remaining nodes.

55. In the case of the surface with 9 nodes, it is clear that this is octadic in one way only: the node 9 cannot form an octad with any 7 of the remaining nodes. But in the case of the surface with 10 nodes, the question arises whether the nodes 9 and 10 may not be such as to form an octad with some six, say with the nodes 1, 2, 3, 4, 5, 6 of the remaining 8 nodes; that is, whether we can have 1, 2, 3, 4, 5, 6, 7, 8 forming an octad, and also 1, 2, 3, 4, 5, 6, 9, 10 forming an octad. I will show that this is impossible if only the points 1, 2, 3, 4, 5, 6 are given points, that is, points assumed at pleasure and not specially related to each other. For this purpose, assuming that the points form 2 octads as above, take through 1, 2, 3, 4, 5, 6, 7, 9 the quadric surfaces $P = 0$, $Q = 0$, then each of these passes through 8, 10; take $R = 0$ any other quadric surface through 1, 2, 3, 4, 5, 6, 7, 8, and $S = 0$ any other quadric surface through 1, 2, 3, 4, 5, 6, 9, 10. Then $P = 0$, $Q = 0$, $R = 0$ intersect in the 1st octad, [*continues in book p. 154; the memoir then resumes at § 60 in the separate retype on book p. 155.*]

(End of chunk: book pages 129–153 of Vol. VII. The memoir on Quartic Surfaces continues with §§ 56–59 on book p. 154, and is then resumed at § 60 in cayley_vol07_paper_445_RETYPE.tex.)

[445] A MEMOIR ON QUARTIC SURFACES. (CONTINUED)

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 19–69.
Read February 10, 1870.]

(Continuation from book p. 153; articles 56–112.)

56. We may through 3 draw the lines LM , QT to meet 14, 26 and 12, 46 respectively; and through 5 the lines RS , NP to meet 14, 26 and 12, 46 respectively. Observe that the points O in the figure are apparent intersections only; viz., NP does not meet QT , nor LM meet RS . In fact, if NP met QT it would be a line in the series of lines meeting 14, QT , 26; or 5 would be situate in a hyperboloid, determined by means of the points 1, 2, 4, 6, 3; viz., 5 would not be an arbitrary point: and so LM does not meet RS . Now the quadrics E , H meet in the lines 14, 26, LM , NP , and the quadrics A , C in the lines 12, 46, QT , RS . Suppose that we had identically $(* \$ A, C, E, H)^2 = 0$; putting therein $E = 0$, $H = 0$, we should have $(* \$ A, C)^2 = 0$, viz.,

$$(A + \lambda C)(A + \mu C) = 0;$$

or there would exist quadrics of the forms $A + \lambda C = 0$ containing the lines 14, 26, LM , NP . Now there is no quadric surface $A + \lambda C = 0$ containing the line NP ; for $A + \lambda C = 0$ is a quadric containing the sides of the quadrilateral $QRST$; the generating lines of the one kind meet each of the lines RS , QT ; those of the other kind neither. Hence NP , which meets RS but not QT , cannot be a generating line of either kind; and we have no identical relation $(A, C, E, H)^2 = 0$.

57. In the octadic surface with 9 nodes: starting with any 7 nodes of the octad, 9 is not the 8th point of the octad, and hence (by the theory of the dianome) it must lie in the dianodal surface of the 7 points; that is, the dianodal surface of the 7 points must pass through 9, viz., through any point whatever of the Jacobian curve of the 7 points, that is, of the octad; or (what is the same thing) the dianodal surface of the 7 points passes through the Jacobian curve of the octad. This is an obvious property of the dianodal surface, the surface $J(P, Q, R, V) = 0$ contains the Jacobian curve $J(P, Q, R) = 0$. But it further appears that, starting with any 6 points of the octad and with the point 9 (that is, any point whatever of the Jacobian curve), the dianodal surface of these 7 points must contain the remaining 2 points of the octad. And in the octadic surface with 10 nodes, starting with any 5 points of the octad and with the points 9 and 10 (that is, any two points on the Jacobian curve) the dianodal surface of these 7 points must contain the remaining three points of the octad. I have not attempted to verify these last properties of the dianodal surface.

Dianomes with 9 or 10 Nodes.

58. I now consider the dianomes with 9 and 10 nodes. Starting from the general form

$$(a, b, c \times P, Q)^2 + \theta \Delta = 0,$$

where $\Delta = 0$ is a particular quartic surface having the 8 nodes, it at once appears that if there is a 9th node, say 9, the node must be a point on the Jacobian curve $J(P, Q, \Delta) = 0$; hence, taking it to be one such point, the number of constants in the resulting equation should be 1. Hence if $P = 0$ be the quadric surface through the 9 points, and $\Delta = 0$ a particular quartic surface having the 9 points as nodes, the general equation is $\alpha P^2 + \theta \Delta = 0$.

59. But we may consider the question somewhat differently. Starting with the 7 given points 1, 2, 3, 4, 5, 6, 7 and with 8 a given point on the dianodal surface of the 7 points; it is clear that 9 must be on the dianodal surface 1234567 and also on the dianodal surface 1234568; the complete intersection is of the order 36, and we have to consider how this breaks up so as to contain a part of itself the dianodal curve of the order 18.

Dianodal Curve of 8 Points.

60. Consider first any 8 points whatever 1, 2, 3, 4, 5, 6, 7, 8 where 8 is not on the dianodal surface 1234567, nor 7 on the dianodal surface 1234568. The two surfaces have in common the 15 lines 12, 13, ..., 56 and the skew cubic 123456, they therefore besides intersect in a curve of the order 18. At the point 1 the tangent cubic cones of the two surfaces intersect in the lines 12, 13, 14, 15, 16 and the tangent to the skew cubic 123456, 6 lines lying in a quadric cone; they therefore besides intersect in 3 lines lying in a plane; that is, the point 1 is on the curve of the order 18 an actual triple point, the 3 tangents lying *in plano*; and the like of course in regard to each of the points 2, 3, 4, 5, 6. But as 7, 8 lie each of them on only one of the two surfaces, the curve of the order 18 does not pass through 7 or 8.

61. If, however, 8 lies on the dianodal surface 1234567, then each of the 8 points will lie on the dianodal surface of the other 7; and in particular 7 will lie on the dianodal surface 1234568. The surfaces intersect as before in a residual curve of the order 18; the only difference is that 7 and 8 are now points on each surface; viz., each of them is on one of the surfaces an ordinary point, and on the other a cubic node; the points 7 and 8 are thus each of them an actual triple point on the curve; and at each of them the 3 tangents are *in plano*. We thus see that the dianodal curve 12345678 is a curve of the order 18, such that each of the 8 points is a triple point on the curve, the tangents at each of them being *in plano*.

Ten Nodes.

62. Suppose there is a 10th node, say 10; starting from the equation $\alpha P^2 + \theta \Delta = 0$ (P = the quadric surface through the 9 points, Δ = a particular quartic surface having the 9 points as nodes), it at once appears that the node must be one of the points $J(P, \Delta) = 0$; hence, taking it to be one of these points, we have 4 equations, which, in virtue of the node being one of the points in question, reduce themselves to a single equation determining the ratio $\alpha : \theta$; we have thus a completely determinate surface, say $\square = 0$ having the 10 points as nodes. The number of points $J(P, \Delta)$, writing in the formula No. 5, $a = 1$, $b = 3$, is obtained as $1 + 3 + 9 + 27 = 40$, but it is to be observed that the surface $P = 0$ passes through each of the 9 nodes of the surface $\Delta = 0$; these count twice among the points $J(P, \Delta) = 0$, and the number of residual points (or say the dianodal centres of the 9 points) is $40 - 18 = 22$; viz., this is the number of positions of the node 10. [The nine points count each three times, and the number of residual points, or positions of the node 10, is thus not $40 - 18 = 22$, but $40 - 27 = 13$.]

Dianodal Centres of 9 Points.

63. In further explanation, observe that 9 is any point on the dianodal curve 12345678; the node 10 must lie on this same curve, and also on the dianodal surface 1234569. Take $P = 0$ the quadric through all the 9 points, $Q = 0$ a quadric through all but the point 9, $R = 0$ through all but the point 8, $S = 0$ through all but the point 7. The dianodal curve 12345678 is $J(P, Q, V) = 0$, and the dianodal surface 1234569 is $J(P, R, S, V) = 0$; the total number

of intersections is $6 \times 18 = 108$; these include the $4 \times 18 = 72$ points of intersection of the dianodal curve $J(P, Q, \Delta) = 0$ with the Jacobian surface $J(P, Q, R, S) = 0$, except the four points $J(P, Q) = 0$, which are the vertices of the 4 quadric cones through 1, 2, 3, 4, 5, 6, 7, 8 (which 4 points are not situate on the curve $J(P, R, S) = 0$), and there are besides 40 points $\{108 = (72 - 4) + 40\}$ which are the before-mentioned points $J(P, \Delta) = 0$; viz., these are the 9 points each twice [three times], and the residual 22 [13] points which are the dianodal centres of the 9 points.

General result as to the Dianomes.

64. We have thus established the theory of the dianome quartic surfaces; viz., we have

- The octodianome, 8 nodes, 7 of them arbitrary, and the 8th an arbitrary point on the dianodal surface (order 6) of the 7 points.
- The enneadianome, 9 nodes, the 9th an arbitrary point on the dianodal curve (order 18) of the 8 points.
- The decadianome, 10 nodes, the 10th any one of the 22 [13] dianodal centres of the 9 points.

And as already mentioned, so long as the first 7 nodes are arbitrary, there cannot be more than 10 nodes in all.

The Symmetroid.

The Lineolar Correspondence of Quartic Surfaces.

65. I consider four equations $S = 0, T = 0, U = 0, V = 0$, lineolar in regard to the two sets of coordinates (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$; viz., each of these equations is of the form

$$(*\$x, y, z, w \times \alpha, \beta, \gamma, \delta) = 0.$$

This implies that the point (x, y, z, w) lies on a certain quartic surface $\Theta = 0$, and the point $(\alpha, \beta, \gamma, \delta)$ on a certain quartic surface $\Delta = 0$, and that the two surfaces correspond point to point to each other. In fact, writing the four equations in the form

$$\begin{aligned} L\alpha + M\beta + N\gamma + P\delta &= 0, \\ L'\alpha + M'\beta + N'\gamma + P'\delta &= 0, \\ L''\alpha + M''\beta + N''\gamma + P''\delta &= 0, \\ L'''\alpha + M'''\beta + N'''\gamma + P'''\delta &= 0, \end{aligned}$$

where $L, \&c.$, are linear functions of (x, y, z, w) , then eliminating $(\alpha, \beta, \gamma, \delta)$, we obtain the equation

$$\Theta = \begin{vmatrix} L & M & N & P \\ L' & M' & N' & P' \\ L'' & M'' & N'' & P'' \\ L''' & M''' & N''' & P''' \end{vmatrix} = 0;$$

and similarly, writing the four equations in the form

$$\begin{aligned} Ax + By + Cz + Dw &= 0, \\ A'x + B'y + C'z + D'w &= 0, \\ A''x + B''y + C''z + D''w &= 0, \\ A'''x + B'''y + C'''z + D'''w &= 0, \end{aligned}$$

where $A, \&c.$, are linear functions of $(\alpha, \beta, \gamma, \delta)$, then eliminating (x, y, z, w) , we obtain the equation

$$\Delta = \begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix} = 0.$$

Moreover, Θ being $= 0$, the four linear equations in $(\alpha, \beta, \gamma, \delta)$ are equivalent to three equations, and give for instance $(\alpha, \beta, \gamma, \delta)$ proportional to the determinants formed with the matrix

$$\begin{vmatrix} L' & M' & N' & P' \\ L'' & M'' & N'' & P'' \\ L''' & M''' & N''' & P''' \end{vmatrix};$$

and similarly, Δ being $= 0$, the four linear equations in (x, y, z, w) are equivalent to three equations, and give for instance (x, y, z, w) proportional to the determinants formed with the matrix

$$\begin{vmatrix} A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix},$$

which establishes the point-to-point correspondence of the two surfaces.

66. It would at first sight appear that any quartic surface $(*\$ \alpha, \beta, \gamma, \delta)^4 = 0$ whatever might have its equation expressed in the foregoing determinant form $\Delta = 0$. This equation seems, in fact, to contain homogeneously as many as 64 constants. But if we multiply the determinant line into line by a constant determinant

$$\begin{vmatrix} a & b & c & d \\ a' & b' & c' & d' \\ a'' & b'' & c'' & d'' \\ a''' & b''' & c''' & d''' \end{vmatrix}$$

and then column into column by another constant determinant, the coefficients, all but one of them, of these constant determinants may be used to specialize the form of the resulting equation, [say they are apoclastic constants]; this equation will really contain $64 - (2 \cdot 16 - 1) = 33$ constants; and in order that the quartic surface $(*\$ \alpha, \beta, \gamma, \delta)^4 = 0$ may have its equation expressible in the form $\Delta = 0$, a single relation must hold good among the coefficients; but this in passing¹.

67. Returning to the quartic surface

$$\Delta = \begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix} = 0,$$

we may connect this not only with the foregoing surface $\Theta = 0$, but in a similar manner with another quartic surface $\Phi = 0$; viz., taking the current coordinates $(\xi, \eta, \zeta, \omega)$, we may form the

¹Applying the same reasoning to a cubic determinant $\Delta = 0$, the number of constants is $36 - (2 \cdot 9 - 1) = 19$; so that a cubic surface is expressible in the form in question. And so for the quadric determinant $\Delta = 0$, the number of constants is $16 - (2 \cdot 4 - 1) = 9$; so that a quadric surface is expressible in the form in question, as is otherwise obvious.

lineolar equations

$$\begin{aligned} A\xi + A'\eta + A''\zeta + A'''\omega &= 0, \\ B\xi + B'\eta + B''\zeta + B'''\omega &= 0, \\ C\xi + C'\eta + C''\zeta + C'''\omega &= 0, \\ D\xi + D'\eta + D''\zeta + D'''\omega &= 0, \end{aligned}$$

which, by the elimination of $(\xi, \eta, \zeta, \omega)$, give $\Delta = 0$, and by the elimination of $(\alpha, \beta, \gamma, \delta)$ a determinant quartic equation $\Phi = 0$ between the coordinates $(\xi, \eta, \zeta, \omega)$; and of course the two surfaces $\Delta = 0$, $\Phi = 0$ have a point-to-point correspondence such as exists between the surfaces $\Theta = 0$, $\Delta = 0$. The relation of the point $(\alpha, \beta, \gamma, \delta)$ on the surface $\Delta = 0$ to the point (x, y, z, w) on the surface $\Theta = 0$, and to the point $(\xi, \eta, \zeta, \omega)$ on the surface $\Phi = 0$, may be conveniently indicated by means of the diagram

x	y	z	w	
A	B	C	D	ξ
A'	B'	C'	D'	η
A''	B''	C''	D''	ζ
A'''	B'''	C'''	D'''	ω

$\Theta / \Delta / \Phi.$

68. It is to be observed that, writing for $A, B, \dots$ their values as linear functions of $(\alpha, \beta, \gamma, \delta)$, we have in all 64 constant coefficients, which we may conceive arranged in the form of a cube, thus

$$\begin{array}{cccc} a & b & c & d \\ a' & b' & c' & d' \\ \vdots & \vdots & \vdots & \vdots \\ a_1 & b_1 & c_1 & d_1 \\ a'_1 & b'_1 & c'_1 & d'_1 \\ \vdots & \vdots & \vdots & \vdots \end{array}$$

and taking these in fours height-wise, (a, a_1, a_2, a_3) , &c., we compose with them the linear functions $a\alpha + a_1\beta + a_2\gamma + a_3\delta$, &c., which enter into the equation $\Delta = 0$; taking them in fours length-wise, (a, b, c, d) , &c., we compose the linear functions $ax + by + cz + dw$, &c., which enter into the equation $\Theta = 0$; and taking them in fours breadth-wise (a, a', a'', a''') , &c., we compose the linear functions $a\xi + a'\eta + a''\zeta + a'''\omega$, &c., which enter into the equation $\Phi = 0$.

69. The process may be indefinitely repeated; we obtain always the same three surfaces over and over again, but on them an indefinite series of corresponding points; viz., we may write

$$\begin{aligned} \dots \Theta, \Delta, \Phi, \Theta, \Delta, \Phi, \Theta, \Delta, \Phi \dots \\ \dots P_1, Q_1, R_1, P, Q, R, P', Q', R' \dots \end{aligned}$$

viz., a point Q on Δ corresponds to a point P on Θ and to a point R on Φ ; R corresponds to Q on Δ and to a new point P' on Θ ; P' to R on Φ and to a new point Q' on Δ , and so on. And in the opposite direction P corresponds to Q on Δ and to a new point R_1 on Φ ; R_1 to P on Θ and to a new point Q_1 on Δ and so on. And of course the correspondence of any two points of the series, whether belonging to the same surface or to different surfaces, is a one-to-one correspondence.

The Symmetrical Case; Symmetroid and Jacobian.

70. I have established the foregoing general theory; but it is only a particular case of it which connects itself with the theory of nodal quartics; viz., the cube of coefficients is a symmetrically arranged cube

$$\begin{array}{cccc} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{array} \quad \begin{array}{ccc} a_1 & h_1 & \dots \\ \vdots & & \end{array}$$

or say its upper face is the symmetrical square matrix

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{vmatrix}$$

and the other horizontal planes, the like squares with the several terms affected by suffixes. The surface $\nabla = 0$ is here a surface of the form

$$\nabla = \begin{vmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & P \end{vmatrix} = 0$$

$[A, B, \&c. \text{ linear functions of } (\alpha, \beta, \gamma, \delta)]$; viz., ∇ is a symmetrical determinant; I call this a *symmetroid*; the surfaces $\nabla = 0$, $\square = 0$ are one and the same surface, the Jacobian of 4 quadric surfaces; moreover the points P and R are one and the same point, and the correspondence R to P' is a reciprocal one; so that, instead of the indefinite series $\dots \Delta, \Theta, \Delta, \Theta, \Delta \dots$, $\dots P', Q, P, Q, P \dots$, we have a surface, the *symmetroid* $\nabla = 0$ which is a surface with 10 nodes, which is clearly not octadic, and which is therefore the decadianome.

71. Consider the quadric surfaces

$$\begin{aligned} S &= (a, b, c, d, f, g, h, l, m, n \mid x, y, z, w)^2 = 0, \\ T &= (a_1, \dots) \quad \quad \quad , \quad y = 0, \\ U &= (a_2, \dots) \quad \quad \quad , \quad y = 0, \\ V &= (a_3, \dots) \quad \quad \quad , \quad y = 0, \end{aligned}$$

and a point $(\alpha, \beta, \gamma, \delta)$ in the same or in a different space, such that the surface

$$\alpha S + \beta T + \gamma U + \delta V = 0$$

is a cone, or say for shortness,

$(\alpha, \beta, \gamma, \delta)$ is said to be the determining point, or determinant of the cone.

And if we establish the equations

$$\begin{aligned} \partial_x(\alpha S + \beta T + \gamma U + \delta V) &= 0, \\ \partial_y(\quad \text{do.} \quad) &= 0, \\ \partial_z(\quad \text{do.} \quad) &= 0, \\ \partial_w(\quad \text{do.} \quad) &= 0, \end{aligned}$$

which express that the surface is a cone, then (x, y, z, w) is the vertex of the cone. We have thus 4 equations linear in (x, y, z, w) and also in $(\alpha, \beta, \gamma, \delta)$, so that the relation between the Σ points is of the nature of that above considered. The relation between (x, y, z, w) is given by the equation

$$J(S, T, U, V) = 0;$$

viz., the locus is the Jacobian of the 4 quadric surfaces. The relation between $(\alpha, \beta, \gamma, \delta)$ is given by the equation

$$\nabla = \begin{vmatrix} a\alpha + a_1\beta + a_2\gamma + a_3\delta, & h\alpha + \dots, & g\alpha + \dots, & l\alpha + \dots \\ h\alpha + \dots, & b\alpha + \dots, & f\alpha + \dots, & m\alpha + \dots \\ g\alpha + \dots, & f\alpha + \dots, & c\alpha + \dots, & n\alpha + \dots \\ l\alpha + \dots, & m\alpha + \dots, & n\alpha + \dots, & d\alpha + \dots \end{vmatrix} = 0;$$

so that the locus is (by the foregoing definition) the symmetroid. And the determinant and the symmetroid thus corresponds to the cone-vertex on the Jacobian.

72. But the Jacobian may be obtained in a different manner; viz., if we establish the equations

$$\begin{aligned} &(\xi\partial_a + \eta\partial_b + \zeta\partial_c + \omega\partial_d)S = 0, \\ &(\quad \quad \quad \text{do.} \quad \quad \quad)T = 0, \\ &(\quad \quad \quad \text{do.} \quad \quad \quad)U = 0, \\ &(\quad \quad \quad \text{do.} \quad \quad \quad)V = 0, \end{aligned}$$

then the elimination of $(\xi, \eta, \zeta, \omega)$ leads to the equation $J(S, T, U, V) = 0$ of the Jacobian. And since each of the equations is symmetrical in regard to (x, y, z, w) , C. VII. 21

73. The points (x, y, z, w) of points of $(\alpha, \beta, \gamma, \delta)$, viz., we may either of these latter ones taking the point (x, y, z, w) on the Jacobian surface, the point $(\alpha, \beta, \gamma, \delta)$ on the symmetroid is related to the point (x, y, z, w) , and conversely the system of 4 points, Q on the symmetroid, whose centres of these 4 points respectively; or the equation of the symmetroid, when referred to the 4 points, Q on the symmetroid, P and P' on the Jacobian. Q. V. S. I.

74. Consider any 5 pairs of points (x_1, y_1, z_1, w_1) , $(\xi_1, \eta_1, \zeta_1, \omega_1)$, &c., related as above; the quartic surfaces $S = 0$, $T = 0$, $U = 0$, $V = 0$ are by the general quadric surface which cuts harmonically the five pairs of points respectively; or say they are quadric surfaces *harmonically related* to the 5 pairs of points respectively. The general quadric surface $\alpha S + \beta T + \gamma U + \delta V = 0$, harmonically related to the 5 pairs, will be a cone if $(\alpha, \beta, \gamma, \delta)$ is on the symmetroid, and the vertex of the cone is then defined by M . Chasles (*Comptes Rendus*, tom. LXI., 1865, p. 1147–62), and shown by him to be a variable surface, is thus identified with the Jacobian of any 4 quartic surfaces; and indicated herein we have the particular case, also considered by him, of the locus of the vertices of the quadric cones which pass through 6 given points, or Jacobian of the 6 given points.

75. It is to be shown that there are 10 systems of values $(\alpha, \beta, \gamma, \delta)$, for which the symmetroid $\nabla = 0$ is satisfied; viz., the number of conditions that there should be a node (x, y, z, w) are sufficient: the equation $\alpha S + \beta T + \gamma U + \delta V = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting with any given symmetrical determinant, the line by a constant determinant, and then column into column by another constant determinant; the coefficients of the constant determinants would in this manner be used, but the resulting product is a symmetrical determinant; and the equation is then with 10 nodes, contains $34 - 10 = 24$ constants.

Symmetroid with given Nodes.

76. A symmetroid can be formed with 7 given points as nodes; but there is no proper symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid, we obtain a system of 2 quartic cones, each of them passing through the 8 points; the curve of intersection is then a sextic C_2 found in the plane of the conditions of having a given point with the 7 points; the additional condition $(x, y, z, w) = 0$ may be expressed as a linear identification with the conditions of nodality; and also that it is the lineo-linear transformation.

77. Applying the conclusion to the system of quadric surfaces $\alpha S + \beta T + \gamma U + \delta V = 0$, we constants. But starting with this shall be a plane-pair implies a threefold relation between the coefficients A, B , &c., and the required number of nodes is equal to the order of this threefold relation. Establishing the conditions A, B , &c., any 6 linear relations whatever, we should have a ninefold relation between the coefficients of the number of solutions would be equal to the order of the threefold relation; and the number of solutions is clearly = 10.

78. The equation $\nabla = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting with any given symmetrical determinant, we may multiply it line into line by a constant determinant, and then column into column by another constant determinant, in such wise that the resulting product is still a symmetrical determinant; and the coefficients of the constant determinant may then be used to specialize the form of the equation. The equation $\nabla = 0$ of the symmetroid contains, therefore, the constant determinant; the term $(\alpha, \beta, \gamma, \delta) \Sigma \sigma x_i = w^2 - 1$, $\Sigma x x_i = 3a - 3$, but not every rational determinant; the number of constants in $\nabla = 0$ may then be used to specialize the form of the equation. The equation $\nabla = 0$ of the symmetroid contains, therefore, $40 - 16 = 24$ constants; this is as it should be, for the symmetroid, qua quartic surface with 10 nodes, contains $34 - 10 = 24$ constants.

Symmetroid with given Nodes.

79. A symmetroid can be formed with 7 given points as nodes; but there is no *proper* symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid with 8 given points as nodes, we obtain a system of 2 quartic cones, each of them passing through the 8 points: the curve of intersection is in fact a quartic curve, C_4 in the plane Π , Π ($r = 4$). It is clear, from this consideration, that any one of the conditions for the nodality of any one of the 8 points is in fact deducible from the others, leaving us with 7 conditions; and a symmetroid with 7 given points as nodes can be formed: but I have not succeeded in effecting this.

80. We have for any node $(\alpha, \beta, \gamma, \delta)$ of the symmetroid,

$$\alpha S + \beta T + \gamma U + \delta V = \text{plane-pair.}$$

If, then, 4 of the given nodes are $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, we must have S , T , U , V each of them a plane-pair. We may without loss of generality assume $S = x^2 + y^2$, $T = x^2 + w^2$; this, however, does not determine the signification of the coordinates (x, y, z, w) ; viz., we have S to determine the meaning of x, y ; and similarly T to determine the meaning of x, w ; and similarly T remain unaltered if we write

$$x \cos \theta + y \sin \theta, \quad x \sin \theta - y \cos \theta \quad \text{for } x, y;$$

and similarly T will remain unaltered if we write

$$x \cos \theta_1 + w \sin \theta_1, \quad x \sin \theta_1 - w \cos \theta_1 \quad \text{for } x, w.$$

Hence, if we go on to assume

$$\begin{aligned} U &= k(x + my + nz + pw)(x + m'z + n'z + p'w), \\ V &= l_1(x + m_1y + n_1z + p_1w)(x + m'_1z + n'_1z + p'_1w), \end{aligned}$$

formulae which contain the 12 constants

$$(k, m, n, p, m', n', p', l_1, m_1, n_1, p_1, m'_1, n'_1, p'_1),$$

this is right, for the symmetroid containing 24 constants, the symmetroid with 4 given nodes contains $(24 - 4 \cdot 3 =) 12$ constants. And each additional condition gives a single one of these constants; and so on.

81. But for any 4 new nodes, the equations may be satisfied by writing therein $u = u', p = -p', m = -m_1, n_1 = n'_1$; viz., they thus assume the form

$$\begin{aligned} S &= x^2 + y^2, \\ T &= x^2 + w^2, \\ U &= (\alpha x + cx)^2 + (by + dw)^2, \\ V &= (\alpha'x + c'x')^2 + (b'y + d'w)^2, \end{aligned}$$

containing 8 constants, which may be determined to make the surfaces have 4 given points. If now with the last mentioned values we form the value of $\alpha S + \beta T + \gamma U + \delta V$, this is a plane-pair if a square if

$$(\alpha + \gamma a^2 + \delta a'^2)(\beta + \gamma c^2 + \delta c'^2) - (\gamma ac + \delta a'c')^2 = 0, \quad \text{say this is } \Lambda = 0,$$

(continued)

82. (continued)

and the second will be a square if

$$(\alpha + \gamma b^2 + \delta b'^2)(\beta + \gamma d^2 + \delta d'^2) - (\gamma bd + \delta b'd')^2 = 0, \quad \text{say this is } \Lambda' = 0;$$

so that the condition

$$\alpha S + \beta T + \gamma U + \delta V = \text{cone}$$

will be satisfied if $\Lambda = 0$, or if $\Lambda' = 0$; that is, the equation of the symmetroid will be $\Lambda\Lambda' = 0$, breaks up into the 2 quadric surfaces $\Lambda = 0, \Lambda' = 0$, each of which is a cone.

82. It is to be further observed that, considering the first mentioned 4 points $(1, 0, 0, 0)$, &c., and any other 4 given points whatever, the equation of the 4 quadric cones through 4 given points 1, 2, 3, 4 and one of the new points, are obtained by considering the values $(\alpha, \beta, \gamma, \delta)$ which satisfy $\Lambda = 0$, viz., they are equal to be of the form

$$(*\$ \beta\gamma, \gamma\alpha, \alpha\beta, \beta\delta, \gamma\delta)^2 = 0;$$

this combined with the first form gives, say through $(1, 0, 0, 0)$, &c., and the 5th point of the symmetroid will be satisfied if $\Lambda' = 0$; the equation of the symmetroid will be one of the 4 cones;

viz., we have to express that one of the 4 cones; viz., we have to express the conditions of the constants b, c in the expressions of the foregoing equations; that is the conditions for 4 given points different therefrom by changing a, b, c into $a + \frac{1}{w}(by - gz)$, $b + \frac{1}{w}(-kx + fz)$, $c + \frac{1}{w}(gx - fy)$ respectively; the equation may therefore be written in the symbolic form

$$w^2 \cdot \exp \frac{1}{w} [(ky - gz) \partial_a + (-kx + fz) \partial_b + x^2(gx - fy) \partial_c] \cdot [(a, b, c)^2] = 0,$$

or, what is the same thing,

$$w^2 \cdot \exp \frac{1}{w} [x(g\partial_b - h\partial_c) + y(h\partial_a - f\partial_c) + z(f\partial_b - g\partial_a)] \cdot [(a, b, c)^2] = 0,$$

where $\exp \theta$ (read exponential) denotes e^θ , and $[(a, b, c)^2]$ represents a determinant as above explained. The equation contains, it is clear, the four terms

$$x^2[(a, -h, g)^2] + y^2[(-h, b, -f)^2] + z^2[(g, -f, c)^2] + w^2[(a, b, c)^2].$$

I am not sure whether this surface of the eighth order has been anywhere considered.

83. In order that the symmetroid may have a 6th given node $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, I observe that the constants may be determined so that $\alpha_1 S + \beta_1 T + \gamma_1 U + \delta_1 V$ shall be equal to the variable factor, writing this as Π^2 and the like for the 5th point; we find $a(\alpha_1 + 5\alpha_1 + 6\alpha_1) + \dots = 0$, viz., the new node being $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, and putting therein $a_i(1 + i\alpha_i + 6\alpha_1) = 0$, &c., for the various values of values which the new nodes can assume the formulae

$$\alpha_i S + \beta_i T + \gamma_i U + \delta_i V = (a_i, b_i, c_i \times x, y, z, w)^2,$$

the new variables given through 4 given nodes $(1, 0, 0, 0)$, &c., and the points $1, 2, \dots, 8$ points; the given plane-pairs at the points $1, 2, \dots, 7$.

Quartics with 11 or more Nodes.

103. I mention two results which, although they relate to quadric surfaces with more than 10 nodes, present themselves in such immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & D \end{vmatrix} = 0$$

of the symmetroid $(A, B, \dots$ linear functions of the coordinates), we have identically $A = 0$, then the surface has evidently a node in the ordinary manner in addition to the usual 10 nodes; an the surface has in all 11 nodes. And so also if (identically in every case) $B = 0$, there are 12 nodes; if $C = 0$, there are 13 nodes. These are, in fact, quartic surfaces with 11, 12, 13, and 14 nodes respectively, mentioned in Kummer's Memoir.

104. We may consider the symmetroid derived from the quadric surfaces which pass through 6 given points: viz., taking as before (see No. 25) the conditions of the 6 points to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, $(1, 1, 1, 1)$, (a, b, c, d) ; the coordinates of the line joining the last-mentioned two points: and, to avoid confusion, taking for the present purpose (X, Y, Z, W) instead of $(\alpha, \beta, \gamma, \delta)$ for the coordinates of a point on the symmetroid, the equation is obtained by arranging in the form of a determinant the coefficients of the quadric form

$$\begin{aligned} & Xx(\quad \quad \quad) + hy(X - Y) + cyW, \\ & h(X - Y) + cyW, \\ & g(Z - X) + byW, \quad f(Y - Z) + axW, \\ & W(axyx + b\beta zx + cyxy \quad \quad \quad); \end{aligned}$$

viz., the equation in question is

$$\begin{vmatrix} \cdot & h(X - Y) + cyW & g(Z - X) + byW & aX \\ h(X - Y) + cyW & \cdot & f(Y - Z) + axW & bY \\ g(Z - X) + byW & f(Y - Z) + axW & \cdot & cZ \\ aX & bY & cZ & \cdot \end{vmatrix} = 0,$$

or, as it may be more simply written,

$$\sqrt{aX[f(Y - Z) + axW]} + \sqrt{bY[g(Z - X) + byW]} + \sqrt{cZ[h(X - Y) + cyW]} = 0.$$

This is, in fact, a surface with 16 nodes. It would appear that additional nodes correspond to the six common nodes that the 4 quadric surfaces having in common 1, 2, 3, 4, 5, or 6 points, the corresponding symmetroid would have 11, 12, 13, 14, 15, or 16 nodes. But I reserve this for future consideration.

105. To find the equation to the quadric surface through the three lines $(\alpha_1, \beta_1, \gamma_1, f_1, g_1, h_1)$, $(\alpha_2, \beta_2, \gamma_2, f_2, g_2, h_2)$, $(\alpha_3, \beta_3, \gamma_3, f_3, g_3, h_3)$, $(\alpha_4, \beta_4, \gamma_4, f_4, g_4, h_4)$, $(\alpha, \beta, \gamma, f, g, h)$. Take the equation of a quadric surface through this line will be of the form

$$\begin{vmatrix} x^2 & y^2 & z^2 & w^2 & yz & zx & xy & yw & zw & xw \\ \alpha^2 & \beta^2 & \gamma^2 & f^2 & \gamma\beta & \gamma\alpha & \alpha\beta & h\gamma & f\beta + g\alpha & f\gamma \\ \alpha_1^2 & \beta_1^2 & \gamma_1^2 & f_1^2 & \gamma_1\beta_1 & \gamma_1\alpha_1 & \alpha_1\beta_1 & h_1\gamma_1 & h_1\alpha_1 & h_1\beta_1 \\ 2\alpha_1\alpha & 2\beta_1\beta & 2\gamma_1\gamma & 2ff_1 & f\gamma_1 + f_1\gamma & f\alpha_1 + f_1\alpha & \frac{1}{2}h(g_1 + gh_1) & f_1h + fh_1 & f_1g + fg_1 & \frac{1}{2}h(g + \beta h_1) \\ \alpha_2^2 & \beta_2^2 & \gamma_2^2 & f_2^2 & \gamma_2\beta_2 & \gamma_2\alpha_2 & \alpha_2\beta_2 & h_2\gamma_2 & h_2\alpha_2 & h_2\beta_2 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{vmatrix} = 0$$

and we form thus a determinant with three of its lines relating to the line 1, three of them to the line 2, and three to the line 3, and the equation of the quadric surface through the lines 1, 2, 3; and it is clear that the determinant of the order $9 \cdot 9$ contains, as a factor, the principal counter-system of the second plane, and the corresponding curves the principal counter-system of the first plane.

106. Taking $(\xi, \eta, \zeta, \omega)$ as plane-coordinates, two quadric surfaces

$$\begin{aligned} & (a, b, c, d, f, g, h, l, m, n) \times (\xi, \eta, \zeta, \omega)^2 = 0, \\ & (a_1, b_1, c_1, d_1, f_1, g_1, h_1, l_1, m_1, n_1) \times (\xi, \eta, \zeta, \omega)^2 = 0 \end{aligned}$$

are said to be intervorts (or interverse) one of the other, when we have between the coefficients the relation

$$(a, b, c, d, f, g, h, l, m, n) \times (a_1, B_1, C_1, F_1, G_1, H_1, L_1, M_1, N_1) = 0,$$

that is,

$$aA + \dots + 2fF + \dots = 0.$$

The condition that the two surfaces may be intervorts of each other is linear in regard to the coefficients of each of them; hence, taking a before explained location, intervorts to two given quadrics A quadric, we have 9 quadrics; intervorts to 5 given quadrics is a given quadric surface, we have between the coefficients $10 - k$ quadrics, that is, $5 - k$ moreover, and the question is easily supplied.

If the quadric $(a, \dots) \times (\xi, \eta, \zeta, \omega)^2 = 0$ be a plane-pair, the plane-coordinates of the planes are by the equations

$$\begin{aligned} (\partial_a + ay + az + pw) \partial_\xi + p_1 \partial_\eta &= 0, \\ hx + \dots, \quad bx + \dots, \quad fx + \dots, \quad mx + \dots &= 0, \\ gx + \dots, \quad fx + \dots, \quad nx + \dots &= 0, \\ lx + \dots, \quad mx + \dots, \quad nx + \dots, \quad dx + \dots &= 0; \end{aligned}$$

viz., the locus is (by the foregoing definition) the symmetroid. And the determinant and the symmetroid corresponds to the cone-vertex on the Jacobian.

107. Taking the lines to be those of $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, b_2, c_2, f_2, g_2, h_2) \dots (a_5, b_5, c_5, f_5, g_5, h_5)$ to be those of three quadric surfaces having no common conditions: But if the conditions $(\xi, \eta, \zeta, \omega)$ belonging to a point, the value is $= 0$.

108. Any 6 points whatever may be regarded as points on a skew cubic; and the coordinates (x, y, z, w) may then be taken to be of the form

$$\begin{vmatrix} x & y & z \\ y & z & w \end{vmatrix} = 0. \quad \text{This being so, the coordinates of the 6 given points may be taken to be } (1, t_1, t_1^2, t_1^3), (1, t_2, t_2^2, t_2^3),$$

the 6 points; that is, of the surface containing the conditions $(1, t_3, t_3^2, t_3^3), (1, t_4, t_4^2, t_4^3)$; and the equation of the Jacobian surface of

109. Representing as before each line by means of its six coordinates, let $(\alpha, \beta, \gamma, f, g, h)$ be the coordinates of the vertex, the coordinates (X, Y, Z, W) be the coordinates of any one of the lines, the equation (a, b, c, f, g, h) the coordinates of any one of the six lines, the equation of the plane through this line and the vertex for shortness

$$\alpha(zW - wY) + b(yW - wY) + c(xW - wX) = 0;$$

or, what is the same thing, putting for shortness

$$\begin{aligned} P &= \cdot - hy - gz + aw, \\ Q &= -hx \cdot + fz + bw, \\ R &= gx - fy \cdot + cw, \\ S &= -ax - by - cz \cdot, \end{aligned}$$

the equation is

$$PX + QY + RZ + SW = 0.$$

The plane in question is a tangent plane to the cone touched by each of the 6 lines. Now when the plane is a tangent plane to the cone touched by each of the 6 lines, the plane in question is a tangent plane to the cone touched by each of the 6 lines. Hence taking the plane $PX + QY + RZ + SW = 0$ on the cone touched by the 6 lines, the equation

$$PX + QY + RZ + SW = 0.$$

To ascertain the form of this, write for a moment $y = 0, x = 0$; the equation is

$$[(aw + bz + bw, \quad yz + ew)^2] = 0,$$

or attending only to the highest and lowest powers of w , this is

$$w^2[(\alpha, b, c)^2] + \dots + w^4[(a, -h, g)^2] \dots$$

and it is thence easy to infer that the whole equation divides by w^2 , so that, omitting this factor, the form of the equation is

$$[(a, b, c \times g, y, z, w)^2 \dots] = 0;$$

viz., this equation is satisfied if z denote any one of the quantities $(\alpha_1, \beta_1, \gamma_1, f_1, g_1, h_1)$, &c., that is, if any one of the 6 lines pass through the assumed point in the direction of any vertex of the cone, and if so, the equation of the cone, regarded as a function of w , omitting this factor, the form of the equation is

$$((a, b, c, f, g, h \times x, y, z, w)^2 = 0,$$

viz., the equation is satisfied if z denote any of the quantities $\alpha_1, b_1, c_1, f_1, g_1, h_1$; and if so, the equation of the cone, regarded as a function of x, y, z, w , is

$$\begin{vmatrix} x, & y, & z \\ a, & b, & c \end{vmatrix} = 0.$$

Locus of the vertex of a Quadric Cone which touches each of Six given Lines.

110. Representing as before each line by means of its six coordinates, let $(\alpha, \beta, \gamma, f, g, h)$ be the coordinates of the vertex, and (X, Y, Z, W) the current coordinates of any one of the lines, then the equation of the plane through this line and the vertex is

$$\alpha(zW - wZ) + b(yW - wY) + c(xW - wX) + f(yZ - zY) + g(zX - xZ) + h(xY - yX) = 0;$$

or, in the same notation as before, for shortness

$$\begin{aligned} P &= \cdot - hy - gz + aw, \\ Q &= -hx \cdot + fz + bw, \\ R &= gx - fy \cdot + cw, \\ S &= -ax - by - cz \cdot, \end{aligned}$$

the equation is

$$PX + QY + RZ + SW = 0.$$

The plane in question is a tangent plane to the cone touched by each of the 6 lines. Hence taking the plane $PX + QY + RZ + SW = 0$ on the cone touched by the 6 lines, attending only to the highest and lowest powers of w , this is

$$w^2[(a, b, c)^2] + \dots + w^4[(a, -h, g)^2] \dots,$$

and it is thence easy to infer that the whole equation divides by w^2 , so that, omitting this factor, the form of the equation is

$$((a, b, c, f, g, h) \times (x, y, z, w))^2 = 0,$$

viz., the equation in question is of the order 8 in the coordinates (x, y, z, w) and of the degree 2 in the coordinates of each of the lines. It would not be very

111. difficult to actually develop the equation; in fact, starting from the term $w^2[(a, b, c)^2]$ the other terms are obtained therefrom by changing a, b, c into $a + \frac{1}{w}(hy - gz)$, $b + \frac{1}{w}(-hx + fz)$, $c + \frac{1}{w}(gx - fy)$ respectively; the equation may therefore be written in the symbolic form

$$w^2 \cdot \exp \frac{1}{w}[(hy - gz) \partial_a + (-hx + fz) \partial_b + (gx - fy) \partial_c] \cdot [(a, b, c)^2] = 0,$$

or, what is the same thing,

$$w^2 \cdot \exp \frac{1}{w}[x(g \partial_b - h \partial_c) + y(h \partial_a - f \partial_c) + z(f \partial_b - g \partial_a)] \cdot [(a, b, c)^2] = 0,$$

where $\exp \theta$ (read exponential) denotes e^θ , and $[(a, b, c)^2]$ represents a determinant as above explained.

112. I am not sure whether this surface of the eighth order has been anywhere considered.

[446] ON THE MECHANICAL DESCRIPTION OF A NODAL BICIRCULAR QUARTIC.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 100–106.]

The ingenious method, devised by Mr S. Roberts (*Proceedings*, vol. II. p. 133) for the description of a nodal bicircular quartic suggests a further investigation. We have a quadrilateral $OAA'O'$, in which the adjacent sides OA, AA' are equal to each other, and the other two adjacent sides $OO', O'A'$ are also equal to each other; O, O' are fixed points; and we have thus a link AA' , the extremities of which are connected by means of the radii $OA, O'A'$ respectively, and consequently describe circles about the centres O, O' respectively, the radius OA of the one circle being equal to the length AA' of the link, and the radius $O'A'$ of the other circle being equal to the distance OO' of the centres. The theorem is, that any point C , rigidly connected with the link AA' , describes a nodal bicircular quartic, that is, a quartic curve with three nodes (or unicursal quartic), two of the nodes being the circular points at infinity. Any such curve is the inverse of a conic, and it is also the antipode of a conic; viz., if at such point of the curve we draw a line at right angles to the radius vector from the node, then envelope of the line passing through this point is a circle. If the equalities $OA = AA', O'A' = OO'$ did not subsist, then to a given position of the link AA' corresponds two positions of A, A' respectively, on the circles; the radius OA of the one circle being equal to the length AA' of the link, and the radius $O'A'$ of the other circle being equal to the distance OO' . The theorem is, that any point C , rigidly connected with the link AA' , describes a nodal bicircular quartic, that is, a quartic curve with three nodes (or unicursal quartic), two of the nodes being the circular points at infinity.

I have a curve AA' : the link; OO' may be called the bar; and the locus of the radius AA' produced; moreover AA' may be called the link; the radius OA of the one circle being equal to

the length AA' , and the radius $O'A'$ of the other circle being equal to the distance OO' of the centres of the two circles respectively. Suppose the link AA' to slide along the bar OO' ; the link AA' remains parallel to itself in this its motion; viz., on this slide-motion the bar OO' moves about its centre, and the link AA' traverses the constant triangle; and, producing OA , $O'A'$ to meet in K , the locus of K is called the constant triangle; and the locus of C will be a higher order than is in the actual problem.

I have called AA' the link; OO' may be called the bar. OA is then the link-radius, $O'A'$ the bar-radius; moreover AA' may be called the constant triangle; and, producing OA , $O'A'$ to meet in K , then AKA' is the constant triangle. Since at any instant the motion of A is normal to KA , and the motion of A' to KA' , it is clear that the motion of any point on KA or on KA' is also normal to KA , KA' respectively; and in like manner the motion of A' , in its position on the line KA' , is in like manner the motion of any point on the line KA' . We have thus an instantaneous rotation about the point K .

Imagine any two positions of the link; say these are $A_1A'_1$, and $A_2A'_2$. Join A_1A_2 and at its mid-point draw a perpendicular thereto; join in like manner $A'_1A'_2$, and at its mid-point draw a perpendicular thereto; and let these two perpendiculars meet in T ; we have the two equal triangles $TA_1 = TA_2$, $TA'_1 = TA'_2$, and the angles $A_1TA'_1 = A_2TA'_2$, so that there is a rotation about the centre T , which conducts $A_1A'_1$ into $A_2A'_2$. Now, in the limit, when the two positions $A_1A'_1$, $A_2A'_2$ are made to coincide with the triangles $A_1A'_1T$ and $A_2A'_2T$ respectively; that is, T will be a node in the locus described by this point, but in the present case, this point K is a fixed point. It is indifferent any point will pass with the link AA' in some such manner, and not indifferent any point of the locus C , of which the same considerations may be applied to itself; it is, for any point of the locus, when the link describes the ordinary points of curvature. And in the linkage describing the bicircular quartic, the centres are at the same distance, a from the centre O of the linkage. The quartic, when described by the two points A_1A_2 describes a circle of the same diameter centred at O of the linkage and centres at the same distance from the centre O of the linkage; the quartic curve, when described by the two points A_1A_2 describes a circle of the same diameter a from the centre O of the linkage and centres at the same distance from O .

Passing now to the analytical investigation, I take the origin at O , the axis of x along the direction from O to O' , and the axis of y , at right angles thereto, upwards from O . The inclinations of OA , AA' , $O'A'$ to the axis Ox are taken to be θ , ϕ , θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and the angles θ , θ' are connected by the relation $\sin \theta' / \sin \theta = a/a'$, and finally $AB = b$, $BC = c$.

Observing that the angle $AA'O'$ is $= \theta$, we have $\theta' = \pi - \theta - \phi$, or $\theta - \phi$ is the deviation from O , and the axis of y , at right angles, upwards from O . The inclinations of OA , AA' , $O'A'$ to the axis Ox are taken to be θ , ϕ , θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and finally $AB = b$, $BC = c$.

Hence, if we go on to assume

$$m = 1 : 1 + m : 1 - m = a' : a : a' + a : 2a' : 2a;$$

and finally $AB = b$, $BC = c$.

Observing that the angle $AA'O'$ is $= \theta$, we have $\theta' = \pi - \theta - \phi$, upwards from O . The inclinations of OA , AA' , $O'A'$ to the axis Ox are θ , ϕ , θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and $OO' = a$, $OO' = a'$; and finally $AB = b$, $BC = c$.

(Algebraic developments.) For the locus of C we have

$$x = a \cos \theta + b \cos \phi + c \sin \phi,$$

$$y = a \sin \theta + b \sin \phi + c \cos \phi,$$

or, instead of θ, ϕ introducing u , we have

$$x = -a \frac{u^2 - 1}{u^2 + 1} + b \frac{u^2 - m^2}{u^2 + m^2} - c \frac{2mu}{u^2 + m^2},$$

$$y = a \frac{2u}{u^2 + 1} + b \frac{2mu}{u^2 + m^2} + c \frac{u^2 - m^2}{u^2 + m^2}.$$

[Further reductions and the curve obtained as an inverse-of-a-conic and antipode-of-a-conic are given in the remainder of the paper.]

(Closing remarks.) There remains to be observed, further, that the equation of the line in question obtained by writing in the foregoing equations, $u = u'$ and eliminating the remaining quantities δx , δy , &c., and the corresponding expressions δx_i , δy_i , &c.; the result of the elimination will be the same as if, writing $u = u_1$, we have in fact only to consider the equation of the line in the form

$$ax' + by' + c = 0 \quad (\text{in fact, for } A, B \text{ or of } (x, y).)$$

The remainder of the paper carries out this elimination and shows that the locus is in fact a circular quartic with the prescribed nodal properties.

[447] ON THE RATIONAL TRANSFORMATION BETWEEN TWO SPACES.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 127–180.
Account of the Paper given at the Meeting 11 March 1869.]

Two figures are rationally transformable each into the other (or, say, there is a rational transformation between the two figures) when to a variable point of each of them there corresponds a single variable point of the other. The figures may be either lost in a space, or loci of any number of dimensions; or they may be lost in plane spaces themselves. Thus the figures may be each a line of one space (or space of one dimension); and there will be a rational transformation between two lines, or they may be each a plane (or space of two dimensions); and there will be a rational transformation between two planes; or, finally, the figures may be each surfaces (or spaces of three dimensions), and there will be a rational transformation between two spaces. The problem of the rational transformation between two planes (or generally between two plane spaces of equal dimensions) is the only one which has been hitherto considered, and it is in fact thereto, that every rational transformation between two planes, and to be more particular still, between two plane spaces, may be reduced. The general problem in three dimensions is here treated of; the most general transformation between two plane spaces of equal dimensions is, in fact, the transformation (which is already mentioned) in the only one peculiar transformation between two plane spaces; and consequently a rational transformation between two planes; viz., in regard hereto I examine the general theory, but attend mainly to what I call the lineo-linear transformation; viz., it is assumed that the coordinates

The General Principle of the Rational Transformation between Two Spaces.

1. In all that follows, the two spaces (lines, planes, or three dimensional spaces, as the case may be), to any rational corresponding loci in the two spaces respectively, are referred to as the first and second figures respectively; viz., between two lines, between two planes, or between two three-dimensional spaces, as the case may be. The notation of the first and second figures and the corresponding loci, when considered, not as superimposed or situate in a common space, but as existing, each independently of the other, but as together mentioned in the only one peculiar way, will be referred to as the first and the second figures and the corresponding loci. Writing as before

$$x : y : z = X : Y : Z,$$

the unicursal correspondence (which is, in fact, the only one between two lines) unicursal correspondence; viz., if as before, we refer to the lines, planes, or three-dimensional spaces, the correspondence between them is essentially unsymmetrical.

2. If, to fix the ideas, we attend to the case of two planes, or take the sets

$$x : y : z = X : Y : Z, \quad x : y : z = X' : Y' : Z',$$

[The opening articles, §§ 2–35 on PDF pages 212–225, develop the general theory of the rational transformation between two spaces, in particular establishing the basic formulæ of the lineo-linear (or, as Cayley calls it, the homographic) transformation between two planes; the construction of the constants of the transformation; the conditions under which a system of equations of the rational correspondence is satisfied; the count of free constants; the various special and degenerate cases (when the transformation breaks up into a product of a linear and a rational correspondence, or when certain equations of correspondence become identically satisfied); the use of these formulæ for the calculation of the number of intersections of a curve in the first plane with its transform in the second plane; the count of free parameters in the transformation; and, finally, the discussion of the system of *principal curves* and *principal counter-curves* (or *réseaux* of Cremona) which arise in the transformation. The detailed algebraic developments occupy pages 189–203 of the book, and the memoir is continued from book page 204 onward in the next chunk of the typesetting.]

(End of chunk: PDF pages 176–225, book pages 154–203.

Paper 445 §§ 56–112 (tail of the Memoir on Quartic Surfaces),

paper 446 in full,

paper 447 §§ 1–35 (opening of the Rational Transformation memoir).)

[450] NOTE ON THE THEORY OF THE RATIONAL TRANSFORMATION BETWEEN TWO PLANES, AND ON SPECIAL SYSTEMS OF POINTS. (*CONCLUDED*)

(*Continuation from the previous chunk; closing pages of paper 450.*)

I believe the better course is to assume (1) and (3) as the fundamental equations, from them deducing (2); and we thus also get over a difficulty presently referred to, but which did not occur to me when the memoir was written.

In fact, starting with the equations

$$x' : y' : z' = \frac{X}{x} : \frac{Y}{y} : \frac{Z}{z}$$

(which are to give $x : y : z = X' : Y' : Z'$), we have in the first instance the equation (1). Moreover, establishing for x', y', z' a linear equation $ax' + by' + cz' = 0$, we have corresponding hereto a curve $aX + bY + cZ = 0$, and the coordinates x, y, z of a point on this curve are proportional to $X' : Y' : Z'$; that is, substituting for z' the value $-\frac{1}{c}(ax' + by')$, they are proportional to rational and integral (homogeneous) functions of (x', y') , that is, to rational and integral functions of the single parameter $x' : y'$; wherefore the curve $aX + bY + cZ = 0$ is unicursal; whence the equation (3). The like change may be made in the theory of the rational transformation between two spaces; and it is in this case a more important one.

The difficulty is as follows: It is not self-evident that we are at liberty to assume

$$a_1 + 3a_2 + 8a_3 + \cdots \leq \frac{1}{2}(n^2 + 3n) - 2;$$

for imagine that we had a system of $(a_1, a_2, a_3, \dots)$ points, such that $a_1 + 4a_2 + \cdots$ being $= n^2 - 1$, and $a_1 + 3a_2 + \cdots$ being $> \frac{1}{2}(n^2 + 3n) - 2$, the points were such that the conditions in question (viz., the condition that the curve passes once through each of the points a_1 , twice through each of the points $a_2, \dots$) should be less than $a_1 + 3a_2 + \cdots$, and in fact $=$ or $< \frac{1}{2}(n^2 + 3n) - 2$; then the functions X, Y, Z would not of necessity be connected by a linear relation $\lambda X + \mu Y + \nu Z = 0$, and the ground for the assumption in question, $a_1 + 3a_2 + \cdots < \frac{1}{2}(n^2 + 3n) - 2$, would no longer exist. And except by the process now adopted of deriving the equation (2) from the equations (1) and (3), I do not know how the impossibility of such a system is to be established; viz., I do not know how we are to prove the following theorem: *There is not any system of $(a_1, a_2, a_3, \dots)$ points, where*

$$\begin{aligned} a_1 + 4a_2 + 9a_3 + \cdots &= n^2 - 1, \\ a_1 + 3a_2 + 6a_3 + \cdots &> \frac{1}{2}(n^2 + 3n) - 2, \end{aligned}$$

such that (for a curve of the order n passing once through each point a_1 , twice through each point $a_2, \dots$) the number of conditions actually imposed on the curve is $=$ or $< \frac{1}{2}(n^2 + 3n) - 2$.

A system of $(a_1, a_2, \dots)$ points such that the number of actually imposed conditions is less than $a_1 + 3a_2 + \cdots$, may be termed a *special* system; we have, of course, the well-known case ($a_1 = n^2$) of a system of n^2 points, such that any curve of the order n passing through $\frac{1}{2}(n^2 + 3n) - 1$ of these passes through all the remaining points {or what is the same thing, where the number of conditions actually imposed is $= \frac{1}{2}(n^2 + 3n) - 1$ }; and we have the following special system, which presented itself to Dr Clebsch, in his researches on the *Abbildung* of a quintic surface with two non-intersecting nodal lines; viz., “we may have $a_1 = 12, a_2 = 2$. We may have 12 points and 2 points such that, for a quintic curve passing once through each of the 12 points and twice through each of the 2 points, the number of conditions actually imposed (instead of being $12 + 3 \cdot 2 = 18$) is $= 17$.” The construction is as follows: viz., starting with the 2 points and any 10 points, we may draw a quartic passing twice through the first of the 2 points, once through the second of them, and through the 10 points; and another quartic passing

twice through the second of the 2 points, once through the first of them, and through the 10 points: the two quartics intersect in the 2 points each twice, in the 10 points, and in 2 new points, forming, with the 10 points, a system of 12 points; and the first-mentioned 2 points and the 12 points form the system in question.

A more complicated case, $a_1 = 10$, $a_2 = 6$, $a_3 = 1$, occurs in Dr Nöther's paper, "Ueber Flächen, welche Schaaren rationaler Curven besitzen," [*Math. Ann.*, t. iii. (1871), pp. 161–227]. Except these two, I do not know any other case of a special system for which $a_2, a_3, \dots$ are not all = 0; the investigation of such systems would, I think, be very interesting.

[A concluding paragraph of seven lines gave some corrections to the "Memoir on the Rational Transformation between Two Spaces," 447, which corrections are made in the present reprint of that paper.]

[451] A SECOND MEMOIR ON QUARTIC SURFACES.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 198–202.
Read December 8, 1870.]

In my Memoir on Quartic Surfaces, *ante* pp. 19–69, [445], although remarking (see No. 79) that the identification was not completely made out, I tacitly assumed that the symmetroid and the decadianome (each of them a quartic surface with 10 nodes) were in fact identical. There is yet a good deal which I cannot completely explain; but the truth appears to be, that the decadianome includes two cases of coordinate generality, say the *sextic decadianome*, and the *bicubic decadianome* = *symmetroid*: viz., in the first of these the circumscribed cone, having for vertex any one of the 10 nodes, is a proper sextic cone with 9 double lines; in the second it is a system of two cubic cones, intersecting, of course, in 9 lines, which are double lines of the aggregate sextic cone: or, in the notation of the Table No. 11, in the case of the sextic decadianome, the circumscribed cones are each of them 6₉; in that of the bicubic decadianome = symmetroid, they are each of them (3, 3). We thus arrive at a very remarkable system of 10 points in space, viz., giving the name "ennead" to any 9 points *in plano*, which are the intersections of two cubic curves, or to any 9 lines through a point which are the intersections of two cubic cones; the 10 points in space are such that, taking any one whatever of them as vertex, and joining it with the remaining points, the 9 lines form an ennead. I purpose in the present short Memoir to consider the theories in question; the paragraphs are numbered consecutively with those of the Memoir on Quartic Surfaces.

Plane Sextic Curve with 9 Nodes.

110. A sextic curve contains 27 constants; and the number of conditions to be satisfied in order that a given point may be a node is = 3. Hence it would at first sight appear that the curve could be found so as to have 9 given nodes; this would be $9 \times 3 = 27$ conditions, or the curve would be completely determinate. But observe that through the 9 given points we have a determinate cubic curve $U = 0$; we have therefore $U^2 =$ a sextic curve, and the only sextic curve with the 9 given nodes; that is, there is not in a proper sense any sextic curve with the 9 given nodes. The number of given nodes is thus = 8 at most.

111. The sextic curve with 8 given nodes should contain $27 - 3 \cdot 8 = 3$ constants. We may through the 8 given points draw the two cubics $P = 0$, $Q = 0$; and we have then $(a, b, c \mid P, Q)^2 = 0$, a bicubic, or improper sextic curve having the 8 nodes, and also a ninth node, viz.,

the remaining point of intersection of the two cubic curves, or say the remaining point of the ennead. Hence if $V = 0$ be any particular sextic curve having the 8 given nodes, we have

$$(a, b, c \mid P, Q)^2 + \theta V = 0,$$

a proper sextic curve having the 8 given nodes; and this, as containing the right number ($= 3$) of constants, will be the general sextic curve having the 8 given nodes.

112. There will be a ninth node if $\theta = 0$; viz., the curve is then $(a, b, c \mid P, Q)^2 = 0$, a bicubic, or improper sextic curve, having for nodes the 9 points of the ennead. Observe that the ninth node is here a point completely and uniquely determined by means of the given 8 nodes. Moreover the number of constants is $= 2$, so that we have here a general (improper) solution of the question of finding a sextic curve with 9 nodes, 8 of them given.

113. But if θ is not $= 0$, then the ninth node must be a point on the curve $J(P, Q, V) = 0$; viz., this is a curve of the order 9, determined by means of the given 8 points; say it is the “dianodal curve” of these 8 points, and, as is easy to see, it has each of these 8 points for a node. The ninth node of the sextic may be any point whatever on the dianodal curve; and regarding it as a given point, the sextic will still contain 1 constant; that is, we have the general solution of the problem of finding a sextic curve with 9 nodes, 8 of them given, and the 9th a given point on the dianodal curve.

114. So long as the 8 points are arbitrary, the dianodal curve does not pass through the 9th point of the ennead, and the two cases above considered are mutually exclusive. It will be noticed how closely analogous this theory of the plane sextic with 9 nodes, is to that of the quartic surface with 8 nodes.

115. Of course, instead of the plane sextic curve, we may have a sextic cone; such a cone has at most 8 given double lines; and if there be a 9th double line, then there are the two cases of coordinate generality; viz., (1), the new double line is the ninth line of the ennead, the cone being in this case not a proper sextic cone, but a bicubic cone; (2), the new double line may be any line whatever on the dianodal cone, (cone of the order 9 determined by the 8 given lines, and having each of these for a double line,) and regarding it as a given line on the dianodal cone, the sextic cone contains 1 constant.

Each circumscribed cone of the Symmetroid is (3, 3).

116. Using (x, y, z, w) as current coordinates of a point of the symmetroid, I take S, T, U, V to be quadric functions of the coordinates $(\alpha, \beta, \gamma, \delta)$; the equation of the symmetroid is therefore given by

$$xS + yT + zU + wV = \text{cone},$$

and the nodes thereof are determined by

$$xS + yT + zU + wV = \text{plane-pair}.$$

Suppose that a node is $(x = 0, y = 0, z = 0)$; the condition for this is $V = \text{plane-pair}$, and we may without loss of generality write $V = \gamma^2 + \delta^2$. Hence, putting for shortness $\Omega = xS + yT + zU$, that is a quadric function

$$(a, b, c, d, f, g, h, l, m, n \mid \alpha, \beta, \gamma, \delta)^2,$$

wherein the coefficients $a, b, \dots$ are arbitrary linear functions of (x, y, z) , but not containing w , the equation of the symmetroid is given by

$$\Omega + w(\gamma^2 + \delta^2) = \text{cone}.$$

117. It follows that the equation is

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c+w, & n \\ l, & m, & n, & d+w \end{vmatrix} = 0;$$

viz., this is

$$V + w(\partial_c + \partial_d)V + w^2\partial_c\partial_dV = 0,$$

where V denotes the foregoing determinant, writing therein $w = 0$. Or, observing that V contains c and d each only linearly, the equation may be written

$$V + w(\partial_cV + \partial_dV) + w^2\partial_c\partial_dV = 0,$$

which is a quartic surface having, as it should have, the point $(0,0,0)$ for one of its ten nodes.

118. The equation of the circumscribed cone is

$$(\partial_cV - \partial_dV)^2 + 4(\partial_cV \cdot \partial_dV - V \cdot \partial_c\partial_dV) = 0,$$

or, what is the same thing, it is

$$(\partial_cV - \partial_dV)^2 + 4(\partial_cV \cdot \partial_dV - V \cdot \partial_c\partial_dV) = 0.$$

But we have identically

$$\partial_cV \cdot \partial_dV - \left(\frac{1}{2}\partial_{cd}V\right)^2 = V \cdot \partial_c\partial_dV;$$

so that the equation is

$$(\partial_cV - \partial_dV)^2 + (\partial_{cd}V)^2 = 0;$$

a sextic cone breaking up into the two cubic cones

$$\partial_cV - \partial_dV \pm i\partial_{cd}V = 0,$$

so that the cone is $(3,3)$. And since clearly the point $(0,0,0)$ may be regarded as representing any one whatever of the 10 nodes, it follows that for any node whatever of the symmetroid, the circumscribed cone is $(3,3)$, so that, as stated above, bicubic decadianome = symmetroid.

Deductions from the foregoing theory.

119. Referring to No. 85 of the original memoir, it appears that, with 6 given points as nodes, we can actually find for the symmetroid an equation containing 6 constants. I cannot discover any ground for doubting that 3 of these may be determined so as to give to the symmetroid a seventh given node; and I therefore assume that with 7 given points as nodes, an equation can be found with 3 constants. The symmetroid is certainly not octadic, hence the eighth node must lie on the dianodal surface of the 7 given points. I can discover no ground for doubting but that two of the constants may be determined so that the eighth node shall be any given point whatever on the dianodal surface of the 7 points; and (this being so) that further the remaining constant may be determined so that the ninth node shall be any given point whatever on the dianodal curve of the 8 points. But if all this be so, the consequence is very remarkable; the tenth node is not any one whatever of the 22 dianodal centres of the 9 points, but it is a uniquely determinate “enneadic centre,” viz., we must have the following theorem:

120. “Take any 7 points; an eighth point at pleasure on the dianodal surface of the 7 points; a ninth point at pleasure on the dianodal curve of the 8 points. In the system of 9 points so determined, take any one as vertex, and joining it with the remaining 8, construct the ninth line of the ennead. Performing this construction with each of the 9 points successively as vertex, we obtain 9 lines passing through the 9 points respectively. These 9 lines meet in a point which is the ‘enneadic centre’ of the 9 points: and further, the 10 points form a completely symmetrical system, so that each one of them is the enneadic centre of the remaining 9.”

121. Assuming that the 9 lines do intersect so as to give rise to an enneadic centre, there is no difficulty in conceiving that the loci, which by their intersection determine the dianodal centres, do each of them pass through the enneadic centre; so that this enneadic centre counts once or more among the dianodal centres, and the number of proper dianodal centres, instead of being $= 22$, will be suppose $= 22 - \omega$; and if, further, the 9 points, together with the enneadic centre, are the nodes of a symmetroid, but the 9 points together with any one of the $22 - \omega$ dianodal centres are the nodes of a sextic decadianome, then we must also have as follows:

122. “Considering any 9 points as above; taking any one as vertex, and joining it with the remaining 8, these 8 lines determine a dianodal ninthic cone. We have thus 9 dianodal cones, which cones pass all of them through the same $22 - \omega$ points.”

123. I am not able to verify these theorems *à posteriori*. It appears to me that the theorem in regard to the enneadic centre subsists for a system of 9 points such as referred to in the statement; but that if by possibility the statement be too general, the theorem must, at all events, subsist for a more special system of 9 points; and that there certainly exist systems of 10 points, such that each 9 of the points have as an enneadic centre the tenth point. (*I have since ascertained that if a quartic surface with 10 nodes has a single node (3,3), the surface is a symmetroid; whence, by what precedes, the remaining nine nodes are each of them (3,3). Added 25 March, 1871.*)

124. I notice, as a subject of investigation, the following system of correspondence; viz., given any 8 points in space: then to every point in space corresponds a line through this point, viz., the ninth line of the ennead obtained by joining the point with the 8 given points respectively; and to each line in space a point or points on the line, viz., the point or points for each of which the line is the ninth line of the ennead obtained by joining the point with the eight given points respectively.

[452] ON AN ANALYTICAL THEOREM FROM A NEW POINT OF VIEW.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 220, 221.
Read February 9, 1871.]

The theorem is a well-known one, derived from the equation

$$(az^2 + 2bz + c)w^2 + 2(a'z^2 + 2b'z + c')w + a''z^2 + 2b''z + c'' = 0;$$

viz., considering this equation as establishing a relation between the variables z and w , and writing it in the forms

$$0 = u = Aw^2 + 2Bw + C = A'z^2 + 2B'z + C' = 0,$$

(where, of course, A, B, C are quadric functions of z , and A', B', C' quadric functions of w ,) we have

$$0 = \frac{du}{dw} dw + \frac{du}{dz} dz = (Aw + B) dw + (A'z + B') dz;$$

but in virtue of the equation $u = 0$, we have $Aw + B = \sqrt{B^2 - AC}$, and $A'z + B' = \sqrt{B'^2 - A'C'}$, and the differential equation thus becomes

$$\frac{dw}{\sqrt{B'^2 - A'C'}} + \frac{dz}{\sqrt{B^2 - AC}} = 0,$$

where $B'^2 - A'C'$ and $B^2 - AC$ are quartic functions of w and z respectively. This is, of course, integrable (viz., the integral is the original equation $u = 0$); and it follows, from the theory of elliptic functions, that the two quartic functions must be linearly transformable into each other; viz., they must have the same absolute invariant $I^3 \div J^2$. It is, in fact, easy to verify, not only that this is so, but that the two functions have the same quadrinvariant I , and the same cubinvariant J .

The new point of view is, that we take the coefficients a, b , &c., to be homogeneous functions of (x, y) , their degrees being such that the equation $u = 0$ is a quartic equation $(*|x, y, z, w)^4 = 0$: viz., this equation now represents a quartic surface having a node (conical point) at the point $(x = 0, y = 0, z = 0)$, and also a node at the point $(x = 0, y = 0, w = 0)$, say, these points are O, O' respectively. The equation $B^2 - AC = 0$ gives the circumscribed sextic cone having O for its vertex, and the equation $B'^2 - A'C' = 0$ the circumscribed sextic cone having O' for its vertex; each of these cones has the line OO' ($x = 0, y = 0$) for a nodal line, as appears geometrically, and also by the equations containing z, w respectively in the degree 4. Considering $B^2 - AC$ as a quartic function of z , its quadrinvariant is a function $(x, y)^4$, and its cubinvariant a function $(x, y)^6$; and similarly, considering $B'^2 - A'C'$ as a quartic function of w , its invariants are functions $(x, y)^4$ and $(x, y)^6$. We have thus, between the two cones, a geometrical relation answering to the analytical one of the identity of the invariants; but the nature of this geometrical relation is not obvious; and it presents itself as an interesting subject of investigation.

[453] ON A PROBLEM IN THE CALCULUS OF VARIATIONS.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 221, 222.
Read February 9, 1871.]

The problem is, $z = \frac{1}{2}(3x - y^2)y$, to find y a function of x such that $\int z dx = \max.$ or $\min.$, subject to a given condition $\int y dx = c$ (the limits of each integral being x_0, x_1 , where these quantities are each positive, and $x_1 > x_0$). The ordinary method of solution gives $y^2 = x + \lambda$, where $(x_1 + \lambda)^{3/2} - (x_0 + \lambda)^{3/2} = \frac{3}{2}c$; so long as c is not less than $(x_1 - x_0)^{3/2}$, there is a real value of λ , but for a smaller value of c there is no real value. The difficulty arising in this last case is somewhat illustrated by replacing the original problem by a like problem of ordinary maxima and minima; viz., $x_1, x_2, \dots, x_n$ being given positive values of x , in the order of increasing magnitude; and if, in general, $z_i = \frac{1}{2}(3x_i - y_i^2)y_i$, then the problem is to find y_i a function of x_i , such that $\Sigma z_i = \max.$ or $\min.$, subject to the condition $\Sigma y_i = c$. We have here $y_i^2 = x_i + \lambda$ where λ is to be determined by the condition $\Sigma y_i = c$; the remainder of the investigation turns on the question of the sign $y_i = +\sqrt{x_i + \lambda}$ or $y_i = -\sqrt{x_i + \lambda}$, to be taken for the several values of i respectively.

[454] A THIRD MEMOIR ON QUARTIC SURFACES.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 234–266.
Read April 13, 1871.]

The present Memoir is a continuation of my former researches on Nodal Quartic Surfaces, [445, 451]. The leading idea is, that for a quartic surface with k -nodes, given the nature of the circumscribed, $(k - 1)$ nodal, sextic cone belonging to any one node of the surface (for instance, $k = 10$, that it is a cone $(3, 3)$ composed of two cubic cones), we thereby determine the equation of the quartic surface, and consequently the nature of the remaining $(k - 1)$ nodes thereof. By means of this general theory I complete, in an essential point, the theory of the Symmetroid; viz., I show that a 10-nodal quartic surface having a single node $(3, 3)$ is a Symmetroid; whence, as appears by my second Memoir, [451], each of the remaining nine nodes is also a node $(3, 3)$; and we have the theory of the remarkable system of ten points in space such that, joining any one of them with the remaining nine, the nine lines thus obtained are the intersections of two cubic cones. A large part of the Memoir is devoted to the consideration of the surfaces with 16, 15, 14, and 13 nodes: this is substantially a reproduction of the results obtained by Kummer in the Memoir “Ueber die algebraischen Strahlensysteme, &c.,” already referred to; but the results in question are brought into connexion with the theory of the present Memoir, and they are, by a change of the constants, exhibited in a form of much greater symmetry and elegance. I attach importance also to the square diagrams by means of which I have exhibited, in a compendious form, the relation between the several nodes and circumscribed sextic cones.

The paragraphs are numbered consecutively with those of the first and second Memoirs.

Preliminary Considerations and Classification.

125. I call to mind that if a quartic surface has a node (conical point), then there is for this node a tangent quadricone and a circumscribed sextic cone; viz., if the surface has $(k - 1)$ other nodes, or in all k nodes, then the sextic cone has $(k - 1)$ nodal lines (passing through the other nodes respectively), and we have thus for the different forms of the sextic the table No. 11; viz., this is

CIRCUMSCRIBED SEXTIC CONE.

Nodes of Surface	Nodal Lines of Cone	Forms
1	0	6
2	1	6 ₁
3	2	6 ₂
4	3	6 ₃
5	4	6 ₄
6	5	6 ₅ , 5
7	6	6 ₆ , 5 ₁
8	7	6 ₇ , 5 ₂ , 4, 2
9	8	6 ₈ , 5 ₃ , 4 ₁ , 2, 4, 1, 1
10	9	6 ₉ , 5 ₄ , 4 ₂ , 2, 4 ₁ , 1, 1, 3, 3
11	10	6 ₁₀ , 5 ₅ , ..., 4 ₃ , 1, 1, 3 ₁ , 3
12	11	5 ₆ , ..., 4 ₃ , 1, 1, 3, 2, 1
13	12	..., 3, 2, 1, 3, 1, 1, 1, 2, 2, 2
14	13	..., 3 ₁ , 1, 1, 1, 2, 2, 1, 1
15	14	..., 2, 1, 1, 1, 1
16	15	1, 1, 1, 1, 1, 1

viz., 6 denotes a proper sextic cone without nodal lines; 6_1 a proper sextic cone with one nodal line; $5, 1$ a proper quintic cone and a plane, &c.

We may distinguish the nodes according to the sextic cones; thus, a node 6 means a node for which the circumscribed cone is a proper sextic cone, $(1, 1, 1, 1, 1, 1)$ a node where the circumscribed cone breaks up into six planes, &c.

126. A 16-nodal surface has 16 nodes $(1, 1, 1, 1, 1, 1)$, and a 15-nodal surface has 15 nodes $(2, 1, 1, 1, 1)$; but, for a 14-nodal surface, the question arises how many nodes are $(3_1, 1, 1, 1)$, and how many $(2, 2, 1, 1)$. It was remarked, No. 13, that the only possible cases were 14, 0; 8, 6; or 2, 12; and that we might, in like manner, limit the number of possible cases for other values of k ; but that the inquiry was not then further pursued. I resume this inquiry, but without obtaining as yet a complete answer.

127. It is to be observed that a line joining any two nodes is not, in general, a line on the surface, but that it may be so; the surfaces for which this is so (viz., any surface which contains upon it a line through two nodes) form, however, a class by themselves, which at present I altogether exclude from consideration. This being so, it will appear in the sequel that there is but one kind of surface having a node $(2, 2, 1, 1)$, and but one kind of surface having a node $(3_1, 1, 1, 1)$. Now there is a surface, Kummer's 14-nodal, the nodes of which are $8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$; wherefore the two kinds are identical, and are each of them Kummer's 14-nodal surface. Similarly, for the 13-nodal surfaces, there is but one kind having a node $(4_1, 1, 1)$, but one kind having a node $(3, 1, 1, 1)$, and moreover but one kind having a node $(3_1, 2, 1)$; and we have Kummer's 13-nodal surface with the nodes $3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$; hence the three kinds are identical with each other and with Kummer's. Moreover, there is but one kind having a node $(2, 2, 2)$; hence all the other nodes must be $(2, 2, 2)$, and we have a surface $13(2, 2, 2)$ not given by Kummer. And in like manner for the 12-nodal surfaces, we have the two kinds given by Kummer, and a third kind $12(4_1, 1, 1)$ not given by him; the arrangement thus far being

No. of Nodes.	Character of Surface.
16	$16(1, 1, 1, 1, 1, 1)$,
15	$15(2, 1, 1, 1, 1)$,
14	$8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$,
$13(\alpha)$	$3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$,
(β)	$13(2, 2, 2)$.
$12(\alpha)$	$12(4_1, 2)$,
(β)	$2(5_1, 1) + 6(3_1, 3_1) + 4(3, 2, 1)$,
(γ)	$12(4_1, 1, 1)$.

128. But in the next following case we have Kummer's surface, viz.

$$11(\alpha) \quad 1(6_{10}) + 10(3_1, 3),$$

and I do not know whether one, two, or three kinds of surface having nodes $(4_1, 1, 1)$, $(4_2, 2)$, and $(5_5, 1)$. And in the next case we have (as will appear) the Symmetroid, viz.,

$$10(\alpha) \quad 10(3, 3),$$

and I do not know how many kinds of surfaces having a node or nodes 6_9 , $(5_4, 1)$, $(4_2, 2)$, $(4, 1, 1)$.

It will be observed that the present division has nothing to do with the octadic and disinodal division in the former Memoir.

129. I consider a conic $A = 0$, and any six tangents thereof, $t_1 = 0, t_2 = 0, t_3 = 0, t_4 = 0, t_5 = 0, t_6 = 0$; we have an identical equation which might be written $AC - B^2 = t_1 t_2 t_3 t_4 t_5 t_6$, but it will be convenient, introducing a constant factor K , to write it

$$AC - B^2 = K t_1 t_2 t_3 t_4 t_5 t_6,$$

B being a cubic function and C a quartic function of the coordinates.

Consider now the series of factors, such as

$$\begin{aligned} lA + mt_1 t_2, \\ sA + mt_1 t_2 t_3, \\ UA + mt_1 t_2 t_3 t_4, \\ VA + mt_1 t_2 t_3 t_4 t_5, \\ WA + mt_1 t_2 t_3 t_4 t_5 t_6, \end{aligned}$$

where m is a constant, l a constant, s a linear function, U, V, W functions of the degrees 2, 3, 4 respectively; and compose with one or more such factors an expression involving the term $K t_1 t_2 t_3 t_4 t_5 t_6$; for instance, such an expression is

$$\frac{mm'}{K} (sA + mt_1 t_2 t_3)(l'A + m't_4 t_5 t_6);$$

this is, of the form $A\Omega + K t_1 t_2 t_3 t_4 t_5 t_6$, viz., $A\Omega + (AC - B^2)$, or $A(\Omega + C) - B^2$, say $A\Gamma - B^2$; or what is the same thing, introducing a new coordinate w , we have a quadric function

$$Aw^2 + 2Bw + \Gamma,$$

the discriminant of which, $A\Gamma - B^2$, is equal to the expression in question.

130. In the sequel (x, y, z, w) are considered as the coordinates of a point in space; $A = 0$ is thus a quadric cone, $t_1 = 0, t_2 = 0, \dots, t_6 = 0$ any six tangent planes thereof; and hence $Aw^2 + 2Bw + \Gamma = 0$ is a quartic surface, having the point $(x = 0, y = 0, z = 0)$ for a node, whereof the circumscribed cone $A\Gamma - B^2 = 0$ breaks up in the assumed manner.

Thus, in order that the circumscribed cone may be as above

$$(sA + mt_1 t_2 t_3)(l'A + m't_4 t_5 t_6),$$

we have only to assume

$$\Gamma = C + \frac{mm'}{K} (sl'A + sm't_4 t_5 t_6 + l'mt_1 t_2 t_3),$$

and so in other cases. Observe that $sA + mt_1 t_2 t_3 = 0$ is a cubic cone, which, so long as m, s are arbitrary, has no nodal line; but establishing a single relation (say s remains arbitrary, but a proper value is assigned to m) it will be a cubic cone having a nodal line. And so $UA + mt_1 t_2 t_3 t_4 = 0$ is a quartic cone without any nodal line, but by particularising the constants it may be made to have one, two, or three nodal lines. Such nodal determinations are obviously required in order that the formula may extend to all the before-mentioned forms of the circumscribed cone. The foregoing analysis is the foundation of the whole theory: I have given it, as above, apart from the theory, in order that the nature of it may be the better perceived; but I have now to bring it into connexion with the theory.

On the Sextic Curves, $AC - B^2 = 0$.

131. I revert to the consideration of plane curves. The equation of a sextic curve $(*|x, y, z)^6 = 0$ cannot be in general expressed in the form $AC - B^2 = 0$, where the degrees of A, B, C are 2, 3, 4 respectively; in fact, the existence of such a form implies that there is a conic $A = 0$ touching the sextic 6 times; and since a conic can only be made to satisfy 5 conditions, there is not in general any such conic.

132. Such conic, when it exists, is said to be *inscribed in* the sextic, and the sextic to be *circumscribed about* the conic, or to be an “amphigram;” and then, $A = 0$ being the equation of the conic, that of the sextic is expressible in the form in question $AC - B^2 = 0$. It is clear that $B = 0$ is a cubic curve passing through the 6 points of contact of the conic with the sextic, and that any such curve may be taken for the curve B ; in fact if a particular cubic through the 6 points is $B = 0$, and the equation of the sextic is $AC - B^2 = 0$, then taking p an arbitrary linear function of the coordinates, the equation of the general cubic is $B' = B + pA = 0$; and then writing

$$\begin{aligned} A &= A, \\ B' &= B + pA, \\ C' &= C + 2Bp + Ap^2, \end{aligned}$$

we have $AC' - B'^2 = AC - B^2$; so that the original form $AC - B^2 = 0$ becomes $AC' - B'^2 = 0$. But the cubic $B = 0$ being assumed at pleasure, the quartic $C = 0$ is a determinate curve.

133. It is to be observed that a sextic curve may be an amphigram in more than one way: certainly in two, three, or four, and possibly in a greater number of ways. For the equation of the curve contains 27 constants, and hence determining the sextic so as to touch 4 given conics each of them 6 times, there are still 3 constants; and the curve will be an amphigram in regard to each of the 4 conics; say it is a quadruple amphigram. But in the sequel we are only concerned with a sextic curve considered as an amphigram in regard to a given conic $A = 0$ (no attention being paid to the other inscribed conics, if any); and then, by what precedes, taking $B = 0$ any cubic whatever through the 6 points of contact, we have a determinate quartic curve $C = 0$, and the equation of the sextic curve assumes the form $AC - B^2 = 0$.

134. The curves $A = 0$, $B = 0$, $C = 0$ contain respectively 5, 9, 14 constants; whence considering the function B as containing an arbitrary constant factor, for the curve $AC - B^2 = 0$, the number of constants is *primâ facie* $5 + 9 + 14 + 1 = 29$; but on account of the arbitrary linear function p , the real number is $29 - 3 = 26$: this is right, for a sextic curve contains 27 constants; and the curve being an amphigram, there is one relation between the constants, $27 - 1 = 26$.

135. Suppose now that the sextic curve $AC - B^2 = 0$ breaks up into two or more separate curves, say into the two curves $P = 0$, $Q = 0$ of the orders f , g respectively ($f + g = 6$). We have

$$AC - B^2 = PQ = 0;$$

and the conic $A = 0$ touching the sextic six times, must, it is clear, touch the curves $P = 0$, $Q = 0$, f and g times respectively. And so when the sextic breaks up into any number of curves, each component curve $P = 0$ of the order f must touch the sextic f times.

136. It follows that if the sextic break up into six lines, say $AC - B^2 = t_1 t_2 t_3 t_4 t_5 t_6 = 0$, then that each of the lines $t_1 = 0$, $t_2 = 0$, $\dots$, $t_6 = 0$ is a tangent to the conic. And conversely, starting with the conic $A = 0$ and any six tangents thereof $t_1 = 0$, $t_2 = 0$, $\dots$, $t_6 = 0$, we have an identity of the form in question. In fact, taking any two of the tangents, say $t_1 = 0$ and $t_2 = 0$, then, if $p = 0$ be the equation of the line joining their points of intersection, the equation of the conic will be of the form $t_1 t_2 + p^2 = 0$, that is, we may write $A = t_1 t_2 + p^2$, or what is the same thing, $t_1 t_2 = A - p^2$. (Considering A as a given quadric function of the coordinates, this of course implies that the implicit constant factors of t_1 , t_2 , p are properly determined.) Similarly, $q = 0$ being the line through the points of contact of t_3 , t_4 , and $r = 0$ that through the points of contact of t_5 , t_6 , we have $t_3 t_4 = A - q^2$ and $t_5 t_6 = A - r^2$; whence, to satisfy the equation

$$AC - B^2 = t_1 t_2 t_3 t_4 t_5 t_6,$$

we have only to assume $B = lA + pqr$, l an arbitrary linear function of the coordinates, and the equation then gives

$$C = l^2 A - A(p^2 + q^2 + r^2) + (q^2 r^2 + r^2 p^2 + p^2 q^2) + l^2 A + 2l pqr.$$

137. It will be observed that the grouping of the six tangents into pairs is arbitrary. By altering this grouping, we merely alter the linear function l , but do not obtain any new solution. Thus, say that the new form is $B' = l'A + p'q'r'$, then, by properly determining the linear function l , we can reduce this to the original form $B = lA + pqr$; viz., we can satisfy identically the equation $(l - l')A + pqr - p'q'r' = 0$; or what is the same thing, $\lambda A + pqr - p'q'r' = 0$, where λ is a linear function of the coordinates. We have, in fact, the conic $A = 0$ and the cubic $pqr = 0$ intersecting in the six points of contact; any other cubic through these six points; and consequently the cubic $p'q'r'$ must be expressible in the form $\lambda A + pqr = 0$, and we have thus the identity in question.

138. We have just seen that the value of B is necessarily of the form $B = lA + pqr$, but we are not concerned with its expression in this particular form. What we require in the sequel is a value of B , and thence one of C , satisfying the identity in question, $AC - B^2 = t_1 t_2 t_3 t_4 t_5 t_6$; or what is the same thing, introducing for convenience a constant factor K , the identity

$$AC - B^2 = K t_1 t_2 t_3 t_4 t_5 t_6.$$

139. Instead of C , I write Γ , and consider the sextic amphigram $A\Gamma - B^2 = 0$ touched by the conic $A = 0$ in the points of contact of the conic with the six tangents $t_1 = 0$, $t_2 = 0$, $\dots$, $t_6 = 0$. Suppose the sextic curve breaks up into factors; if one of these factors is a line, it is one of the six tangents, say the tangent $t_6 = 0$. If there is a conic factor, this is a conic touching the conic $A = 0$ at its points of contact with two of the tangents, say the equation is $lA + mt_1 t_2 = 0$. Similarly, if there is a cubic, quartic, or quintic factor, then the equation hereof is $sA + mt_1 t_2 t_3 = 0$, $UA + mt_1 t_2 t_3 t_4 = 0$, or $VA + mt_1 t_2 t_3 t_4 t_5 = 0$. Or going on to the next case of a sextic factor (being of course the whole curve), we may say that this is $WA + mt_1 t_2 t_3 t_4 t_5 t_6 = 0$. (Observe that since $AC - B^2 = K t_1 t_2 t_3 t_4 t_5 t_6$, this means only that the equation of the sextic amphigram is of the assumed form $A\Gamma - B^2 = 0$.)

140. By what precedes we can, for a sextic amphigram which breaks up in any assigned manner, determine the value of Γ . For instance, let the amphigram break up into two cubic curves; say these are $sA + mt_1 t_2 t_3 = 0$, $s'A + m't_4 t_5 t_6 = 0$. Assume

$$A\Gamma - B^2 = \frac{mm'}{K} (sA + mt_1 t_2 t_3)(s'A + m't_4 t_5 t_6),$$

then this equation is

$$A\Gamma - B^2 = \frac{mm'}{K} [ss'A^2 + (sm't_4 t_5 t_6 + s'mt_1 t_2 t_3)A] + AC - B^2;$$

that is, we have

$$\Gamma = C + \frac{mm'}{K} (ss'A + sm't_4 t_5 t_6 + s'mt_1 t_2 t_3),$$

and so in any other case.

I have already adverted to the question of the “nodal determination” of the formulae, and it might be properly here considered; viz., the question is as to the determination of the constants in such manner that, for instance, $sA + mt_1 t_2 t_3 = 0$ may be a nodal cubic, $UA + mt_1 t_2 t_3 t_4 = 0$ a nodal, binodal, or trinodal quartic, &c.; but I defer it for the moment in order first to apply the theory to the quartic surfaces.

Application to Quartic Surfaces.

141. If a quartic surface has a node or nodes, we may take for a node the point $x = 0, y = 0, z = 0$; the equation of the quartic surface is then of the form

$$Aw^2 + 2Bw + \Gamma = 0,$$

where A, B, Γ are functions of x, y, z of the degrees 2, 3, 4 respectively. $A = 0$ is the tangent quadricone at the node in question; and the circumscribed cone is $A\Gamma - B^2 = 0$. By what precedes, this is an amphigram touching the quadricone along six generating lines thereof; say the tangent planes of the cone $A = 0$ along these six lines respectively are $t_1 = 0, t_2 = 0, \dots, t_6 = 0$. We have then an identical equation

$$AC - B^2 = Kt_1t_2t_3t_4t_5t_6,$$

viz., regarding for a moment this equation as an equation for the determination of B , and B' as any particular solution thereof, then its general solution is $B = B' + tA$, where t is an arbitrary linear function of (x, y, z) , and the B in the equation of the surface is properly $= B' + tA$. But by the substituting $w - t$ in place of w , the B of the equation of the surface would then be made $= B'$; and it thus appears that we may, without loss of generality, take the B of this equation to be any particular value B satisfying the identity in question; and then, B having such particular value, C is a quartic function of (x, y, z) completely determined by the same identity. And we then, by what precedes, at once determine Γ so that the circumscribed cone $A\Gamma - B^2 = 0$ may be a cone breaking up in any assigned manner; for instance, if it be a cone $(3, 3)$, then, as just mentioned, the two cubic cones are $sA + mt_1t_2t_3 = 0, s'A + m't_4t_5t_6 = 0$; and Γ has the value

$$\Gamma = C + \frac{mm'}{K}(ss'A + l's't_4t_5t_6 + ls't_4t_5t_6),$$

above obtained.

On the Nodal Determination.

142. I am not able to discuss with much completeness the question of nodal determination. We have to consider a cubic curve $sA + mt_1t_2t_3 = 0$, a quartic curve $UA + mt_1t_2t_3t_4 = 0$, &c., as the case may be, and to determine the constants so that this shall have a node or nodes. Consider for a moment the form $PA + t_1\Omega = 0$, where Ω denotes the product $mt_2t_3 \dots$ of all or any of the tangents $t_2, \dots, t_n$; the orders of $PA, t_1\Omega$ are of course equal, that is, the order of P is less by unity than that of Ω . I say that, by establishing a single relation between the constants, this may be made to have a node at the point of contact $A = 0, t_1 = 0$. In fact, writing $\Delta = \lambda\partial_x + \mu\partial_y + \nu\partial_z$, where λ, μ, ν are arbitrary, there will be a node at any point if for that point $\Delta(PA + t_1\Omega) = 0$. But for the point $A = 0, t_1 = 0$ this becomes $P\Delta A + \Omega\Delta t_1 = 0$; moreover, if t be any other tangent of the conic $A = 0$, and if $p = 0$ be the line joining the points of contact of the tangents t, t_1 , then we may write $A = tt_1 - p^2$, and thence (since at the point in question, $A = 0, t_1 = 0$, we have also $p = 0$) we find $\Delta A = t\Delta t_1$, and the foregoing equation thus becomes $(tP + \Omega)\Delta t_1 = 0$; viz., this equation is satisfied irrespectively of the values of λ, μ, ν , if only at the point in question (that is, for the values of the coordinates which belong to the point $t_1 = 0, A = 0$) we have $tP + \Omega = 0$, which is a single relation between the constants.

143. In particular the cubic curve $sA + mt_1t_2t_3 = 0$ may be made to have a node at the point of contact of any one of the three tangents; the quartic curve $UA + mt_1t_2t_3t_4 = 0$, a node, or two or three nodes, at the point or points of contact of any one, two, or three of the four tangents; and so in other cases. These are not the only solutions, and they are in fact solutions which (as afterwards explained) I propose to reject, attending in each case only to the remaining or proper solutions of the problem.

144. To obtain in a different manner the foregoing result, consider again the cubic curve $sA + mt_1t_2t_3 = 0$; regarding this as a given curve, the conic $A = 0$ is a conic determined (not of course completely) as a conic having therewith 3 points of 2-pointic intersection; viz., if the cubic has a node, then the cone $A = 0$ is either a conic passing through the node and besides touching the curve twice, or else it is a conic touching the curve 3 times; the former is of course the above mentioned case where there is a node at one of the points of contact on the conic $A = 0$; the latter is regarded as the proper solution. So in the case of a quartic curve $UA + mt_1t_2t_3t_4 = 0$, regarding this as a given curve, the conic $A = 0$ is a conic having therewith four points of 2-pointic intersection; viz., if the quartic curve has one, two, or three nodes, then the conic is either a conic passing through one, two, or three nodes, and besides touching the quartic thrice, twice, or once; or else it is a conic touching the quartic four times. The former is the above mentioned case where there is a node or nodes at a point or points of contact with the conic $A = 0$; the latter is regarded as the proper solution.

145. To fix the ideas, and at the same time obtain a result which will be afterwards useful, I work out the formulae for the cubic curve $sA + mt_1t_2t_3 = 0$, taking this equation under the form

$$s\left(\frac{x}{\lambda} + \frac{y}{\mu} + \frac{z}{\nu}\right) + \frac{m}{\lambda\mu\nu}(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) + \frac{m}{\lambda\mu\nu}xyz = 0.$$

This may have a node in two different ways; viz.,

1°. At the point of contact of one of the tangents $x = 0, y = 0, z = 0$ with the conic $A = 0$; say at the point of contact of $x = 0$, that is, the point $x = 0, y = z = 0$. The value of m is $\frac{4}{\mu} + \frac{4}{\nu}$; hence $\frac{m}{s} = \frac{4}{\mu} + \frac{4}{\nu}$; and, substituting, the equation of the curve becomes

$$\frac{s}{\lambda} \left[\left(\frac{x}{\lambda} \right) + \frac{1}{\mu\nu} \{ x^2 - 2x(y+z) + (y-z)^2 \} \right] + \frac{1}{\lambda\mu\nu} m (x+y-z)^2 = 0,$$

which has obviously a node at the point in question.

2°. The node may be at a point not on the conic $A = 0$, viz. the value of $\frac{m}{s}$ is $-\frac{4\lambda}{\mu\nu}$; the equation is

$$s\left(\frac{x}{\lambda} + \frac{y}{\mu} + \frac{z}{\nu}\right) - \frac{m}{\lambda\mu\nu}(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) + \frac{m}{\lambda\mu\nu}xyz = 0.$$

In fact, writing for shortness

$$-\lambda + \mu + \nu = L,$$

$$\lambda - \mu + \nu = M,$$

$$\lambda + \mu - \nu = N,$$

$$\lambda + \mu + \nu = P,$$

the node is at the point $x : y : z = L\lambda : M\mu : N\nu$; which is at once verified, if we remark that, writing for convenience $x, y, z = L\lambda, M\mu, N\nu$, then we have

$$-x + y + z = MN,$$

$$x - y + z = NL,$$

$$x + y - z = LM,$$

$$\frac{x}{\lambda} + \frac{y}{\mu} + \frac{z}{\nu} = P, \quad x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = -LMNP (= A).$$

For, of the three equations for the coordinates of a node, the first is

$$-\frac{s}{\lambda}LMNP + P(-2MN) + \frac{m}{\lambda\mu\nu}MN\mu\nu = 0,$$

that is, for the values in question,

$$-LMNP - 2\lambda MNP + P^2MN = 0,$$

or, finally, $-P - 2\lambda + P = 0$, which is satisfied; and similarly the other two equations are satisfied.

Quartic Surfaces resumed.

146. Passing now from curves to cones, and to the theory of the quartic surface, suppose that there is a component cone having a nodal line, say the cubic cone $sA + mt_1t_2t_3 = 0$: if the remaining factor is $t_4t_5t_6$, then we have

$$\Gamma = C + \frac{m}{K} s t_4 t_5 t_6.$$

Suppose the nodal line is a line of contact with the cone $A = 0$, say its equations are $t_1 = 0, p = 0$ (p a linear function), then $sA + mt_1t_2t_3$ is a quadric function $(*|t_1, p)^2$ (of course with variable coefficients); hence $A\Gamma - B^2$ is a quadric function; and B being a linear function $(*|t_1, p)$, it follows that Γ is a linear function, and thence that Γ is also a linear function; that is, A, B, Γ are each of them a linear function $(*|t_1, p)$, or the line in question (viz. the line of contact $A = 0, t_1 = 0$) is a line on the quartic surface $Aw^2 + 2Bw + \Gamma = 0$. As already mentioned, *I exclude from consideration the surfaces which have upon them a line through two nodes*; that is, I exclude from consideration the case in question where any component cone, or say where the sextic cone, has a nodal line which is a line on the tangent cone $A = 0$.

147. Now, excluding the case just referred to, *I assume as a postulate* that there is but one way in which the cubic cone $sA + mt_1t_2t_3 = 0$ can be made to have a nodal line, or the quartic cone $UA + mt_1t_2t_3t_4 = 0$ one, two, or three nodal lines &c., as the case may be. It is to be understood that this does not mean that the constants are in any of these cases completely determined, but that there is between them a relation or relations constituting a general solution which includes in itself every particular solution whatever. I have no doubt that as regards the cubic cone at least the assumption is correct. This being so, the character of a single node determines the nature of the surface; for instance, if there is a node $(3_1, 3)$, then taking this as the point $(x = 0, y = 0, z = 0)$ the equation of the surface is $Aw^2 + 2Bw + \Gamma = 0$, where

$$\Gamma = C + \frac{mm'}{K}(ss'A + l's t_4 t_5 t_6 + ls't_1 t_2 t_3),$$

a surface of a determinate nature; so that the character of all the remaining nodes is completely determined.

148. The point to be attended to is, that if for instance there were two essentially distinct ways of giving the cubic cone $sA + mt_1t_2t_3 = 0$ a nodal line (such as there would be if the excluded case were considered admissible), then the foregoing equation of the surface would or might include two distinct forms of equation applying to different kinds of surface. The conclusion is that there is but one kind of quartic surface having a node $(3_1, 3)$. Admitting this, and similarly that there is but one kind of quartic surface having a node 6_{10} , it follows that if (as the fact is) there is a surface having the nodes $1(6_{10}) + 10(3_1, 3)$ (Kummer's 11-nodal surface), then that the two first-mentioned kinds are in fact each of them this last-mentioned kind of surface; and it was in this manner that I arrived at the enumeration given near the beginning of the present Memoir.

149. The reasoning is, of course, in place of a direct demonstration which would consist in showing that a surface having a node $(3_1, 3)$ has 9 other like nodes, and also a node 6_{10} ; and that a surface having a node 6_{10} has 10 other nodes $(3_1, 3)$; and that, starting from either form of equation, we could, by passing to a node of the other kind, obtain the other form of equation.

Enumeration of the Cases.

150. I collect the results as follows: I call to mind that we have always the identical equation $AC - B^2 = Kt_1t_2t_3t_4t_5t_6$, that the equation of the surface is $Aw^2 + 2Bw + \Gamma = 0$, and that the

circumscribed cone is $A\Gamma - B^2 = 0$. The equation of a surface having different kinds of nodes will assume different forms according as the origin (or point $x = 0, y = 0, z = 0$) is taken to be at a node of one or other of these kinds; these forms of the equations are distinguished as “node-forms,” viz., we speak of the node-form $(3_1, 3)$ when the origin is a node $(3_1, 3)$, and so in other cases.

The 16-nodal surface $16(1, 1, 1, 1, 1, 1)$.

node-form $(1, 1, 1, 1, 1, 1)$,

cone is $t_1 t_2 t_3 t_4 t_5 t_6 = 0$,

and $\Gamma = C$;

viz., equation is $Aw^2 + 2Bw + C = 0$.

The 15-nodal surface $15(2, 1, 1, 1, 1, 1)$.

node-form $(2, 1, 1, 1, 1, 1)$,

cone is $(lA + mt_1 t_2) t_3 t_4 t_5 t_6 = 0$,

and $\Gamma = C + \frac{Kl}{m} t_3 t_4 t_5 t_6$.

The 14-nodal surface $8(3_1, 1, 1, 1, 1) + 6(2, 2, 1, 1, 1)$.

node-form $(3_1, 1, 1, 1, 1, 1)$,

cone is $(sA + mt_1 t_2 t_3) t_4 t_5 t_6 = 0$,

where $sA + mt_1 t_2 t_3 = 0$ is a nodal cubic 3_1 ,

and $\Gamma = C + \frac{Ks}{m} t_4 t_5 t_6$.

node-form $(2, 2, 1, 1, 1, 1)$,

cone is $(lA + mt_1 t_2)(l'A + m't_3 t_4) t_5 t_6 = 0$,

and $\Gamma = C + \frac{mm'}{K}(ll'A + lm't_3 t_4 + l'mt_1 t_2) t_5 t_6$.

The 13(α)-nodal surface $3(4_1, 1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$.

node-form $(4_1, 1, 1, 1)$,

cone is $(UA + mt_1 t_2 t_3 t_4) t_5 t_6 = 0$,

where $UA + mt_1 t_2 t_3 t_4 = 0$ is a trinodal quartic 4_1 ,

and $\Gamma = C + \frac{KU}{m} t_5 t_6$.

node-form $(3, 1, 1, 1, 1)$,

cone is $(sA + mt_1 t_2 t_3) t_4 t_5 t_6 = 0$,

and $\Gamma = C + \frac{Ks}{m} t_4 t_5 t_6$.

node-form $(3_1, 2, 1)$,

cone is $(sA + mt_1 t_2 t_3)(l'A + m't_4 t_5) t_6 = 0$,

where $sA + mt_1 t_2 t_3 = 0$ is a nodal cubic 3_1 ,

and $\Gamma = C + \frac{mm'}{K}(sl'A + m's t_4 t_5 + ml't_1 t_2 t_3) t_6$.

The 13(β)-nodal surface $13(2, 2, 2)$.

node-form $(2, 2, 2)$,

cone is $(lA + mt_1 t_2)(l'A + m't_3 t_4)(l''A + m''t_5 t_6) = 0$,

and

$$\begin{aligned} \Gamma = C + \frac{mm'm''}{K^2} [& ll'l''A^2 + (l'l''mt_1 t_2 + l''lm't_3 t_4 + ll'm''t_5 t_6)A \\ & + lm'm''t_3 t_4 t_5 t_6 + l'm''m t_5 t_6 t_1 t_2 + l''mm't_1 t_2 t_3 t_4]. \end{aligned}$$

The 12(α)-nodal surface $12(4_1, 2)$.

node-form $(4_2, 2)$,

cone is $(UA + mt_1 t_2 t_3 t_4)(l'A + m't_5 t_6) = 0$,

where $UA + mt_1t_2t_3t_4 = 0$ is a trinodal quartic 4_3 ,
and $\Gamma = C + \frac{mm'}{K}(l'UA + m'Ut_5t_6 + l'mt_1t_2t_3t_4)$.

The $12(\beta)$ -nodal surface $2(5_1, 1) + 6(3_1, 3_1) + 4(3, 2, 1)$.

node-form $(5_1, 1)$,

cone is $(VA + mt_1t_2t_3t_4t_5)t_6 = 0$,

where $VA + mt_1t_2t_3t_4t_5 = 0$ is a 6-nodal quintic 5_1 ,

and $\Gamma = C + \frac{KV}{m}t_6$.

node-form $(3_1, 3_1)$,

cone is $(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0$,

where $sA + mt_1t_2t_3 = 0$ and $s'A + m't_4t_5t_6 = 0$ are each of them a nodal cubic 3_1 ,

and $\Gamma = C + \frac{mm'}{K}(ss'A + m'st_4t_5t_6 + ms't_1t_2t_3)$.

node-form $(3, 2, 1)$,

cone is $(sA + mt_1t_2t_3)(l'A + m't_4t_5)t_6 = 0$,

and $\Gamma = C + \frac{mm'}{K}(sl'A + m'st_4t_5 + l'mt_1t_2t_3)t_6$.

The $12(\gamma)$ -nodal surface, $12(4_1, 1, 1)$.

node-form $(4_1, 1, 1)$,

cone is $(UA + mt_1t_2t_3t_4)t_5t_6 = 0$,

where $UA + mt_1t_2t_3t_4 = 0$ is a binodal quartic 4_3 ,

and $\Gamma = C + \frac{KU}{m}t_5t_6$.

The $11(\alpha)$ -nodal surface, $1(6_{10}) + 10(3_1, 3)$.

node-form (6_{10}) ,

cone is $WA + mt_1t_2t_3t_4t_5t_6 = 0$,

where this is a 10-nodal sextic 6_{10} ,

and $\Gamma = C + \frac{KW}{m}$.

node-form $(3_1, 3)$,

cone is $(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0$,

where $sA + mt_1t_2t_3 = 0$ is a nodal cubic 3_1 ,

and $\Gamma = C + \frac{mm'}{K}(ss'A + m'st_4t_5t_6 + ms't_1t_2t_3)$.

Other 11-nodal surfaces.

node-form $(5_1, 1)$,

cone is $(VA + mt_1t_2t_3t_4t_5)t_6 = 0$,

where $VA + mt_1t_2t_3t_4t_5 = 0$ is a 5-nodal quintic 5_1 ,

and $\Gamma = C + \frac{KV}{m}t_6 = 0$;

node-form $(4_2, 2)$,

cone is $(UA + mt_1t_2t_3t_4)(l'A + m't_5t_6) = 0$,

where $UA + mt_1t_2t_3t_4 = 0$ is a binodal quartic 4_2 ,

and $\Gamma = C + \frac{mm'}{K}(l'UA + m'Ut_5t_6 + l'mt_1t_2t_3t_4)$;

node-form $(4, 1, 1)$,

cone is $(UA + mt_1t_2t_3t_4)t_5t_6 = 0$,

where $UA + mt_1t_2t_3t_4$ is a nodal quartic 4_1 ,

and $\Gamma = C + \frac{KU}{m}t_5t_6$,

but whether these node-forms belong to the same or to different surfaces is not ascertained.

The enumeration is not extended to the 10-nodal surfaces, but I consider one case of these surfaces.

The $10(\alpha)$ -nodal surface $10(3, 3)$.

151. I assume only that there is a single node $(3, 3)$: taking the cone to be

$$(sA + mt_1t_2t_3)(s'A + m't_4t_5t_6) = 0;$$

then for the equation of the surface, in the node-form $(3, 3)$ in question, we have

$$\Gamma = C + \frac{mm'}{K}(ss'A + sm't_4t_5t_6 + s'mt_1t_2t_3).$$

But I present this result under a different form, as follows: I write

$$A = p^2 + ft_1t_2 = q^2 + gt_3t_4 = r^2 + ht_5t_6;$$

where f, g, h are constants, and, as before, $p = 0, q = 0, r = 0$ are the lines joining the points of contact of $t_1, t_2; t_3, t_4; t_5, t_6$ respectively: we have

$$sA + mt_1t_2t_3 = sA + mt_3\left(\frac{A - q^2}{g}\right), \quad \text{and} \quad s'A + m't_4t_5t_6 = s'A(\cdots),$$

or in place of s, s' introducing new linear functions σ, σ' , the cubic curves may be taken to be $\sigma A - \frac{m}{g}q^2t_3, \sigma' A - \frac{m'}{h}r^2t_5$, so that we have

$$\frac{mm'}{K}\left(\sigma A - \frac{m}{g}q^2t_3\right)\left(\sigma' A - \frac{m'}{h}r^2t_5\right) + \cdots = A(\cdots) - \frac{mm'}{fgh}(\cdots);$$

whence $B = \frac{mm'}{fgh}(pqr + tA)$, where t is a linear function of the coordinates; and we then have

$$\Gamma = C + \frac{mm'}{K}[\sigma\sigma'A + \cdots],$$

where A may be considered as standing for $q^2 + gt_3t_4$. The equation $Aw^2 + 2Bw + \Gamma = 0$ of the surface, substituting throughout for A its value, is therefore

$$\left(\frac{1}{g}\right)(q^2 + gt_3t_4)w^2 + 2\{pqr + t(q^2 + gt_3t_4)\}w + (\cdots) = 0,$$

where the cone is $\cdots = 0$.

152. Writing in the equation of the surface $w\sqrt{f}$ instead of w , it becomes

$$(q^2 + gt_3t_4)w^2 + 2[pqr + t(q^2 + gt_3t_4)]w + \frac{fgh}{mm'}\left[\frac{m'}{f}(q^2 + gt_3t_4) + 2tpqr\right] = 0,$$

and then writing $\frac{m}{f}\sigma$ and $\frac{m'}{h}\sigma'$ for σ and σ' respectively, this is

$$(q^2 + gt_3t_4)w^2 + 2pqrw + 2tw(q^2 + gt_3t_4) + g[\cdots] + t^2(q^2 + gt_3t_4) + 2tpqr = 0.$$

We may consider t_3, t_4 as denoting not the functions originally so represented, but these functions each multiplied by a suitable constant, and thereupon write $g = -1$; viz., $t_3 = 0, t_4 = 0$, will

now denote any two tangents to the conic $A = 0$, the implicit factors being so determined that $A = q^2 - t_3 t_4$. The equation of the surface is

$$(q^2 - t_3 t_4) w^2 + 2pqrw + 2tw(q^2 - t_3 t_4) - \sigma\sigma'(q^2 - t_3 t_4) + \sigma r^2 t_5 + \sigma' p^2 t_3 + p^2 r^2 + t^2(q^2 - t_3 t_4) + 2tpqr = 0;$$

viz., this is

$$-(q^2 - t_3 t_4)[(w + t)^2 - \sigma\sigma'] + 2pqr(w + t) + \sigma r^2 t_5 + \sigma' p^2 t_3 + p^2 r^2 = 0,$$

the sextic cone being

$$-[\sigma(q^2 - t_3 t_4) - p^2 t_3] \cdot [\sigma'(q^2 - t_3 t_4) - r^2 t_5] = 0.$$

153. But the foregoing equation of the surface is

$$\begin{vmatrix} -\sigma', & w + t, & \cdot, & r \\ w + t, & -\sigma, & p, & \cdot \\ \cdot, & p, & -q, & t_3 \\ r, & \cdot, & t_3, & -q \end{vmatrix} = 0,$$

as is at once seen by developing the determinant; the functions $w + t, \sigma, \sigma', p, q, r, t_3, t_5$ are all of them linear; and the determinant is thus a symmetrical quartic determinant the terms whereof are linear functions of the coordinates; viz. the surface is a symmetroid. That is, a surface having a single node (3, 3) is a symmetroid; but I have shown (Second Memoir, No. 116) that a symmetroid has each of its ten nodes (3, 3); wherefore the surface having a single node (3, 3) is the 10(α)-nodal surface, nodes 10(3, 3).

154. Start from two cubic cones $U = 0, V = 0$, having each the same vertex ($x = 0, y = 0, z = 0$); we may in a variety of ways determine the two cones $\alpha U + \beta V = 0, \gamma U + \delta V = 0$, having a common inscribed quadric cone $A = 0$ (viz., $\alpha : \beta$ being assumed at pleasure, then $\gamma : \delta$ will be determined; not, I believe, uniquely, but I do not know what the multiplicity is). This being so, the quadric cone $A = 0$ is uniquely determined; and then, assuming at pleasure the plane $w = 0$, the 10(α)-nodal surface $Aw^2 + 2Bw + \Gamma = 0$ is uniquely determined: consequently the remaining nine nodes are determinate points on the nine lines $U = 0, V = 0$ respectively. And we have thus a system of ten points in space such that, joining any one of them with the remaining nine, the nine lines so obtained are the intersections of two cubic cones, or say that they are an ennead of lines.

Notation for the Cases afterwards considered.

155. I proceed to further develop the theory of some of the different surfaces. The same node-form of equation will, of course, assume different shapes according to the actual expressions in terms of the coordinates (x, y, z) of the several functions A, B , &c., which enter into it. I have found it convenient to attribute to A and B certain specific values which are not in every case those of the coefficients of w^2, w in the equation of the surface: this means that we must, in the equation of the surface, substitute new symbols for these coefficients, and write the equation say in the form $A'w^2 + 2B'w + \Gamma' = 0$; the change of notation, when it occurs, will be duly explained.

156. It is in general (but not always) convenient to take the equation of the tangent cone to be $x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0$; for then any plane $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 0$, where $\alpha + \beta + \gamma = 0$, will be a tangent plane; so that six tangent planes may be represented by $x = 0$, $y = 0$, $z = 0$, and by three equations of the form just referred to. And in reference to this assumed form of the equation of the tangent cone, and to what follows, I write

$$\begin{aligned}\alpha + \beta + \gamma &= 0, \\ \alpha' + \beta' + \gamma' &= 0, \\ \alpha'' + \beta'' + \gamma'' &= 0, \\ P &= \frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma}, \\ P' &= \frac{x}{\alpha'} + \frac{y}{\beta'} + \frac{z}{\gamma'}, \\ P'' &= \frac{x}{\alpha''} + \frac{y}{\beta''} + \frac{z}{\gamma''},\end{aligned}$$

$$\begin{aligned}X &= \alpha(\gamma'\gamma''y - \beta'\beta''z), \\ Y &= \beta(\alpha'\alpha''z - \gamma'\gamma''x), \\ Z &= \gamma(\beta'\beta''x - \alpha'\alpha''y), \\ X' &= \alpha'(\gamma''\gamma y - \beta''\beta z), \\ Y' &= \beta'(\alpha''\alpha z - \gamma''\gamma x), \\ Z' &= \gamma'(\beta''\beta x - \alpha''\alpha y), \\ X'' &= \alpha''(\gamma\gamma'y - \beta\beta'z), \\ Y'' &= \beta''(\alpha\alpha'z - \gamma\gamma'x), \\ Z'' &= \gamma''(\beta\beta'x - \alpha\alpha'y),\end{aligned}$$

$$A = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy,$$

$$B = \alpha\alpha'\alpha''(y^2z - yz^2) + \beta\beta'\beta''(z^2x - zx^2) + \gamma\gamma'\gamma''(x^2y - xy^2) + Mxyz,$$

$$C = (\alpha\alpha'\alpha''yz + \beta\beta'\beta''zx + \gamma\gamma'\gamma''xy)^2 - \frac{1}{4}\dots$$

where

$$\begin{aligned}M &= (\beta - \gamma)\alpha'\alpha'' + (\gamma - \alpha)\beta'\beta'' + (\alpha - \beta)\gamma'\gamma'', \\ &= (\beta' - \gamma')\alpha''\alpha + (\gamma' - \alpha')\beta''\beta + (\alpha' - \beta')\gamma''\gamma, \\ &= (\beta'' - \gamma'')\alpha\alpha' + (\gamma'' - \alpha'')\beta\beta' + (\alpha'' - \beta'')\gamma\gamma',\end{aligned}$$

$$\frac{1}{2}\dots = [(\beta - \gamma)(\beta' - \gamma')(\beta'' - \gamma'') + (\gamma - \alpha)(\gamma' - \alpha')(\gamma'' - \alpha'') + (\alpha - \beta)(\alpha' - \beta')(\alpha'' - \beta'')];$$

also

$$A^2 = 4\alpha\alpha'\alpha''\beta\beta'\beta''\gamma\gamma'\gamma'';$$

and we have identically [an identity follows from this notation].

The 16-nodal Surface 16(1, 1, 1, 1, 1, 1).

157. Kummer starts from an irrational equation, which is readily converted into the following

$$\sqrt{x(X - w)} + \sqrt{y(Y - w)} + \sqrt{z(Z - w)} = 0,$$

and then, rationalizing, we have

$$Aw^2 + 2Bw + C = 0,$$

where as above

$$AC - B^2 = KxyzPP'P''.$$

This agrees with the foregoing theory; viz., the point $(x = 0, y = 0, z = 0)$ being a node, the rationalized equation must, of course, be in the node-form $(1, 1, 1, 1, 1, 1)$, (being the only node-form); and the symmetry of the formulae enables us at once to write down the equations of the 16 singular planes, and thence to deduce the coordinates of the 16 nodes; viz.,

the singular planes are and the nodes are

(1) $x = 0$,	(1) $(0, -\beta, \gamma, \alpha'\alpha''\beta\gamma)$,
(2) $y = 0$,	(2) $(\alpha, \cdot, -\gamma, \beta'\beta''\gamma\alpha)$,
(3) $z = 0$,	(3) $(-\alpha, \beta, \cdot, \gamma'\gamma''\alpha\beta)$,
(4) $w = 0$,	(4) $(\alpha'\alpha'', \beta'\beta'', \gamma'\gamma'', \cdot)$,
(5) $X - w = 0$,	(5) $(1, \cdot, \cdot, \cdot)$,
(6) $Y - w = 0$,	(6) $(\cdot, 1, \cdot, \cdot)$,
(7) $Z - w = 0$,	(7) $(\cdot, \cdot, 1, \cdot)$,
(8) $P = 0$,	(8) $(\cdot, \cdot, \cdot, 1)$,
(9) $X' - w = 0$,	(9) $(\cdot, -\beta', \gamma', \alpha''\alpha\beta'\gamma')$,
(10) $Y' - w = 0$,	(10) $(\alpha', \cdot, -\gamma', \beta''\beta\gamma'\alpha')$,
(11) $Z' - w = 0$,	(11) $(-\alpha', \beta', \cdot, \gamma''\gamma\alpha'\beta')$,
(12) $P' = 0$,	(12) $(\alpha''\alpha, \beta''\beta, \gamma''\gamma, \cdot)$,
(13) $X'' - w = 0$,	(13) $(\cdot, -\beta'', \gamma'', \alpha\alpha'\beta''\gamma'')$,
(14) $Y'' - w = 0$,	(14) $(\alpha'', \cdot, -\gamma'', \beta\beta'\gamma''\alpha'')$,
(15) $Z'' - w = 0$,	(15) $(-\alpha'', \beta'', \cdot, \gamma\gamma'\alpha''\beta'')$,
(16) $P'' = 0$,	(16) $(\alpha\alpha', \beta\beta', \gamma\gamma', \cdot)$,

where the nodes and planes are numbered as by Kummer; and by means of his (differently arranged) diagram of the relation between the several nodes and planes, I was enabled to form the following square diagram, which exhibits this relation in, I think, the most convenient form. To explain this, observe that in the upper and left-hand margins, the numbers refer to the nodes; in the body of the table, and in the right-hand margin to the planes, the table shows that for the node 1, the circumscribed cone is made up of the planes 1, 6, 7, 8, 9, 13; and that the remaining 15 nodes are situate on the nodal lines of this cone, the node 2 on the intersection of the planes 7, 8; the node 3 on the intersection of the planes 6, 8, and so on; and the like as regards the other lines of the table.

[The original page 283 here gives a 16×16 square diagram (Table No. 12) exhibiting the incidence of the 16 nodes with the 16 singular planes of the 16-nodal surface. Each row i shows the planes through node i (the six planes whose intersection is the node), and the body of the table records, in row i column j , the pair of planes whose intersection at node j lies on the sextic cone of node i . The right-hand margin lists the planes meeting at the node of that row: 1... 1, 6, 7, 8, 9, 13; 2... 2, 6, 8, 10, 14, ...; etc. (See PDF page 283 of the volume for the full table.)]

158. The before mentioned irrational equation may be written

$$\sqrt{1 \cdot 5} + \sqrt{2 \cdot 6} + \sqrt{3 \cdot 7} = 0,$$

and by symmetry we see that also

$$\sqrt{1 \cdot 9} + \sqrt{2 \cdot 10} + \sqrt{3 \cdot 11} = 0,$$

$$\sqrt{1 \cdot 13} + \sqrt{2 \cdot 14} + \sqrt{3 \cdot 15} = 0;$$

viz., these are three equations each containing the planes 1, 2, 3, which are three of the planes belonging to the node 1; the other three planes in any such equation (for instance, the planes 5, 6, 7, in the first equation) being three planes belonging to another node. Instead of the planes 1, 2, 3, we may have any other three planes belonging to the node 1; and instead of the node 1, any other node; but each equation belongs to two nodes: the number of equations is thus

$$\frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3} \times 16 \times 3 \div 2, = 480.$$

159. To obtain the planes belonging to any such equation, combine any two of the outside right-hand lines of the diagram, these contain in common two numbers the places of which are interchanged; striking these out, we have four columns, and taking out of these any three columns, we have the corresponding sets of planes. For instance, lines 1 and 2 contain 7 8 and 8 7 respectively; striking these out, the lines are

$$1, \quad 9, 13, \quad 6;$$

$$2, \quad 10, 14, \quad 5;$$

whence we have the sets (1, 9, 13) and (2, 10, 14); viz., there is an irrational equation of the form

$$\sqrt{1 \cdot 2} + \sqrt{9 \cdot 10} + \sqrt{13 \cdot 14} = 0,$$

but it is probably necessary to introduce constant factors along with the products $1 \cdot 2$, $9 \cdot 10$, and $13 \cdot 14$ respectively. There are $\frac{1}{2}16 \cdot 15, = 120$ pairs of lines, and each line gives 4 equations; in all $120 \times 4, = 480$ equations, as above.

160. I stop to remark that Kummer gives for his 13-nodal surface an equation containing three arbitrary constants, say λ, μ, ν , such that, putting one of these $= 0$, we have the 14-nodal surface; putting two of them each $= 0$, the 15-nodal surface; putting all three of them $= 0$, the 16-nodal surface. The equations for the 16-nodal surface and that for the 14-nodal surface, made use of by Kummer, are, in fact, those deduced as above from the equation of the 13(α)-nodal surface; and the like form might have been used for the 15-nodal surface. But the form actually used by Kummer, as presently appearing, is an equivalent form not thus deducible from the equation of the 13(α)-nodal surface.

The 15-nodal Surface 15(2, 1, 1, 1, 1).

161. Kummer's equation is readily converted into the following:

$$Aw^2 + 2Bw + C + \frac{Kl}{m}xyzP = 0,$$

the circumscribed cone being thus

$$(lA + mP'P'')xyzP = 0,$$

and the equation being in the node-form (2, 1, 1, 1, 1).

The formulae for the 15 nodes and the 10 singular planes depend upon a quadric equation, for the symmetrical expression of which I write

$$\omega = \alpha'\alpha'' - \beta'\gamma'' = \beta'\beta'' - \gamma''\alpha' = \gamma'\gamma'' - \alpha''\beta',$$

$$\varpi = \alpha'\alpha'' - \beta'\gamma'' = \beta'\beta'' - \gamma'\alpha'' = \gamma'\gamma'' - \alpha'\beta'';$$

so that

$$\omega - \varpi = \frac{\beta'\gamma'' - \beta''\gamma'}{\alpha} = \frac{\gamma'\alpha'' - \gamma''\alpha'}{\beta} = \frac{\alpha'\beta'' - \alpha''\beta'}{\gamma},$$

$$\omega + \varpi = \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'' + \cdots;$$

the equation in question then is

$$(\rho - \omega)(\rho - \varpi) - \frac{Kl}{4\alpha\beta\gamma m} = 0;$$

so that, calling the roots of it ρ_1, ρ_2 , we have

$$\rho_1 + \rho_2 = \omega + \varpi, \quad \rho_1\rho_2 = \omega\varpi + \frac{Kl}{4\alpha\beta\gamma m};$$

or we may write

$$\rho_1 = \frac{1}{2}(\omega + \varpi + \sqrt{\Omega}), \quad \rho_2 = \frac{1}{2}(\omega + \varpi - \sqrt{\Omega}),$$

if for shortness

$$\Omega = (\omega - \varpi)^2 - \frac{Kl}{\alpha\beta\gamma m}.$$

162. I write also for shortness

$$\begin{aligned} \mathfrak{a} &= (\beta - \gamma)\alpha'\alpha'' + \alpha(\beta'\beta'' - \gamma'\gamma''), \\ \mathfrak{b} &= (\gamma - \alpha)\beta'\beta'' + \beta(\gamma'\gamma'' - \alpha'\alpha''), \\ \mathfrak{c} &= (\alpha - \beta)\gamma'\gamma'' + \gamma(\alpha'\alpha'' - \beta'\beta''); \end{aligned}$$

and I say that the singular planes are

$$\begin{aligned} (1) \quad & x = 0, \quad (2) \quad y = 0, \quad (3) \quad z = 0, \\ (9) \quad & -\frac{\gamma'\beta'}{\alpha} p_1(Z' - w) + (\rho_1 - \omega)\left(\frac{x}{\beta} + \frac{y}{\gamma}\right) = 0, \\ (10) \quad & -\frac{\beta'\alpha'}{\gamma} p_1(Y' - w) + (\rho_1 - \omega)\left(\frac{z}{\alpha} + \frac{x}{\beta}\right) = 0, \\ & \dots \\ & P = 0; \end{aligned}$$

[The full lists of 15 nodes and 10 singular planes, with their pairings to Kummer's numbering, occupy book pages 285–286. The formulae are routine consequences of the symmetric functions $\rho_1, \rho_2, \mathfrak{a}, \mathfrak{b}, \mathfrak{c}$ above. See the PDF pages 307–308 of the scan for the literal expressions.]

163. The small reference numbers are those used by Kummer. It is, I think, better to retain the reference numbers belonging to the case of the 16-nodal surface; viz., there are here given, large, 1, 2, 3, 8, 9, 10, 11, 13, 14, 15 for the planes, and 1, 2, 3, 5, 6, ... to 16 for the nodes. Belonging to each node (that is, with the node as vertex) there is a quadric cone passing through 8 other nodes; and each node lies (exclusively of the cone whose vertex it is) in 8 such cones. We have thus the following square diagram:

[The original page 287 here gives a 15×15 square diagram exhibiting, for the 15-nodal surface, the incidence of nodes with planes and quadric cones C , where the cone C replaces a pair of the planes of the 16-nodal table. See PDF page 287 for the table.]

164. The arrangement is the same as in the 16-nodal square diagram; only, in the right-hand margin, a bracket (6, 7) denotes that instead of the planes (6, 7) we have a quadric cone; which cones are, in the body of the table, denoted by C . Thus for the node 1 the sextic cone is made up of the planes 1, 8, 9, 13 and of a quadric cone (6, 7), = C : the remaining 14 nodes lie on the nodal lines of the sextic cone, viz., the node 2 on an intersection of the cone C with the plane 8, the node 3 on an intersection of the same cone and plane, the node 5 on the intersection of the planes 9, 13, and so on.

The Equation of the 15-nodal Surface, as deduced from that of the 13(α)-nodal.

165. If, in the equation hereafter given for the 13-nodal surface, we write $\nu = 0$, $\mu = 0$, or (what is the same thing) in that of the 14-nodal surface we write $\mu = 0$, the form is

$$[A + 4\lambda yz] w^2 + 2w[B - 2\lambda yzX] + C = 0.$$

The circumscribed sextic cone is thus $(2, 1, 1, 1, 1)$,

$$(\alpha\alpha''\beta\beta''yz + \beta\gamma\alpha''zx + \gamma\alpha\beta''xy + \alpha\beta xz) yz (P' - p_1P'')(P' - p_2P'') = 0,$$

where p_1, p_2 now denote the roots of the equation

$$PP' (p\alpha - \alpha''\gamma' - \alpha'\gamma'') (\dots) + (\dots) p\alpha'\alpha'' = 0.$$

The singular planes are

$$\begin{aligned} w &= 0, \\ P' - p_1P'' &= 0, \\ P' - p_2P'' &= 0, \\ z &= 0, \\ y &= 0, \\ \frac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma} p_2 (Z' - w) - (Z'' - w) &= 0, \\ \frac{\gamma'\alpha'\beta''}{\beta\alpha''\gamma} p_1 (Z' - w) - (Z'' - w) &= 0, \\ \frac{\beta'\gamma''}{\alpha'\alpha''} p_2 (Y' - w) - (Y'' - w) &= 0, \\ \frac{\beta'\gamma''}{\alpha'\alpha''} p_1 (Y' - w) - (Y'' - w) &= 0, \\ Z - w &= 0, \end{aligned}$$

and we have then the same square table as before: the coordinates of the 15 nodes may be obtained without difficulty.

166. The form is really equivalent to that first considered in regard to the 15-nodal surface. To show that this is so, we have only to arrange according to powers of x ; viz., the equation thus becomes

$$\begin{aligned} x^2 [(\gamma'\gamma'')^2 y^2 + (\beta'\beta'')^2 z^2 + w^2 - 2w\beta'\beta''z + 2w\gamma'\gamma''y + 2\beta'\beta''\gamma'\gamma''yz] \\ + 2x[\dots] + [a\alpha'\alpha''yz + wy - wz]^2 - 4\lambda yzw(X - w) = 0, \end{aligned}$$

where, if for a moment A denotes the coefficient of x^2 , we have $y = 0, z = 0, w = 0, X - w = 0$, four tangent planes of the quadric cone $A = 0$.

14-nodal Surface $8(3_1, 1, 1, 1) + 6(2, 2, 1, 1)$, **Node-form** $(3_1, 1, 1, 1)$.

167. In the equation hereafter given for the 13-nodal surface, writing $\nu = 0$, the sextic cone becomes

$$4z(\lambda\alpha y^2 + \mu\beta zx + \beta'\gamma''xy + \gamma\alpha xy^2 + \alpha\beta xyz) (P' - p_1P'')(P' - p_2P'') = 0;$$

viz., this is of the form in question $(3_1, 1, 1, 1)$; and the equation of the surface is

$$w^2 \{A + 4(\lambda y + \mu x)z - 4\lambda\mu z^2\} + 2w \{B - 2(\lambda yzX + \mu zxY)\} + C = 0.$$

The singular planes are

$$w = 0, \quad (4)$$

$$P' - p_1 P'' = 0, \quad (12)$$

$$P' - p_2 P'' = 0, \quad (16)$$

$$z = 0, \quad (3)$$

$$\frac{\gamma' \alpha' \beta''}{\beta \alpha'' \gamma} p_2 (Z' - w) - (Z'' - w) = 0, \quad (15)$$

$$\frac{\gamma' \alpha' \beta''}{\beta \alpha'' \gamma} p_1 (Z' - w) - (Z'' - w) = 0, \quad (11)$$

and the nodes are

$$(0, 0, 0, 1), \quad (1, 0, 0, 0), \quad (0, 1, 0, 0), \quad (0, 0, 1, 0),$$

together with the 11 other nodes corresponding (with Kummer's small numbers in parentheses) to the table on page 289 of the scan.

168. In these formulae, p_1, p_2 denote the roots of the equation

$$\begin{aligned} & [\lambda(\beta' \gamma'')^2 + \mu(\alpha \beta'')^2] (p \alpha \beta \gamma)^2 \\ & + [\lambda(\beta' \gamma')^2 + \mu(\gamma' \alpha')^2] (\alpha'' \beta' \gamma'')^2 \\ & - 2[\lambda \beta' \beta'' + \mu \alpha' \alpha'' + 2\gamma' \gamma'' (\omega - \varpi)^2] p \alpha \beta \gamma \alpha'' \beta' \gamma'' = 0. \end{aligned}$$

169. The relation of the nodes and sextic cones is given by means of the square diagram on the opposite page:

[The original page 291 here gives a 14×14 square diagram of the 14-nodal surface, with notations C for the nodal cubic cone (a cubic replacing planes 5, 6, 8), D, D' for two quadric cones (replacing the planes 2, 5 and 9, 13 respectively), and $\times$ denoting the nodal line of the cubic. Vacant squares are understood as (D, D') . See PDF page 291 for the full table.]

170. The equation of the surface may be presented in the irrational form

$$\begin{aligned} & \sqrt{(P' - p_1 P'')} \left\{ \frac{\gamma' \alpha' \beta''}{\beta \alpha'' \gamma} p_2 (Z' - w) - \gamma' \alpha'' \beta'' (Z'' - w) \right\} \\ & + \sqrt{(P' - p_2 P'')} \left\{ \frac{\gamma' \alpha' \beta''}{\beta \alpha'' \gamma} p_1 (Z' - w) - \gamma' \alpha'' \beta'' (Z'' - w) \right\} \\ & + (p_2 - p_1) \mu \alpha' \alpha'' \beta' \sigma \sqrt{\dots} (z_0 + \dots) zw = 0. \end{aligned}$$

In fact the norm of the left-hand side is

$$\begin{aligned} & (p_1 - p_2)^2 (\omega - \varpi)^2 \{ w^2 [A + 4z(\lambda y + \mu x) - 4\lambda \mu z^2] \\ & + 2w [B - 2z(\lambda y X + \mu x Y)] \\ & + C \}. \end{aligned}$$

To partially verify this, observe that, writing the equation under the form $w\sqrt{R} + \sqrt{S} + \sqrt{T} = 0$, on writing therein $w = 0$, we ought to have

$$(R - S)^2 = (p_1 - p_2)^2 (\omega - \varpi)^2 (\alpha \alpha' \alpha'' yz + \beta \beta' \beta'' zx + \gamma \gamma' \gamma'' xy)^2.$$

But writing $w = 0$, we have

$$\begin{aligned}
 R - S &= (P' - p_2 P'') (\gamma' \alpha' \beta'' p_1 Z' - \gamma' \alpha'' \beta'' Z'') \\
 &\quad - (P' - p_1 P'') (\gamma' \alpha' \beta'' p_2 Z' - \gamma' \alpha'' \beta'' Z'') \\
 &= (p_2 - p_1) (\gamma' \alpha'' \beta'' P'' Z' - \gamma' \alpha' \beta'' P' Z''), \\
 &= (p_2 - p_1) [\gamma'' (\beta' \gamma' x + \gamma' \alpha'' y + \alpha' \beta' z) (\beta \gamma \beta x - \alpha \alpha'' y) \\
 &\quad - \gamma' (\beta'' \gamma'' x + \gamma'' \alpha'' y + \alpha'' \beta'' z) (\beta \beta' x - \alpha \alpha' y)],
 \end{aligned}$$

which is easily found to be

$$= -(p_2 - p_1)(\omega - \varpi)(\alpha \alpha' \alpha'' y z + \beta \beta' \beta'' z x + \gamma \gamma' \gamma'' x y);$$

and thus $(R - S)^2$ has the value in question.

171. In further verification, observe that, writing $x, y, z = \alpha' \alpha'', \beta' \beta'', \gamma' \gamma''$, and therefore $P' = 0, P'' = 0$, we ought to have

$$\frac{1}{4}(p_1 - p_2)^2 (\alpha' \beta' \gamma')^2 (\dots) z^2 w^2 = (p_1 - p_2)^2 (\omega - \varpi)^2 w^2 [A + 4z(\lambda y + \mu x) - 4\lambda \mu z^2],$$

observing that, for the values in question, B, X, Y, C all vanish.

This is

$$(p_1 - p_2)^2 (\alpha' \beta' \gamma')^2 \left(\frac{\lambda}{\alpha} + \frac{\mu}{\beta} \right) (\gamma' \gamma'')^2 z^2 w^2 = (\omega - \varpi)^2 [(\omega - \varpi)^2 + 4\gamma' \gamma'' (\lambda \beta' \beta'' + \mu \alpha' \alpha'' - \lambda \mu \gamma' \gamma'')],$$

which is in fact the value of $(p_1 - p_2)^2$ obtained from the equation in p .

The 13(α)-nodal Surface $3(4_1, 1, 1) + 1(3, 1, 1, 1) + 9(3_1, 2, 1)$.

172. The equation, node-form $(4_3, 1, 1)$, is

$$\begin{aligned}
 &w^2 \{A + 4(\lambda y z + \mu z x + \nu x y) - 4(\mu \nu x^2 + \nu \lambda y^2 + \lambda \mu z^2)\} \\
 &\quad + 2w \{B - 2(\lambda y z X + \mu z x Y + \nu x y Z)\} + C = 0;
 \end{aligned}$$

for the circumscribed cone we have

$$\begin{aligned}
 &\{A + 4(\lambda y z + \mu z x + \nu x y) - 4(\mu \nu x^2 + \nu \lambda y^2 + \lambda \mu z^2)\} C \\
 &\quad - \{B - 2(\lambda y z X + \mu z x Y + \nu x y Z)\}^2 \\
 &= 4\{\lambda \alpha \gamma z^2 + \mu \beta z x^2 + \nu \gamma x y^2 + \beta \gamma \alpha x y^2 + \gamma \alpha \beta y z^2 + \alpha \beta \gamma z^2 x y\} \\
 &\quad \cdot \left\{ -\lambda \frac{X^2}{\alpha^2} - \mu \frac{Y^2}{\beta^2} - \nu \frac{Z^2}{\gamma^2} + \frac{1}{\alpha' \alpha'' \beta' \beta'' \gamma' \gamma''} P' P'' \right\},
 \end{aligned}$$

where, on the right-hand side, the first factor, equated to zero, represents a trinodal quartic cone, the nodal lines whereof are $(y = 0, z = 0)$, $(z = 0, x = 0)$, $(x = 0, y = 0)$.

173. As regards the second factor, it is to be observed that, writing as above

$$\omega - \varpi = \frac{\beta' \gamma'' - \beta'' \gamma'}{\alpha} = \frac{\gamma' \alpha'' - \gamma'' \alpha'}{\beta} = \frac{\alpha' \beta'' - \alpha'' \beta'}{\gamma},$$

we have identically

$$\begin{aligned}
 \alpha' P' - \alpha'' P'' &= \frac{\dots}{\beta' \beta''} x - \dots, \\
 \beta' P' - \beta'' P'' &= \dots, \quad \gamma' P' - \gamma'' P'' = \dots,
 \end{aligned}$$

so that the second factor is

$$\frac{1}{(\omega - \varpi)^2} \left[-\lambda(\beta' \beta'' \gamma' \gamma'')^2 (\alpha' P' - \alpha'' P'')^2 - \mu(\gamma' \gamma'' \alpha' \alpha'')^2 (\beta' P' - \beta'' P'')^2 \right. \\ \left. - \nu(\alpha' \alpha'' \beta' \beta'')^2 (\gamma' P' - \gamma'' P'')^2 + (\omega - \varpi)^2 \alpha' \alpha'' \beta' \beta'' \gamma' \gamma'' P' P'' \right];$$

or, what is the same thing,

$$\frac{1}{(\omega - \varpi)^2} \alpha' \alpha'' \beta' \beta'' \gamma' \gamma'' \left\{ \dots (\alpha' \beta' \gamma' P')^2 + \dots (\alpha'' \beta'' \gamma'' P'')^2 + 2[\dots] \alpha' \beta' \gamma' P' \alpha'' \beta'' \gamma'' P'' \right\},$$

so that, equating to zero, we have a pair of planes, each passing through the line $P' = 0, P'' = 0$, and which it is clear must be the tangent planes from this line to the quadric cone $A' = 0$. I will presently return to this, but I consider first the foregoing identical equation in regard to the circumscribed cone.

174. In verification hereof, observe first that, if $x = 0$, the equation becomes

$$\left\{ (y - z)^2 + 4\lambda yz - 4\lambda(\beta' \gamma y^2 + \mu z^2) \right\} (\alpha \alpha')^2 y^2 z^2 \\ - [(\alpha \alpha') (y^2 z - y z^2) - 2\lambda yz \alpha(\beta' \beta'' y - \gamma' \gamma'' z)]^2 \\ = 4\lambda y^2 z^2 [-\lambda \alpha^2 (\beta' \beta'' y - \gamma' \gamma'' z)^2 - (\alpha \alpha')^2 (\nu y^2 + \mu z^2) - \alpha^2 \alpha' \alpha'' (\gamma' y + \beta' z)(\gamma'' y + \beta'' z)],$$

viz., omitting terms which destroy each other, this is

$$[(y - z)^2 + 4\lambda yz] (\alpha \alpha')^2 y^2 z^2 - [\alpha \alpha' (y^2 z - y z^2) - 2\lambda yz \alpha(\beta' \beta'' y - \gamma' \gamma'' z)]^2 \\ = 4\lambda y^2 z^2 [-\lambda \alpha^2 (\beta' \beta'' y - \gamma' \gamma'' z)^2 - \alpha^2 \alpha' \alpha'' (\gamma' y + \beta' z)(\gamma'' y + \beta'' z)].$$

Or again, this is

$$4\lambda yz (\alpha \alpha')^2 y^2 z^2 + 4\lambda yz \alpha^2 \alpha' \alpha'' (y^2 z - y z^2)(\beta' \beta'' y - \gamma' \gamma'' z) \\ - 4\lambda^2 y^2 z^2 \alpha^2 (\beta' \beta'' y - \gamma' \gamma'' z)^2 = -4\lambda yz \alpha^2 \alpha' \alpha'' (\gamma' y + \beta' z)(\gamma'' y + \beta'' z);$$

viz., this is

$$\alpha' \alpha'' yz + (y - z)(\beta' \beta'' y - \gamma' \gamma'' z) = (\gamma' y + \beta' z)(\gamma'' y + \beta'' z),$$

which is at once seen to be true.

175. Again, compare the terms which contain $x^2 yz$. On the right-hand side, we have

$$(-K\nu - \mu - \nu - \frac{1}{4} \dots) 4\beta \gamma yz \dots \text{ (a term in } x^2 yz \text{ of } \alpha' \alpha'' \beta \beta' \gamma \gamma'' P' P'')^2,$$

viz., the coefficient is

$$4\beta \gamma \{ -\mu(\gamma'' \gamma')^2 - \nu(\beta' \beta'')^2 + \beta' \beta'' \gamma' \gamma'' \}.$$

On the left-hand side, that is in $A'C - B'^2$, the only terms which give rise to the terms in question are

$$\text{in } A', (1 - 4\mu\nu)x^2; \text{ in } C, 2\beta\beta'\beta''\gamma\gamma'\gamma'' x^2 yz;$$

and in B^2

$$x^2 z (-\gamma \gamma' \gamma'' - 2\mu \gamma' \beta' \beta'') - x^2 y (\beta \beta' \beta'' - 2\nu \beta \gamma' \gamma''),$$

whence the coefficient is

$$-2\beta\beta'\beta''\gamma\gamma'\gamma''(1 - 4\mu\nu) + 2(\gamma\gamma'\gamma'' + 2\mu\beta'\beta'')(\beta\beta'\beta'' + 2\nu\beta\gamma'\gamma''),$$

which is in fact

$$= 4\beta\gamma \{ -\mu(\gamma' \gamma'')^2 - \nu(\beta' \beta'')^2 + \beta' \beta'' \gamma' \gamma'' \},$$

which is right; and the verification may be completed without difficulty.

176. The singular planes are

$$w = 0, \quad (4)$$

$$P' - p_1 P'' = 0, \quad (12)$$

$$P' - p_2 P'' = 0, \quad (16)$$

$$z = 0, \quad (3)$$

and the nodes are

$$(0, 0, 0, 1), \quad (8)$$

$$(1, 0, 0, 0), \quad (5)$$

$$(0, 1, 0, 0), \quad (6)$$

$$(0, 0, 1, 0), \quad (7)$$

$$\left(\frac{\alpha\alpha'\alpha''}{\alpha' - \alpha p_1}, \frac{\beta\beta'\beta''}{\beta' - \beta p_1}, \frac{\gamma\gamma'\gamma''}{\gamma' - \gamma p_1}, 0 \right), \quad (16)$$

$$\left(\frac{\alpha\alpha'\alpha''}{\alpha' - \alpha p_2}, \frac{\beta\beta'\beta''}{\beta' - \beta p_2}, \frac{\gamma\gamma'\gamma''}{\gamma' - \gamma p_2}, 0 \right), \quad (12)$$

$$(\alpha'\alpha'', \beta'\beta'', \gamma'\gamma'', 0), \quad (4)$$

$$\text{three nodes on line } P' - p_1 P'' = 0, \quad z = 0 \quad (13, 14, 15)$$

$$\text{three nodes on line } P' - p_2 P'' = 0, \quad z = 0 \quad (9, 10, 11)$$

where the large numbers are those referring to the 16-nodal surface, and are here adopted. In the foregoing formulae p_1, p_2 are the roots of

177.

$$\begin{aligned} & [\lambda(\beta'\gamma'')^2 + \mu(\gamma'\alpha'')^2 + \nu(\alpha'\beta'')^2](p\alpha'\beta'\gamma')^2 \\ & + [\lambda(\beta''\gamma'')^2 + \mu(\gamma''\alpha'')^2 + \nu(\alpha''\beta'')^2](\alpha''\beta''\gamma'')^2 \\ & - 2[\lambda\beta'\beta''\gamma'\gamma'' + \mu\gamma'\alpha'\gamma''\alpha'' + \nu\alpha'\beta'\alpha''\beta'' + \frac{1}{4}(\omega - \varpi)^2] p\alpha'\beta'\gamma'\alpha''\beta''\gamma'' = 0. \end{aligned}$$

We have the square diagram in the following page:

[The original page 296 here gives a 13×13 square diagram for the $13(\alpha)$ -nodal surface, with notations including the cubic cone C (or G) replacing the planes 5, 6, 7 at node 4, the trinodal quartic cone (8; 1, 2, 3) at node 8, the nodal cubic (2, 3; 5) at node 5, &c. The vacant places are understood to be CC . See PDF page 296 for the full table.]

where for greater clearness I have omitted the symbol CC , which is to be understood as occupying each of the vacant squares.

The arrangement is the same as before; the right-hand margin shows the sextic cone; viz., for the node 4 this is made up of the singular planes 4, 12, 16 and of a cubic cone represented by (5, 6, 7) (as replacing the planes 5, 6, 7 in the case of the 16-nodal surface). Similarly for the node 5, the sextic cone is made up of the singular plane 4, the nodal cubic cone (2, 3; 5), and the quadric cone (9, 13) (the numbers in these last symbols indicating the planes in the case of the 16-nodal surface, which are here replaced by cones). So for the node 8, the sextic cone is made up of the singular planes 12, 16 and of the trinodal quartic cone (8; 1, 2, 3). As regards a nodal cubic cone, for example (2, 3; 5), the semicolon is used to indicate that the nodal line replaces the intersection of the planes 2, 3; the other intersections 2, 5 and 3, 5 having disappeared. And so for a trinodal quartic cone (8; 1, 2, 3), the semicolon is used to indicate that the nodal lines replace the intersection (1, 2), the intersection (1, 3), and the intersection (2, 3) respectively; the other intersections 1, 8; 2, 8; and 3, 8 having disappeared. Finally, in the body of the table, C

is used to denote the cubic or the quartic cone (as the case may be); $\times$ to denote a nodal line of either of these cones; and C' the quadric cone; as already mentioned, the vacant places are considered to be CC .

The reading of the table is then as follows; viz., for the node 4, the remaining twelve nodes lie on the nodal lines of the sextic cone 4, 12, 16, (5, 6, 7), as shown; viz., 5, 6, 7 are each of them on the intersection of the cubic cone with the plane 4; 8 is on the intersection of the planes 12 and 16; and so on.

I reserve for another Memoir the discussion of the $13(\beta)$ -nodal surface, and the surfaces with less than 13 nodes.

[455] ON PLÜCKER'S MODELS OF CERTAIN QUARTIC SURFACES.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 281–285.
Read June 8, 1871.]

The Society possesses a series of 14 wooden models of surfaces, constructed under the direction of the late Prof. Plücker, in illustration of the theory developed in his posthumous work, “*Neue Geometrie des Raumes gegründet auf die Betrachtung der geraden Linie als Raumelemente*,” Leipzig, 1869. These all of them represent, I believe, Equatorial Surfaces; viz., models 1 to 8 represent cases of the 78 forms of equatorial surfaces “*deren Breiten-Curven eine feste Axenrichtung besitzen*,” vol. ii. pp. 352–363, the remaining models, Nos. 9–14, I have not completely identified. I propose to go into the theory only so far as is required for the explanation of the models.

In a “Complex,” or triply infinite system of lines, there is in any plane whatever a singly infinite system of lines enveloping a curve; and if we attend only to the curves the planes of which pass through a given fixed line, the locus of these curves is a “complex surface.” Similarly, there is through any point whatever a single infinite series of lines generating a cone; and if we attend only to the cones which have their vertices in the given fixed line, then the envelope of these cones is the same complex surface. In the case considered of a complex of the second degree, the curves and cones are each of them of the second order; the fixed line is a double line on the surface, so that (attending to the first mode of generation) the complete section by any plane through the fixed line is made up of this line twice, and of a conic; the surface is thus of the order 4: it is also of the class 4; the surface has, in fact, the nodal line, and also 8 nodes (conical points), and we have thus a reduction = 32 in the class of the surface.

In the particular case where the nodal line is at infinity, the complex surface becomes an equatorial surface; viz., (attending to the first mode of generation) we have here a series of parallel planes each containing a conic, and the locus of these conics is the equatorial surface.

It is convenient to remark that, taking a, b, h to be homogeneous functions of (x, w) of the order 2; f, g of the order 1; and c of the order 0 (a constant); then the equation of a complex surface (see vol. i. p. 162) is

$$\begin{vmatrix} \cdot, & y, & z, & 1 \\ y, & a, & h, & g \\ z, & h, & b, & f \\ 1, & g, & f, & c \end{vmatrix} = 0;$$

and that, writing $w = 1$, or considering $a, h, b; f, g; c$ as functions of x of the orders 2, 1, 0 respectively, we have an equatorial surface.

A particular form of equatorial surface is thus $bcy^2 + caz^2 + ab = 0$, or taking $c = 1$, this is $by^2 + az^2 + ab = 0$, where a, b are quadric functions of x .

The surface is still, in general, of the fourth order: it may however degenerate into a cubic surface, or even into a quadric surface; the last case is however excluded from the enumeration. The section by any plane parallel to that of yz is a conic; the section by the plane $y = 0$ is made up of the pair of lines $a = 0$, and of the conic $z^2 + b = 0$; that by the plane $z = 0$ is made up of the pair of lines $b = 0$, and of the conic $y^2 + a = 0$; the last-mentioned planes may be called the principal planes, and the conics contained in them principal conics. The surface is thus the locus of a variable conic, the plane of which is parallel to that of yz , and which has for its vertices the intersections of its plane with the two principal conics respectively. And we have thus the particular equatorial surfaces considered by Plücker, vol. ii. pp. 346–363 (as already mentioned), under the form

$$\frac{y^2}{Ex^2 + 2Ux + G} + \frac{z^2}{Fx^2 + 2Rx + B} + 1 = 0,$$

and of which he enumerates 78 kinds; viz., these are

- 1 to 17 Principal conics each proper.
 18 „ 29 One of them a line-pair.
 30 „ 32 Each a line-pair.
 33 „ 39 Principal conics, each proper, but having a common point.
 40 „ 43 One of them a line-pair, its centre on the other principal conic.
 44 „ 61 One principal conic a parabola.
 62 „ 73 One principal conic a pair of parallel lines.
 74 „ 76 Principal conics each a parabola.
 77 and 78 Principal conics, one of them a parabola, the other a pair of parallel lines.

The models 1 to 8 correspond hereto, as follows:

Mod. 1 is 13,	i.e. 2 is Mod. 9,
„ 2 9,	„ 3 40,
„ 3 40,	„ 4 34,
„ 4 34,	„ 5 2,
„ 5 2,	„ 6 3,
„ 6 3,	„ 7 4,
„ 7 4,	„ 8 32,
„ 8 32,	„ 13 1.

Mod. 5 is 2: the form of the equation is here

$$\frac{y^2}{l^2[(x - \alpha)^2 + \beta^2]} + \frac{z^2}{l'^2[(x - \alpha')^2 + \beta'^2]} = 1;$$

viz., the principal conics are one of them a hyperbola, the other imaginary; hence the generating conic has always two, and only two, real vertices, viz., it is always a hyperbola; there are no real lines.

Mod. 6 is 3: the form of the equation is

$$\frac{y^2}{l^2[(x - \alpha)^2 + \beta^2]} + \frac{z^2}{l'^2[(x - \alpha')^2 + \beta'^2]} = 1;$$

viz., the principal conics are each of them a hyperbola; the generating conic has four real vertices, viz., it is always an ellipse: there are no real lines.

Mod. 7 is 4: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2[(x-\alpha')^2 + \beta'^2]} + 1 = 0.$$

The principal conics are one of them an ellipse, the other imaginary; for values of x between γ and δ , the variable conic has two real vertices or it is a hyperbola; for any other values it is imaginary, so that the surface lies wholly between the planes $x = \gamma$, $x = \delta$: the surface contains the real lines $y = 0$, $x = \gamma$ and $y = 0$, $x = \delta$.

Mod. 2 is 9: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta')} + 1 = 0,$$

where, say the values γ , δ lie between the values γ' , δ' : the principal conics are each of them an ellipse, the vertices (on the axis or line $y = 0$, $z = 0$) of the one ellipse lying between those of the other ellipse. The variable conic for values of x between γ , δ has four real vertices, or it is an ellipse; for values beyond these limits, but within the limits γ' , δ' , say from γ to γ' , and from δ to δ' , there are two real vertices, or the conic is a hyperbola; and for values beyond the limits γ' , δ' , the variable conic is imaginary.

There are four real lines ($y = 0$, $x = \gamma$), ($y = 0$, $x = \delta$), ($z = 0$, $x = \gamma'$), ($z = 0$, $x = \delta'$). The surface consists of a central pillow-like portion, joined on by two conical points to an upper portion, and by two conical points to an under portion, the whole being included between the planes $x = \gamma'$, $x = \delta'$.

Mod. 1 is 13: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta')} + 1 = 0;$$

the values γ' , δ' lying between γ , δ ; the principal conics are one of them a hyperbola, the other an ellipse, the vertices (on the axis or line $y = 0$, $z = 0$) of the hyperbola lying between those of the ellipse.

The variable conic, for values of x between γ' , δ' , has two real vertices, or it is a hyperbola; for the values, say, from γ' to γ , and δ' to δ , there are four real vertices, or the conic is an ellipse; for values beyond the limits γ , δ , there are two real vertices, and the conic is a hyperbola. There are the four real lines ($y = 0$, $x = \gamma$), ($y = 0$, $x = \delta$), and ($z = 0$, $x = \gamma'$), ($z = 0$, $x = \delta'$). The surface consists of 8 portions joined to each other by 8 conical points, but the form can scarcely be explained by a description.

Mod. 8 is 32: the form of the equation is

$$\frac{y^2}{l^2(x-\gamma)^2} + \frac{z^2}{l'^2(x-\gamma')^2} = 1;$$

viz., the principal conics are each of them a line-pair, the variable conic is always an ellipse.

There are the two real nodal lines ($y = 0$, $x = \gamma$) and ($z = 0$, $x = \gamma'$), each of these being in the neighbourhood of the axis crunodal, and beyond certain limits acnodal; the surface is a scroll, being, in fact, the well-known surface which is the boundary of a small circular pencil of rays obliquely reflected, and consequently passing through two focal lines.

Mod. 4 is 34: the equation is

$$\frac{y^2}{l^2(x-\gamma)(x-\delta)} + \frac{z^2}{l'^2(x-\gamma')(x-\delta)} + 1 = 0,$$

where $x = \delta$ is not intermediate between the values $x = \gamma$ and $x = \gamma'$; say the order is δ, γ, γ' . The surface is thus a cubic surface; the principal conics are ellipses having on the axis a common vertex at the point $x = \delta$, and the remaining two vertices on the same side of the last-mentioned one. The variable conic for values between δ and γ has four real vertices, or it is an ellipse; for values between γ and γ' two real vertices, or it is a hyperbola; and for values beyond the limits δ, γ' it is imaginary. There are on the surface the two real lines ($y = 0, x = \gamma$) and ($z = 0, x = \gamma'$). The surface consists of a finite portion joined on by two conical points to the remaining portion.

Mod. 3 is 40: the form of equation is

$$\frac{y^2}{l^2(x - \gamma)(x - \delta)} + \frac{z^2}{l'^2(x - \delta)^2} + 1 = 0.$$

The surface is thus a cubic surface; the principal conics are, one of them an ellipse, the other a pair of imaginary lines intersecting on the ellipse; for values of x between γ and δ , the variable conic has thus two real vertices, and it is a hyperbola; for values beyond these limits it is imaginary, and the whole surface is thus included between the planes $x = \gamma$ and $x = \delta$. There are the two real lines ($y = 0, x = \gamma$) and ($z = 0, x = \delta$).

Taking $l^2 = l'^2 = 1$, the surface is

$$\frac{y^2}{(x - \gamma)(x - \delta)^2(x - \delta)^2} + \frac{z^2}{\dots} + 1 = 0,$$

which is a particular case of the parabolic cyclide.

As already mentioned, I have not completely identified the remaining models 9 to 14, but I will say a few words about them.

The equatorial surfaces, not included in the preceding 78 cases, Plücker distinguishes (vol. ii. p. 363) as “gedrehte” or “tordirte,” say as twisted equatorial surfaces; the equation of such a surface is

$$by^2 + 2hyz + az^2 + ab - h^2 = 0,$$

where

$$b = Fx^2 - 2Rx + B,$$

$$a = Ex^2 + 2Ux + G,$$

$$h = Kx^2 - Ox - G \text{ (or in particular } = -Ox - G).$$

Mod. 13 is such a surface, being a twisted form of model 2.

Mod. 9 and Mod. 14 belong, I think, to the case $a = 0$; viz., the form of the equation is $by^2 + 2hyz - h^2 = 0$. The variable conic is a hyperbola, the direction of one of the asymptotes being constant (vol. ii. p. 368).

There are moreover (p. 372) equatorial surfaces in which the variable conic is always a parabola, and where there are on the surface four real or imaginary lines.

Mods. 10, 11, and 12 seem to represent such surfaces.

[456] NOTE ON THE DISCRIMINANT OF A BINARY QUANTIC.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. X. (1870), p. 23.]

It is well known that the discriminant of a binary quantic $(a, b, c, d, \dots | t, 1)^n$ is of the form

$$Ma + Nb^2,$$

but it is further to be remarked that if $b = 0$, then the form is

$$a(Ma + Nc^3),$$

if $b = 0$, $c = 0$, the form is

$$a^2(Ma + Nd^4),$$

and so on, until only the lowest two coefficients are not put $= 0$. Or, what is the same thing, if in the discriminant of the original function we put $a = 0$, then the discriminant divides by b^2 ; if $b = 0$, the discriminant divides by a , and, omitting this factor, if we then write $a = 0$, it divides by c^3 ; if $b = 0$, $c = 0$, the discriminant divides by a^2 , and omitting this factor, if we then write $a = 0$, it divides by d^4 ; and so on, until as before.

Thus if $b = 0$, the discriminant of $(a, 0, c, d, e | t, 1)^4$, divides by a , and omitting this factor it is

$$\begin{aligned} & a^2e^2 \\ & - 18ace^2 \\ & + 54acd^2e \\ & - 27ad^4 \\ & + 81c^4e \\ & - 54c^3d^2, \end{aligned}$$

which for $a = 0$ has the factor c^3 ; if $b = 0$, $c = 0$, the discriminant of $(a, 0, 0, d, e | t, 1)^4$ has the factor a^2 , and omitting this factor it is

$$ae^4 - 27d^4,$$

which for $a = 0$ has the factor d^4 ; the series of theorems here terminates, since the lowest two coefficients d, e are not to be put $= 0$.

(End of chunk: PDF pages 276–325, book pages 254–303.

Paper 450 (tail, two pages),

paper 451 in full,

paper 452 in full,

paper 453 in full,

paper 454 in full,

paper 455 in full,

paper 456 opening (one page).)

[457] ON THE QUARTIC SURFACES $(*|U, V, W)^2 = 0$.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. X. (1870), pp. 24–34.]

I propose to myself for investigation the quartic surfaces represented by an equation

$$(*|U, V, W)^2 = 0,$$

where U, V, W are quadric functions of the coordinates.

Such a surface has 8 nodes (conical points), viz., these are the points of intersection of the quadric surfaces $U = 0, V = 0, W = 0$. It is to be observed, that not every quartic surface with 8 nodes is included under the above form; in fact the equation of a quartic surface contains (homogeneously) 35 coefficients, or say 34 arbitrary parameters; in order that a given point may be a node, 4 conditions must be satisfied, and it is consequently possible to find a quartic surface having 8 given points as nodes (and having in its equation $(34 - 8 \cdot 4 =) 2$ arbitrary parameters): but 8 given points are not in general the intersections of three quadric surfaces, and such a quartic surface is therefore not in general included under the above form. I think, however, that it may be assumed that the above form includes all the quartic surfaces having 8 nodes, points of intersection of three quadric surfaces. It will presently appear that, included in the form, we have surfaces where (instead of the 8 nodes) there is a nodal or cuspidal conic; and that these are the most general forms of such quartic surfaces.

A quartic surface has at most 16 nodes, and the general form with 8 nodes must admit of being particularised so that the surface shall acquire any number not exceeding 8 of additional nodes. This does not show, but it is probable, that the above special form with 8 nodes can be particularised so that the surface shall in like manner acquire any number not exceeding 8 of additional nodes. Similarly, a quartic surface with a nodal conic may have besides 1, 2, 3, or 4 nodes; and it will be shown in the sequel how the form, particularised so as to give a nodal conic, may be further particularised so as to give the 1, 2, 3, or 4 nodes. So a quartic surface with a cuspidal conic may besides have 1 node, and it will be shown how the form, particularised so as to give a cuspidal conic, may be further particularised so as to give 1 node.

Starting from the equation $(*|U, V, W)^2 = 0$, we may, by substituting for U, V, W linear functions of these expressions, transform the equation precisely in the manner of a conic, and therefore into any of the forms under which the equation of a conic can be exhibited; for instance, in the forms

$$aU^2 + bV^2 + cW^2 = 0, \quad fVW + gWU + hUV = 0, \quad UW - V^2 = 0, \quad \&c.$$

I attend at present only to the last-mentioned form $UW - V^2 = 0$, which, it thus appears, is equally general with the original form $(*|U, V, W)^2 = 0$.

The quartic surface

$$UW - V^2 = 0,$$

where U, V, W are any quadric functions of the coordinates, may be considered as the envelope of the quadric surface

$$(U, V, W|\theta, 1)^2 = 0,$$

where θ is an arbitrary parameter. And it thus appears that it is very easy to reciprocate (in regard to any given quadric surface) the quartic surface. For the reciprocal of the quartic surface is clearly the envelope of the reciprocal of the variable quadric surface; this reciprocal is itself a quadric surface, and the reciprocal of the quartic surface is thus given in the same form as the original surface, viz., as the envelope of a quadric surface the equation whereof contains rationally the variable parameter θ ; the equation of the reciprocal surface is consequently obtained by equating to zero the discriminant in regard to θ , of the equation of the reciprocal quadric surface.

It is to be observed that, inasmuch as the equation of the reciprocal quadric surface is of the third degree in the coefficients of the original quadric, it is in general of the degree 6 in the parameter; we have thus a sextic function of θ , the coefficients whereof are quadric functions of the coordinates; and the discriminant is a function of the order 10 in these coefficients, that is, of the order 20 in the coordinates. The reciprocal of the quartic surface is thus a surface of the order 20; this is right, for in a general quartic surface the order of the reciprocal surface is 36, and the 8 nodes reduce the order by 16; $36 - 16 = 20$.

In the equation $UW - V^2 = 0$, if U reduce itself to the square of a linear function, $= P^2$, the equation becomes $V^2 - P^2W = 0$, which is the general form of the quartic surface having the nodal conic $V = 0, P = 0$. And if, moreover, W be the product of this same linear function P by another linear function Q , $W = PQ$, then the form is $V^2 - P^3Q = 0$, which is the general form of the quartic surface having the cuspidal conic $V = 0, P = 0$.

Writing for greater convenience x, y in the place of P, Q respectively, we have the quartic

$$V^2 - x^3y = 0, \quad (\text{AA})$$

having the cuspidal conic $V = 0, x = 0$; and which has besides the conic of plane contact $V = 0, y = 0$. In virtue of the cuspidal conic the reciprocal surface should be of the order 6; and by the foregoing method of obtaining the equation of the reciprocal surface, I will verify that this is so. To effect this as simply as possible, I fix the remaining coordinates z, w as follows. The line $x = 0, y = 0$ is not in general a tangent to the surface $V = 0$; it therefore meets this surface in two points, and we may take $z = 0, w = 0$ to be the equations of the tangent planes at these two points respectively; we have thus $V = ax^2 + 2hxy + by^2 + 2nzw$. Introducing for convenience the numerical factor 2, and taking the equation of the surface to be

$$(ax^2 + 2hxy + by^2 + 2nzw)^2 - 2x^3y = 0,$$

this is the envelope of the quadric surface

$$\theta^2 \cdot x^2 + 2\theta(ax^2 + 2hxy + by^2 + 2nzw) + 2xy = 0,$$

which is a surface $(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{f}, \mathbf{g}, \mathbf{h}, \mathbf{l}, \mathbf{m}, \mathbf{n} | x, y, z, w)^2 = 0$, where $\mathbf{a} = \theta^2 + 2a\theta$, $\mathbf{b} = 2b\theta$, $\mathbf{h} = 2\theta h + 1$, $\mathbf{n} = 2\theta n$, and where all the other coefficients vanish. Assuming, as usual, that the reciprocation is effected in regard to the surface $x^2 + y^2 + z^2 + w^2 = 0$, the general equation is

$$\begin{aligned} & x^2 \cdot \mathbf{d}(\mathbf{bc} - \mathbf{f}^2) - \mathbf{cm}^2 - \mathbf{bn}^2 + 2\mathbf{fmn} \\ & + y^2 \cdot \mathbf{d}(\mathbf{ca} - \mathbf{g}^2) - \mathbf{an}^2 - \mathbf{cl}^2 + 2\mathbf{gnl} \\ & + z^2 \cdot \mathbf{d}(\mathbf{ab} - \mathbf{h}^2) - \mathbf{bl}^2 - \mathbf{am}^2 + 2\mathbf{hlm} \\ & + w^2 \cdot \mathbf{abc} - \mathbf{af}^2 - \mathbf{bg}^2 - \mathbf{ch}^2 + 2\mathbf{fgh} \\ & + 2yz \cdot \mathbf{d}(\mathbf{gh} - \mathbf{af}) + \mathbf{l}^2\mathbf{f} + \mathbf{amn} - \mathbf{hnl} - \mathbf{glm} \\ & + 2zx \cdot \mathbf{d}(\mathbf{hf} - \mathbf{bg}) + \mathbf{m}^2\mathbf{g} + \mathbf{bnl} - \mathbf{fhn} - \mathbf{hmn} \\ & + 2xy \cdot \mathbf{d}(\mathbf{fg} - \mathbf{ch}) + \mathbf{n}^2\mathbf{h} + \mathbf{clm} - \mathbf{gmn} - \mathbf{fnl} \\ & + 2xw \cdot -\mathbf{l}(\mathbf{bc} - \mathbf{f}^2) - \mathbf{m}(\mathbf{hf} - \mathbf{bg}) - \mathbf{n}(\mathbf{fg} - \mathbf{ch}) \\ & + 2yw \cdot -\mathbf{m}(\mathbf{ca} - \mathbf{g}^2) - \mathbf{l}(\mathbf{fg} - \mathbf{ch}) - \mathbf{n}(\mathbf{gh} - \mathbf{af}) \\ & + 2zw \cdot -\mathbf{n}(\mathbf{ab} - \mathbf{h}^2) - \mathbf{m}(\mathbf{gh} - \mathbf{af}) - \mathbf{l}(\mathbf{hf} - \mathbf{bg}), \end{aligned}$$

(I write down this general result as it will be useful for reference in other cases); in the present case this becomes simply

$$x^2 \cdot \mathbf{bn}^2 + y^2 \cdot \mathbf{an}^2 + 2xy \cdot \mathbf{nh}^2 + 2zw(-\mathbf{nab} + \mathbf{nh}^2) = 0,$$

where $a, b, \mathfrak{n}, \mathfrak{h}$ have the foregoing values; the equation is thus only of the order 4 in regard to θ ; but it in fact divides by $\mathfrak{n} (= 2\theta n)$ and thus reduces itself to the third order, viz. it becomes

$$\mathfrak{n}(bx^2 + ay^2) - 2\mathfrak{h}^2xy + 2(ab - \mathfrak{h}^2)zw = 0,$$

or, substituting for $\mathfrak{a}, \mathfrak{h}, \mathfrak{n}, \mathfrak{b}$ their values, this is

$$x^2 \cdot 4bn\theta^3 + 2y^2(2n\theta^3 + 4an\theta^2) + 2zw(2b\theta^2 + 4ab\theta) - 2(xy + zw)(2\theta h + 1)^2 = 0,$$

say this is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} | \theta, 1)^3 = 0,$$

where $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ are each of them a quadric function of the coordinates; it being observed that $\mathfrak{C}$ and $\mathfrak{D}$ are respectively numerical multiples of the same function $(xy + zw)$. Hence equating the discriminant to zero, we have

$$\mathfrak{A}^2\mathfrak{D}^2 + 4\mathfrak{A}\mathfrak{C}^3 + 4\mathfrak{B}^3\mathfrak{D} - 3\mathfrak{B}^2\mathfrak{C}^2 - 6\mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D} = 0,$$

which equation, inasmuch as every term contains either $\mathfrak{C}$ or $\mathfrak{D}$ as a factor, divides by $xy + zw$, and thus becomes an equation of the order 6 in the coordinates: that is, the order of the reciprocal surface is 6. Multiplying by $\frac{1}{8}$ to avoid fractions, the actual values of $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ are

$$\begin{aligned}\mathfrak{A} &= bn^2 + 3(ny^2 + 2hzw), \\ \mathfrak{B} &= 2\{hnx^2 + any^2 + 2ahzw - 2h^2(xy + zw)\}, \\ \mathfrak{C} &= -4fh(xy + zw), \\ \mathfrak{D} &= -3(xy + zw),\end{aligned}$$

or say $\mathfrak{A} = 3\alpha, \mathfrak{B} = 2\beta, \mathfrak{C} = -4h\gamma, \mathfrak{D} = -3\gamma$; where $\gamma = xy + zw$; substituting these values and omitting the factor 8γ , the equation is

$$27\alpha^3\gamma - 256h^4\alpha\gamma^3 - 32\beta^3 - 64h^3\beta^2\gamma^2 - 144h\alpha\beta^2\gamma = 0,$$

which is an equation of the form $(* | \alpha, \beta, \gamma)^3 = 0$. The sextic surface has thus singular points $\alpha = 0, \beta = 0, \gamma = 0$, viz. these are the two points $(x = 0, y = 0, z = 0), (x = 0, y = 0, w = 0)$ each four times. The further discussion of the sextic surface is reserved for another occasion.

I do not at present attempt to enumerate the particular cases of the surface $V^2 - x^3y = 0$, but content myself with the discussion of a particular case in which the order of the reciprocal surface is $= 3$. Suppose that $V = 0$ is a cone, $y = 0$ a tangent plane to the cone (so that the conic $y = 0, V = 0$ breaks up into a line twice repeated), $x = 0$ an arbitrary plane (so that we have still the proper cuspidal conic $x = 0, V = 0$). Any other tangent plane of the cone may be taken for the plane $z = 0$; the plane containing the lines of contact of the two tangent planes for the plane $w = 0$; the equation of the conic then is $V = dw^2 + 2fyz = 0$; and the equation of the surface is

$$(dw^2 + 2fyz)^2 - x^3y = 0. \tag{AB}$$

For convenience of comparison, I change x, y, w, z into y, w, z, x , and assign numerical values to the coefficients, writing the equation under the form

$$27(4aw + z^2)^2 - 64y^3w = 0. \tag{AB}$$

The quartic is here the envelope of the quadric surface

$$\theta^2 \cdot 4yw + \theta \cdot 9(4aw + z^2) + 12yz = 0,$$

viz., comparing with the general form

$$0 = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n} | x, y, z, w)^2,$$

we have $\mathfrak{b} = 12$, $\mathfrak{c} = 9\theta$, $\mathfrak{l} = 18\theta$, $\mathfrak{m} = 2\theta^2$, all the other coefficients vanishing. The reciprocal equation is

$$\mathfrak{d}\mathfrak{m}^2 + x^2(-\mathfrak{c}\mathfrak{m}^2) + y^2(-\mathfrak{c}\mathfrak{l}^2) + z^2(-\mathfrak{b}\mathfrak{l}^2) + 2xw \cdot \mathfrak{m}^2\mathfrak{g} + 2xw(-\mathfrak{l}\mathfrak{b}\mathfrak{c}) = 0,$$

or substituting for $\mathfrak{b}$, $\mathfrak{c}$, $\mathfrak{l}$, $\mathfrak{m}$ their values, this is found to be

$$\theta(\theta x - 9y)^2 + 108(z^2 + xw) = 0.$$

Representing this by $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} | \theta, 1)^3 = 0$, the discriminant in regard to θ would, in virtue of the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, contain $\mathfrak{D}$ as a factor; the reason of this appears from the original form; in fact, forming the derived equation in regard to θ , this is found to be $(\theta x - 9y)(\theta x - 3y) = 0$; the value $\theta x - 9y = 0$ gives as a factor of the discriminant $z^2 + xw$; the value $\theta x - 3y = 0$ gives $3y(-6y)^2 + 108x(z^2 + xw)$, that is the factor $y^3 + x(z^2 + xw)$; the complete value of the discriminant as obtained by substitution of the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ being $x^2(z^2 + xw)\{y^3 + x(z^2 + xw)\}$; the equation of the reciprocal surface is

$$y^3 + x(z^2 + xw) = 0,$$

viz. this is a cubic surface. Prof. Schläfli's Case XX., having a uniplanar point $x = 0$, $y = 0$, $z = 0$ reducing the class by 8, and so giving a reciprocal surface of the order $(12 - 8) = 4$, viz. the surface $27(4xw + z^2)^2 - 64y^3w = 0$. See the Memoir, Schläfli, "On the distribution of surfaces of the third order into species in reference to the absence or presence of singular points and the reality of their lines," *Phil. Trans.*, vol. CLIII. (1853), pp. 193–241.

I pass to the case of a surface

$$V^2 - P^2U = 0$$

having a nodal conic $V = 0$, $P = 0$, but not having in general any nodes. And I propose to show how the constants may be determined so that the surface shall have 1, 2, 3, or 4 nodes. It is to be remarked that in the above equation the plane $P = 0$ is a determinate plane, but the quadric surface $U = 0$ is not a determinate quadric, we may in fact substitute for it the quadric $U + \lambda P^2 = 0$, writing the equation under the form

$$V^2 + 2\lambda VP^2 + \lambda^2 P^4 = P^2(U + \lambda^2 P^2),$$

so that we may without loss of generality, by means of the disposable constant λ , subject the surface $V = 0$ to any single condition; for instance, take it to be a cone, or to pass through a given point, &c.

Taking the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$ to be arbitrary planes, the implicit constant factors in these equations may be determined in such wise that the equation of the given plane $P = 0$ shall be $x + y + z + w = 0$. The equation of the surface will then be

$$\begin{aligned} & \{(a, b, c, d, f, g, h, l, m, n | x, y, z, w)^2\}^2 \\ & = (x + y + z + w)^2 \cdot (a', b', c', d', f', g', h', l', m', n' | x, y, z, w)^2, \end{aligned}$$

and we may assume that the node or nodes (if any) lie at a vertex or vertices of the tetrahedron $x = 0$, $y = 0$, $z = 0$, $w = 0$, say at the points A , B , C , D . The conditions for a node at each of these points are at once found to be

Node at A ,	Node at B ,	Node at C ,	Node at D ,
$2a^2 = a'$,	$2bh = h' + b'$,	$2cg = g' + c'$,	$2dl = l' + d'$,
$2ah = h' + a'$,	$2b^2 = b'$,	$2cf = f' + c'$,	$2dm = m' + d'$,
$2ag = g' + a'$,	$2hf = f' + h'$,	$2c^2 = c'$,	$2dn = n' + d'$,
$2al = l' + a'$,	$2bm = m' + b'$,	$2cn = n' + c'$,	$2d^2 = d'$.

The first set of equations gives

$$a' = a^2, \quad h' = 2ah - a^2, \quad g' = 2ag - g^2, \quad l' = 2al - a^2.$$

If the first and second sets are satisfied simultaneously, we have $2(a - b)h = a^2 - b^2$, that is $a = b$, or else $h = \frac{1}{2}(a + b)$; that is, the two sets may be satisfied in two different ways according as a and b are equal or unequal. Similarly the first, second, and third sets may be satisfied in three different ways and the four sets in five different ways according as there are or are not any equalities between a, b, c , and between a, b, c and d respectively. The several solutions are shown in the annexed table, viz., in the line I no set is satisfied; in the line II only the first set; in the lines III and IV the first and second sets; in the lines V to VII the first, second, and third sets; and in the lines VIII to XII the four sets.

[See next page for this Table, which should come in here.]

I is the general case, $V^2 = P^2U$, of a quartic and a nodal conic but without nodes. (AC).

II is the case of a single node; writing, as without loss of generality we may do, $a = 0$, the equation is

$$\{(0, b, c, d, f, g, h, l, m, n | x, y, z, w)^2\}^2 = (x + y + z + w)^2 \cdot (b, c, d, n, m, f | y, z, w)^2,$$

viz., the quadric $U = 0$ is here a cone having its vertex on the quadric $V = 0$. (AD).

III and IV are two cases each of them with two nodes, viz. III, the equation is

$$\begin{aligned} &\{(ax + by)(x + y) + cz^2 + 2nzw + dw^2 + 2x(gz + lw) + 2y(fz + mw)\}^2 \\ &= (x + y + z + w)^2 [(ax + by)^2 + c'z^2 + 2n'zw + d'w^2 \\ &+ 2x\{(2ag - a^2)z + (2al - a^2)w\} + 2y\{(2bf - b^2)z + (2bm - b^2)w\}], \quad (AE) \end{aligned}$$

where it is to be observed that the line $z = 0, w = 0$ joining the two nodes ($y = 0, z = 0, w = 0$) and ($w = 0, z = 0, x = 0$) is a line on the surface. Writing, as we may do, $a = 0$, the equation assumes the more simple form

$$\begin{aligned} &\{by(x + y) + cz^2 + 2nzw + dw^2 + 2x(gz + lw) + 2y(fz + mw)\}^2 \\ &= (x + y + z + w)^2 [b^2y^2 + c'z^2 + 2n'zw + d'w^2 + 2y\{(2bf - b^2)z + (2bm - b^2)w\}]. \quad (AE) \end{aligned}$$

[Page 310 of the original carries here a full-page sideways table classifying the twelve cases I–XII according to the equalities between a, b, c, d and the consequent values of $a', b', c', d', h', g', l', f', m', n'$. The table is reproduced in schematic form below.]

Line	Conditions on a, b, c, d	Number of nodes
I	all distinct, no set satisfied	0
II	1st set only; $a = 0$	1
III	1st and 2nd sets; a, b distinct	2
IV	1st and 2nd sets; $a = b$	2
V	1st, 2nd, 3rd sets; a, b, c all distinct	3
VI	1st, 2nd, 3rd sets; one equality	3
VII	1st, 2nd, 3rd sets; $a = b = c$	3
VIII	all four sets; a, b, c, d all distinct	4
IX	all four sets; one pair equal	4
X	all four sets; two pairs equal	4
XI	all four sets; three equal	4
XII	all four sets; $a = b = c = d$	4

In IV the equation is

$$\begin{aligned} & \{a(x^2 + y^2) + 2hxy + cz^2 + dw^2 + 2nzw + 2x(gz + lw) + 2y(fz + mw)\}^2 \\ &= (x + y + z + w)^2 [a^2(x^2 + y^2) + 2(2ah - a^2)xy + c'z^2 + 2n'zw + d'w^2 \\ & \quad + 2x\{(2ag - a^2)z + (2al - a^2)w\} + 2y\{(2af - a^2)z + (2am - a^2)w\}], \quad (AF) \end{aligned}$$

where it will be observed that the line $z = 0, w = 0$ joining the two nodes is not a line on the surface.

Writing, as we may do, $a = 0$, the equation becomes

$$\begin{aligned} & \{2hxy + cz^2 + 2nzw + dw^2 + 2x(gz + lw) + 2y(fz + mw)\}^2 \\ &= (x + y + z + w)^2 (c'z^2 + 2n'zw + d'w^2), \quad (AF) \end{aligned}$$

viz. the form is $V^2 = P^2QR$, the quadric surface $U = 0$ breaking up into the two planes $Q = 0, R = 0$; and the nodes being situate at the intersections of the line $Q = 0, R = 0$ with the surface $V = 0$.

V, VI, VII are apparently cases with three nodes, but in fact VI is the only case of a proper quartic surface with three nodes. For in V the equation is

$$\begin{aligned} & \{(ax + by + cz)(x + y + z) + dw^2 + 2w(lx + my + nz)\}^2 \\ &= (x + y + z + w)^2 [(ax + by + cz)^2 + d'w^2 \\ & \quad + 2w\{(2al - a^2)x + (2bm - b^2)y + (2cn - c^2)z\}], \end{aligned}$$

which is satisfied by $w = 0$, and the surface thus breaks up into the plane $w = 0$ and a cubic surface.

And in VII the equation is

$$\begin{aligned} & \{a(x^2 + y^2 + z^2) + dw^2 + 2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw\}^2 \\ &= (x + y + z + w)^2 [a^2(x^2 + y^2 + z^2) + d^2w^2 \\ & \quad + 2\{(2af - a^2)yz + (2ag - a^2)zx + (2ah - a^2)xy \\ & \quad + (2al - a^2)xw + (2am - a^2)yw + (2an - a^2)zw\}], \end{aligned}$$

which putting $a = 0$ is

$$(dw^2 + 2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw)^2 = (x + y + z + w)^2 d'^2 w^2,$$

viz., this is a pair of quadric surfaces.

In the remaining case VI the equation is

$$\begin{aligned} & \{a(x^2 + y^2) + 2hxy + cz^2 + dw^2 + (a + c)(yz + zx) + 2w(lx + my + nz)\}^2 \\ &= (x + y + z + w)^2 [a^2(x^2 + y^2) + c'^2z^2 + d'w^2 + 2acz(x + y) + 2(2ah - a^2)xy \\ & \quad + 2w\{(2al - a^2)x + (2am - a^2)y + (2cn - c^2)z\}], \quad (AG) \end{aligned}$$

which putting therein $a = 0$ is

$$\begin{aligned} & \{2hxy + dw^2 + cz(x + y + z) + 2w(lx + my + nz)\}^2 \\ &= (x + y + z + w)^2 \{c'^2z^2 + d'w^2 + 2(2cn - c^2)zw\}, \quad (AG) \end{aligned}$$

which is a surface having the nodes A, B, C ; and it is to be observed that the lines CA, CB , but not the line AB , are lines on the surface.

IX, X, XI, XII are apparently cases with four nodes, but it is only XI which is a proper quartic with four nodes. In fact IX is

$$\{(ax+ay+cz+dw)(x+y+z+w)+2(h-a)xy\}^2 = (x+y+z+w)^2\{(ax+ay+cz+dw)^2+4a(h-a)xy\},$$

which is satisfied if $x = 0$ or if $y = 0$; that is, the surface breaks up into the two planes $x = 0$, $y = 0$, and a quadric surface.

X is

$$\begin{aligned} &\{a(x^2 + y^2 + z^2) + dw^2 + 2fyz + 2gzx + 2hxy + (a + d)(xw + yw + zw)\}^2 \\ &= (x + y + z + w)^2 \{a^2(x^2 + y^2 + z^2) + d^2w^2 \\ &\quad + 2(2af - a^2)yz + 2(2ag - a^2)zx + 2(2ah - a^2)xy + 2ad(xw + yw + zw)\}, \end{aligned}$$

which putting therein $a = 0$ is

$$\{dw(x + y + z + w) + 2fyz + 2gzx + 2hxy\}^2 = (x + y + z + w)^2 d^2w^2,$$

and thus breaks up into two quadrics.

And XII is

$$\begin{aligned} &\{a^2(x^2 + y^2 + z^2 + w^2) + 2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw\}^2 \\ &= (x + y + z + w)^2 [a^2(x^2 + y^2 + z^2 + w^2) \\ &\quad + 2(2af - a^2)yz + 2(2ag - a^2)zx + 2(2ah - a^2)xy \\ &\quad + 2(2al - a^2)xw + 2(2am - a^2)yw + 2(2an - a^2)zw], \end{aligned}$$

which putting $a = 0$ is

$$(2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw)^2 = 0,$$

and is thus a quadric surface twice repeated.

There remains XI, and here the equation is

$$\begin{aligned} &\{(ax + ay + cz + cw)(x + y + z + w) + 2(h - a)xy + 2(n - c)zw\}^2 \\ &= (x + y + z + w)^2 \{(ax + ay + cz + cw)^2 + 4a(h - a)xy + 4c(n - c)zw\}, \end{aligned}$$

or writing $h + a$, $n + c$ in place of a , c respectively, this is

$$\begin{aligned} &\{(ax + ay + cz + cw)(x + y + z + w) + 2hxy + 2nzw\}^2 \\ &= (x + y + z + w)^2 \{(ax + ay + cz + cw)^2 + 4ahxy + 4cnzw\}, \quad (AH) \end{aligned}$$

or putting herein $a = 0$ it is

$$\{c(z + w)(x + y + z + w) + 2hxy + 2nzw\}^2 = c(x + y + z + w)^2 \{c(z + w)^2 + 4nzw\}, \quad (AH)$$

which may also be written

$$c(x + y + z + w)\{hxy(z + w) - nzw(x + y)\} + (hxy + nzw)^2 = 0, \quad (AH)$$

the equation of a quartic surface with the four nodes A , B , C , D ; it is to be observed that the lines AC , AD , BC , BD are, the lines AB and CD are not, lines on the surface.

A more simple form may be given to the equation as follows; using the second of the above forms, multiplying the equation by 4, and writing therein

$$\begin{aligned} p &= \sqrt{c}(x + y + z + w), \\ qr &= c(z + w)^2 + 4nzw, \\ st &= c(x + y)^2 - 4hxy, \end{aligned}$$

$q, r,$ and s, t being the linear factors of the two quadric functions respectively, we have

$$qr - st = c(x + y + z + w)(-x - y + z + w) + 4hxy + 4nzw,$$

and thence

$$p^2 + qr - st = 2c(z + w)(x + y + z + w) + 4hxy + 4nzw,$$

wherefore the equation is

$$\frac{1}{4}(p^2 + qr - st)^2 = st \cdot \frac{1}{4}p^2 - qr, \quad (AH)$$

or, what is the same thing,

$$p + \sqrt{qr} + \sqrt{st} = 0, \quad (AH)$$

where p, q, r, s, t are any linear functions of the coordinates; this is the equation of a quartic surface having the nodal conic $p = 0, qr - st = 0$; and the four nodes ($q = 0, r = 0, p - \sqrt{st} = 0$) and ($s = 0, t = 0, p - \sqrt{qr} = 0$). It includes the Cyclide, the equation of which may be written

$$\theta = \sqrt{(ax - ek)^2 + b^2z^2} + \sqrt{(ex - ak)^2 - b^2z^2}.$$

I remark that Prof. Kummer in his most valuable Memoir, “Ueber die Flächen vierten Grades auf welchen Schaaren von Kegelschnitten liegen,” *Crelle*, t. LXVI. (1864), pp. 66–76, has considered several of the cases of a quartic surface with a nodal conic, viz. no node, (AC); a single node, (AD); two nodes (the case AF); and four nodes, (AH); but he has not considered two nodes, the case (AE); nor three nodes, (AG).

In reference to the general case of a quartic surface with a nodal conic, some most interesting properties have recently been obtained by Prof. Clebsch, see *Berl. Monatsh.*, April 30, 1868, where it is shown that there are on the surface 16 right lines forming 20 systems of double-fours, analogous in some respect to the 27 lines and 36 systems of double-sixes of a cubic surface.

[458] ON THE ANHARMONIC-RATIO SEXTIC.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. X. (1870), pp. 56, 57.]

Mr Walker’s equation is

$$4A(\lambda^2 + \lambda + 1)^3 - 27I^2(\lambda^2 - \lambda)^2 = 0;$$

changing the sign of λ , and also the numerical multipliers of I, A (so as to convert the discriminant equation into its standard form $\Delta = I^3 - 27J^2$), the equation is

$$4\Delta(\lambda^2 + \lambda + 1)^3 - 27J^2(\lambda^2 + \lambda)^2 = 0.$$

I remark that this is most readily obtained as follows; writing

$$\begin{aligned} A &= (a - d)(b - c), \\ B &= (b - d)(c - a), \\ C &= (c - d)(a - b), \end{aligned}$$

then we have $A + B + C = 0$,

$$\begin{aligned} I &= \frac{1}{6}(A^2 + B^2 + C^2) = -\frac{1}{3}(BC + CA + AB), \\ J &= \frac{1}{216}(B - C)(C - A)(A - B), \\ \sqrt{\Delta} &= \frac{1}{12}ABC, \end{aligned}$$

see my Fifth Memoir on Quantics, *Phil. Trans.*, vol. CXLVIII. (1858), pp. 429–460, [156].

And observe also, that in virtue of the relation $A + B + C = 0$, we have

$$12I = A^2 + AB + B^2 = A^2 + AC + C^2 = B^2 + BC + C^2.$$

Hence writing

$$u = \frac{A}{B} + \frac{1}{4} \frac{B}{A} + \frac{1}{4},$$

when λ has any one of the values $\frac{A}{B}$, $\frac{B}{A}$, $\frac{A}{C}$, $\frac{C}{A}$, $\frac{B}{C}$, $\frac{C}{B}$, we see that u assumes only the values A , B , C , and u is thus determined by the equation

$$u^2 - 12Iu - 16\sqrt{\Delta} = 0.$$

Eliminating u , we obtain

$$16\sqrt{\Delta} \left\{ \frac{4\Delta}{27J^2} \left(\lambda + \frac{1}{4} \cdot \frac{1}{\lambda} + 1 \right)^2 - \left(\lambda + \frac{1}{4} \cdot \frac{1}{\lambda} + 1 \right) - 1 \right\} = 0,$$

or, what is the same thing,

$$4\Delta (\lambda^2 + \lambda + 1)^3 - 27J^2 \lambda^2 (\lambda + 1)^2 = 0,$$

that is

$$4\Delta (\lambda^2 + \lambda + 1)^3 - 27J^2 (\lambda^2 + \lambda)^2 = 0,$$

the required equation.

[459] ON THE DOUBLE-SIXERS OF A CUBIC SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. X. (1870), pp. 58–71.]

The 27 lines on a cubic surface include, and that in 36 different ways, a double-sixer; viz. a system of two sets of six lines 1, 2, 3, 4, 5, 6; 1', 2', 3', 4', 5', 6', such that every line of the one set intersects all the non-corresponding lines of the other set, thus

	1	2	3	4	5	6
1'	.	•	•	•	•	•
2'	•	.	•	•	•	•
3'	•	•	.	•	•	•
4'	•	•	•	.	•	•
5'	•	•	•	•	.	•
6'	•	•	•	•	•	.

there being in all 30 intersections.

Any line say 4, of the one set, intersects five lines 1', 2', 3', 5', 6' of the other set; and these six lines being given the double-sixer may be constructed; viz. (besides the line 4) we have a line 1 meeting the lines 2', 3', 5', 6'; a line 2 meeting the lines 3', 5', 6', 1'; a line 3 meeting the lines

$5', 6', 1', 2'$; a line 5 meeting the lines $6', 1', 2', 3'$; and a line 6 meeting the lines $1', 2', 3', 5'$; and then the lines 1, 2, 3, 5, 6 are all of them met by a single line $4'$, which completes the system.

We may, if we please, consider the lines 4, 2 as given, and then $1', 3', 5', 6'$ will be any four lines each of them meeting the two given lines 4, 2; $2'$ will be any line meeting 4; and we have to determine a line $4'$ meeting 2, such that there may exist the lines 1, 3, 5, 6, completing the system as above. Or what is the same thing, we have a skew quadrilateral $1', 2, 3', 4$; $5'$ and $6'$ meet 2 and 4; $2'$ meets 4, and $4'$ meets 2; 5 and 6 meet $1'$ and $3'$; 1 meets $3'$ and 3 meets $1'$; and the two sets $2', 4', 5', 6'$ and 1, 3, 5, 6 meet thus

	1	3	5	6
$2'$	.	.	•	•
$4'$	.	.	•	•
$5'$	•	•	.	.
$6'$	•	•	.	.

Hence, starting with the skew quadrilateral $1'23'4$, and taking $x = 0, y = 0, z = 0, w = 0$ for the equations of the four planes $41', 1'2, 23', 3'4$ respectively; or what is the same thing $x = 0, y = 0$ for the equations of the line $1'$; $y = 0, z = 0$ for those of the line 2; $z = 0, w = 0$ for those of the line $3'$; and $w = 0, x = 0$ for those of the line 4; the several lines may be determined, each of them by means of its six coordinates, as follows:

	a	b	c	f	g	h
$1'$	.	.	.	.	.	1
2	.	.	.	1	.	.
$3'$	.	.	.	.	1	.
4	1	.	.	.	.	.
$2'$	A_1	B_1	C_1	.	G_1	H_1
$4'$	.	B_2	C_2	F_2	G_2	H_2
$5'$	.	B_3	C_3	F_3	G_3	H_3
$6'$	.	B_4	C_4	F_4	G_4	H_4
1	a_1	b_1	c_1	f_1	g_1	.
3	.	.	.	f_2	g_2	.
5	.	.	.	f_3	g_3	.
6	a_4	b_4	.	f_4	g_4	.

where

$$\begin{aligned} B_1G_1 + C_1H_1 &= 0, \\ B_2G_2 + C_2H_2 &= 0, \\ B_3G_3 + C_3H_3 &= 0, \\ B_4G_4 + C_4H_4 &= 0, \end{aligned}$$

the conditions in regard to the lines $2', 4', 5', 6'$ and 1, 3, 5, 6 are formed by means of the diagram:

	1	3	5	6
$2'$	f_1	.	.	.
$4'$	$b_1g_2 - g_1b_4$	.	.	.
$5'$	.	.	.	.
$6'$	.	.	.	.

viz. we have the equations

first set,

$$\begin{aligned} f_1 A_1 + g_1 B_1 + h_1 C_1 + a_1 F_1 + b_1 G_1 + c_1 H_1 &= 0, \\ g_1 B_1 + a_1 F_1 + b_1 G_1 + c_1 H_1 &= 0, \\ g_1 B_1 + b_1 G_1 + c_1 H_1 &= 0, \\ g_1 B_1 + b_1 G_1 + c_1 H_1 &= 0; \end{aligned}$$

second set,

$$\begin{aligned} f_2 A_1 + g_2 B_1 + h_2 C_1 + b_2 G_1 + c_2 H_1 &= 0, \\ g_2 B_1 + c_2 H_1 &= 0, \\ g_2 B_1 + c_2 H_1 &= 0, \\ g_2 B_1 + c_2 H_1 &= 0; \end{aligned}$$

third set,

$$\begin{aligned} f_3 A_1 + g_3 B_1 + h_3 C_1 + b_3 G_1 + c_3 H_1 &= 0, \\ g_3 B_1 + b_3 G_1 + c_3 H_1 &= 0, \\ g_3 B_1 + b_3 G_1 + c_3 H_1 &= 0; \end{aligned}$$

fourth set,

$$\begin{aligned} f_4 A_1 + g_4 B_1 + h_4 C_1 + a_4 F_1 + b_4 G_1 + c_4 H_1 &= 0, \\ g_4 B_1 + b_4 G_1 + c_4 H_1 &= 0, \\ g_4 B_1 + b_4 G_1 + c_4 H_1 &= 0; \end{aligned}$$

and it is to be shown, that taking as given the coordinates of $2', 5', 6'$, that is $(A_1, B_1, C_1, G_1, H_1)$, $(B_3, C_3, F_3, G_3, H_3)$, $(B_4, C_4, F_4, G_4, H_4)$, we can find the coordinates of the remaining lines $4', 1, 3, 5, 6$.

The first set of equations gives

$$g_1 b_1 = b_1 \cdot g_1; \quad \begin{vmatrix} B_1 & C_1 & \cdot \\ B_3 & C_3 & H_3 \end{vmatrix};$$

viz. g_1, b_1, a_1 are proportional, but so only the ratios are material, they may be taken equal to the determinants $G_1 H_3 - C_1 H_3, B_3 H_1 - B_1 H_3, B_1 C_3 - B_3 C_1$, &c.; that is to signify these values respectively, the first equation gives $f_1 A_1$, and the second equation gives $a_1 F_1$; multiplying together these values, and writing $a_1 f_1 = k_1 g_1$, we find

$$(B_1 B_3 \cdot G_1 G_3 \cdot H_1 H_3 \cdot G_1 H_3 \cdot G_3 H_1) \cdot k_1 (A_1, B_1, C_1, F_1, G_1, H_1) + B_3 (B_1 C_3 - B_3 C_1) F_2 \cdot A_1 + B_1 (B_1 C_3 - B_3 C_1) F_2 = 0.$$

Proceeding in a similar manner with the second set of equations, we have first

$$g_2, b_2, b_2 = \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix} \quad (\text{observe } k_2 = G_1 B_3 - G_3 B_1, = -c_1),$$

and then

$$(B_3 B_1 \cdot G_1 C_1 \cdot G_4 C_4 \cdot G_4 G_1 \cdot C_4 C_1 \cdot G_1 G_4 \cdot G_4 G_1 + A_1 F_1 \cdot B_3 C_2 + B_1 C_2 \cdot G_3 \cdot B_3 G_1) k_2 b_2 = 0.$$

The third set gives in like manner with the second set of equations, this becomes

$$g_3^2 B_1 B_3 + g_3 b_3 (B_1 G_3 + B_3 G_1 + A_1 F_2) + b_3^2 G_1 G_3 = 0,$$

or since $g_3 \cdot b_3 = G_1 \cdot -B_1$, this is

$$G_1^2 B_1 B_3 - G_1 B_1 (B_1 G_3 + B_3 G_1 + A_1 F_2) + B_1^2 G_1 G_3 = 0,$$

and similarly, the fourth set gives

$$g_4^2 B_1 B_3 + g_4 b_4 (B_1 G_3 + B_3 G_1 + A_1 F_2) + b_4^2 G_1 G_3 = 0,$$

or since $g_4 \cdot b_4 = G_1 \cdot -B_1$, this is

$$G_1^2 B_1 B_3 - G_1 B_1 (B_1 G_3 + B_3 G_1 + A_1 F_2) + B_1^2 G_1 G_3 = 0,$$

and these last two results lead to the values of the ratios of $B_3 B_4$, $B_3 G_4 + B_4 G_3$, $G_3 G_4$ in the values containing the common factor $B_1 G_1$, viz. these are proportional to expressions containing the common factor $B_3 G_3 - B_4 G_4$, and taking them equal instead of merely proportional to the resulting expressions (which is allowable, since the absolute values are not material), we have

$$B_3 B_4, B_3 G_4 + B_4 G_3 + A_1 F_2, G_3 G_4 = B_1 B_3, B_1 G_3 + B_3 G_1, G_1 G_3,$$

$$(B_3 B_4, G_3 G_4, G_3 B_4 + B_3 G_4; p_4, q_4, r_4) = (B_1 B_3, G_1 G_3, B_1 G_3 + G_1 B_3; H_3, G_1, -H_1);$$

but the terms containing g_1, b_1 are $(B_1 g_1 + G_1 b_1)(B_4 g_2 + G_4 b_4) = -H_1 c_1 \cdot -H_4 c_4$, that is $H_1 H_4 c_1 c_4$; the whole equation is thus divisible by c_1 , and omitting this factor, it becomes

$$g_1(H_1 B_4 + H_4 B_1) + h_1(G_1 H_4 + G_4 H_1) + a_1(H_1 F_4 + H_4 F_1) = 0.$$

Proceeding in like manner with the result obtained from the second set of equations, this becomes

$$(B_1 B_3, C_2 C_4, C_1 G_1, C_1 G_4 + C_4 G_1, B_2 G_4 + B_4 G_2, B_4 G_4 + B_2 G_4 + B_2 C_1(G_1, b_4, c_4)) = 0,$$

where the terms containing g_3, b_3 are $(B_2 g_3 + C_2 b_3)(B_4 g_2 + C_4 b_4)$, viz. the whole equation divides by h_3 , and it then becomes

$$g_3(B_3 C_4 + B_4 C_3) + h_3(C_3 G_4 + C_4 G_3) + b_3(G_3 B_4 + G_4 B_3) = 0.$$

Considering B_3, G_3, F_3 as given by the equations

$$B_3 B_4 = B_1 B_3, \quad G_3 G_4 = G_1 G_3, \quad B_3 G_4 + G_3 B_4 = A_1 F_3 = B_1 G_3 + B_3 G_1,$$

But we have

$$\begin{aligned} g_1 B_1 B_4 - b_1 H_4 H_1 H_4 &= B_4 H_1 (G_1 H_4 - G_4 H_1) - B_1 H_4 (G_1 H_4 - G_4 H_1), \\ &= B_4 H_4 (B_3 G_1 + G_3 B_1) - B_1 H_1 (B_3 G_4 + G_3 B_4) + C_1 B_3 = 0, \end{aligned}$$

and similarly

$$g_1 C_1 C_4 - b_1 G_4 G_1 G_4 = 0,$$

$$g_1 C_1 B_4 - b_1 H_4 G_1 G_4 = 0,$$

$$h_1 C_1 G_4 - g_1 B_4 H_1 G_4 = 0.$$

Moreover, writing $p_4 B_4 + q_4 G_4 + r_4 H_4$, we have

$$p_4 B_4 G_4 = b_4 (G_4^2 - G_1 G_4),$$

$$q_4 G_4 G_1 = b_4 (G_1 G_4 - G_4^2),$$

$$r_4 G_4 = -b_4 (G_4 - G_1),$$

and the few terms of the equation in question thus separately vanish; and the equation is consequently verified.

We may collect the results as follows:

Data are lines $1', 2, 3', 4; 5', 6'$;

and then, for the remaining lines $1, 3, 5, 6$ at the common factor omitted in each case, the coordinates are as follows:

For 4',

$$\begin{aligned} B_1 B_4 &= B_1 B_3, \quad G_3 G_4 = G_1 G_3, \quad B_3 G_4 + G_3 B_4 = A_1 F_3 = B_1 G_3 + B_3 G_1, \\ A_1 F_2 &= B_1 G_1 \cdot B_3 G_3 - B_3 G_1 \cdot B_1 G_3, \quad A_1 = 0, \\ \begin{vmatrix} B_1 & C_1 & H_1 \\ B_3 & C_3 & H_3 \end{vmatrix}, \quad \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix}, \quad H_1 H_3, \quad B_1 B_3, \quad C_1 C_3, \quad G_1 G_3; \\ \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix} + \begin{vmatrix} C_1 & G_1 & H_1 \\ C_3 & G_3 & H_3 \end{vmatrix} &\quad (B_1 G_3 + B_3 G_1 = -2A_1 F_2, \text{ identity}). \end{aligned}$$

For 1,

$$\begin{aligned} (g_1, b_1, a_1) &= \begin{vmatrix} B_1 & C_1 & H_1 \\ B_3 & C_3 & H_3 \end{vmatrix}, \quad A_1 F_1 = \begin{vmatrix} B_3 & C_3 & H_3 \\ B_1 & C_1 & G_1 \end{vmatrix}, \\ &\quad (a_1 f_1 + k_1 g_1, \text{ identity}). \end{aligned}$$

For 3,

$$\begin{aligned} (g_2, b_2, b_2) &= \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix}, \quad A_1 F_2 = \begin{vmatrix} B_3 & C_3 & G_3 \\ B_1 & C_1 & C_1 \end{vmatrix}, \\ &\quad (a_2 f_2 + k_2 g_2, \text{ identity}). \end{aligned}$$

For 5,

$$\begin{aligned} g_3 &= G_3, \quad b_3 = B_3, \quad A_1 F_3 = B_1 G_3 - B_3 G_1, \\ c_3 &= G_3, \quad b_3 = 0, \quad a_3 = \frac{A_1 F_3 G_3 + b_3 (B_1 G_1 + C_1 H_1)}{B_3 G_1 + b_1 H_3}, \quad k_3 = 0. \end{aligned}$$

For 6,

$$\begin{aligned} g_4 &= G_4, \quad b_4 = -B_4, \quad A_1 F_4 = B_1 G_4 - B_4 G_1, \\ c_4 &= 0, \quad b_4 = 0, \quad a_4 = 0, \end{aligned}$$

and, for actual calculation, it is convenient to remark that as only the ratios are material, we may suppose without loss of generality that $c_1 = 1$, &c., respectively.

But results may be further reduced. Writing

$$(g_1, b_1, c_1) = \begin{vmatrix} B_3 & C_3 & H_3 \\ B_4 & C_4 & H_4 \end{vmatrix},$$

we have

$$-(g_1 B_4 + b_1 G_4 + h_1 C_4) = C_1 \left(\frac{B_3 G_3}{B_1 G_1} \right) + h_4 C_4$$

and many further simplifications follow, leading to compact final forms for C_1 , H_3 , etc.

[The remaining algebraic deductions on book pp. 321–325 develop in the same manner the explicit formulas for the coordinates of 2', 4', 5', 6' and 1, 3, 5, 6, the closing display exhibiting the four quantities

$$C_1 = \frac{C_1 G_4 p_4 B_4 + b_4 G_4 \cdot k_4 + B_4 G_4}{B_4 \cdot p_4 B_4 + h_4 b_4 + k_4 a_4},$$

together with the parallel expressions for G_1 , F_1 , H_1 , H_3 , and the like for the alternative subscripts. The explicit verification is carried out on book p. 325.]

I have thought it worth while to effect the numerical calculations for enabling the construction of a drawing or model. For this purpose taking X , Y , Z as ordinary rectangular coordinates, I write

$$\begin{aligned} x &= X + Y + Z - 10, \\ y &= Z, \\ z &= -X + Y + Z - 2, \\ w &= Y, \end{aligned}$$

that is, I take $1'$ and 2 to be lines in the plane of XY , defined by the equations $X + Y = 10$ and $X - Y = -10$ respectively, and X and $3'$ to be lines in the plane of YZ defined by the equations $Z = 10$, $Z + Z - 2 = 12$ respectively. And I take $5'$ to be the line joining the points $(9, 0, 7)$ and $(-9, 2, 0)$; $4'$ the line joining the points $(9, 0, 7)$ and $(-2, 8, 0)$; $5'$ the line joining the points $(9, 0, 7)$ and $(-2, 8, 0)$; $6'$ the line joining the points $(9, 0, 7)$ and $(-2, 8, 0)$. We calculate then for each of the lines the upper coordinates of two points, and thence the six coordinates of the line; thus

	x	y	z	w	p	yw	A	B	C	F	G	H
$1'$	0, 0, -1, 0	-10, 0, 0, 1	0, 0, 0, 0	0, 0, 0, 0			0,	10,	50,	0,	0,	-10
2	0, 0, -1, 0	-10, 0, 0, 1	0, 0, 0, 0				0,	10,	50,	0,	0,	-10
$3'$	0, 0, 1, 0	-10, 0, 0, 1					0,	10,	50,	0,	0,	-10
4	0, 0, 1, 0	-10, 0, 0, 1					76,	30,	0,	0,	10,	0

and, effecting the calculations for the remaining lines, we have

	A	B	C	F	G	H
$2'$	$\cdot$	24	$\frac{144}{15}$	$\cdot$	$\cdot$	$\frac{8}{15}$
1	$\frac{380}{15}$	60	35	$\frac{380}{15}$	0	3
3	127500	-720	0	$\frac{21}{15}$	140	-30
5	304	-54	6	0	7	0
6	$\frac{333}{5}$	-10	0	$\frac{12}{5}$	0	0

or reducing to integers, the values are

	A	B	C	F	G	H
$2'$	0	48	36	0	7	8
$4'$	76	26	2	0	0	1
1	1444	13452	6720	16443	-1131	0
3	463384	-2738	0	5	581	-114
5	9772	-552	0	21	132	0
6	5772	-3033	0	84	226	0

The line $(a, b, c; f, g, h)$ is given as the intersection of the two of the four planes

$$\begin{vmatrix} \cdot & h & -g & a \\ -h & \cdot & f & b \\ g & -f & \cdot & c \\ -a & -b & -c & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0,$$

or substituting for x, y, w the values $X + Y + Z - 10, Z, -X + Y + Z - 2, Y$, these become

$$\begin{vmatrix} \cdot & h & -g & a \\ -h & \cdot & f & b \\ g & -f & \cdot & c \\ -a & -b & -c & \cdot \end{vmatrix} \begin{Bmatrix} X + Y + Z - 10 \\ Z \\ -X + Y + Z - 2 \\ Y \end{Bmatrix} = 0,$$

viz., what is the same thing,

$$\begin{vmatrix} \cdot & 2y - a - c & 2y - f - h & \cdot & -2y \\ -2y + a + c & \cdot & a + b + c + f - h & \cdot & -c \\ -2y + f + h & -a - b - c - f + h & \cdot & \cdot & -f - h \\ 2y & c + c & f + h & \cdot & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0.$$

And substituting, we have the equations of the several lines, viz.

$$\begin{aligned}
 (1') \quad & X + Y = 10, \quad Z = 0; \\
 (2) \quad & -X + Y = 10, \quad Z = 0; \\
 (3') \quad & -X + Y = 2, \quad Z = 10; \\
 (4) \quad & X + Y = 10, \quad Z = 0; \\
 (2') \quad & \begin{vmatrix} \cdot & \cdot & 49 & 5 \\ -49 & \cdot & \cdot & -36 \\ -5 & 36 & \cdot & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0; \\
 (4') \quad & \begin{vmatrix} \cdot & -55 & 26 & -7 \\ -55 & \cdot & \cdot & -3 \\ 26 & \cdot & \cdot & -1 \\ 7 & 26 & -3 & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0;
 \end{aligned}$$

and so on for (5'), (6'), (1), (3), (5), (6) with the corresponding numerical matrices given on book p. 328.

[Page 329 of the original reproduces the numerical coordinates of the various intersection points 11', 12', 13', ..., in fractional form (numerators and denominators displayed on separate lines), and adds:]

viz., the coordinates of 12' (intersection of lines 1 and 2') are $(\frac{611}{132}, \frac{-554}{132}, \frac{-216}{132})$, and so in other cases; where there is no divisor the coordinates are integers. I find however, on laying down the figure, that the lines 3 and 4, 5' and 6 come so close together, that the figure cannot be obtained with any accuracy.

[460] NOTE ON MR FROST'S PAPER ON THE DIRECTION OF LINES OF CURVATURE IN THE NEIGHBOURHOOD OF AN UMBILICUS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. X. (1870), pp. 111–113.]

I remark as follows:

1. In regard to a quartic surface, it is not, I think, correct to say that the generation through an umbilic (= curve of curvature) notwithstanding that, in fact in the present case there will generate 5 in one plane and consequently intersect. The way in which these generatrices as quasi-curves-of-curvature present themselves is as follows:

The curve of curvature satisfy a certain differential equation, the complete integral of which gives those curves as the intersections of the given quadric surface by the series of confocal quadrics

$$\frac{x^2}{a^2 - h} + \frac{y^2}{b^2 - h} + \frac{z^2}{c^2 - h} = 1,$$

h being the constant of integration. The singular solution of the differential equation, or envelope of the series of curvatures denoted as above, gives the umbilical generatrices.

2. In regard to a surface in general, I think it must be considered, not that there pass through the umbilic three distinct curves, but that there pass through the umbilic a single point, or rather a point at which there are in general three distinct directions of the curve. The umbilicar curve of curvature is in fact present itself as the curve belonging to a certain value of the constant of integration h , in order that the curve of curvature may pass through a given point on the

surface; a most simple example will, in order that there are in general distinct directions of the curve.

But, as an umbilic is a point for which (as in effect shown, p. 81, for the particular case of a quadric surface) the two values of h become equal; that is, through the umbilic only a singular curve of curvature; but $\frac{dy}{dx}$ is determined by a cubic equation, and the umbilic is thus (as just mentioned) a point at which there are in general three distinct directions of the curve.

3. Some researches on the subject are contained in my paper “On Differential Equations and Umbilici,” *Phil. Mag.*, vol. XXVI. (1863), pp. 373–379 and 441–452, [230]. It is noticeable that in the integral equations which I have there obtained for the differential equation $dy(p^3 - 1) + (a - c)pq = 0$, and the more general form $(de + p)(p + q) = 0$, and the umbilic itself, the differential equation breaks up into three distinct curves; and the same is the case with the cubic equation, that the umbilic does not occur in the surface $xyz = 1$ presently referred to.

4. In the paper “Mémoire sur les surfaces orthogonales,” *Liouv.*, t. XII. (1847), pp. 241–254, M. Serret has given two very remarkable cases of three systems of surfaces intersecting each other at right angles, and consequently in the curves of curvature of each system. It was only on referring to this paper, in connexion with that of Mr Frost, that I perceived an obvious enough simplification of M. Serret’s formulae, whereby it appears that the curves of curvature on the surface $xyz = 1$ are given as the intersection of this surface with the series of surfaces

$$h = (x^2 + xy^2 + xz^2)^{\frac{1}{3}} + (x^2 + xy^2 + xz^2)^{\frac{1}{3}\omega},$$

where ω is an imaginary cube root of unity; the rationalised equation is of the twelfth order in (x, y, z) , and for the particular value $h = 0$ is $(y^3 - x)^3 + (x^3 - y^2)^3 + (xy^3 - 1)^3$. The point $x = y = z = 1$ is obviously an umbilic on the surface $xyz = 1$, and the corresponding value of h being $h = 0$, the equation just obtained determines the umbilicar curves of curvature, viz., combining therewith the equation $xyz = 1$, we infer that the surface have here hyperbolic curves

$$(y = z, xy^2 = 1), \quad (z = x, yz^2 = 1), \quad (x = y, zx^2 = 1).$$

[461] ON THE GEOMETRICAL INTERPRETATION OF THE COVARIANTS OF A BINARY CUBIC.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. X. (1870), pp. 148–149.]

Consider the binary cubic $U = (a, b, c, d|x, y)^3$, and the discriminantal (invariant)

$$\nabla = a^2d^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2,$$

the Hessian

$$H = (ac - b^2)x^2 + (ad - bc)xy + (bd - c^2)y^2,$$

and the cubicovariant

$$\begin{aligned} \Phi = & (a^2d - 3abc + 2b^3)x^3 \\ & + (3abd + 6ac^2 - 9b^2c)x^2y \\ & - (3acd + 6b^2d - 9bc^2)xy^2 \\ & - (a^2d - 3acd + 2c^3)y^3, \end{aligned}$$

connected by the identical relation

$$\Phi^2 = -\nabla U^2 - 4H^3.$$

Then if we regard (a, b, c, d) as the coordinates of a point in space, but (x, y) as variable parameters, the equation

$$\nabla = 0$$

represents a quartic torse, having for its cuspidal curve the plane cubic $ad - bc = 0$, $bd - c^2 = 0$, the equation

$$U = 0$$

is that of the tangent plane to the torse along the line $ad + 2bc + cy = 0$, $bd^2 + 2acy + dy^2 = 0$; the line is the cuspidal curve in the above considered as line coordinates; viz. $a : b : c : d = ay^3 : -y^2 : y : -1$, the equation

$$H = 0$$

is that of a quadric surface having the first point of the oses point in regard to the torse.

The equation $\Phi^2 = \nabla U^2 - 4H^3$, gives $\Phi = -4H^3$, a result which implies that $\Phi = 0$ is a certain torse required twice, and that $U = 0$, $\Phi = 0$ is the same equator repeated twice. The curve in question is at any rate on the surface $\Phi = 0$; and conversely, the curve is on the surface $\Phi = 0$ regarded as the envelope of the plane $U = 0$, viz. in $(2, 3)$ -plane, &c., what is the same thing.

The equation $\Phi^2 + \nabla U^2 + 4H^3 = 0$, gives $\Phi^2 = -4H^3$, a result which implies that $\Phi = 0$ is a certain torse repeated twice, that is, $H = 0$, $\Phi = 0$, the whole equation in this divisible by c , and omitting this factor, it becomes

$$g(H_1B_4 + H_4B_1) + h(G_1H_4 + G_4H_1) + a(H_1F_4 + H_4F_1) = 0.$$

Proceeding in like manner with the result obtained from the second set of equations, this becomes

$$(B_1B_3, C_2C_4, C_1G_1, C_1G_4 + C_4G_1, B_2G_4 + B_4G_2, B_4G_4 + B_2G_4) = 0.$$

[End of paper 461; the manipulations on book p. 333 verify the geometrical interpretation by reducing to the form $MU = (a^2, b, c, d \mid x, y)^3 \cdot M$, $\Phi = U$ and complete the discussion of the cubicovariant.]

[462] A NINTH MEMOIR ON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXI. (for the year 1871), pp. 17–50. Received April 7. — Read May 19, 1870.]

It was shown not long ago by Professor Gordan that the number of the irreducible covariants of a binary quantic of any order is finite (see his memoir “Beweis dass jede Covariante und Invariante einer binären Form eine ganze Function mit numerischen Coefficienten einer endlichen Anzahl solcher Formen ist,” *Crelle*, t. LXIX. (1869), Memoir dated 8 June 1868), and in particular that for a binary quintic the number of irreducible covariants (including the quintic and the invariants) is = 23, and that for a binary sextic the number is = 26. From the theory given in my “Second Memoir on Quantics,” *Phil. Trans.*, 1856, [141], I derived the conclusion, which, as it now appears, was erroneous, that for a binary quintic the number of irreducible covariants was infinite. The theory requires, in fact, a modification, by reason that certain linear relations, which I had assumed to be independent, are really not independent, but, on the contrary, linearly connected together; the interconnexion in question does not occur in regard to the quadric, cubic, or quartic; and for these cases respectively the theory is as it stands; for the quintic the interconnexion first presents itself in regard to the degree 8 in the coefficients and order 14 in the

variables, viz. the theory gives correctly the number of covariants of any degree not exceeding 7, and also those of the degree 8 and order less than 14; but for the order 14 the theory as it stands gives a non-existent irreducible covariant $(a, \dots | x, y)^{14}$, viz. of the form in question $10 - 6, = 4$ aszygetic covariants; but the number of aszygetic covariants being $= 5$, there is left, according to the theory, 1 irreducible covariant of the form in question. The fact is that in applying the theory, the composite covariants are equivalent to 5 independent syzygies only, the composite covariants are equivalent to $10 - 5, = 5$, the full number of the aszygetic covariants. And similarly the theory as it stands gives a non-existent of the aszygetic covariants.

[Article No. 326 continues on book pp. 334–335 with Cayley's revised treatment of the issue; thence Articles 327–365 develop, on book pp. 335–353, the systematic proof. The articles fall into the following sections.]

Article Nos. 328 to 332. *Reproduction of my original Theory as to the Number of the Irreducible Covariants.*

328. I reproduce to some extent the considerations by which, in my Second Memoir on Quantics, I endeavoured to obtain the number of the irreducible covariants of a given binary quantic $(a, b, \dots | x, y)^n$.

Considering in the first instance the covariants as functions of the coefficients $(a, b, c, \dots)$ without regarding the variables (x, y) , and attending only to the following properties — 1°, without regarding the variables¹, a covariant is a rational and integral homogeneous function of the coefficients $(a, b, c, \dots)$ of the properties — 1°, a covariant is a rational and integral homogeneous function of the coefficients $(a, b, c, \dots)$; and 2°, the coefficients of a given degree μ are linearly independent of the coefficients of a less degree; and that a covariant not connected by linear relations as above; and it is assumed that we know the number of the *irreducible* covariants; and it is in the same manner that we know the number of the aszygetic covariants in regard to a given quantic, and that as the degree μ is increased successively to 1, 2, 3, ... there is a certain function $P, Q, R, \dots$ for each degree, and that a function not so obtained must be an *irreducible* covariant; and it is assumed that we know the number of the aszygetic covariants of the degree μ as a given function P_μ of the coefficients $a_i, \dots$.

[The text continues through Articles 329–346 on book pp. 336–345, treating successively: the binary cubic with its three syzygies; the binary quartic with its three covariants and one syzygy; the quintic with its 23 irreducible covariants (Table No. 87, identifying the 23 in terms of Gordan's notation $A, B, C, \dots, W$); Table No. 88, giving the actual generation, by successive transvectants, of each of the 23 forms (degrees 1 through 8); the related Tables Nos. 89 and 90 exhibiting the relations and complete linear expressions of the syzygies among the products of the covariants. The reasoning is throughout that of Gordan's Beweis but cast in the notation $(f, f), (f, (f, f)), \dots$ of the Second Memoir.]

Article Nos. 347 to 365. *Sketch of Professor Gordan's proof for the finite Number, = 23, of the Covariants of a Binary Quintic.*

347. I propose to reproduce the leading points of Professor Gordan's proof that the binary quintic $(a, b, c, d, e, f | x, y)^5$ has a finite system of 23 covariants, viz. a system such that every other covariant whatever is a rational and integral function of these 23 covariants.

348. DERIVATION. Consider for a moment any two binary quantics ϕ, ψ of the same or different orders, and which may be either independent or, as will be the case here, identical or connected

¹For this case of covariants, α_1 is of course $= 1$; but in the investigation the term covariant stands for any function satisfying the conditions 1° and 2°.

by some relation f . We may form the derivatives

$$\begin{aligned}(\phi, \psi) &= \phi\psi, \\(\phi, \psi)' &= 1 \cdot 2 \phi\psi, \partial_x \phi \cdot \partial_y \psi, -\partial_y \phi \cdot \partial_x \psi, \\(\phi, \psi)'' &= 1 \cdot 2 \phi\psi, \partial_x \phi, \partial_y \phi \cdot \partial_x \psi, -2\partial_x \partial_y \phi \cdot \partial_x \partial_y \psi + \partial_y^2 \phi \cdot \partial_x^2 \psi,\end{aligned}$$

where, however, there is no occasion to use the notation $(\phi, \psi)'$ (as this is simply the product $\phi\psi$), and the succeeding derivatives may (when there is no risk of ambiguity) be written more shortly $(\phi\psi)$, $(\phi\psi)'$, &c.; in all that follows the term “derivative” (Gordan’s *Ueberschiebung*) is to be understood in this special sense.

349. The degree of the derivative $(\phi\psi)^r$ is the sum of the degrees of ϕ, ψ ; the order of the derivative is the sum of the orders of the constituents ϕ, ψ ; the order of the derivative is less than this by $2r$; it being understood throughout that the next degree refers to the coefficients, and the order to the variables.

350. THEOREM A. The covariants of a quantic f of a given degree m can be all of them obtained by derivation from f and the covariants of the next inferior degree $(m-1)$.

In particular for the degree 1 the only covariant is the quantic f itself; the degree 2 the covariants of the third degree consist of the quantic f itself in succession, the covariants of the third degree are each of them obtained, by derivation from f and the covariants of the second degree.

351. Suppose that the covariants of the second degree $(f, f)^r, (f, f, f)^r, (f, (f, f))^r, \dots$ are in this order represented by $\beta_1, \beta_2, \beta_3, \dots$, then the covariants of the third degree written in the order

$$(\beta_1 f)^r, (\beta_1 f)^{r'}, \dots, (\beta_2 f)^r, (\beta_2 f)^{r'}, \dots, (\beta_3 f)^r, (\beta_3 f)^{r'} \dots$$

may be represented by $\delta_1, \delta_2, \dots$, and the covariants of the fourth degree written in the order

$$(\gamma_1 f)^r, (\gamma_1 f)^{r'}, \dots, (\gamma_2 f)^r, (\gamma_2 f)^{r'}, \dots, (\gamma_3 f)^r, (\gamma_3 f)^{r'}, \dots$$

may be represented by $\delta_1, \delta_2, \dots$, and so on.

Observe that each term μ_r is a derivative $(\lambda f)^r$, the derivatives of λ with f being assumed to be an earlier term; the derivative with a less index of derivation is earlier than that with a greater index of derivation, or, what is the same thing, those, those are earlier which have a less index r , and which have a greater index of the higher order.

352. The series $\mu_1, \mu_2, \mu_3, \dots$ is not asyzygetic; we make this so by considering in succession whether the several terms $\mu_1, \mu_2, \dots$ respectively are expressible as linear functions of the other terms, or omitting every term which is so expressible. The reduced series obtained is a series of terms which is in fact a complete asyzygetic series of covariants of the degrees in question.

353. The quadricovariants $(a, b, c | x, y)^2$ are proportional, but as only the ratios are material, they may be taken equal to the determinants $G_1 H_3 - G_3 H_1, B_3 G_1 - B_1 G_3, B_1 C_3 - B_3 C_1$, &c. They form a system T of T -series, and the second equation gives $\alpha_1 f_1$; multiplying together these values, and writing $a_1 f_1 = k_1 g_1$, we find

$$(B_1 B_3 \cdot G_1 G_3 \cdot H_1 H_3 \cdot G_1 H_3 \cdot G_3 H_1) \cdot k_1 (A_1, B_1, C_1, F_1, G_1, H_1) = 0.$$

354. For the cubic $(a, b, c, d | x, y)^3$ the number of asyzygetic covariants $a^\mu x^\nu$ is

$$\text{coeff. } a^\mu x^\nu \text{ in } \frac{1 - \frac{1}{x^2}}{(1-a)(1-ax)\left(1-\frac{a}{x}\right)},$$

or transforming as before, this is

$$\text{coeff. } a^\mu x^\nu \text{ in } \frac{1 - x^2}{(1-ax)(1-ax^3)(1-a^3)}.$$

The development is

$$\begin{array}{l|l}
 1 & -\frac{1}{x^2} \\
 +ax^3 & +a \cdot \left(\frac{1}{x}\right) \\
 +a^2(x^2+1) & +a^2\left(\frac{1}{x}+1\right) \\
 +a^3(x^3+x+1) & +a^3\left(\frac{1}{x}+\frac{1}{x^3}\right) \\
 +a^4(x^2+x^4+1) & +a^4\left(\frac{1}{x}+\frac{1}{x^3}+1\right) \\
 \vdots &
 \end{array}$$

which is

$$= A(x) - \frac{1}{x^2} A\left(\frac{1}{x}\right),$$

where

$$A(x) = \frac{1}{(1-ax^3)(1-ax)},$$

and, since $\frac{1}{x^2} A\left(\frac{1}{x}\right)$ contains only negative powers, the required number is

$$= \text{coeff. } a^\mu x^\nu \text{ in } \frac{1}{(1-ax^3)(1-ax)},$$

indicating that the covariants are powers and products of $(ax^3$ and $ax)$, the quadric itself, and the discriminant. Compare Second Memoir, No. 40, according to which, writing therein a for a , the number of aszygetic covariants is

$$= \text{coeff. } a^\mu x^\nu \text{ in } \frac{1}{(1-ax)(1-x)}.$$

[Article numbers 355 onwards (New formula for the number of Aszygetic Covariants) continue on book pp. 338–345 in the same vein, expressing the number of aszygetic covariants of the quadric, cubic, quartic and quintic as the coefficient of $a^\mu x^\nu$ in a generating fraction with a quadric or quartic Hessian-like denominator; these articles culminate in the explicit verification (book p. 344) that for the quintic the number is finite and $= 23$, and in the introduction (book p. 345) of **Article No. 337. Identification of the 23 Fundamental Covariants**, with Table No. 87 relating each of the 23 irreducible covariants to its Gordan symbol $A, B, C, \dots, W$.]

[Tables Nos. 87, 88 and the supplementary Table No. 89 (giving the syzygies and interconnexions of the syzygies among the products of the 23 covariants) occupy book pp. 341–346. They are intricate purely typographical exhibits whose intelligent reproduction is deferred to a later chunk; the present chunk closes with the opening of Articles 347–365, “Sketch of Professor Gordan’s proof for the finite Number, $= 23$, of the Covariants of a Binary Quintic,” as begun above.]

ARTICLE NO. 363. For U equal to any one of the last eleven values, the form is $(Q, S)R$ which is $= R(QS) + S(QR)$, and is thus a function of covariants of a lower degree; there remain the derivatives formed with two of the functions f, ϕ, i, j, p, τ , or of one of these with α or γ . But these are all U other than the derivatives

$$(fj), (\phi j), (\phi p), (\phi \tau), (p\tau); (f\alpha), (\phi\alpha), (j\alpha), (p\alpha); (f\gamma), (\phi\gamma), (j\gamma), (p\gamma), (\tau\gamma),$$

and since $\gamma = (\tau\alpha)$, the derivatives containing γ will depend upon covariants of a lower degree; there remain therefore only $(fj), (\phi j), (\phi p), (\phi \tau), (p\tau); (f\alpha), (\phi\alpha), (j\alpha), (p\alpha)$: each of these can be actually calculated in the form $F(U)$.

Hence finally, assuming that every covariant of a degree inferior to m is $F(U)$, it follows that every covariant of the degree m is $F(U)$; whence every covariant whatever is $F(U)$, viz. it is a rational and integral function of the 23 covariants U .

364. It will be observed that, writing A, B, C for P, Q, i , the proof depends on the theorems

$$\begin{aligned} &((AB), C), \quad \text{a linear function of } A(BC)', B(CA)', C(AB)', \\ &(AB, C)' \quad \text{do. do. do.} \\ &((AB), C)' \quad \text{do. do. } ((AC)', B), ((BC)', A), B(AC)', C(AB)', \end{aligned}$$

which are theorems relating to any three functions A, B, C whatever.

365. I remark upon the proof that the really fundamental theorem seems to be that which I have called theorem A . As to the forms W it is difficult to see *à priori* why such forms are to be considered, or what the essential property involved in their definition is; and in fact in a more recent paper, “Die simultanen Systeme binären Formen” (*Math. Annalen*, t. II. (1869), see p. 256), Professor Gordan has modified the definition of the forms W by omitting the condition that the order of the function shall exceed n ; if it were possible further to omit the condition of at least one index being $=$ or $< \frac{1}{2}n$, and so only retain the conditions $-n - i, n - j$, &c., each of them > 0 , then the essential property of the forms W would be that any such form was a rational and integral function of the special covariants formed, as above, by means of the quantic of the next inferior order. And moreover, as regards the theorem B , there seems something indirect and artificial in the employment of such a property; one sees no reason why, when a system of irreducible covariants is once written down, it should not be possible to show that the derivatives of $F(U)$ with the original quantic are each of them $F(U)$, instead of having to show this in regard to the derivatives of $F(U)$ with the several covariants χ : as regards the quintic, where there is a single covariant χ , the quadric function i , there is obviously a great abbreviation in this employment of i in place of f ; but for the higher orders, assuming that the proof could be conducted by means of the quantic itself, it does not appear that there would be even an abbreviation in the employment in its stead of the several covariants χ . The like remarks apply to the proof in the last-mentioned paper. I cannot but hope that a more simple proof of Professor Gordan’s theorem will be obtained — a theorem the importance of which, in reference to the whole theory of forms, it is impossible to estimate too highly.

(End of chunk: PDF pages 326–375, book pages 304–353.

Paper 457 in full,

paper 458 in full,

paper 459 in full,

paper 460 in full,

paper 461 in full,

paper 462 opening (Articles 326–365, book pp. 334–353).)

445. (continued)

A MEMOIR ON QUARTIC SURFACES.

(continuation, articles 60–112)

[From the Proceedings of the London Mathematical Society, vol. iii. (1869–1871), pp. 19–69. Read February 10, 1870.]

Dianodal Curve of 8 Points.

60. Consider first any 8 points whatever 1, 2, 3, 4, 5, 6, 7, 8; the dianodal surface 1234567, our 7 on the dianodal surface 1234568. The two surfaces have in common the line through the points 1, 2, 3, 4, 5, 6, twice; this would imply a reduction 25, viz. we should have $5 \cdot 6 = 30 - 25$. It is not understood this, and I drop the question.

61. However, the line 12 is in involution 6 given quadrics, there should be a reduction 23, viz. we should have

$$(a, b, c, d, f, g, h, l, m, n \times P, Q, R, S, T, U, V, W, X, Y)^2 = 0$$

of a fourth quadric surface to touch the line, in order that the dianodal curve $5 \cdot 6 = 30 - 23$.

62. It may be observed, in regard to the foregoing solution, that if we take

$$123 = 0, \quad 456 = 0, \quad 78 = 0,$$

the line 12, ... in involution with the 6 given quadrics; the relation between $(a, b, c, d, \dots)$, 4 equations; and similarly, if we attempt to find m_1, m_2, m_3, m_4 , such that

$$m_1(P_1 Q_1, \dots) + m_2(P_2 Q_2, \dots) + m_3(P_3 Q_3, \dots) + m_4(P_4 Q_4, \dots) = 0;$$

we have between the coefficients (b, c) 4 equations; and similarly, if we attempt to find m_1, m_2, m_3, m_4 such that

$$\omega_1(1, b_1, c_1 \times P_1, Q_1)P + \dots + \omega_4(1, b_4, c_4 \times P_4, Q_4)P = 0,$$

we have between the coefficients (b, c) 4 equations; viz., these may be determined and the system give 6 lines for another given axis; we have more equations between the coefficients (b, c) ; viz., these will be determined by 5 equations of the form (b, c) 4 equations; and the result should be of this nature, that if at any point of the axis a quadric were to pass through 4 of the given lines, the quadric surface generated by the lines we have a close cubic meeting the axis 3 times.

63. It may be observed, in working this out, to consider H, K as standing, in the first instance, for $(x, y, z, w), (x', y', z', w')$, three random points respectively. Writing also, for shortness, take the abovementioned points H, H' to denote the points on the absolute conic of one of the 4 quadrics cutting harmonically the system of the 4 points, Q on the symmetroid, P and P' on the Jacobian, Q' on the symmetroid; but in what follows I make no use of this last point Q' .

The Symmetroid: Symmetroid and Jacobian.

I have established in the foregoing general theory; but it is a particular case of it which connects itself with the theory of nodal quartics; viz., the cube of coefficients is a symmetrically arranged cube

$$\begin{array}{cccc} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{array}$$

or say its upper face is the symmetrical square matrix

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix}$$

and the other horizontal planes, the like square with the several letters affected by suffixes.

The surface $\nabla = 0$ is here a surface of the form

$$\nabla = \begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

($A, B, \&c.$ linear functions of (x, y, z, w)), viz., ∇ is a symmetrical determinant; I call this a symmetroid; the surface $\nabla = 0, \Theta = 0$ are not the same, and the same point, and the correspondence R to P is a reciprocal one; or that, instead of the Jacobian being the locus of points, we have only 1 point on the surface ∇ , and 2 points P, P' on the surface $\Theta = 0$; viz., the diagram is

$$\begin{array}{c} \Delta, \Theta, 0, 0; 0, 0, \dots, \Delta \\ \dots, Q, P, Q, P, P', Q, \dots \end{array}$$

moreover the symmetroid surface $\nabla = 0$ is a surface with 10 nodes, which is clearly not octadic, and which is therefore the decadianome.

The Decadianome.

64. The decadianome has thus 10 nodes; and it is a surface of the form

$$(a, b, c, d, f, g, h, l, m, n \times P, Q, R, S)^2 = 0,$$

where $\Delta = 0$ is a particular quartic surface having the 8 nodes, it at once appears that the 9th node must be on the dianodal surface of the 8 points $\nabla J(P, Q, R, S) = 0$, and on the Jacobian; and then the formulae No. 5, this is a curve of the order 16; the node may be any point whatever on this curve, and taking it to be a given point on the curve, the number of constants in the resulting equation should be 2. Hence if $P = 0$ be the quadric surface through the 9 points, and $\Delta = 0$ is a particular quartic surface having the 9 nodes as nodes, the general equation is $aP^2 + \Delta\Phi = 0$.

65. But we may consider the question somewhat differently. Starting from the equation

$$(a, b, c, d, f, g, h \times P, Q, R, S, T, U)^2 = 0,$$

also on the dianodal surface 1234567; it is clear that 9 must be on the dianodal surface 1234568; the complete intersection is of the order 36, and we have to consider how this breaks up so as to contain as part of itself the dianodal curve of the order 18.

Dianodal Curve of 9 Points.

66. Consider first any 9 points whatever 1, 2, 3, 4, 5, 6, 7, 8, 9; the dianodal surface 1234567, our 7 on the dianodal surface 1234568. The two surfaces have in common the line through the points 1, 2, 3, 4, 5, 6 twice; this would imply a reduction $4 \cdot 10 = 40$, but it is clear that the line 123 is in involution to the dianodal surface, the complete intersection is of the order 36, and we have therefore by what precedes the value $4 \cdot 10 - 18 = 22$, viz. the curve we are in search of is of the order 22.

Pseudosymmetrical Cyzo: the Hessian of a Cubic.

67. Reverting to the general equation

$$aS + \beta T + \gamma U + \delta V = \text{cone},$$

which connects the symmetroid and Jacobian, it is evident that if S, T, U, V are the derivatives, in regard to the coordinates, of a single cubic function $U_c = (* \times x, y, z, w)^3$, then the symmetroid and the Jacobian become one and the same surface; viz., this is the Hessian surface $H = 0$ derived from the given cubic surface. The two corresponding points P on the surface have a coincidence in this case; and the equation $\Phi = 0$ vanishes identically when S, T, U, V are derivatives of a single cubic function. Hence, in either case, we may use as a node on the symmetroid, viz., the Hessian of the cubic surface, of which the equation is

$$xyzw \left(\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} + \frac{\delta}{w} \right) = 0,$$

a form which puts in evidence the properties just referred to.

Quartics with 11 or more Nodes.

68. I mention two results which, although they relate to quartic surfaces with more than 10 nodes, present themselves in such immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

of the symmetroid ($A, B, \dots$ linear functions of the coordinates), we have identically $A = 0$, then the surface has a node $H = 0, G = 0, L = 0$; or the surface has in all 11 nodes. And so also if (identically in every case) B is $= 0$, there are 12 nodes; if C is $= 0$, there are 13 nodes; and if D is $= 0$, 14 nodes respectively, mentioned in Kummer's Memoir with 11, 12, 13, and 14 nodes respectively.

69. We may consider the symmetroid derived from the quartic surfaces which pass through 6 given points; viz., taking as before (see No. 25) the coordinates of the 6 points, the dianodal surface of these 6 is 7, on the dianodal surface 1234568, the surface has 11 nodes; viz., we have, in fact, quartic surfaces with 11, 12, 13, and 14 nodes respectively. But I reserve this for future consideration.

Two Nodes.

70. Suppose there is a 10th node, say 10; starting from the equation $aP^2 + \Delta\Phi = 0$ ($P = 0$ the quadric surface through the 9 points, $\Delta = 0$ a particular quartic surface having the 9 points), we have to express that the surface has 10 also for a node; the conditions are, that on the curve $P = 0, \Delta = 0$, we have

$$aP^2 + \Delta\Phi = 0, \quad a(P^2)_x + (\Delta\Phi)_x = 0, \quad \&c.,$$

or, as this is more conveniently written,

$$(\alpha \dots) \chi(x, y, z, w) = 0;$$

if for a moment $(\alpha, \dots)$ denote the coefficients of the quadric function $aP + \beta T + \gamma U + \delta V$, then $(P, P', \dots)$ are at the point 10.

The Lineodianic Correspondence of Quartic Surfaces.

71. Consider the quartic surfaces

$$S = 0, \quad T = 0, \quad U = 0, \quad V = 0,$$

lineliner in regard to the two sets of coordinates (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$; viz., each of these expressions is of the form

$$(* \times y, z, w \times \beta, \gamma, \delta).$$

This implies that the point (x, y, z, w) lies on a certain quartic surface $\Theta = 0$, and the point $(\alpha, \beta, \gamma, \delta)$ on a certain quartic surface $\Theta = 0$, and that the two surfaces correspond point to point to each other. In fact, writing the four equations in the form

$$\begin{aligned} La + Mb + Nc + Pd &= 0, \\ L'a + M'b + N'c + P'd &= 0, \\ L''a + M''b + N''c + P''d &= 0, \\ L'''a + M'''b + N'''c + P'''d &= 0, \end{aligned}$$

where $L, \&c.$ are linear functions of (x, y, z, w) , we obtain the equation

$$\Theta = \begin{vmatrix} L, & M, & N, & P \\ L', & M', & N', & P' \\ L'', & M'', & N'', & P'' \\ L''', & M''', & N''', & P''' \end{vmatrix} = 0;$$

and similarly, writing the four equations in $(\alpha, \beta, \gamma, \delta)$ we obtain

$$\begin{vmatrix} A\alpha + B\beta + C\gamma + D\delta = 0, \\ A'\alpha + B'\beta + C'\gamma + D'\delta = 0, \\ A''\alpha + B''\beta + C''\gamma + D''\delta = 0, \\ A'''\alpha + B'''\beta + C'''\gamma + D'''\delta = 0, \end{vmatrix}$$

where $A, \&c.$ are linear functions of $(\alpha, \beta, \gamma, \delta)$, we obtain the equation

$$\Theta = \begin{vmatrix} A, & B, & C, & D \\ A', & B', & C', & D' \\ A'', & B'', & C'', & D'' \\ A''', & B''', & C''', & D''' \end{vmatrix} = 0.$$

and similarly, Δ being $= 0$, the four linear equations in (x, y, z, w) are equivalent to three equations, and give for instance (x, y, z, w) proportional to determinants formed with the matrix

$$\begin{vmatrix} A, & B, & C, & D \\ A', & B', & C', & D' \\ A'', & B'', & C'', & D'' \\ A''', & B''', & C''', & D''' \end{vmatrix}$$

which establishes the point-to-point correspondence of the two surfaces.

72. It would at first sight appear that any quartic surface $(a, b, c, d, f, g, h \times P, Q, R, S, T, U, V)^2 = 0$ might have its equation expressed in the foregoing determinant form $\Delta = 0$. This equation seems, in fact, to contain homogeneously as many as 64 constants. But if we multiply the determinant first into line by a constant determinant, the equation will really contain $64 - (2 \cdot 16 - 1) = 32$ constants; and in order that the quartic surface $(a \times P, Q, R, S, T, U, V)^2 = 0$ may have its equation expressible in the form $\Delta = 0$, a single relation must hold good among the coefficients: but this is passing(?).

73. Returning to the quartic surface

$$\Delta = \begin{vmatrix} A, & B, & C, & D \\ A', & B', & C', & D' \\ A'', & B'', & C'', & D'' \\ A''', & B''', & C''', & D''' \end{vmatrix} = 0,$$

we may convert this not only with the foregoing surface $\Theta = 0$, but in a similar manner with another quartic surface $\Theta = 0$, viz., taking the new coordinates $(\xi, \eta, \zeta, \omega)$, we may form the lineliner equations

$$\begin{aligned} A\xi + B\eta + C\zeta + D\omega &= 0, \\ A'\xi + B'\eta + C'\zeta + D'\omega &= 0, \\ A''\xi + B''\eta + C''\zeta + D''\omega &= 0, \\ A'''\xi + B'''\eta + C'''\zeta + D'''\omega &= 0, \end{aligned}$$

which, by the elimination of $(\xi, \eta, \zeta, \omega)$, gives $\Delta = 0$; and by the elimination of (A, B, C, D) a different determinant, $\Theta = 0$, which is to the points (x, y, z, w) for a cone, or say for shortness,

$$aS + \beta T + \gamma U + \delta V = \text{plane-pair};$$

viz., the values of $(a, \beta, \gamma, \delta)$, which satisfy this condition, belong to a threefold relation between the coefficients of aS , having previously explained location, we may say—intervene to a given point of the symmetroid, and that the line on the Jacobian is the axis of the plane-pair.

74. I have a fish lay 4 cuspe, with then to the plane-pair implies a threefold relation between aS , having previously explained location, but the required number of nodes is equal to the order of the threefold relation; hence the order of $aS + \beta T + \gamma U + \delta V$ is equal to the order of the threefold relation; and the number of solutions would be equal to the order of the threefold relation; and the number of solutions would be equal to the order of $(* \times \dots)^4$, viz., as a further abbreviation referring aS , instead of $(\alpha) aS$, to denote the abovementioned determinants, the equation is

$$\begin{aligned} & ab\,abc + a\,abc - ach + bch - bda - dab\,c \\ & + b'\,abc \cdot bgh - h\,bcg + b\,bcf - ach + bch\,c - bda - dab\,c \\ & + b'\,abc \cdot bgh - h\,bcg + b\,bcf - ach + bch\,c - bda - dab\,c \\ & + \dots \end{aligned}$$

The final result contains 184 terms; viz., as a further abbreviation referring abc , instead of (abc) abc , to denote the abovementioned determinants, the equation is symbolically given by

$$abc \cdot abg - hfg \cdot abc + bcg \cdot abf - acf \cdot bch + chg \cdot abd = 0.$$

75. The final result contains 184 terms; viz., 6 terms in the multiplier abc of aS , T , U , V , W , viz., as a further abbreviation referring abc , instead of (abc) abc , to denote the abovementioned determinants, the equation is symbolically given by

$$acf \cdot bgh - dh\,bcg + a\,cdf - bf\,ach = 0.$$

I am not sure whether the surface of the eighth order has been anywhere considered.

Symmetroid with given Nodes.

76. A symmetroid can be formed with 7 given points as nodes; but there is no proper symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid, we obtain, in answer to put it differently: any 7 points must be the points of a single system; if 8 nodes are required, then it is necessary that the exterior region contain $\Delta\Delta$ a posteriori identification with the decadianome, it would be necessary to construct a $\Delta\Delta$ relation containing 7 given points, and a 9th point anywhere on the dianodal curve; but I have not succeeded in effecting this.

Symmetroid with given Lines.

77. To find the equation to the quadric surface through the three lines

$$(a_1, b_1, c_1, f_1, g_1, h_1), \quad (a_2, b_2, c_2, f_2, g_2, h_2), \quad (a_3, b_3, c_3, f_3, g_3, h_3).$$

Take for the equation to the quadric surface through the three lines

$$\begin{vmatrix} x^2, & y^2, & z^2, & w^2, & xz, & xy, & xw, & yw, & zw \\ a_1^2, & b_1^2, & c_1^2, & f_1^2, & g_1^2, & h_1^2, & a_1g_1, & b_1f_1, & c_1h_1 \\ 2a_1^2, & 2b_1^2, & 2c_1^2, & 2f_1^2, & 2g_1^2, & 2h_1^2, & a_1g_1 + a_1c_1, & b_1f_1 + b_1c_1, & c_1h_1 + c_1d_1 \\ \vdots & & & & & & & & \end{vmatrix} = 0;$$

and if we form thus a determinant with three of its lines relating to the line 1, three of them to the line 2, and three to the line 3, we have the equation of the quadric surface through the three lines $(a, b, c, f, g, h)_1, (a, b, c, f, g, h)_2, (a, b, c, f, g, h)_3$. Take to denote the equation; the order of the quadric surface through the three lines becomes indeterminate (in fact, the plane of the intersecting lines, and throwing out these factors, the condition should be of the order 2 in regard to the coordinates of each line; and throwing out these factors, the condition becomes of the order 2; viz., we have the condition for this is

$$\begin{vmatrix} \dots, & 12, & 13, & 14, & 15 \\ 21, & \dots, & 23, & 24, & 25 \\ 31, & 32, & \dots, & 34, & 35 \\ 41, & 42, & 43, & \dots, & 45 \\ 51, & 52, & 53, & 54, & \dots \end{vmatrix} = 0,$$

being, as it should be, of the order 2 in regard to the coordinates of each line.

Condition that 7 given lines shall lie on a Quartic Surface.

78. Taking the lines to be $(a_1, b_1, c_1, f_1, g_1, h_1), \dots, (a_7, b_7, c_7, f_7, g_7, h_7)$, then in precisely the same way we form a determinant of the order $\frac{1}{2}5 \cdot 4 = 10$ (in regard to the columns of each line, and 4 (in regard to the columns of each line), with the several factors $(\dots, \cdot, \cdot, \cdot, \dots)$ in regard to the coordinates of each line.

The Jacobian of 6 points.

79. Any 6 points whatever may be regarded as points on a skew cubic; and the coordinates (x, y, z, w) may be such, $b_1, \dots, b_5$ are linear functions of (x, y, z, w) , and the coordinates (x, y, z, w) are

$$\begin{vmatrix} x & y & z \\ y & z & w \\ b_1 & b_2 & b_3 \end{vmatrix} = 0.$$

This being so, the coordinates of the 6 points through the skew cubic are taken to be $(1, b_1, c_1, c'_1), (1, b_2, c_2, c'_2), \dots, (1, b_5, c_5, c'_5)$; and the equation of the Jacobian surface of the 6 points is the Jacobian surface of the 6 points.

80. We may now consider the symmetroid derived from the quartic surfaces which pass through 6 given points; viz., taking as before (see No. 25) the coordinates of the 6 points, putting in evidence the passage of the surface through the skew cubic, viz., writing

$$p_1 = \Sigma\tau_1, \quad p_2 = \Sigma\tau_2, \quad p_3 = \Sigma\tau_3, \quad p_4 = \Sigma\tau_4, \quad p_5 = \Sigma\tau_5;$$

moreover,

$$\Sigma\Sigma' = \frac{1}{6}(7xyzw - 4xx' - 4yy' + 2pp' - aba'b'),$$

and therefore

$$\begin{aligned} L_1 &= \dots - xx' - \Sigma x + 2yzw, \\ L_2 &= \dots - 4xx' - 6yyz + 6yzw, \\ L_3 &= \dots - 4xx' - 6yyz + 6yzw, \\ L_4 &= \dots - xx' - 2y' + 4yzw; \end{aligned}$$

then the equation of the Jacobian surface is

$$\begin{vmatrix} 3(p_1 & +xy, & -2y, & \frac{1}{2}\square) \\ +(& 2pp, & -xx, & \frac{1}{2}\square) \\ +(-xy, & -3p, & , & \frac{1}{2}\square) \\ +3(2xy & -xx, & -xyz, & 6) \end{vmatrix} = 0.$$

There is not much difficulty in the direct investigation; but a simple verification may be obtained by showing that the surface contains upon it the 15 lines 12, 13, ..., 56. Write in the equation

$$(x, y, z, w) = (1, b, c, c'),$$

the value L_1 becomes = ..., the value L_2 becomes = ..., the value L_3 becomes = ..., the value L_4 becomes = ...; and the equation becomes

$$\frac{1}{4}(\sigma\pi - \sigma'\pi')(a - 3a' + 3(2(p_2 - p_3))) \cdot \frac{1}{2}(c - c'\pi + \frac{1}{2}c\pi') = 0;$$

viz., relations of the form $\sigma + cb + c = 0$ at the vertices, and the which is the equation are

$$\begin{aligned} -2\Sigma(1, p_1, p_2, p_3, p_4 \times \sigma_1, \pi_1) &= 0, \\ 2\Sigma(1, p_1, p_2, p_3, p_4 \times \sigma_1, \pi_1) \cdot 7 &= 0, \end{aligned}$$

viz., this equation is satisfied if σ denote any one of the quantities $(1, b_1, c_1, c'_1)$, and 1 any one of the same 4 quantities; that is, the equation of the surface is satisfied when (x, y, z, w) are the coordinates of a point on the line joining any 2 of the 6 points.

Locus of the vertex of a Quadric Cone which touches each of Six given Lines.

81. Representing as before each line by means of its six coordinates, let (x, y, z, w) be the coordinates of the vertex, and (X, Y, Z, W) current coordinates. Suppose this (a, b, c, f, g, h) are the coordinates of any one of the lines, the equation of the plane through this line and the vertex is

$$\begin{aligned} & a(wY - wZ) + b(yW - wY) + c(zX - xZ) \\ & + f(yZ - zY) + g(zX - xZ) + h(xY - yX) = 0; \end{aligned}$$

or, what is the same thing, writing for shortness

$$P = \dots, \quad by - gz + dw,$$

$$Q = -hx, \quad +fz - bw,$$

$$R = gx - fy, \quad +cw,$$

$$S = -ax - by - cz,$$

the equation is

$$PX + QY + RZ + SW = 0;$$

and forming in like manner the equations belonging to each of the given lines, the condition that the 6 cones may unite into a single cone, is

$$(\sigma, b, c, f, g, h) \times (P, Q, R)^2 = 0,$$

where the left-hand side represents a determinant of 6 lines, the several lines being respectively $(P, Q, R)_1, (P, Q, R)_2, \dots, (P, Q, R)_6$, also S denoting that the equation by

$$(P, Q, R)^2 = 0.$$

To ascertain the form of this, write for a moment $y = 0, z = 0$; the equation becomes $\{(au, -hx + dw, gx - fx)^2 = 0\}$; or attending only to the highest and lowest powers of x , this is

$$x^2(a, a, -h, g)^2 = 0,$$

and it is thence easy to infer that the whole equation divides by w^4 , we thus, omitting this factor, the form of the equation is

$$(\sigma, b, c, f, g, h) \times (x, y, z, w)^4 = 0,$$

viz., the equation is satisfied if x denote any one of the 6 coordinates (a, b, c, f, g, h) , and 1 any one of the same 6 quantities (a, b, c, f, g, h) ; that is, the equation is satisfied when (x, y, z, w) are the coordinates of any one of the 6 lines, and the degree in the coordinates (x, y, z, w) is 4. It would not be very easy to find the value of the discriminant of the equation.

Quartic Surface through three given Lines.

82. To find the equation of the quartic surface through the three lines

$$(a_1, b_1, c_1, f_1, g_1, h_1), \quad (a_2, b_2, c_2, f_2, g_2, h_2), \quad (\alpha', \beta', \gamma', \delta'),$$

we may form the determinant

$$\begin{vmatrix} x^4 & y^4 & z^4 & w^4 & yz & xz & xy & xw & yw & zw \\ x^3y & xy^3 & yz^3 & y^3z & \dots & \dots & \dots & \dots & \dots & \dots \\ 2x^3 & 2y^3 & 2z^3 & 2w^3 & a\beta' & a\gamma' & a\delta' + a\beta'\gamma' & b\gamma' + a\beta'\delta' & \dots & \dots \\ a^4 & \beta^4 & \gamma^4 & \delta^4 & \beta\gamma & a\beta & a\delta & a\gamma & \beta\delta & \gamma\delta \\ a'^4 & \beta'^4 & \gamma'^4 & \delta'^4 & \beta'\gamma' & a'\beta' & a'\delta' & a'\gamma' & \beta'\delta' & \gamma'\delta' \end{vmatrix} = 0;$$

and if we form thus a determinant with three of its lines relating to the line 1, three to the line 2, and three to the line 3, we have the equation of the quartic surface through the three lines.

Condition that five given lines may lie on a Cubic Surface.

83. Taking the lines to be $(a_1, b_1, c_1, f_1, g_1, h_1), \dots, (a_5, b_5, c_5, f_5, g_5, h_5)$, and (X, Y, Z, W) the coordinates of the vertex, the equation of the cubic surface through the 5 lines is taken to be $(1, b_1, c_1, c'_1), \dots, (1, b_5, c_5, c'_5)$; and the equation of the Jacobian surface of the 5 points

and hence it appears that, forming a determinant of 20 lines, wherein four lines relate to the four required condition. But the condition so obtained is of the order $(\frac{1}{2} \cdot 4) = 6$ in regard to the coordinates of each line; and, as for the quadric, it is satisfactorily contains the several factors which lead to zero, we leave the required relation between the five lines that they shall all of them be cut by 4 or 5 lines; and moreover that, writing $a_i f_i + b_i g_i + c_i h_i = 0$, &c., then every one of the diagonal determinants of the order 4 vanishes; the condition for this is

$$\begin{vmatrix} \cdot, & 12, & 13, & 14, & 15 \\ 21, & \cdot, & 23, & 24, & 25 \\ 31, & 32, & \cdot, & 34, & 35 \\ 41, & 42, & 43, & \cdot, & 45 \\ 51, & 52, & 53, & 54, & \cdot \end{vmatrix} = 0,$$

being, as it should be, of the order 2 in regard to the coordinates of each line.

Condition that 7 given lines shall lie on a Quartic Surface.

84. Taking the lines to be $(a_1, b_1, c_1, f_1, g_1, h_1), \dots, (a_7, b_7, c_7, f_7, g_7, h_7)$, then in precisely the same way we form a determinant of the order 7, which sufficiently shows the conditions; the order of the surface is 4, and so similarly any number of lines on a given Quartic Surface have many to be reduced 4; the more the equation it would be in fact be needed.

General result as to the Dianomes.

85. We have thus established the theory of the dianome quartic surfaces; viz., we have

- The octadianome, 8 nodes, 7 of them arbitrary, and the 8th on an arbitrary line passing through it.
- The enneadianome, 9 nodes, the 9th on the dianodal curve of the 8 points.
- The decadianome, 10 nodes, the 10th any one of the 22 [13] dianodal points of the 9 points.

And as already mentioned, so long as the first 7 nodes are arbitrary, there cannot be more than 10 nodes in all.

The Jacobian with given Lines.

86. The Jacobian contains 24 constants; obviously it is uniquely determined if 4 of the plane-pairs thereof are given; and it is also uniquely determined if 4 of the nodes of the symmetroid can be considered as corresponding to given plane-pairs, or lines of the Jacobian. Take as given any 4 nodes of the symmetroid: viz., if 4 of the plane-pairs are given, the corresponding 4 plane-pairs are taken to be given plane-pairs; and, for the corresponding 4 nodes of the symmetroid would be considered as given, viz., 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1; the given plane-pairs $P_1Q_1 = 0$, $P_2Q_2 = 0$, $P_3Q_3 = 0$, $P_4Q_4 = 0$ of values of values such that we have

$$(S, T, U, V) = (1, 0, 0, 0)P_1Q_1 + (0, 1, 0, 0)P_2Q_2 + (0, 0, 1, 0)P_3Q_3 + (0, 0, 0, 1)P_4Q_4 = \text{plane-pair};$$

and $(1, 1, 1, 1)$ is of the symmetroid; we have to assume

$$(S, T, U, V) = (l_1 P_1Q_1 + l_2 P_2Q_2 + l_3 P_3Q_3 + l_4 P_4Q_4 = \text{plane-pair}),$$

and $(1, 1, 1, 1)$ is of the symmetroid, we have to assume

$$(S, T, U, V) = (l_1 P_1Q_1, l_2 P_2Q_2, l_3 P_3Q_3, l_4 P_4Q_4) = \text{plane-pair of given axis.}$$

87. Suppose, however, that on the Jacobian we have given, not 4 plane-pairs, but 6 lines lying $(l_1, l_2, l_3, l_4, l_5, l_6)$. As before $P_1Q_1 = 0, \dots, P_4Q_4 = 0$ are the first instance undetermined. If we attempt to find $l_1, l_2, l_3, l_4, l_5, l_6$, any one of these lines, l_4 , can be a plane-pair of values such that we have

$$l_1 P_1Q_1 + l_2 P_2Q_2 + l_3 P_3Q_3 + l_4 P_4Q_4 = \text{plane-pair of given axis,}$$

we have between the coefficients (l, l_1, l_2, l_3, l_4) 4 equations; and similarly, if we attempt to find m_1, m_2, m_3, m_4 , such that

$$m_1 P_1Q_1 + m_2 P_2Q_2 + m_3 P_3Q_3 + m_4 P_4Q_4 = \text{plane-pair of another given axis,}$$

we have more equations between the coefficients (b, c) ; viz., these will be determined by 6 equations of the form (b, c) 4 equations; and the result should be of this nature, that on the axis of the assumed plane-pair, the cubic should be in 6 nodes on the surface generated by the lines we have $S = c\xi T$ — at axis 3 times.

Correspondence in the Jacobian: Lines and Skew Cubics.

88. I consider the correspondence of two points on the Jacobian; it is to be shown that when one of the points is on a line of the Jacobian, the corresponding point will be on it. Call the plane-pairs of the system of quadric surfaces 1, 2, 3, ..., 10; selecting any 4 of these (on the dianodal surface of the system of quadric surfaces 1, 2, 3, ..., 10), the corresponding 6 nodes (on the symmetroid) will be on a plane on this plane, the equation of the plane through these would be required correspondence. Conversely, if we have an axis of one of the plane-pairs, say of the plane pair 1, the point P , in the plane pair 1 in regard to these 4 plane-pairs will meet in a point; and observe that, in regard to any 4 plane-pairs, the polar plane of a point in regard to these 4 plane-pairs meet in a point. Call this point P' ; moves along the line, the corresponding point moves along a curve, viz., in particular, the corresponding curve to a skew cubic; that is, the corresponding curve to a line of the Jacobian is a skew cubic, cutting plane-pair twice. In particular, the corresponding curve to a line of the Jacobian; we will have therefore 4, 5, ..., 10 each have a corresponding skew cubic, cutting the line 1, 2; on it; we have therefore 6 skew cubics cutting the line 1, 2 of the Jacobian.

89. I cut a moment to prove the subsidiary theorem assumed in the foregoing investigation. Let the 3 plane-pairs be $PQ = 0$, $RS = 0$, $TU = 0$, the line joining the points (x, y, z, w) and (x', y', z', w') ; the coordinates of any point on the line are $(\lambda x + \mu x_1, \lambda y + \mu y_1, \lambda z + \mu z_1, \lambda w + \mu w_1)$; and hence the equation may be written as

$$[(\lambda x + \mu x_1) \&c. \dots] PQ + (\lambda x + \mu x_1) \&c. RS = 0;$$

viz., the equation may be written as

$$\lambda(P, Q + \mu(P, Q + \dots)) = 0;$$

forming the like equations in regard to $RS = 0$ and $TU = 0$, and eliminating λ, μ , we obtain the required locus

$$\begin{vmatrix} P_1 Q_1 + PQ, & R_1 S_1 + RS, & T_1 U_1 + TU \\ P_1 Q + PQ_1, & R_1 S + RS_1, & T_1 U + TU_1 \end{vmatrix} = 0,$$

forming the like equations in regard to RS and TU , and eliminating λ, μ , we obtain the required locus

$$\begin{vmatrix} (P Q_1 + Q P_1) \delta_{1\nu} \cdot \delta_{1\mu} + (P, Q_1 \delta_{1\nu} \cdot \delta_{1\mu}), & (\dots), & (\dots) \\ \cdot \delta_{1\nu} \cdot R P, & \cdot \delta_{1\nu} \cdot S P, & \cdot \delta_{1\nu} \cdot T P \end{vmatrix} = 0,$$

viz., this equation is satisfied if a denote any one of the quantities (σ, ρ, q, r) , and 1 any one of the same 4 quantities; that is, the equation of the surface is satisfied when (x, y, z, w) are the coordinates of a point on the line joining any two of the points.

a skew cubic; and on writing therein $P = 0, Q = 0$, the equations become

$$\begin{vmatrix} RS_1 + R_1S, & TU_1 + T_1U \\ RS, & TU \end{vmatrix} = 0,$$

viz., the line ($P = 0, Q = 0$) meets the skew cubic in the points where the line meets the quadric surfaces $RS_1 = 0, TU_1 = 0$ corresponding to it on the Jacobian.

90. To determine a skew cubic cutting twice each of 6 given lines, I proceed as follows. Let the lines be 1, 2, 3, 4, 5, 6 and correspondingly therein respectively the 10 skew cubics 1, 2, ..., 9, 10, where each of the skew cubics cuts the line 1, 2, taking the cubics except that denoted by the same number; we may, for instance, write down the 4 equations of the cubics, and from the equations of the 4 quadrics, the conditions among the 4 plane-pairs, since the points P and P' are on the same line through the vertex, P' are reciprocal to one another, P' to P , on which corresponds to the vertex, the corresponding ones to one another, we have therefore

$$J(P, T, U, V) = 0;$$

viz., the locus is the Jacobian of the foregoing definitions of the symmetroid. And the determinant gives in the equation expressed by the corresponds to the curve curves.

91. But the Jacobian may be obtained in a different manner; viz., if we establish the equations

$$(2P_1 + aQ_1 + bR_1)\delta_{1\nu} = 0,$$

$$(\quad)P_1 + \delta_{1\nu} = 0,$$

$$(\quad)Q_1 = 0,$$

$$(\quad)V = 0,$$

then the elimination of $(\xi, \eta, \zeta, \omega)$ from these is the equation of the Jacobian. And since each of the equations is symmetrical in regard to $(x, y, z, w), (\xi, \eta, \zeta, \omega)$, the equation $J(P, T, U, V) = 0$ of the Jacobian is symmetrical in regard to $(x, y, z, w), (\xi, \eta, \zeta, \omega)$.

the coefficients of U , V , we have between the points p , q , r a twofold relation, which may be represented as follows:

$$\begin{vmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ p & p & p & p & q & q & q \\ p & p & p & p & q & q & q \\ p & p & p & p & q & q & q \\ p & p & p & p & q & q & q \end{vmatrix} = 0,$$

it being understood that the p 's, q 's represent constants, and the p 's, q 's, r 's linear functions of these variables respectively. The two several equations of the system, regarding therein p , q , r as coordinates, represent each of them a quartic curve; any 2 of these are coresidual; and the number of points common to all the curves is $10 - 3 = 7$. There are corresponding included among the 10 points, but they are not in general included among the 10 points to be regarded as proper solutions; the others are extraneous solutions, accordingly the line $J(P, Q, R, V) = 0$ of the equations respectively; it is at once seen that any common solution to the foregoing equations corresponds to a value $p = q = r$ common to all the curves of the system; this solution counts, I believe, once only, and the number of relevant solutions is $7 - 1 = 6$.

92. It may be observed, in regard to the foregoing solution, that if we take $a = 0$, $b = 0$, $c = 0$ for the equation of the quadric surfaces through the points 1, 2, 3, and as the dianodal surfaces $U = 0$ and $V = 0$ may be taken to be other cones, the line through the singularity in the surfaces.

93. The conclusion is, that touching twice each of the six lines, we have one solution

$$\lambda 123 + \mu 124 + \nu 125 + \rho 126 = 0,$$

$$\lambda' 123 + \mu' 134 + \nu' 135 + \rho' 136 = 0,$$

respectively, the coefficients of the two surfaces being in evidence. And it is clear that for $\mu = \mu'$, $\nu = \nu'$, $\rho = \rho'$, the equations become respectively the equation of the quadric surface through the points 1, 2, 3, 4, and the equation of the quadric surface through the points 1, 2, 3, 4, that is the result.

94. The conclusion is, that touching twice each of the six lines 5, 6, 7, 8, 9, 10, we have 15 systems of lines 7, 8, 9, 10, and therefore 15 Jacobian surfaces.

Further Investigations as to the Jacobian, &c.

95. Taking $(\xi, \eta, \zeta, \omega)$ as plane-coordinates, the two quadric surfaces

$$(a, b, c, d, f, g, h, l, m \times \xi, \eta, \zeta, \omega)^2 = 0$$

and

$$(A, B, C, D, F, G, H, L, M, N \times \xi, \eta, \zeta, \omega)^2 = 0$$

are said to be intervex (or intervex) one of the other, when we have between the coefficients the relation

$$(a, b, c, d, f, g, h, l, m \times A, B, C, D, F, G, H, L, M, N) = 0,$$

that is

$$aA + \dots + 2fF + \dots = 0.$$

The condition that the two surfaces may be intervex of each other is linear in regard to the coefficients of each surface separately; hence, as before explained, intervex to two given quadrics σ quadrics; or generally, that intervex to $10 - k$ quadrics. And, moreover, if the quadrics be $L = 0, M = 0$, &c., of two given quadrics σ quadrics, we have between them the relation

$$(a, b, c, d, f, g, h, l, m, n \times L, M) = 0,$$

that is,

$$\alpha L + \dots = 0.$$

The condition that two surfaces are harmonic in regard to the pair of planes is linear in regard to the coefficients of each surface separately; hence, as before explained, intervex to two given quadrics σ quadrics; or generally, that intervex to $10 - k$ quadrics.

The Symmetroid in connection with the Jacobian.

96. If the equation $\nabla = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting from the only with the foregoing symmetrical determinant of the order 4 may multiply by a constant determinant, that is by a symmetrical determinant in the same way; or starting in such ways we may add to a symmetrical determinant. Their column or rows are interchanged by the determinant column, the constant determinants 40 constants; and the coefficients of the constant determinant may be used to be a constant determinant column; and the coefficients of the constant determinant may be used to be a symmetrical determinant column; or say each quartic surface $\nabla = 0$ of the form

Symmetroid with given Nodes.

97. A symmetroid can be formed with 7 given points as nodes; but there is no proper symmetroid with 8 given points as nodes. If we endeavour to form a symmetroid with 7 given points as nodes; if 8 nodes are required, then it is necessary that the resulting determinant must be the resulting condition that the 8th node may, however, be any point on the symmetroid; and the dianodal surface through the 7 points; and a 9th point anywhere on the dianodal curve through them; this will be probably nowhere on the surface. I have not succeeded in effecting this.

means of the surfaces 1 and 2 determine the two planes of the plane-pair, and then these will be harmonics in regard to the surface 3. It thus appears that the two planes 1, 2, 3, to the surfaces 1, 2, 4, the surfaces 1, 2, 5, &c. are intervex in regard to 1, 2, are the same as those which are intervex in regard to the involution in regard to the surfaces 1, 2, 6, 1, 2, 7, and we have therefore 6 nodes, since we always come round upon the surface (quoad the plane-pair).

98. It is substantially the same thing, but it is rather easier, to consider the whole question under the reciprocal form; viz., instead of a plane-pair and a quadric surface represented by the lines of intersection of the line $\nabla J(P, Q, R, V) = 0$ given quadric surfaces; viz., this is equivalent to the involution in regard to the given quadric surfaces, the involution in regard to the given quadric surfaces, viz., we have therein a line which is a (quoad the lines passes) in involution in regard to the three quadric surfaces. Conversely, when the axis of a plane-pair represented by the dianodal surface through the 9 points, considered as an involution in regard to the involution in regard to the 9 quadric surfaces.

99. Consider a line which is in involution 5 given quadric surfaces $(a, b, c, f, g, h, l, m, n \times P, Q, R, S, T)^2 = 0$; the plane P of the line through the same line through a 6th quadric surface, that we should have an indefinite series of identities of the form

$$(\alpha, \dots \times a, b, c, f, g, h, l, m, n \times P, Q, R, S, T) = 0;$$

viz., this expresses that the line a has, with the 5 given quadric surfaces, $aS + \beta T + \gamma U + \delta V + \epsilon W = 0$.

100. The numerical relations are as before. The transversal of any 6 lines, regarded as an involution 6 lines, is the dianodal surface through the 6 lines; and there are clearly 8 lines in involution to 8 quadric surfaces, the lines are intervex to 8 quadric surfaces, the lines are interpret as a system of $8 - 4 = 4$ lines; and similarly any 5 of the 8 quadric surfaces, the lines are 1, 2 of the 8 quadric surfaces. Let the equation of the 8th quadric surfaces be 7, 8, 9, 10, of the 5 quadric surfaces, the lines having $5 - 1 = 4$ involution in regard to 1, 1 of the 5 quadric surfaces have therefore the relation

$$\lambda 125 + \mu 124 + \nu 123 + \rho 126 = 0,$$

$$\lambda' 135 + \mu' 134 + \nu' 135 + \rho' 136 = 0,$$

respectively, the coefficients of the two surfaces being in evidence. And it is clear that for $\mu = \mu'$, $\nu = \nu'$, $\rho = \rho'$, the equation becomes respectively the equation of the quadric surface through the points 1, 2, 3, 4 and the equation of the quadric surface through the points 1, 2, 3, 4, that is the result.

101. The conclusion is, that touching twice each of the six lines, we have 15 systems of lines 7, 8, 9, 10, and therefore 15 Jacobian surfaces.

and conversely. But the value 20 (which represented without difficulty. For the question may be transformed as follows: It a single quartic surface having 5 given quadrics, how many of the axes of the lines of these (any) 4 are clear at any of those points one of the system there are quadrics, the same number which are connected to those quadrics, the same problem comes upon the form $\nabla = 0$, $\Theta = 0$ of the threefold relations

$$aS + \beta T + \gamma U + \delta V = 0, \quad (\sigma, \beta, \gamma, \delta) = 0,$$

where the coefficients $(\sigma, \beta, \gamma, \delta)$ are arbitrary; and the question becomes: how shall we have between the equations satisfied points common to the threefold relation $aS + \beta T + \gamma U + \delta V = 0$? we need not actually establish the equations; we have only to find a value λ such that we shall have between the equations satisfied points common to the three equations; we shall have

$$J(P, T, U, V) = 0;$$

and as the question becomes, what is the order of the system formed by the lines of intersection of the 4 quadric surfaces? But each pair of quadric surfaces will give a curve of the eighth order; and there are 6 such pairs, and the total order is $6 \cdot 8 = 48$. We have to subtract from this the order of the curves which lie on more than one such pair of quadric surfaces; we have to subtract from this the order of the curves which lie on more than one such pair of quadric surfaces; and the value obtained is 20.

102. Returning to the equation $\nabla = 0$ of the symmetroid, where ∇ is a symmetrical determinant of the order 4, we have between any two of the rows the relation

$$\nabla S + \nabla' T + \nabla'' U + \nabla''' V = \text{plane-pair};$$

viz., the values of $(\alpha, \beta, \gamma, \delta)$ which satisfy this condition belong to a threefold relation between the coefficients of aS , having previously explained location, so that the rows of the determinant have the threefold relations; and the the the threefold relation, the rows of the determinant have the property that, considering each line independent—the line $aS + \beta T + \gamma U + \delta V$ is, by the determinant relation, a plane-pair; and the threefold relations, the line is also a plane-pair.

103. In converse, in working this out, to consider H, K as standing, in the first instance, for $(x, y, z, w), (x', y', z', w')$, but in the other manner, w, w', z, z' , we have, with the constants a, b, c, d, e, f, g , &c., of the constant w, z , the constants a, c , of the constant. But what follows make no use of this last quartic.

104. Reverting to the equation $\nabla = 0$ of the symmetroid, where ∇ is a symmetrical determinant of the order 4 it has already explained, viz.,

$$(\sigma, \beta, \gamma, \delta) \times (\sigma', \beta', \gamma', \delta') \cdot xyz + \nu \sigma' \beta' \gamma' \delta' (yz - yz')^2 + \dots = 0,$$

where the second factor is

$$\begin{aligned} & (abc)(a'y' - y) + a'b'c \cdot c'a + a'b'c'(\sigma\delta' - \delta\sigma') + abc(\beta y' - y\beta') \\ & = b\sigma' b'\sigma(\sigma\delta + \beta\gamma) - z\sigma b'(\sigma\delta' + \beta\gamma') - z'\sigma' b(\sigma'\delta + \beta'\gamma) - z'\sigma'b'(\sigma'\delta' + \beta'\gamma'), \end{aligned}$$

or, omitting the factor $= ab$, the term in (xz) is $z(zb - h bz)$.

105. The final result contains 184 terms; viz., 6 terms in the multiplier a of aS, T, U, V, W , viz., $abc \cdot abf - hfbc + bcg \cdot abf$, &c., $abc \cdot abf$, &c. The final result may be exhibited in the same form

$$\begin{aligned} \Sigma = & + 2\{abc \cdot abf - hfbc + bcg \cdot abf - acf \cdot bgh + cgh \cdot bdh\} \\ & + 2\{bcd \cdot abf - lf cgh + df g \cdot bch - dgh \cdot abc + df g \cdot bch\} \\ & + 2\{efg \cdot abc + glg \cdot bcg + fch b - acg + chg \cdot bdh\} \\ & + 2\{bch \cdot abf + abc \cdot bgh - abc \cdot bgh + abc \cdot bgh\} \\ & + \dots \\ & = 0. \end{aligned}$$

Paragymmetrical Cyzo: the Hessian of a Cubic.

106. Reverting to the general equation

$$aS + \beta T + \gamma U + \delta V = \text{cone},$$

which connects the symmetroid and Jacobian, it is evident that if S, T, U, V are the derivatives, in regard to the coordinates, of a single cubic function, $U_c = (* \times x, y, z, w)^3$, then the symmetroid and the Jacobian become one and the same surface; viz., this is the Hessian surface derived from the given cubic surface. The two corresponding points P on the surface have a coincidence in this case; and the equation $\Phi = 0$ vanishes identically when S, T, U, V are the derivatives, in regard to the coordinates, of a single cubic surface; we obtain

$$\nabla = \begin{vmatrix} S_x & T_x & U_x & V_x \\ S_y & T_y & U_y & V_y \\ S_z & T_z & U_z & V_z \\ S_w & T_w & U_w & V_w \end{vmatrix} = 0,$$

which is the Hessian; and the equation may equally be regarded as a Jacobian equation

$$J(S, T, U, V) = 0,$$

a form which puts in evidence the properties just referred to.

Quartics with 11 or more Nodes.

107. I mention two results which, although they relate to quartic surfaces with more than 10 nodes, present themselves in such immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

of the symmetroid ($A, B, \dots$ linear functions of the coordinates), we have identically $A = 0$, then the surface has a node $H = 0, G = 0, L = 0$; or the surface has in all 11 nodes. And so also if (identically in every case) B is $= 0$, there are 12 nodes; if C is $= 0$, there are 13 nodes; and if D is $= 0$, 14 nodes respectively, mentioned in Kummer's Memoir.

108. We may consider the symmetroid derived from the quartic surfaces which pass through 6 given points; viz., taking as before (see No. 25) the coordinates of the 6 points, present themselves in the same immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A, & H, & G, & M \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

of the symmetroid ($A, B, \dots$ linear functions of the coordinates), we have identically $H = 0$ (and not identically) every case) A is $= 0$, there are 11 nodes; if B is $= 0$ (and not identically) every case) A is $= 0$ there are 12 nodes; if C is $= 0$, there are 13 nodes; and if D is $= 0$, there are 14 nodes respectively, mentioned in Kummer's Memoir with 11, 12, 13, and 14 nodes respectively. But I reserve this for future consideration.

Quartic Surface through three given Lines.

109. I take the opportunity of mentioning some results which have a connexion, although not an immediate one, with the subject of the present Memoir.

Quartic Surface through three given Lines.

110. To find the equation to the quartic surface through the three lines

$$(a_1, b_1, c_1, f_1, g_1, h_1), \quad (a_2, b_2, c_2, f_2, g_2, h_2), \quad (\alpha', \beta', \gamma', \delta'),$$

the points $(\sigma, \beta, \gamma, \delta)$, $(\sigma', \beta', \gamma', \delta')$ the coordinates of a quadric surface through this line are of the form

$$\begin{vmatrix} x^4 & y^4 & z^4 & w^4 & yz & xz & xy & xw & yw & zw \\ x^3y & xy^3 & yz^3 & y^3z & yzw & xzw & xyw & xyz & y^2zw & yz^2w \\ 2x^3 & 2xy & 2yz & 2zw & a\beta' & a\gamma' & a\delta' + a\beta'\gamma' & b\gamma' + a\beta'\delta' & c\delta' + a\gamma'\beta' & d \\ \sigma^4 & \beta^4 & \gamma^4 & \delta^4 & \beta\gamma & a\sigma & a\delta & a\gamma & \beta\delta & \gamma\delta \\ \sigma'^4 & \beta'^4 & \gamma'^4 & \delta'^4 & \beta'\gamma' & a'\beta' & a'\delta' & a'\gamma' & \beta'\delta' & \gamma'\delta' \end{vmatrix} = 0;$$

and if we form thus a determinant with three of its lines relating to the line 1, three to the line 2, and three to the line 3, we have the equation of the quartic surface through the three lines 1, 2, 3.

Condition that five given lines may lie on a Cubic Surface.

111. Taking the lines to be $(a, b, c, f, g, h)_1, \dots, (a, b, c, f, g, h)_5$, and (x, y, z, w) the coordinates of any point on the line, the equation of the cubic surface through the 5 lines, taken to be $(1, b_1, c_1, c'_1), \dots, (1, b_5, c_5, c'_5)$; and the equation of the Jacobian surface of the 5 points.

112. The complete development of the equations above written, in the form best suited for actual application, is a calculation of some length, and would require to be performed with considerable care. The other terms were obtained therefrom by changing a, b into $a + \frac{1}{w}(\beta y - p)$, $b + \frac{1}{w}(\alpha \delta - p\beta)$, respectively; the equation may therefore be written in the symbolic form

$$\omega^4 \cdot \exp \frac{1}{\omega} [h(bg - p)\delta_a + (-h\sigma + fz)\delta_b + (p\omega - fp)\delta_c] \cdot [(a, b, c, d)^4] = 0,$$

or, what is the same thing,

$$\omega^4 \cdot \exp \frac{1}{\omega} [a\{ph(p\beta - bh\delta) + g(hb\delta - p\beta_1) + z(p\beta_2 - pb)\}\delta_a + \dots] = 0,$$

where $\exp$ (read exponential) denotes e^z , and $\{(a, b, c, d)\}$ represents a determinant as above explained. The equation contains, it is clear, the four terms

$$\omega^4 [(a, -b, p)\beta_2 + y(b, -h, p)\delta_a\beta + \dots + z'(c, \delta_a c'') + \omega'(a, b, c'')] = 0.$$

I am not sure whether the surface of the eighth order has been anywhere considered.

[End of paper 445.]